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CURRENCY EQUIVALENTS

Currency	=	Ugandan Shilling (UGS)
U$1.00	=	1,622.50 UGS

FISCAL YEAR
July 1 – June 30

WEIGHTS AND MEASURES
Metric System

ABBREVIATIONS AND ACRONYMS

AfDB	Africa Development Bank
AG	Auditor General
ASYCUDA	Automated System for Customs Declaration
BFP	Budget Framework paper
BOU	Bank of Uganda
CAS	Country Assistance Strategy
CEM	Country Economic Memorandum
CET	Environmental and Technical Services
CFAA	Country Financial Accountability Assessment
CGE	Computable General Equilibrium
CIT	Corporate Income Tax
COMESA	Common Market for Eastern and Southern Africa.
CPAR	Country Procurement Assessment Report
CPI	Consumer Price Index
DFID	Department for International Development
EAC	East African Community
EASPR	Education Sector Annual Performance Review
EMIS	Education Management Information System
EPRC	Economic Policy Research Center
ESAOR	Education Sector Annual Performance Report
EU	European Union
FDI	Foreign Direct Investment
FIAS	Foreign Investment Advisory Service
FSA	Fiscal Sustainability Analysis
GCR	Gross Compliance Rate
GDP	Growth Domestic Product
GFCF	Gross Fixed Capital Formation
GFS	Government Financial Statistics
GOU	Government of Uganda
HR	Human Resources
ICOR	Incremental Capital-Output Ratio
ICT	Information and Communications Technology
IFC	International Finance Corporation
IFMS	Integrated Financial Management System
IMF	International Monetary Fund
ITA	Income Tax Act
ITAS	Integrated Tax Administration System
LAC	Latin America and Caribbean Department
LIC	Low Income Country
LTEF	Long-term Expenditure Framework
LTO	Large Taxpayer office
MAMS	Maquette for MDG Simulation
MDB	Multilateral Development Bank
MDG	Millennium Development Goals
MDRI	Multilateral Debt Relief Initiative
MIGA	Multilateral Investment Guarantee Agency
MoES	Ministry of Education and Sports
MOFPED	Ministry of Finance, Planning and Economic Development
MTEF	Medium-Term Expenditure Framework
NER	
NGO	Non governmental Organization
NSSF	National Social Security Fund
NWSC	National Water and Sewerage Corporation
ODA	Overseas Development Assistance
OLS	Ordinary Least Squares
OPS	Occupational Pension Scheme
PA	Poverty Assessment
PAF	Poverty Action Fund
PAYE	Pay As You Earn
PEAP	Poverty Eradication Action Plan
PER	Public Expenditure Review
PETS	Public Expenditure Tracking Surveys
PIP	Public Investment program
PIT	Personal Income Tax

PLE	Primary Leaving Examination
PMA	Plan for Modernization of Agriculture
PPP	Projects with Private Participation
PPPs	Public Private Partnerships
RAFU	Road Agency Formation Unit
RMSM-X	Revised minimum Standard Model-Extended
SACMEQ	Southern & Eastern Africa Consortium for Monitoring Education Quality
SME	Small and Medium Enterprise
SNO	Second National Operator's license
SOE	State-Owned enterprises
SSA	Sub-Saharan Africa
STR	Student Teacher Ratio
TFP	Total factor productivity
TIMSS	Trends in International Mathematics & Science Study
TSC	Teachers' Service Commission
UCS	Uganda Computer Services
UDES	Uganda Ed-data survey
UEDCL	Uganda Electricity Distribution Company Limited
UEGCL	Uganda Electricity Generation Company Limited
UETCL	Uganda Electricity Transmission Company
UK	United Kingdom
UNEB	Uganda National Examinations Board
UNHS	Uganda National Household Survey
UPE	Universal Primary Education
URA	Uganda Revenue Authority
USAID	United States Agency for International Development
USE	Universal Secondary Education

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This chapter is split into four sections.

A. The first skims what we know about fiscal policy and growth from theory and global experience[1].

B. The second assesses what all this means for Uganda; ie the implications of Uganda's fiscal policy for growth (as set out in chapter 8 of the PEAP).

C. Section C traces Uganda's public spending priorities for growth and estimates the public investment ***needs*** for growth of 6 and 7 percent in Uganda,

D. Section D summarizes results from a macro model (MAMS) to discuss where Uganda should look in the budget to find additional resources for growth priorities.

[1] This is topically referred to as the "fiscal space" agenda.

A. FISCAL POLICY AND GROWTH: WHAT DO WE KNOW FROM THEORY AND GLOBAL EXPERIENCE?

1.1 The effect of fiscal policy on growth has been a topic of long-standing controversy in economic theory, empirical research, and policymaking. Economic policymaking entails the pursuit of growth, full employment with price stability and a stable balance of payments. Since fiscal policy is the most accessible tool for economic policy makers facing these multiple objectives, it has become embroiled in the trade offs between growth, equity and stability.

Box 1.1. Lessons From World Bank Experience with Fiscal Policy, Stabilization and Growth

The World Bank provided its perspective on "fiscal space" in an interim report, and has prepared a more detailed report on country experience as a sequel which will be discussed by Finance Ministers at the Development Committee's Spring Meetings in April 2007. The Bank's perspective is that:

(a) macroeconomic stability was a necessary but not a sufficient condition for economic growth and development,

(b) macroeconomic stability has often been achieved through fiscal adjustments, and particularly cuts in productive public expenditure, that may have undercut public infrastructure expansion necessary for long term growth,

(c) design of fiscal policy should explicitly seek to take account of both objectives – of macroeconomic stability and long term growth,

(d) the design of an appropriate growth-oriented fiscal policy for any country should factor in the initial conditions in that country, including its fiscal history and reputation, its' public finance potential and its growth and development needs,

(e) a key link between fiscal policy and growth or other development objectives would derive from an understanding of the implications for public expenditure levels, composition and efficiency, and

(f) that the role of institutions and governance are critical to ensuring that public spending has the desired impact on policy objectives, including growth.

(Extract from report to World Bank / IMF Development Committee at 2006 Annual Meetings)

1.2 **In many developing countries high deficits, rising debt and rapid inflation in the 1980s and 1990s led fiscal policy to focus largely on stabilization**. Correspondingly, the growth and poverty reduction objectives were under-emphasized[2]. As an anchor for economic stability the budget became principally a tool through which to pursue short-term macro economic stabilization. Early in the structural adjustment process emphasis was placed upon protecting social sector priority spending from cuts in order to limit the short-term costs of stabilization on poor people (equity objective)[3]. Whereas adjustment reforms were necessary and appropriate, the thinking of the day was that they were also sufficient. Policy makers anticipated that a supply side response to policies of stabilization, liberalization and privatization would ensure growth in

[2] Fiscal Policy for Growth and Development: An Interim Report." WB/IMF Development Committee Meeting background report, April 2006.
[3] This was raised in the "Social Dimensions of Adjustment" initiative in the early 1990s.

the medium-term. For some countries, including Uganda, the economy *did* respond to these policies with growth. For many others it didn't for a whole array of different reasons[4].

1.3 **Global experience suggests that poor countries should not ignore the long-run implications of short run policy in taxation and government expenditure**. In rich countries fiscal policy *is* a short-term instrument; it is typically used to mitigate short-run fluctuations of output and employment against a long-term growth trend; a trend which is determined by technological innovation and labor productivity. Even if fiscal policy *is* regarded as principally a short-term policy instrument in richer countries, several fiscal measures are known to exhibit long-run growth effects. Indeed, theoretical macro models of endogenous growth have introduced public expenditure categories such as research, education and public infrastructure as drivers of economic growth. A standard micro economic rationale for public expenditure is that it corrects for market failure to improve the efficiency of resource use in an economy, i.e. that it is growth-enhancing. In a developing country context where basic public goods provision is inadequate, market failure is more common, and coordination gaps abound, fiscal policy has a stronger role to play in longer-term growth and poverty reduction. However, in many poor countries fiscal policy during the last two decades was used as a short-term measure to stabilize the economy through expenditure reduction.

1.4 **The success of fiscal policy in achieving stabilization in some developing countries seems to have come at the cost of these beneficial longer-term effects.** Today, many countries are re-assessing the growth orientation of their fiscal stance. Initial debate on possible "Limits to Stabilization"[5] started when Latin American Finance Ministers argued that fiscal deficit targets (in particular, the overall fiscal balance) in stabilization programs had led to cuts in infrastructure spending. In the absence of compensating private investment in infrastructure they argued, these cuts had an adverse long term growth impact. Global opinion amongst economists these days is that macro stability was a necessary but insufficient condition for economic growth (see boxes 1.1 ,, 1.2, 2.2 and 2.3)

1.5 **Controversy in empirical studies is strongest over the evidence of a link between aggregate public expenditure and growth**. This is not surprising. To the extent that public spending is on activities that could be more productively undertaken in the private sector, or is financed from taxing the private sector with inefficient taxes[6], or by borrowing from the private sector (whether directly or indirectly through crowding out), the beneficial impact of the spending may be offset by the costs of financing it. Deficits which are domestically financed are generally growth debilitating. Similarly, if beneficial public spending leads to macro instability, it may have a cost in terms of growth. Conversely if spending cuts result in lower inflation and income growth, this could limit revenue collections and compromise the initial fiscal contraction. Aside from these practical pitfalls, there are insufficient years of consistent GDP data in Uganda to test the link from fiscal policy to growth. We therefore do not attempt to do so in this report.

[4] Rodrik, D, (2004) "Re-thinking Growth Policies In the Developing World", Harvard University.

[5] Easterly and Serven (2004), "The Limits to Stabilization" provides the most compelling review of Latin America's Experience, and experience which could be highly germane to Uganda.

[6] Taxes which create disincentives to work or save are notoriously inefficient for growth.

Box 1.2. The Experience of Latin America

Evidence from Latin America over the past decade has shown the crowding out of infrastructure spending by governments in favor of entitlement spending, revenue sharing, and in some cases debt service, with a resulting low level of economic growth performance.

Most Latin American countries reduced their fiscal deficits during the 1990s.

But in most countries fiscal discipline had a major cost in terms of public infrastructure spending. There is strong evidence from Latin America's adjustment that the expenditure category suffering the greatest reductions in order to achieve fiscal stabilization objectives was public capital formation. (Ref Estache, 2005)

Expected private investments in infrastructure failed to materialize. (Even at its peak private investment was insufficient to cover needs).

Furthermore, there was some evidence of public-private complementarity, not only substitution between public and private infrastructure:

- Countries which maintained higher public investment attracted more private investment (Chile, Bolivia, Colombia);
- This varied across sectors; for instance in the telecom sector there was a substitution of private capital for public, but in the transport sector there was clear complementarity.

In the absence of an adequate offsetting private sector response, compression of public infrastructure had a harmful effect on growth. Popular support for private sector participation has subsequently fallen.

Low investment in infrastructure caused an infrastructure gap relative to East Asia which according to Easterly and Serven, explains 22% of the difference in output growth between LAC and faster-growing East Asia.

In sum, Latin American countries in the 1980s stopped thinking about tomorrow in their determination to balance the fiscal books and reduce debt. Growth eventually slowed down, and tax revenues stagnated. Governments became unable to service the recurrent (in many cases unproductive) spending they had shied away from reducing. Deficits started to rise again, and hard fought macro stability started to become undone. By failing to spend on productive investments with the belief that private investment would take care of infrastructure, and by failing to cut unproductive spending, Governments eventually became fiscally insolvent. Today, many Finance Ministers in Latin America and the Caribbean are re-thinking their macro and fiscal strategies, and are making up infrastructure deficits.

1.6 **What is uncontroversial is that reducing unproductive spending, reducing inefficient taxes, and increasing productive spending are most likely to be growth enhancing.** This would be especially true when spending is on outputs which remove growth constraints. The level of public spending seems to be less important for growth than its composition, how it is governed, and how it is financed[7]. When there is macro economic stability with growth, the level of public spending is probably in the right range if growth constraints from public service gaps are not emerging. The composition of spending is then more important than the level in determining whether growth can be further accelerated. We return to discuss efficiency of spending in chapter 2 and we deal with composition in chapter 4. In section C of this chapter we consider what the priority productive expenditures might be for Uganda, and how much investment is likely to be needed. Chapter 5 goes into more detail on allocative efficiency

[7] See review of experience in ECA region (World Bank forthcoming).

in infrastructure spending. Then in section D we discuss where Uganda should look in the budget to find finance for them. First we outline Government's fiscal strategy as outlined in the PEAP, and assess it against global experience with fiscal policy and growth.

B. UGANDA'S FISCAL POLICY & ITS GROWTH IMPLICATIONS

What Does Global Experience Mean For Uganda?

1.7 This section assesses what lessons global experience offers on the implications of Uganda's fiscal policy for growth (as set out in chapter 8 of the PEAP).

1.8 **Uganda's fiscal policy as outlined in chapter 8 of the PEAP puts stability before growth.** The PEAP uses a long term financial programming framework[8] (called the LTEF) to set expenditures consistent with an inflation target and a reduction in external debt. The central objective of macro management is inflation control (below 5%). This is assumed to support private sector-led growth, with private investment programmed to rise to 22% of GDP by 2013/14 (from 17% in 2004/05). Exports are also assumed to rise from 12.6 to 15.4% GDP, with imports increasing to 30.4% of GDP, resulting in a wider trade deficit. Since aid is assumed to fall as a share of a widely rising current account balance, foreign direct investment and remittances are assumed to fill the gap.

1.9 **Fiscal policies are "subordinated" to the objectives of price stability and private credit expansion**[9]. The PEAP suggests that public expenditure should be restricted to a level compatible with low inflation and private credit expansion by controlling the deficit *before* grants. The target is 6.5 percent of GDP by 2009/10, and constant thereafter. This entails a reduced dependence on donor aid (down from 12.1% of GDP to 8.5%), and a reduction in the external debt burden[10].

1.10 **The logic for controlling the deficit *before* grants is that this will control interest rates not by reducing the domestic borrowing requirement (there is none), but by reducing the costs of sterilizing aid-financed spending**. The deficit for domestic financing has in fact been negative for several years because aid in the form of grants, concessional borrowing and debt relief, more than finances Uganda's deficit before grants. Limiting foreign exchange sales and sterilizing through selling Government debt is proposed to avoid appreciation of the real exchange rate. The PEAP argues that too high a deficit could increase inflation, drive up interest rates, and crowd out private investment[11]. The PEAP also notes that controlling the deficit will also reduce Uganda's vulnerability to aid cuts[12].

1.11 **The deficit is to be reduced by raising taxes and by cutting spending, particularly delaying new investments**. The PEAP envisages that domestic revenues will rise to 16% of GDP by 2013/14 (up from its current level of around 13 percent) and the LTEF assumes it will rise to 18% of GDP by 2015/16. No mention is made of *how* and from who tax revenue will be raised. We discuss underlying elasticities in chapter 6. The revenue and deficit targets mean that

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[8] The financial programming framework is generally applied for 3 years, but in Uganda's Finance Ministry, in the absence of a dynamic growth model, it is used to project macro and fiscal variables for 10 years.
[9] PEAP section 3.2 second paragraph on page 34.
[10] The PEAP was written before Uganda received MDRI which has mostly written off Uganda's debts.
[11] There is no empirical evidence presented to support this assumption. In the CEM using monthly and quarterly data with lags, the World Bank could find none.
[12] Gelb and Eiffert (2006) suggest that building reserves (which Uganda has done) is another effective strategy for insuring against aid volatility.

expenditure as a share of GDP must fall from 23.1% in 2002/03, to 22.5% in 2013/14. Adequate maintenance of existing assets and protecting expenditures in the PAF take precedence over new investments.

1.12 **Infrastructure spending is to be slowed down during the early years of the LTEF**[13], ie to "delay the speed of improvement rather than interrupt existing service provision". Investments in infrastructure are be accelerated only in the later years of the LTEF. Initially – presumably in the interests of stability - the priority is to meet outstanding obligations, including settling pension arrears. A shift is proposed in favor of allocations to sectors that make the best contribution to tackle poverty reduction challenges and specifically away form the Justice, Law and Order sector. A list of under-funded PEAP priorities is then highlighted, including:

- Agricultural research,
- Agricultural advisory services (which receive special consideration),
- Disease control,
- Preventative health measures,
- Support for agricultural marketing and cooperatives,
- District and community forests,
- Business development services,
- Industrial parks,
- Rural financial services,
- Rural electrification (which like rural micro finance is to be leveraged in),
- Community roads,
- Urban and community infrastructure,
- Conflict affected areas,
- Targeted support to poor families for secondary and tertiary education.

1.13 **A healthy emphasis is placed on efficiency measures in the PEAP, although the targets and implementation arrangements for these are not clearly specified**. Efficiency is to be attained in all areas, principally by reviewing public structures to avoid duplication (also called "waste"). Limiting the functions of commissions and autonomous agencies, and getting rid of duplication are stated as priorities to reduce waste, along with lower use of expensive foreign technical assistance. The need to consider a rapid increase in unit costs of construction is highlighted. Sector programs with spending channeled through GoU budget are seen as a more effective and efficient means of delivering public services than through donor projects.

1.14 **Sector measures are outlined to improve efficiency**. The most notable are; (i) real per capita recurrent expenditures on primary health and primary education are to be maintained, cost saving measures need to be found in these programs to scale up; (ii) The pupil/classroom ratio in education will be kept roughly constant by building classrooms; and (iii) higher road maintenance will be targeted in the transport sector envelope.

1.15 **Finally a number of public finance reforms are highlighted to improve value for money**. Regulations on project procedures are to be issued by the Ministry of Finance, Planning and Economic Development under the Public Finance and Accountability Act. Sector budgeting is to be strengthened. More use is to be made of tracking studies, and the publication of financial flows, and formal accountability – budget reporting – is to be strengthened.

[13] PEAP section 8.4 bullet 2 on page 193.

1.16 **Our assessment is that this fiscal strategy should be strengthened from a growth perspective.** We strongly endorse the emphasis on value for money, and in chapters 2 and 3 we suggest how efficiency savings could be sought in practice to improve the composition of spending. We would however caution against raising tax revenue in ways which would compromise economic efficiency and growth, and in the medium-term we would caution against imposing too fast a fiscal consolidation in this case. We most strongly urge caution against assuming that reducing the overall level of spending – especially if this is achieved by delaying infrastructure spending – will crowd in private investment and stimulate growth. This argument simply holds no empirical weight in Uganda's current circumstances of power cuts and road congestion. In summary, our view is that;

1. **Getting better value for money is critical**. Since the largest build up in spending in recent years has been in employee costs, public administration and PAF spending, we suggest: (i) an early functional review of Ministries, Departments and Agencies in accountability, security, justice, defense and public administration sectors, with a view to reducing employee costs and overheads in public administration, and (ii) conduct in-depth efficiency analysis of decentralized service delivery (similar to the analysis presented of education in chapter 3) for health, agriculture, water, and rural roads.

2. **The planned postponement of infrastructure investments seems unwise** and given analysis of Uganda's binding constraint to growth, seems unlikely to yield the growth targets set in the PEAP – a point we take up in chapter 4.

3. **The chosen fiscal consolidation strategy can be shown to be sub-optimal.** Raising taxes and lowering infrastructure investment is less growth-enhancing than relying on aid and improving the composition of spending to finance additional infrastructure for growth – a point addressed in recent CGE modeling scenarios for the CEM[14] which are summarized in section D.

4. **The rationale for reducing the deficit *before grants* is flawed in the medium-term from a growth perspective.** Suggesting that this strategy will crowd-in private investment seems mistaken for three main reasons;

    (i) The deficit in Uganda is fully financed by aid and concessional loans and with continued improvement in service delivery, donors likely to be willing to continue to support a deficit of the current size.

    (ii) When spent on import-intensive infrastructure, large donor inflows are less likely to be inflationary and, therefore, do not need to be sterilized to the same extent as fiscal spending with low import content. Thus, the government should focus more on ensuring a higher import content of fiscal spending and less on fiscal consolidation. Examples of such spending include import-intensive infrastructure projects which remove binding growth constraints, for instance paving trunk roads, building power stations, constructing the petroleum pipeline from Eldoret, and fixing the railway. In the medium-term, other approaches to sterilizing aid could be considered, along with their implications for growth and the deficit target[15].

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[14] Diaz-Bonilla, C. and Lofgren, H. "Patterns of Growth and Public Spending in Uganda: Alternative Scenarios for 2003-2020", background paper to the World Bank Country Economic Memorandum (2007).
[15] See Bevan and Adam (2006) for the Country Economic Memorandum, and also Morrisey (2004).

(iii) The size of the current fiscal deficit does not seem to crowd out private investment. In particular, the bulk of private investment in Uganda is not financed from commercial bank lending, and the interest rates for commercial lending seem independent of the interest rates on Treasury Bills and the stock issued. Commercial banks' spreads are high in large part because the administration costs of banking (notably wages, ICT and electricity costs) are high relative to the banks' shallow markets[16].

5. Finally, in the long-term, the assumption of **increased tax revenue collections to 16 or 18 percent of GDP seems at the high end of what is feasible** – a point we take up again in Chapter 6. Furthermore, Uganda's tax revenue collection depends heavily upon trade taxes, and in particular, taxes on fuel which would be harmful to growth. Between 1999/00-2003/04 tax revenue increased by 1.3 percentage points of GDP. Taxes on trade accounted for over 47 percent of the increase, with petroleum excises alone accounting for 11.2 percent of the change. Taxes raised from the transport sector are reckoned to comprise 40% of road transport costs per vehicle kilometer, and expenditure on roads in 2004/05 were estimated at just 60 percent of fuel duties collected[17]. Taxes on new trucks rose recently with the EAC CET. In addition, Uganda imposes VAT at 18 percent on the duty inclusive price of trucks. VAT is then returned over the life of the truck. This creates a tax on one of the most important elements of the private capital stock in Uganda. More recently mobile telephony taxes have been raised, making them amongst the highest in the world when air time taxes are added to duties on mobile 'phones. Recent household survey data show the importance of mobile telephony to rural people in Uganda. Innovations in agricultural pricing information and banking payment for traders accentuate the importance for rural growth of increasing mobile outreach.

C. INVESTMENT NEEDS FOR GROWTH, GROWTH DIAGNOSTIC/CONSTRAINTS TO GROWTH

1.17 This section estimates the public investment requirements for growth of 6 and 7 percent in Uganda, sets out some macro scenarios for long-run growth, and looks at the sensitivity of growth projections to alternative returns to public capital investment. We then consider what it would take to raise the investment response to higher savings, and the responsiveness of private investment to public investment.

1.18 **When reforms began in the early 1990s, growth in Uganda was largely driven by improvements in total factor productivity.** Structural transformation – the movement of labor out of agriculture and the emergence of new crops and products account for much of this, but in the early 1990s so too did favorable terms of trade. Since 2000, as structural transformation has slowed down, physical capital has improved, but mostly in private structures investment (reflecting the growth in services, mostly retail and trade). With the contribution of TFP reducing compared with the preceding period, physical and human capital accounted for greater shares of growth in output per worker in the period 2000-2005.

[16] World Bank (2007) Country Economic Memorandum Chapter 7 "Making Finance Work Harder for Growth". NB there *is* a strong correlation between reductions in the share of commercial banks' asssets in Government stock and increases in the share of their assets in loans to foreign banks.
[17] Johanson, F (2005), "Uganda Transport Review", background paper for Uganda CEM.

1.19 **Growth analysis by the World Bank suggests Uganda's economy is becoming infrastructure constrained**[18]. Flat returns to education over time, high rates of underemployment amongst the limited number of secondary graduates, and high returns to private capital in formal business seem to signal that Uganda is physical capital constrained, and that – although still low - human capital is relatively abundant compared with equipment[19]. Congestion on main roads around the capital and power shortages are symptomatic that growth is outstripping road and energy infrastructure, at least in the main business corridor in the South of the country. Indeed national accounts data suggest that the road transport sector grew in real terms at the average growth of real GDP in the 5 years from 1999/00 to 2003/04 (at 5.7 percent), whereas national accounts data for road construction show annual national growth of just 3.3 percent in real terms.

1.20 **Within infrastructure sectors, Uganda is lagging other countries in the stock of roads and energy** infrastructure. Relative to other countries, Uganda's stock of paved roads is very limited – a particular problem for a landlocked and rural country[20]. In addition to poor access to paved roads, Uganda's transport costs are inflated by high cost petroleum and diesel. Household access to electricity is much lower than the average for Sub-Saharan Africa, especially outside of Kampala. The costs of un-served electricity to the formal economy are estimated to be high suggesting higher investment in energy is needed for Uganda to continue a path of structural transformation.

1.21 **How much infrastructure investment does Uganda need to sustain growth in line with PEAP targets, and how** should **this be financed?** We go about estimating Uganda's public infrastructure from a "top-down" growth accounting perspective (see box 1.3). Moreover, chapter 55 proceeds to prioritize needs from a sectoral perspective.

[18] World Bank (2007) Country Economic Memorandum, "Uganda: Investment and Behavior Change for Growth, Vol II, chapter 3 "Growth Diagnostic"

[19] This is a very common finding in literature on manufacturing in sub-Saharan Africa. See for example any of the recent publications by Francis Teal.

[20] World Bank (2007) Country Economic Memorandum, op cit, chapter 6 "Infrastructure for Growth", and Butterfield, W. and Briceno-Garmendia, C. (2006),

Box 1.3. Estimating Investment Needs "Top Down" for Fiscal Analysis

Ever since Kaldor, macro economists have noted that in industrialized economies the capital to labor ratio and output per worker rise over time. Applying this observation, in the simplest neo-classical growth model (Harrod-Domar) investment "needs" are derived from a target growth rate times the incremental capital to output ratio (ICOR). Extending this simple Harrod-Domar growth model, Solow and Swan introduced human capital, and the term technological progress (TFP) to their exogenous growth model, which Solow then used in growth accounting estimations.

In our projections of Uganda's investment needs we applied an enhanced version of Solow's exogenous growth 'model', including a labor quality index (following Collins and Bosworth,(1996), with returns to an additional year's schooling (S below) of 13 percent derived for Uganda from Mugume and Canagarajah (2005)). We derived the existing capital stock by the perpetual inventory method using Penn World Tables and we equated the labor force with the population of working age using UN data and population projections. We assumed TFP would follow the recent average of 1.4 for sub-Saharan Africa; ie implicitly we assume Uganda's period of exceptional 'catch-up' TFP-driven growth in response to policies and post-conflict rebound is over.

With the labor force growing at 3.8 percent, assuming growth in average years of schooling from 3.4 in 2003 to about 7 in 2015 (a constant growth rate of about 6 percent per annum – which could be a reasonable growth rate given the emphasis on the UPE and USE), we set targets for output per worker to give annual growth of 6 and 7 percent and derived growth in the capital stock as follows:

$y = Ak^{0.35}H^{0.65}$ where $y = Y/L$ and $k = K/L$

$H = Exp(E * S)$ where E is average years of schooling for the workforce and S =0.13

In growth rates:

$$g_y = g_A + 0.35g_k + 0.65 \times 0.13\dot{E}$$

For given $g_y, g_A, \dot{E}$, g_k can be derived as $g_y - g_A - 0.65 \times 0.13\dot{E}$ and thus we derive the capital stock as:

$K_t = K_{t-1}(1+n)(1+(g_y - g_A - 0.65 \times 0.13\dot{E}))$ where n is the labor force growth rate. Aggregate investment at any given time t can be derived as $K_{t+1} - 0.96K_t$ assuming a rate of 4% for capital depreciation.

Projections

The derived average growth in capital stock per worker required would be about 3.3%. This requires real investment growth averaging about 6 percent per annum. Assuming no change in the share of public investment in total investment (25%) as in the last 7 years (1998-2004), this capital stock growth would translate into a public investment growth requirement of 8% per annum on average for the period 2004-20145. If TFP was assumed to be a 4-year moving average (ie to follow Uganda's recent slowdown in structural transformation) , the public investment growth holding other factors constant would still be 9%. To meet a growth target of 7 percent, public investment would need to rise annually by 11 percent.

Top-Down Macro estimation of investment needs

1.22 **It is possible to calculate a rough estimate of investment needs for the macro economy and use this estimate to illustrate the broad public investment requirements[21].** This is an imperfect science, however, because the value of public capital is notoriously difficult to estimate for developing countries, and so therefore are returns to capital to the macro economy[22]. Public spending does not necessarily equate with public capital (Pritchett (1996). Even national accounts data on public gross fixed capital formation is not an exact measure of "infrastructure" because it includes non-productive investments (Arestoff and Hurlin (2005)). Private gross fixed capital formation (GFCF) may increasingly contain infrastructure, as countries (like Uganda) have started opting for private participation in infrastructure (Calderon, Easterly and Serven (2004)).

1.23 **Nevertheless, we derive infrastructure investment requirements for the next 10 years using growth accounting (see box 1.3).** Assuming no change in total factor productivity growth, we estimate that for Uganda to attain GDP growth of 6 percent, real investment would need to increase by an average of 8 percent annually. To achieve 7 percent real GDP growth, our macro estimation is that real investment would need to rise by 11 percent per annum.[23] In chapter 4, we further show that a likely implication of this is that the stock of public infrastructure will have to grow by approximately $120 million pear in the coming decade.

1.24 **Prioritization of needs from sector perspective:** Because the macro number may be misleading in terms of infrastructure requirements, we compared Uganda's infrastructure spending and stocks to other low income countries and other sub-Saharan African countries to look for clues[24]. The logic is that if we accept that infrastructure is likely to become a binding constraint, Government needs to increase allocations to infrastructure in areas where Uganda is clearly lagging behind, and which are, by their nature, important for economic growth in the country. Compared to other countries:

- **Uganda needs to significantly improve power generation and access to electricity**. (chapter 5); and
- **Uganda needs to significantly increase the stock of paved roads and roads around Kampala** (see chapter 5).

1.25 **How to raise the private returns to public infrastructure spending**. As chapter 5 will show, public infrastructure spending could be more efficient. In order to raise the private investment response to public infrastructure, Uganda should:

- develop analytical techniques for planning enhanced infrastructure zones;

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[21] This calculation is merely illustrative. We are not subscribing to the Harrod-Domar convention that there is for Uganda a financing gap between savings and investment which if filled will translate into investment and growth in some linear fashion (for the critique see Easterly (1999) "The Ghost of the Financing Gap". The point of this exercise is to illustrate that with a rapid growing workforce, the capital to labor ratio in Uganda could quickly decline unless investment increases.

[22] Calculations of returns to capital are more straightforward at the project level, through Cost/Benefit Analysis.

23 These estimates are based upon quite generous assumptions about improvements in human capital (see annex 1.1).

24 Instead we use an indicators approach, as advocated in the forthcoming Development Committee Paper, and we imply compile a macro estimate of public infrastructure needs for Uganda which we compare with other countries.

- thoroughly screen new infrastructure investments on the basis of rates of return is good (Chile's approach is regarded as best practice);
- ensure implementation is carried out on time;
- set up maintenance monitoring systems and check their integrity with community-based monitoring (Indonesia provides and example);
- investigate alternative implementation arrangements at District level; and
- Reduce leakages by improving procurement practice.

D. WHERE SHOULD UGANDA LOOK TO FIND FISCAL RESOURCES FOR GROWTH?

Modeling Optimal Fiscal Strategies

1.26 **What is an appropriate fiscal policy for growth?** One possibility is simply to spend more on infrastructure. To spend more, more revenue, more aid grants, or more borrowing would need to be mobilized. An alternative strategy is to spend public resources more productively in terms of achieving objectives. That is, getting more output and outcomes for the same amount of money. For this the Government would need to become more efficient in spending.

1.27 **Government spending can become more productive in two related ways**. First, the government could reallocate resources from wasteful activities, i.e., activities that do not contribute to the achievement of its objectives – an example would be government officials who administer programs that impose regulations with little (or a negative impact) on economic performance. Another would be reducing the overheads of central policy and "back office" functions and making more resources available to service providers. Second, the government may be able to produce larger output quantities with unchanged input quantities at the level of service provision (or the same output quantities using smaller input quantities).

1.28 **CGE modeling of alternatives suggests the optimal fiscal strategy for Uganda is to seek efficiency savings in public administration, education and health** (the 3 biggest sectors). Using the MAMS CGE model, Lofgren and Diaz-Bonilla[25] simulated increased spending financed by: (i) aid, (ii) tax revenue, (iii) debt, (iv) reduced growth in general government administration, and defense spending, and (v) reduced requirements of labor, capital and other inputs per unit of out produced throughout the government. In all cases, public spending was increased in all high priority areas (education, health, economic sectors and infrastructure), adding to growth in GDP, private consumption, and leading to more rapid progress across a full range of development targets, also in human development.

1.29 **As modeled, the Government's fiscal consolidation strategy set out in the PEAP was found to be sub-optimal compared with the efficiency and aid alternatives**. Lofgren and Diaz-Bonilla modeled the impact on growth and the development targets for Uganda, of increased priority spending financed by higher direct taxes with a decreased reliance on aid. The increase in taxes significantly reduces growth in private consumption and savings, private capital stocks and GDP relative to the baseline. Real growth in household consumption is lower than the "with aid" scenario, poverty is higher, and progress towards development targets is slower. From a macro perspective, these results point to the fact that, if government consumption and/or

[25] Lofgren, H and Diaz-Bonilla, C. (2007), "Patterns of Growth and Public Spending in Uganda: Alternative Scenarios for 2003-2020", WOorld Bank, Background Paper for Uganda Country Economic Memorandum.

investment are to expand in the absence of an increase in the trade deficit (aid absorption) or an increase in GDP growth through productivity gains, then private consumption and/or investment have to decrease. This confirms that, in the absence of more foreign aid or productivity improvements, a government that pursues multiple objectives faces difficult trade-offs.

1.30 **The development** returns **to generating more efficient public spending are strong** By illustration, the simulations show that after 17 years (from 2004-2020) with a 50% cut in growth against the baseline for other government services (from 6.2% to 3.1% per year) or, alternatively, with 1% of additional efficiency growth for the government:

- the poverty rate declines by 2-3%-age points (an 8-11% decrease in the number of poor);
- the net primary school completion rate increases by 15 %-age points (a 20% increase in the number of within-cohort primary school graduates);
- gross enrollment rates increase at higher levels of education;
- the under-five mortality rate decreases by 1.4-1.6 %-age points (15-18% decline in the rate); and
- access to an improved water source increases by 11%-age points (17-18% increase in the number of people with access).

1.31 **Towards a more growth-oriented fiscal strategy**. To sum up, an optimal fiscal strategy for growth and poverty reduction in Uganda is one in which Government changes the composition of spending in favor of infrastructure, stems the rise in wages and salaries in the public sector, seeks efficiency savings in sector spending and in the medium-term absorbs and spends available aid,. This gets more favorable results than tax-financed public spending increases, both because reduced reliance on foreign savings means domestic absorption needs to decrease, and because private contributions in the form of out-of-pocket fees from disposable income are a significant and rising share of total spending. As we shall see in chapter 2, ring-fencing spending priorities in the PAF against spending cuts further exposes infrastructure spending to delays and cuts. Thereafter, in chapters 2 and 3 we illustrates a methodology for quantifying the size of inefficiency is in the education primary sub-sector, and shows that in 2005/06 waste amounted to 21 percent of recurrent spending. In chapter 4, we use a newly developed macro and fiscal model to contrast two fiscal scenarios with one another: a scenario where inefficiencies are not addressed and current spending trends continue; "**business-as-usual**", and a scenario where waste is reduced and the freed up resources are used to address infrastructure bottlenecks.

2. IN SEARCH OF THE HOLY GRAIL: ACHIEVING "VALUE FOR MONEY" IN PUBLIC SPENDING IN UGANDA

"The success of this PEAP requires a coherent strategy which can deliver improvements in the efficiency of public expenditure and ensure that expenditure allocations are focused on priorities."[26]

2.1 Given the orders of magnitude in the MAMS simulation results discussed in the previous chapter, it is important to first see whether additional space for infrastructure spending can be created *within* the current level of public spending in Uganda. If Uganda can increase the efficiency of spending and use the savings to increase spending which removes critical infrastructure constraints, the growth effects could *in theory* be sudden, strong, and lasting[27]. Specifically, can non-infrastructure sectors achieve more outputs (e.g. number of graduates that can read and write, number of healthy Ugandans etc) without spending more? That is, can the technical and allocative efficiency of public spending be improved? Chapter 3 looks at technical efficiency in the education sector and concludes with a categorical "yes". Section B of this chapter sets out an operational approach to looking for ways to quantify inefficiency and improve the technical efficiency of a sector's spending. Section C considers ways to protect productive infrastructure in the Budget. We start in section A with a review of Government's existing policy for improving the efficiency of public spending.

A. THE PEAP APPROACH TO EFFICIENCY AND VALUE FOR MONEY

2.2 The PEAP approach to tackling public spending efficiency has two prongs:

i. shift resources to sectors and to programs within sectors which have the largest impact on poverty reduction, and

ii. improve efficiency "in all areas of public expenditure, so that better value for money, in terms of the quality and quantity of services, can be achieved with the scarce resources available to Government".

2.3 **Heavy emphasis is placed on the actions of sector working groups.** The agents for improving efficiency are the "sector working groups". They are variously tasked with; (a) avoiding duplication, (b) streamlining agencies, (c) increasing the share of resources being spent on service delivery, (d) reducing the proliferation of donor projects and the earmarking of donor funds, (e) reducing counterpart fund requirements and (f) improving allocations for maintenance for on-going projects rather than starting up new ones, and (g) calculating unit costs, (h) removing absorptive capacity constraints. A principle set of tools for ensuring spending has been translated into service delivery is proposed as; (i) tracking studies, (ii) publication of financial flows in the newspaper, and (iii) strengthened formal accountability (audit reports).

2.4 **Cross cutting issues are also identified.** One important priority is to rationalize wage costs through the public sector under pay reforms. Another is to increase implementation capacity. The roads sector is singled out as a sector in which construction costs are rising rapidly, cost over-runs are

[26] Republic of Uganda (2004), "Poverty Eradication Action Plan, 2004/5-2007/8" chapter 8 para 1.

[27] Assuming that public spending on infrastructure is efficient in producing infrastructure.

common, and donor funds are slow to disburse. Some sectors (public administration and political governance) are said to already be too large for their mission.

2.5 **PEAP chapter 8 sets out criteria for public expenditure allocation.** Sectors whose plans are based on the application of these criteria could expect to see a larger share of the budget than sectors which do not comply. In summary, these criteria are:

- role for the public sector articulated;
- clear link to the PEAP set out;
- public spending gets prioritized according to returns;
- precise output targets and realistic outcome targets are set;
- fully cost all activities with realistic unit costs;
- define responsibilities with well defined mandates;
- taken full consideration of the impact of spending on future costs;
- take account of the impact on the poor in spending programs;
- do not create new structures with spending.

2.6 **In our view the main components of the PEAP strategy for achieving better efficiency are appropriate and the strategy correctly identifies two broad elements in achieving efficiency: an allocative and a technical dimension.**[28].[29] While the dichotomy between technical and allocative is useful to ensure that different sets of questions are being asked, there is often significant overlap between the two. For instance, a pure allocative efficiency analysis may try to assess whether more funds should be spentt on primary versus secondary education (by looking at a range of data, including demands for primary versus secondary school graduates). A pure technical efficiency study would look at whether the primary sub-sector's resources could be utilized in a different way to produce more primary students who obtain the desired level of proficiency in, say, numeracy and literacy (see chapter 3 for an example of such analysis). However, a technical efficiency study would often include elements of allocative efficiency. For instance, should more government primary education funds be spent on wages versus non-wages, or towards vouchers to private schools rather than paying government teachers? In short, the distinction between the two types of efficiency is not a pure dichotomy.

Is the PEAP approach actually achieving more efficiency in public spending?

2.7 **We think there is room for improvement in the implementation of the strategy for technical efficiency, and in the design of the mechanism for allocating between sectors.** In chapter 3, we present detailed evidence of the former by showing that roughly 21 percent of total recurrent spending on primary education is being wasted, yet the sector working group has not to date adopted a systematic approach to identifying the priority spending areas to focus on to reduce waste in education, nor have they piloted cost-effective measures to reduce it. It is not unreasonable to suspect an equally large amount of waste in other sectors. We suspect that more systematic and detailed technical efficiency analysis will help place more focus on this problem in sector planning. Below, we first document that we believe the PEAP strategy for achieving allocative efficiency is not working. Thereafter, we propose a methodology for measuring efficiency and addressing problems of technical inefficiency.

[28] The two dimensions: are we spending our money on the *right things* (i.e. allocative efficiency) versus *are we spending our money right*? (i.e. technical efficiency)?

[29] The two dimensions: are we spending our money on the *right things* (i.e. allocative efficiency) versus *are we spending our money right*? (i.e. technical efficiency)?

2.8 **Uganda's budget** strategy **is framed by the fiscal consolidation target and ring-fencing of priority expenditures in the Poverty Action Fund (PAF).**

> *"The Poverty Action Fund (PAF) is a subset of the budget considered to contribute directly to poverty reduction". PAF receives privileges:*
> - *Higher priority for spending increases;*
> - *Ring-fenced from spending cuts*

2.9 Stricter reporting and monitoring of PAF spending was designed to protect "priority" poverty-reducing expenditures from cuts whilst military expenditure and aid were simultaneously rising, ie to ensure some allocative efficiency. The PAF's genesis was as much to make the case to donors for budget support for poverty reduction priorities as it was to strengthen the pro-poor orientation of Uganda's Budget. It began in 1997 at a time when health and education spending in Uganda was well below a desirable share of the budget.

2.10 **In our view, the combination of the fiscal consolidation strategy and PAF is likely to compress the infrastructure spending which Uganda needs for growth**. Global experience since the fiscally austere decade of the 1980s predicts that, left unprotected, large scale public infrastructure will suffer most under expenditure pressure (or "fiscal consolidation"). In Latin America and the Caribbean, governments reduced their deficits in the 1980s, but at the cost of annual public infrastructure spending from an average of around 3% of GDP to 0.75%. In Western Europe, 9 countries were exceeding the Maastricht deficit limit in 1992. Of these, 8 had managed to meet the criteria by 1997. In all 8 of them, infrastructure spending had been cut[30]. Since in Uganda non-PAF expenditure is effectively a residual in which infrastructure priorities compete with military spending, interest repayments, other statutory expenditures like pensions, and more political priorities at State House, infrastructure is far more likely to feel the effects of the expenditure squeeze caused by fiscal consolidation and lower than expected revenue mobilization.

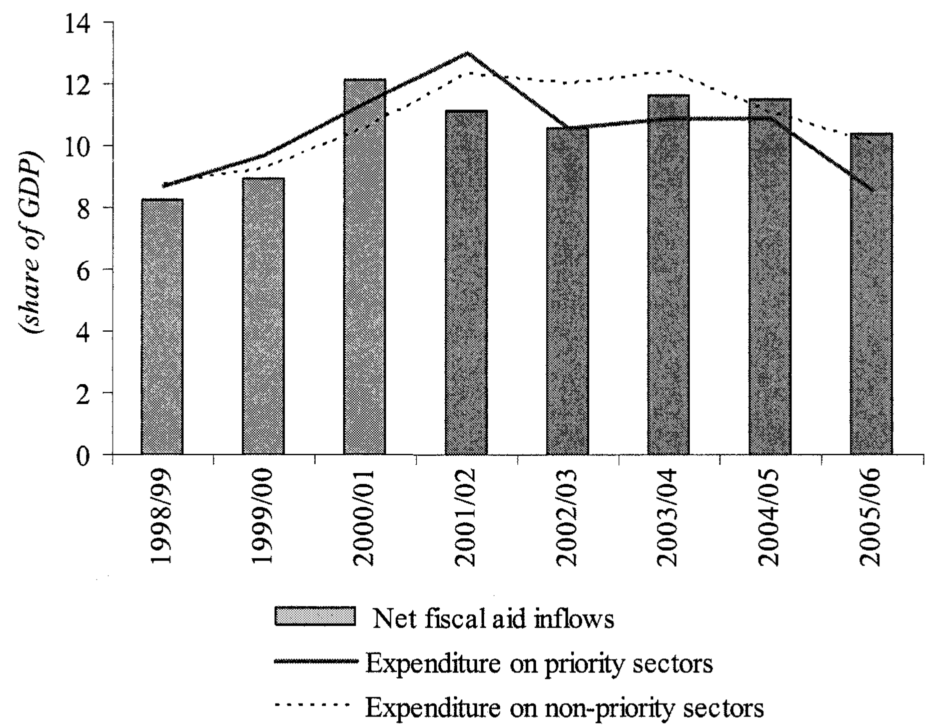
2.11 **In short, PAF does not automatically ensure that spending is allocated most appropriately from an** allocative **efficiency perspective.** Here are some examples of this:

- First, **PAF excludes some infrastructure categories which now seem critical for industrialization and growth (ie** urban infrastructure, energy and trunk roads). Social sectors and rural infrastructure are favored. Indeed with the exception of rural feeder roads and the PMA, the PAF's focus is mainly on enhancing human capital rather than physical capital or growth. With Uganda's labor force set to double in 15 years, education and health investments will continue to be important for the productivity of the future workforce. But they are insufficient for growth. There is a very real risk that without public investments which stimulate private investment in machinery and equipment the fast rising workforce will limit structural transformation in the economy, and so reduce growth in the capital per worker ratio in the Ugandan economy. If this happens, labor productivity and so growth in per capita income could slow down
- Second, if unrestrained, **salaries could erode non-salary PAF spending**. The high composition of salaries in recurrent expenditure in the PAF in the absence of restrictions on salary shares may mean that expenditures *within* the PAF discriminate against public physical capital.

[30] See Easterly and Serven (2003), and Agenor et al. (2005)(2005)

- Third, there appears to be **less scrutiny of technical efficiency in PAF priority sectors** than there might otherwise be, whilst monitoring and accountability of the PAF – including through tracking surveys - has distracted from other areas of the budget.
- In addition, as stated in the PEAP, "*in-year protection (of PAF priorities) has reduced budget flexibility as have conditional grants*". The PEAP further notes (p.188) that approximately 80% of the Government Budget is effectively protected from cuts because they are either statutory payments, or wages, or PAF expenditures. This will squeeze infrastructure.
- Finally, by itself, PAF does not seem in practice to ring-fence to re-priority sectors from cuts. The composition of the PAF has changed over time, so it is less straightforward to judge whether it has protected what is in it[31]. Yet as Figure 2.1 shows, broadly defined spending on non-priority sectors (everything excluding health, education, infrastructure, including water and sanitation) responded roughly in the same way to the aid surge in 2001/02 and during the years of fiscal consolidation since.

Figure 2.1. Have Resources Shifted from Non-Priority to Priority Service Sectors to Promote Allocative Efficiency Since the PAF Came Into Being?



Source: MTEF and World Bank staff calculations

[31] New analysis by officials in MOFPED suggests the rising share of PAF in discretionary spending is due to the inclusion of new spending items in the PAF eligibility criteria rather than by increases in spending on those items which were originally eligible. This suggests the level rather than the composition of PAF has been the variable of concern.

How Could Uganda Improve Allocative Efficiency in the Budget Across Functional and Economic Classifications?

2.12 **Regular efficiency reviews (discussed in the next section) could help improve budget efficiency sector-by-sector.** Subjecting sectors to careful efficiency reviews is usually a good starting point to achieving more outputs for less (or the same) resources, sector-by-sector. But how could Uganda improve the composition of the budget from an economic and functional perspective?

2.13 **Ensuring an efficient inter-sectoral allocation involves difficult trade-offs and requires a regular diagnosis of public spending priorities.** For instance, at what point should a country shift from focusing on access or coverage of services to focusing on quality? At what point should a country focus less on addressing immediate needs and instead try to increase its capacity to better meet needs in the future? Sometimes it is possible to do both at the same time but often a trade off is involved.

2.14 **Regular diagnosis of the macro framework, growth and inequality can help identify the need to** shift **resource allocations.** The World Bank's recent growth diagnostic for Uganda is an example of the type of analysis which can be used to make inferences about the appropriateness of overall budget allocations[32]. It shows that infrastructure is likely to be the next binding constraint to growth. The public sector does not have the sole responsibility to remove this constraint. But given Uganda's underdeveloped capital market to finance large scale investments, and the Ugandan private sector's still limited capacity to initiate large projects, the public sector is likely to continue to play an important role in infrastructure finance in the near term.

2.15 **Are there tools or rules to protect infrastructure spending?** The PAF by itself is unlikely to be sufficient to ensure sufficient resources are spent on infrastructure. It is tempting to simply propose expanding the PAF to include key infrastructure spending. However, as shown in Figure 2.1, even with the PAF in place, non-priority spending has closely tracked priority spending in the past decade.[33] International experience suggests that as stress within the Budget increases, infrastructure is likely to be squeezed (for the reasons set out in Box 2.1). [34]

[32] World Bank (2006), Country Economic Memorandum.
[33] It should be noted that we do not have the counter-factural: we don't know what would have happened to spending on priority sectors had the PAF **not** been in place.
[34] Serven (2004), "Fiscal Discipline, Public Investment and Growth", presentation to World Bank economists, September 2004.

Box 2.1. Why Infrastructure Needs Special Protection Under Fiscal Stress – Lessons from the Experience of Latin America

1. It is politically **easier to defer investments than cut public sector salaries** or pensions.
2. A **cash deficit rule ignores the assets created by investment** – it only considers their acquisition costs; it therefore has an in-built anti-investment bias. Whereas unproductive consumption and productive public investment spending have the same effect on today's fiscal deficit, they have very different impacts on tomorrow's.
3. **Large infrastructure projects widen the deficit up-front.** Infrastructure projects tend to have a negative cash flow over the first 3-6 years before turning positive. A fiscal strategy focused just on liquidity (the cash deficit) discourages such projects, even though they are consistent with good public economics, *unless* they are fully financed by higher tax (which is bad public economics) and have very rapid breakthrough returns. This can also lead to inappropriate user-cost pricing to get faster revenues.
4. **Fiscal solvency requires that the inter-temporal budget constraint balances.** A spending program today which has recurrent spending implications for tomorrow must generate a revenue stream for tomorrow to pay for its recurrent implications. (Infrastructure projects can generate such a benefit stream. Box 2.2 contains a list of further considerations for infrastructure project selection and Box 2.3 presents some examples of fiscal rules).
5. The problem of **fiscal insolvency could be acute for an economy with large unmet infrastructure needs** and a rapidly rising workforce, which requires structural transformation to create employment.

2.16 **A first option would be to have the Macro Unit of MOPFED introduce the concept of fiscal** solvency **to the LTEF planning framework.** This could help improve the focus on productive public capital spending in the macro framework. Fiscal targets focused narrowly on liquidity, the cash deficit, and public debt to the exclusion of growth, inter-temporal fiscal solvency and public net worth were responsible for the decline in public investment and the ensuing growth slow down in Latin America. This is bad fiscal economics. Unless additional taxes are raised to pay for them, the focus on in-year effects of spending on the cash deficit discourages investment projects with a negative cash flow in the initial years, even though they generate a longer term benefit stream. This is because the asset created by investment is not counted under a conventional cash deficit fiscal rule; only the acquisition cost (the liability) gets counted because only today's deficit is counted. Consumption and investment spending are therefore treated equally, and this leads to anti-infrastructure bias because infrastructure investments have a lumpy in expenditure profile; infrastructure projects tend to have a negative cash flow over the first 3-6 years before turning positive. As Serven points out[35], this type of narrow fiscal rule is particularly severe on countries with large infrastructure needs. One alternative is to consider solvency in fiscal planning; ie to consider that the present value of current and future primary surpluses should be greater or equal to public debt. Infrastructure spending raises primary spending today, but can also raise revenues tomorrow (directly through fees and indirectly through taxes on the growth that is unleashed). Theoretical growth models tend to show that the optimal level of net public investment occurs when the public investment to GDP ratio is equivalent to the elasticity of output with respect to public capital. In our fiscal scenarios at the end of chapter 4, we included assumptions on the elasticity of exports and growth to public investment to illustrate this link between public investment and growth[36]. To recognize the solvency concern, the Ministry of Finance, Planning and Economic

[35] Serven (2004), op cit.

[36] Another example of how to incorporate growth in the financial programming framework is set out in Estache and Morenen (2007), forthcoming.

Development should also consider some form of link between infrastructure and growth in the LTEF. At present there is none.

2.17 **Second, the Ministry of Finance, Planning and Economic Development should better track, prioritize and manage infrastructure investments.** In section C of chapter 1 we outline a core list of priority infrastructure investments for growth in Uganda. These should be screened to ensure they are consistent with the concept of fiscal solvency, integrated into sector BFPs, programmed in the budget, and implemented effectively and on time. The redundance of the Public Investment Program (PIP) may have reduced somewhat the attention given in MOFPED to critical infrastructure investments. The Ministry of Finance, Planning and Economic Development should consider whether to reinstate the PIP as part of the MTEF process. Some guidelines to consider in screening projects are listed in Box 2.2.

2.18 **Are there other ways in which PAF could be revised to improve overall allocative efficiency?** One option to which Government could give serious thought is to reduce the pressure imposed on infrastructure priorities from the PAF, and increase pressure on non-teacher, non-health worker wages and salaries by reclassifying the PAF to exclude employee remuneration.

2.19 **Another option in theory would be to to adopt some form of fiscal rule**. Other countries have relied on different fiscal rules to help ensure better fiscal outcomes. Universally, more sophisticated and useful fiscal rules require better fiscal data and transparency than what Uganda currently has (see chapter 4 for a description of some of the expenditure data weaknesses). Box 2.3 discusses some alternative fiscal rules which countries have adopted[37]. In advance of opting for fiscal rules, we propose strengthening fiscal reporting.

Box 2.2. Guidelines for Screening Whether Investment Projects Improve Fiscal Solvency

For fiscal solvency, a share of the growth benefits from infrastructure improvements must be captured through either user charges or through growth-buoyant taxes. The choice of sectors, the choice of projects, and the choice of financing are important up-front considerations for an enhanced PIP process to improve fiscal solvency.

To avoid fiscal insolvency;

- The financial rate of return (direct and indirect) must exceed user interest and depreciation costs;
- Financial returns must be captured either directly through user charges or indirectly through higher growth and buoyant taxation;
- A Shilling's worth of public investment spending must create close to a Shilling's worth of productive public capital. For this:
  - Government must ensure that infrastructure is built on time at lowest cost;
  - Government must select and finance the right priorities for growth.
- Operations and maintenance needs must be met to keep the asset productive (generating benefits);
- The terms of borrowing should not exceed the stream of benefits accruing to the exchequer;
- Public Private Partnerships can be used to ease the fiscal burden, but these should be transparent with regard to public guarantees, contingent obligations, and long-term commitments; and
- There should be export as well as GDP effects. This is not necessary for fiscal solvency per se, but in Uganda's case it is important for taxable growth because traditional rural sectors are hard to tax.

[37] Adapted from Serven (2006) op cit and Agenor, P, (2006).

Box 2.2. Guidelines For Screening Whether Investment Projects Improve Fiscal Solvency (cont'd)
These factors can be expressed in an equation of the marginal impact on *net worth* (*nw*) of a deficit-financed rise in the *public investment ratio* (*i*):

$$\frac{d\ nw}{d\ i} \propto \left\{ \tau\left(\frac{\partial Y}{\partial K}\right) + \theta - m - p_K(r+\delta) \right\}$$

τ = tax ratio, θ = User charge on public services, m = Unit maintenance cost,
p_K = Purchase cost of capital (including corruption, waste etc…)
r = Marginal cost of borrowing (real interest rate)
$\partial Y/\partial K$ = Marginal product of public capital

Box 2.3. Promoting Infrastructure: New Fiscal Rules and Exceptions to Existing Rules

Option 1: Adopt new fiscal targets

(a) **The golden rule**: Target the current balance – finance infrastructure investments separately on their merits.
- Allows borrowing (UK) up to a ceiling to finance investment, but not current spending.
- Does not guarantee solvency unless the investments are financially sound.
- Encourages misclassification of current expenditure as investment
- Not helpful for productive current expenditure (e.g. operations and & maintenance).
- Can be modified to include other aspects of productive spending.

(b) **The permanent balance rule:** Target net worth itself (as in New Zealand).
- Considers both assets and liabilities – i.e. future revenues and expenditures.
- Does not distort spending composition.
- Measuring net worth is complex – requires information about future income and expenditure.
- Requires strong technical capacity and fiscal institutions.

Option 2 : Grant Exceptions: special treatment to *some* projects likely to be financially profitable:

(a) **Investment by "commercially run" public enterprises**:
- those that do not undertake significant quasi-fiscal activities.
- those that do not rely on implicit or explicit government financial support.
- Hard to specify objective criteria and define appropriate quantitative thresholds in practice.

(b) **Projects with private participation (PPPs)**
- Popular in many countries for being 'hidden' off-budget.
- But no accepted accounting standards for their potential fiscal repercussions.
- Scope for PPPs more limited than originally thought.

(c) **Projects (co-) funded by MDBs**
- Could allow a role for social [not only financial] return.
- Project evaluation by MDBs taken as assurance of financial soundness -- but money is fungible: MDBs would have to evaluate the entire project portfolio.
- May encourage biases towards the "special" project category – e.g.:
    i. Invest only in oil if the public oil company is "commercial"
    ii. Only adopt projects that attract private sector involvement or fit MDB priorities

B. PROPOSED METHODOLOGY FOR IMPROVING EFFICIENCY OF PUBLIC SPENDING

2.20 **Achieving better efficiency in public spending means getting more of a desired output for the same or less amount of resources.** The easiest areas to examine from a technical efficiency perspective are the areas where (i) outputs can be narrowly defined and measured (e.g. a primary school graduates who obtain certain competencies in numeracy and literacy); (ii) inputs can be easily linked to outputs and outputs are primarily changing as a result of inputs which are measurable (e.g. primary teachers and primary classrooms are clear inputs to producing a primary graduate); (iii) there is little doubt as to whether the government should be involved in producing the particular output (i.e. making the "allocative" efficiency question less important); (iv) the private sector plays a minor role from a financial perspective and in terms of producing similar outputs (i.e. observed outputs are less distorted by changes in factors outside of the control of the government).

2.21 **In chapter 3 of this PER, we picked a relatively easy area to examine from an efficiency perspective to demonstrate a methodology, and to get a first quantification of the magnitude of the inefficiency problem.** However, albeit technically more difficult in many cases, the same methodology can be applied to other areas of government spending. Below, we highlight the methodology used, using the terminology from the education sector to make the inputs and outputs more concrete.

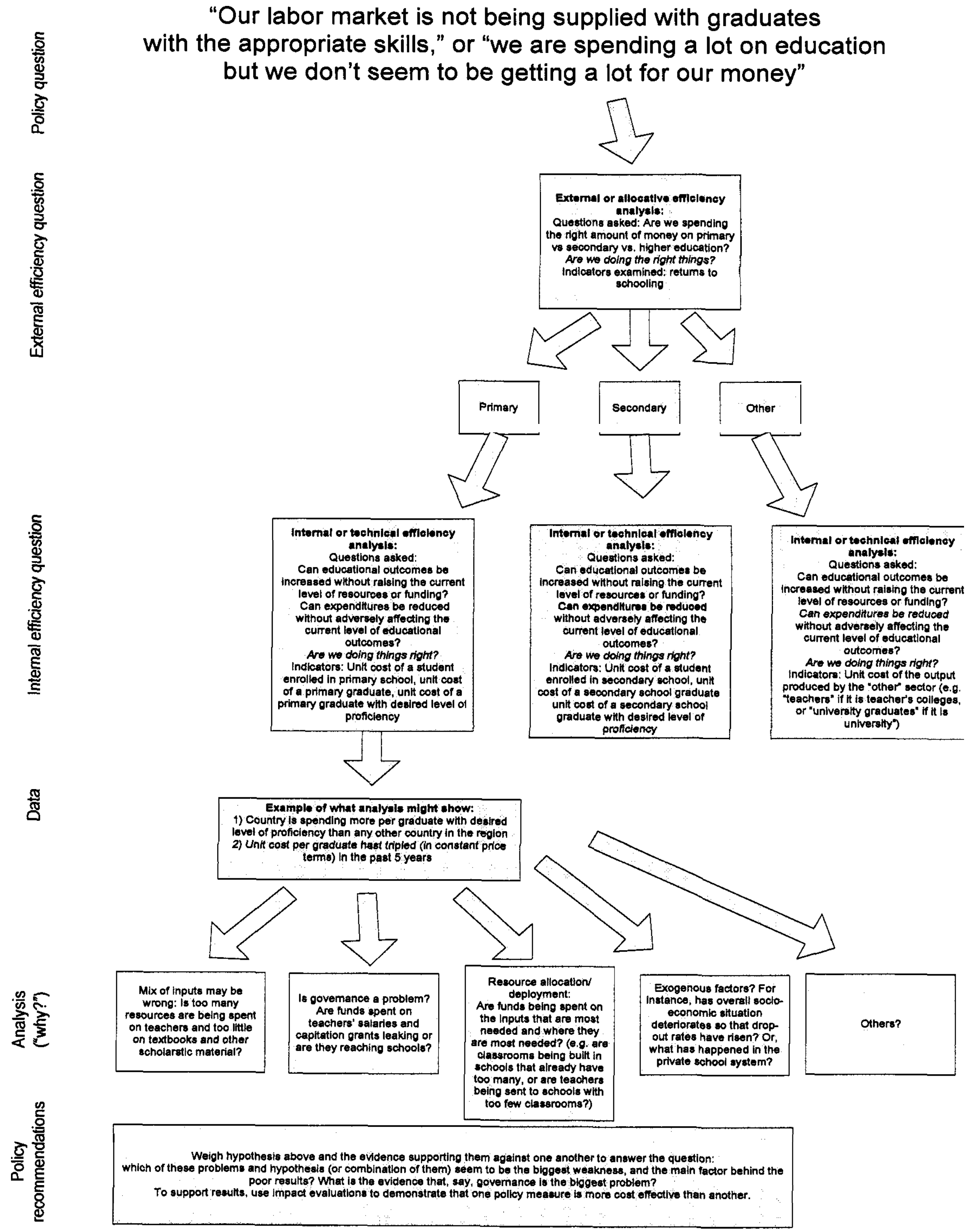
Methodology applied to education sector

2.22 **Efficiency analysis starts by reviewing data on inputs and describing how these inputs are related to outputs observed**. Getting more outputs for the same amount of money involves asking two distinctly different sets of questions: an allocative and a technical set. Regarding the allocative set of questions, governments make expenditures at all levels of education, so two of the most basic efficiency questions are whether government is spending the appropriate amount on each level of education, and whether government is making the appropriate choices on quantity versus the quality of education. These questions can be answered by looking at the success (wages and employment) that graduates of different levels of education have in the labor market relative to the costs of their education. This is called the **allocative (or external) efficiency** of the educational system.

2.23 **By contrast, technical (or internal, or operational) efficiency of an education system concerns the cost of tangible inputs used to produce a unit of output or outcome of education (eg the** cost **per graduate)**. Assessments of technical efficiency are typically done for a specific level of education, say primary education, and the simplest indicator of technical efficiency is the **unit cost** of producing one unit of educational output, which may be a student enrolled, a graduate of that level of education, or a student who has attained some minimum level of knowledge (see further discussion of unit costs below). Other things equal, an education system which can produce a unit of output at lower unit cost than another education system is more efficient. Figure 2.2 shows the relationship between allocative and technical efficiency, and the types of questions asked by the analysis of each.

2.24 **Both technical and allocative efficiency analysis are important to ensure that public resources are spent efficiently**. However, it is convenient to make a distinction between the two types of potential inefficiencies because the responsibility of reducing them often falls on different people in the public administration. The decisions on "allocative efficiency" are in the domain of elected officials in the executive (Ministers of State), whereas "technical efficiency" is often delegated (partially or fully) to public servants and administrators in the sector.

Figure 2.2. Relationship Between Allocative (External) and Technical (Internal) Efficiency



Source: Authors' analytical framework

2.25 **The first step in a technical efficiency analysis is defining what is meant by "inputs, outputs and outcomes" and how they are related and going to be measured.** Figure 2.3 is a schematic representation of how the term "inputs", "outputs" and "outcomes" is used here for primary education, and how they are related. The figure is split into three:

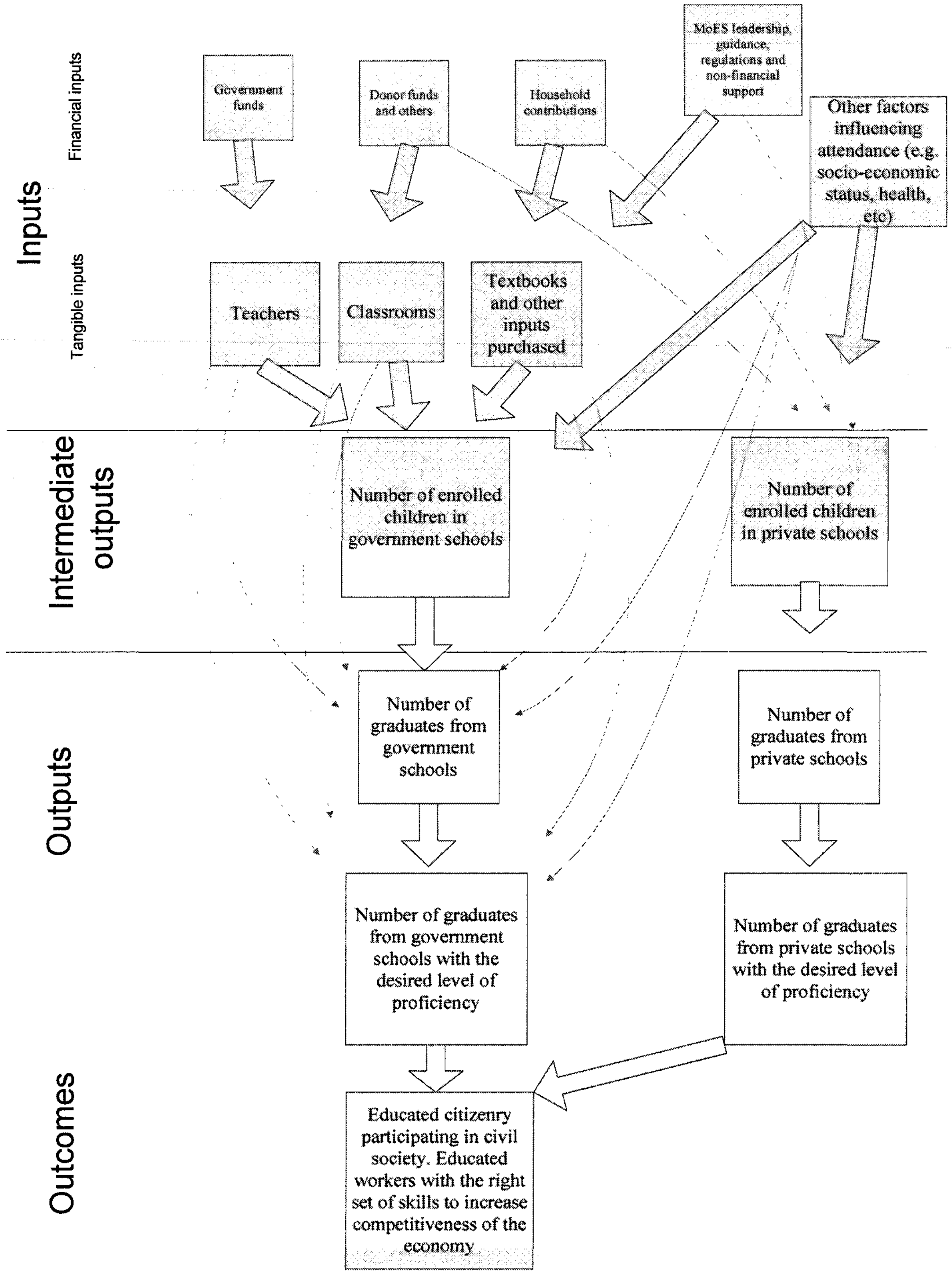
(a) **inputs** – which have been split into financial and tangible inputs (with "MoES guidance" and "other household factors" floating somewhere in between the two);

(b) **intermediate outputs** and

(c) **outputs and outcomes**.

2.26 **Outputs and outcomes are what stakeholders should be interested in achieving, inputs are the means to getting there, and intermediate outputs are milestones along the way**. Towards the bottom of figure 2.3, the distinction between "outputs" and "outcomes" is purposely blurred. For instance, a graduate who has obtained a desired level of proficiency in, say, math and literacy could be said to be either an output or an outcome of the sector. However, at the extreme, obtaining an "educated citizenry that participates actively in civil society to ensure a stable and well-functioning society" is clearly an outcome rather than an output; it is a long-term aspiration and investing in education is one tool to help achieve it.

Figure 2.3: Relationship Between Inputs, Outputs and Outcomes in Education



Source: Authors' schematic representation of relationship between inputs and outputs and outcomes

2.27 **Inputs can be related to outputs via "unit costs" which should be compared over time and across districts and schools.** Perhaps the most difficult task in studying internal efficiency in primary education is to decide on which unit cost to look at, varying by the type of output (and outcome) they measure[38]. Table 2.1 below lists several, alternative output measures in order of complexity. Some of these measures must be used with great care. Almost no one would advocate reducing unit costs of enrolling students if that would also reduce quality. On the other hand, if we observe two regions of the country, and one has lower unit costs and higher achievement than the other, it may very well be more efficient. Also, if low unit costs translate into lower quality, higher repetition, and higher dropout rates, the result may be higher unit costs of attaining literacy or graduation. Thus what appears to be efficient, or low cost, in terms of enrollment is actually inefficient, or high cost, in terms of the "real" or desired output of the sector (i.e. a graduate who can read and write). Perhaps the most useful efficiency indicator is the unit cost of graduates having demonstrated some minimum competency in core subjects. Indeed, government may be able to reduce the unit costs of this measure by increasing the unit costs of enrollment in order to raise quality. [39]

Table 2.1. Indicators Used to Track Technical Efficiency in Education over Time and Across Districts and Schools

Output/Outcome	Unit cost measure to be used
Enrollment	Cost per student enrolled
Increased Enrollment	Cost per additional student enrolled
Literacy	Cost per student completing grade 4
Graduate	Cost per primary school graduate
Achievement	Cost per student achieving at least some minimum level of knowledge in a specified grade
Learning, or increased achievement	Cost per additional knowledge or achievement gained
Graduate with Minimum Achievement	Cost per primary school graduate demonstrating some minimum achievement level

2.28 **By examining *trends over time* in unit costs, it is possible to assess whether increases in spending have been matched by increases in outputs.** If not, this raises new questions: have additional expenditures been spent on purchasing the appropriate additional inputs, too much of one input versus another, or squandered?

2.29 **Analysis of unit costs across districts and across schools helps establish whether existing levels of spending is appropriate.** For instance, the unit costs of government schools located in different districts within Uganda can be compared with one another to assess whether the degree of variation around the mean seems normal, and whether higher spending generates better results.

[38] In general, output refers to the concrete results and products which contribute to educational outcomes, such as, an enrolled child, a child graduating from school, a child graduating with some defined competency. An output is different from an outcome, the societal or ultimate goal of your education policy): for instance: build a competitive workforce; ensure that you have the foundation of a democratic society; empower citizens, etc. In some instances, the distinction between output and outcome is unclear. For instance, "graduating a student with a specified level of proficiency" can be thought of as both an output and an outcome.

[39] Calculated from fiscal data and the annual MoES Statistical Education Abstract.

2.30 **The second step in a technical efficiency analysis is to identify possible leakages or "waste".** Such analysis should identify leakages of resources between the government and the school, and leakages within the school itself.

2.31 **The reason for starting with the identification of waste is that not everything that is notionally spent on an output generates a tangible input** (which can be used to generate outputs). In other words, a possible reason for not achieving as fast progress as desired is that some resources are outright "wasted", i.e. wasted in the sense that money spent on a particular input disappears along the way to purchasing that input – or when the input paid for is either not delivered, or is of a lesser value than what was paid for it.

2.32 **One way to look for leakages is to track the flow of financial resources**. This is the methodology used in the Public Expenditure Tracking Surveys (PETS) which track the flow of funds between different layers of government (e.g. from the central government's treasury account to the district or to the school).

2.33 **Another way to look for leakages and waste is to look at whether goods and services which are being paid for are actually being delivered, and are being delivered to the quality that was paid for**. For instance, are the teachers being paid real teachers or "ghost teachers", and are full-time qualified teachers showing up for their jobs, or are they absent when unannounced school visits are conducted? How many of the classrooms paid for are actually being constructed and built to the standards that were paid for them?[40] Are contractors of classroom construction completing classrooms for the price they quoted at procurement, or could they be making profits from bogus cost over-runs? Are officials taking bribes to accept inflated contracts?

2.34 **Table 2.2 lists a number of surveys and data sources which were used to document the magnitude of waste in the education sector.** For the past decade, starting with the first PETS in 1995, Uganda has pioneered surveys to measure the extent to which financial resources are being wasted. A number of the sources of information listed in table 21 could likely be used to measure waste in other sectors. For instance, the Auditor General's report includes "questionable expenditures" for all central ministries and agencies. Moreover, the payroll clean-up exercise report ghosts for all types of workers. Finally, the 2002/03 survey of absenteeism visited health facilities in addition to schools.

[40] It is important to stress that leakages are not synonymous with theft. PETS-type leakages may result from cash-strapped and finally overstretched districts diverting education funds to, say, health spending. While this is inappropriate and contrary to regulations, it is not theft. Similarly, an absent teacher may have legitimate reasons for being absent (e.g. training, sick leave, vacation etc).

Table 2.2. Surveys and Other Information Used to Estimate Size of waste in Education Sector

Problem area:	Source:	Early 1990s	00/01	01/02	02/03	03/04	04/05	05/06	06/07
Ghost workers	Payroll clean-up exercises	20%	..	..	9%	..	..	4%	..
"Questionable expenditures"	Auditor General's Reports	..	..	..	..	13%	5%	..	..
UPE capitation grant leakages	PETS (capitation grant)	86%	18%	..	..	..	..	16%	..
Teacher absent from school	Teacher absenteeism study	..	..	..	27%	..	..	..	19%
Teacher absent from classroom (and/or school)	Teacher absenteeism study	..	..	..	..	..	..	..	79%

2.35 **The third step in a technical efficiency analysis is to describe how resources that are not wasted may not be productively used.** Ideally, this step should be an analysis of how inputs can best be combined to achieve the maximum amounts of outputs. In the education chapter for this year's PER, several data constraints made this third step impossible to conduct. First, we were unable to obtain school level data on test scores (even for a single year). Second, the Ugandan school system is a system undergoing rapid transformation with schools who have seen enrollments double in the past decade, and the facilities rapidly expanding. In such an environment, relating inputs to outputs (such as test scores of grade 6 students at a particular school or at a particular district) is a formidable task. For a start, since resources have changed so substantially, year-by-year (as a result of the rapid expansion), one would need to track input flows school-by-school (or district-by-district) over the previous six years to accurately relate inputs to outputs. Third, given the large leakages (and dispersion across schools and districts) discovered, inputs would need to be "cleaned" of waste before trying to relate them to outputs. If not, the implicit assumption would be that waste was uniformly distributed across schools (or districts) and constant over time. Our results show that both assumptions would be grossly misleading.

2.36 **Although data constraints may make an exact analysis of how inputs are related to outputs impossible, it is still possible to draw inferences about efficiency.** Specifically, it is possible to rely on other research and international experience to suggest how resources are best allocated (e.g. on class sizes in lower versus higher grades) and use such information to document potential problem areas of resource allocation. Chapter 3 provides an example of how such analysis can be undertaken with imperfect data.

Using results from efficiency analysis to design policies to improve efficiency

2.37 **An efficiency analysis is a tool to highlight problem areas, not an end in itself.** Simply knowing the size of waste or potential areas of resource inappropriate resource allocation does not result in better sector outcomes. Often extensive additional research will be needed to document *why* the documented problems exist. For instance, *why* are teachers absent? Or why are government schools spending most of their resources on wages, leaving virtually no resources for textbooks?

2.38 **In designing policies to improve efficiency, government should distinguish waste (or leakages) from inefficiency**. This is not semantics. Corrective action on waste and leakage can be dealt with in a different way from inefficiency; usually by exerting stronger control systems, better incentives against malpractice, and better accountability. Public service inefficiency which stems from a policy, technology or management choice is harder to detect, requires more thought, and demands more knowledge of the service or sector in question. Dealing with inefficiency is likely to involve building champions for change to offset the entrenched interests which support it.

2.39 **How exactly is "waste" different from inefficiency?** Inefficiency could arise because changing circumstances mean the optimal choice or combination of inputs or the appropriate type of output may be changing over time. In contrast, "waste" occurs when a completely unnecessary and avoidable cost is borne by the public sector. Waste cannot be defended with integrity. It can arise out of system and capacity weaknesses, low levels of management accountability and corruption. Examples of waste might include the following.

i. Duplication in administrative functions; either between central Ministries and local governments, or between Ministries and Agencies;
ii. Unnecessary delays and contract disputes in project implementation which lead to cost-overruns;
iii. Poor asset maintenance leading to an early replacement date for physical capital;
iv. Leakages such as those shown in table 2.2;
v. The existence of idle cash balances which could either earn interest, redeem debt, or be used to generate productive infrastructure assets.

Develop initiatives to measure efficiency in service delivery:

2.40 As part of their Budget Framework Paper (BFP), the Budget Division of the Ministry of Finance, Planning and Economic Development should require key spending Ministries to:

- define their main outputs in their sector. A "main output" should take into consideration two aspects: (i) the size of the sector's budget which is allocated towards achieving this input; (ii) the relative importance which the sector is attaching to the particular output in relation to the overall objective which the sector is trying to achieve (e.g. spending on textbooks make up a very small share of the education sector's budget but an efficiency review of textbooks may be warranted because of its importance in achieving improvements in learning outcomes);

- set out the main inputs (and their costs) used by these institutions to achieve these outputs;

- clarify how they would measure and benchmark the unit costs of producing these outputs at the service delivery point (we do this for the education sector in the next chapter); and

- clarify the data sources and sample surveys which they would use to monitor unit costs at delivery points (eg school, clinic, feeder road, water system, agricultural extension agent).

Develop initiatives to reduce waste and leakage[41]

2.41 **Sector working groups should undertake efficiency reviews and set out plans to tackle problem areas highlighted under budget framework papers.** The Ministry of Finance, Planning and Economic Development should develop a schedule of major spending programs to subject to full efficiency scrutiny over the coming years, as part of the PER process. The complexity of the tasks mean it will be necessary to tackle efficiency studies Ministry by Ministry and expenditure component by expenditure component (salaries, other recurrent, capital), and off-budget (transfers to Local Governments which are not consolidated).

2.42 Also under Budget Framework Paper (BFP), the Budget Division of the Ministry of Finance, Planning and Economic Development should require Ministries to:

- Set out their main inputs;
- Set out a timetable for periodic surveys of staff deployment and staff absenteeism;
- Set out a timetable to track the delivery of non-salary recurrent spending or items to service points;
- Set out a timetable to track infrastructure spending;
- Set out a timetable to take stock of main infrastructure assets, their condition, and their maintenance;
- Set out outputs and input requirements for all agencies and public bodies that they fund; this would also set out how the agency or public body relates to other similar bodies finance by the public sector;
- Set out a summary of physical and financial progress of all key infrastructure projects contracted or sub-contracted by the sector in its annual program of work.

2.43 **Independent of the sector working groups, develop sample service delivery surveys to monitor efficiency.** The education efficiency review (in chapter 3) highlights the importance of complementing administrative data with independent survey data, especially in an environment with high degrees of waste. Therefore, the Ministry of Finance, Planning and Economic Development, working with the local donor economists group, should develop sample service delivery surveys to monitor efficiency outside of sectors' management information systems. In addition to information on unit costs, such surveys could collect information on factors believed to be proximate determinants of better service outcomes.

[41] Some Ministries already use tracking surveys to monitor the extent to which conditional grants of various types reach their intended beneficiaries,

3. OPERATIONAL GUIDE TO MEASURING EFFICIENCY AND WASTE TO GET "VALUE FOR MONEY" IN PUBLIC SPENDING ON PRIMARY EDUCATION

3.1 Improving efficiency of spending in education is ultimately about getting more educated graduates with the right set of skills for life and work for the same amount of money, or less. In short, it is about getting more of what you want without spending more.

3.2 This chapter was developed as a substantial input to the Education Sector-wide Efficiency Study to demonstrate a practical methodology for estimating waste and inefficiency which the Ministry of Finance, Planning and Economic Development could apply to all sectors through the Budget Call Circular. It is split into 4 sections.

A. The **introduction and summary** sets out the scope of the analysis, the main conclusions, and a list of recommendations.

B. The second section starts with **background information** on primary education in Uganda, focusing on **trends in inputs, outputs and unit costs.**

C. The third section presents the efficiency analysis, relying on the methodological framework presented in chapter 2, data obtained from the Education Management Information System, and a unique school survey undertaken as part of this PER.[42]

D. Section D presents conclusions and recommendations.

A. INTRODUCTION, SCOPE OF ANALYSIS AND SUMMARY

3.3 **This chapter provides an example of how to apply the definitions in chapter 2 to identify key measures to improve value for money in the primary education sub-sector**. A review of how a sector spends its resources and what is gets for those resources is particularly important for a low-income country with limited resources which is receiving high aid inflows. As our preceding chapter discusses, this is not a straight-forward task; it requires careful thought and analytical work. It also requires systematic collection of key data over a period of time. Here we provide an example of how such work could done by reviewing unproductive spending and efficiency in Uganda's primary education sub-sector. Our work has only been possible because of outstanding cooperation with the Ministry of Education and Sports (MoES)'s Planning Department. They had already commissioned a consultant to review efficiency sector-wide. This chapter draws from the sector wide efficiency review. The findings on primary education are common to both pieces. Even within a particular sub-sector of education, it is difficult to cover all aspects of efficiency. This chapter focuses on the primary education sub-sector (accounting for more than 60 percent of total spending on education), on government schools, and focuses only on recurrent spending.[43]

[42] The survey was financed by the Bank Netherlands Partnership Program Trust Fund as part of work undertaken in preparation of a wider study on creating fiscal space for which Uganda was a case study.
[43] A further discussion of the scope of this study is included as an annex.

3.4 **An efficiency study for primary education is timely - coming almost a decade after Uganda's policy of Universal Primary Education was rapidly implemented.** UPE led, at least in the short run, to much larger class sizes, higher percentages of unqualified teachers, and fewer school supplies and materials to students. While the Government has attempted to systematically address each of these resource problems, there is still a gap between desired goals and outturns on the ground. Policies and programs that are implemented rapidly often lead to waste and inefficiency, so it is unsurprising that our analysis finds waste and inefficiency in how resources intended to deliver primary education get used.

3.5 **A primary conclusion is that there is substantial scope to reduce waste in Uganda's primary education recurrent budget.** Our estimate is that about a fifth (21 percent) of recurrent budget expenditures does not generate a tangible input into teaching children. The most significant cause of waste is teacher absenteeism. Wages for teachers do not seem to be the binding constraint on teacher presence in schools, because teachers in the private sector earn less but show up at school at least as frequently[44]. Some level of waste is inevitable in service based organizations, but 21 percent seems high, and cannot be tolerated if in the tight fiscal environment discussed in chapter 1, Uganda is also to afford improvements in the quality of primary education and expansion of secondary education. To reduce waste, we suggest that Uganda should experiment with programs to reduce teacher absenteeism, including by building on social accountability mechanisms and working with school management committees. Moreover, we also propose that the Ministry of Finance, Planning and Economic Development discuss future financing in relation to measurable improvements in governance to reduce waste.

3.6 **The main source of *waste* by far in public schools is teacher absenteeism.** If tackled, preliminary analysis suggests this alone could save Uganda up to a maximum of Shillings 52 billion. A reduction in absenteeism by one fifth – a reduction in absenteeism which has been achieved in neighboring Kenya as a result of targeted interventions – could save in the region of Shillings 10 billion. Moreover, the continued existence of ghost workers costs the sector Shillings 11 billion. Together these results suggest that on any given day 23 percent of primary teachers receiving salary payments from the Ugandan Treasury are not present in their place of work. Uncorrected capitation grant leakages lose around Shillings 5 billion while other "questionable"[45] expenditures waste a further Shillings 1 billion. Waste or unproductive spending due to a lack of *compliance* with rules and behavior norms therefore comes to around Shillings 70 Billion – about 21 percent of the total recurrent budget for primary education.

3.7 **Teacher deployment is also the main source of inefficiency in primary education.** Across schools and Districts the poorest areas with the highest drop out and repetition rates have the fewest teachers per student. Within schools, P1 and P2 average 90 students per teacher nationwide, compared with about 35 in P6 and P7. Large class sizes of this size make it virtually impossible for teachers to teach effectively and are, in turn, a likely cause of Uganda's very high drop out rates, high repetition and low completion rates. These factors drive up the costs of producing a completer. Within schools we also find that many schools have an excess of teachers over streams, and an excess of classrooms over streams. In short, teachers and classrooms do not seem to be being used optimally in Uganda.

3.8 In summary, **key specific findings in this chapter** are:

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[44] A follow-up school survey being undertaken in June 2007 will visit a significant number of private schools in order to draw comparisons between public and private schools.

[45] See Auditor General's Report to Parliament (2004/05).

- Teacher absenteeism is high: on a given day, almost 25,000 teachers receiving payment by the Government of Uganda (or 19 percent of the teaching workforce) are likely to be absent from government primary schools;
- Absenteeism among head teachers is particularly high (27 percent);
- Less than 20 percent of paid teachers who *are* present at school are in a class room teaching;
- As a result, a large number of classes are either taught without a teacher (approximately 20 percent) or get cancelled;
- Waste amounted to Ush 70 billion, or 21 percent of recurrent spending on primary education in 2005/06; and
- Waste as a share of spending has been roughly constant for the past decade, suggesting Government needs to review the incentives and accountability framework governing primary education.

3.9 In terms of its size and severity, waste is the most serious problem facing the sector from an efficiency perspective. However, this chapter documents other problems related to efficiency in the sector.

- Whereas enrollment has risen rapidly with UPE, completion rates remain low, with drop-outs and repetition high. The sector is already shifting more focus to quality and completion rates (rather than access and enrollment rates). This move is encouraging and should be further strengthened. We suggest focusing even more on "final" (as opposed to "intermediate") outputs (i.e. graduates and graduates which have achieved a desired level of proficiency in numeracy and literacy as opposed to enrolled students);
- Unit costs have been rising, driven mostly by wage increases for teachers which have outstripped nominal GDP (ie wages rose by more than inflation plus economic growth);
- As a result of high drop-out and repetition rates, the unit cost of a graduate is more than twice what it could be with more efficiency. Poor learning in classrooms implies even higher unit costs of a graduate with acquired level of proficiency in numeracy and literacy;
- Higher monthly salaries to teachers have not translated into more outputs (i.e. more primary graduates);
- Teacher salaries are 60 percent higher in government schools compared with private schools. In turn, government schools can afford fewer teachers per student than private schools;
- The deployment of teachers within a school is such that primary one and two on average have more than 100 students while primary seven have less than 40 students. Most educators would suggest the resource deployment, with more teachers allocated in early grades; and
- This deployment problem could – in theory at least – be quickly fixed with double-shifting.

3.10 **We do not provide detailed policy recommendations to address each of the key findings**. These need to be designed by the Ministry of Education in discussion with the sector working group based upon the sector wide efficiency study. However both this report and the sector wide efficiency report it accompanies, outline examples from global experience of how Uganda might go about addressing the largest problems of waste and inefficiency.

3.11 **In terms of the severity of the problems identified, waste is clearly the number one problem facing the sector from an efficiency perspective.** Therefore, we propose the following steps to combat the waste problems identified:

- The sector working group should develop time-bound action plan to achieve progress in reducing teacher absenteeism, ghost workers, UPE capitation grant leakages, and "questionable expenditures";
  - Such a plan would include commitment to regularly monitor problem areas (including repeating random school visits to measure teacher absenteeism, using internationally recognized methodology to measure absenteeism).
  - The proposed interventions to combat the problem areas (e.g. teacher absenteeism) should be tested first using randomized control group experiments to measure their impact.
- Governance weaknesses leading to high amount of waste should be discussed in the Education sector's annual performance report (ESAPR), including;
- Discussion of progress or lack thereof in reducing teacher absenteeism, ghost workers, UPE capitation grant leakages, and "questionable expenditures";
- Given the documented history of governance problems, we propose that the assumption in the sector working group (and vis-à-vis MOFPED and donors) be that the magnitude of the problems documented in this chapter on ghosts, UPE capitation grant leakages, absenteeism and questionable expenditure is unchanged until more up to date information shows otherwise; and
- The Ministry of Finance, Planning and Economic Development should discuss increases in funding in relation to measurable improvements in reducing teacher absenteeism, ghost workers, UPE capitation grant leakages, and "questionable expenditures".

3.12 To ensure more focus on better utilization of resources in the future, future EASPRs should:[46]

- Focus *even more* on graduates and graduates with a desired level of proficiency and less on enrollments; and
- Focus more on what it costs to produce outputs (i.e. unit costs) and why these unit costs have evolved as they have.

3.13 While the challenges facing the sector will not be fixed in the short-term by articulating an action plan to combat waste and focusing more on *real* outputs such as the number of graduates (rather than intermediate outputs such as the number of enrolled students), we believe these represent first important steps.

[46] The Education Sector Annual Performance Review already includes a discussion on quality so the point made here is one of relative importance. As discussed below, the review appears to be placing relatively more emphasis on enrollment as opposed to the number of graduates acquiring the desired level of proficiency in numeracy and literacy.

B. Background: Overall Financing and Trends in Inputs, Outputs and Unit Costs in Primary Education in Uganda

Background: Financing Education in Uganda

3.14 **Ugandans continue to value education highly**. As a society, Uganda invests over seven percent of GDP in formal primary, secondary, and tertiary education, excluding the income foregone by students. As shown below in Table 3.1, in aggregate this investment is funded almost equally by government and private households. However, at the primary level government bears the larger financing burden, and at the secondary level households bear the larger financing burden. (Annex 1 discusses the results of analysis of household spending on primary and secondary education from the 2002/03 and 2005/06 Uganda National Household Surveys (UNHS)).

Table 3.1. Estimate of Uganda Education Expenditure (share of GDP)

Level/Source	Government/Donor	Household	Total
Primary	2.27	1.32	3.59
Secondary	0.63	1.88	2.51
Tertiary	0.55	0.48	1.03
Total	3.45	3.68	7.13

Source: Calculated on basis of UNHS 2006 and fiscal data

3.15 **Most primary schools in Uganda are government-aided schools and more than 90 percent of students are enrolled in these schools.** As table 3.2 shows, the primary education system is denominated by the government-aided schools, both in terms of numbers of schools but more prominently in terms of enrollments. According to EMIS (2005), 91 percent of students were enrolled in a Government-aided schools. As will be discussed in more details in Box 3.2 below, the EMIS numbers should be treated with some caution. In particular, enrollment numbers are likely overestimated (due to incentives of head masters to inflate numbers) and the response rate of government schools to the annual school census (on which the EMIS is compiled) is higher than the response rate of private schools.

Table 3.2. Number of Government and Non-Government Schools and Enrollments by Ownership

	No. schools	% of total	Enrollments	% of total
Total	13,595	100%	7,223,879	100%
Government	11,332	83%	6,609,677	91%
Private	1,509	11%	406,165	6%
Other	754	6%	208,037	3%

Source: EMIS (downloaded in October 2006)

3.16 **In 1996 Uganda adopted a policy of Universal Free Primary Education by abolishing school fees in public schools**. As shown in Table 3.3, school fees are far lower at the primary level than other levels of education, but at UShs 14,254 (on average in a government school) they are above zero. However, for most students in public schools, fees are low. Of course, school fees are only part of the financing burden facing families. On average, households with students in primary school pay as much for uniforms, transportation and school supplies as they do for school fees.

Table 3.3. Average per Pupil Household Expenditure on Education by Level of Education in 2005/2006 (in UShs)

Level of Education	School Fees	Other School Expenditures	Total
Primary	14,254	14,240	28,494
Secondary	161,432	76,512	237,944
Post-Secondary	129,693	179,844	309,537

Source: Calculated from UNHS 2006.

Government Expenditure on Education

3.17 **To assess trends, we first compiled fiscal tables which are amenable for efficiency analysis (Table 3.4).** The breakdown of education spending in the MTEF (see Table 3) does not lend itself to analytical work on efficiency of spending. Specifically, it does not clearly show how much is spent on primary education vs. secondary education[47]. We therefore split total government expenditure from 1998/99 – 2005/6 into three categories: spending on primary, secondary and "others" (see Table 3.5). This break-up was done by first allocating district spending (which is clearly labeled as primary and secondary) into primary and secondary. Second, we split MoES spending (vote 013) into primary and secondary and others by assuming that 60 percent of MoES spending goes to primary, 30 percent goes to secondary while the remaining goes to others. Itemized budget estimates were consulted for the year 2005/06 to verify that this break-down was valid. To calculate unit costs in real terms, we deflated table 6 by the CPI.

Table 3.4. Government Budget for Education in 2005/06 (MTEF classification) (bn Ush)

		Recurrent expenditure		Development expenditure		
	Total recurrent	**Wage**	**Non-Wage**	**Domestically financed**	**Donor-financed**	**Total expenditure**
Uganda Management Institute	0.4	0.0	0.4	0.0	0.0	**0.4**
Education and Sports (incl Prim Educ)	46.9	7.3	39.6	19.3	39.5	**105.7**
Makerere University	33.4	0.0	33.4	0.1	17.6	**51.1**
Mbarara University	6.4	4.0	2.4	0.4	0.0	**6.8**
Kyambogo University	11.0	6.1	4.9	0.3	0.0	**11.3**
Education Service Commission	2.2	0.5	1.8	0.1	0.0	**2.3**
Makerere University Business School	4.2	0.0	4.2	0.0	0.0	**4.2**
Gulu University	3.2	0.0	3.2	1.2	0.0	**4.4**
District Primary Educ incl SFG	287.5	254.0	33.5	51.0	0.0	**338.5**
District Secondary Education	82.8	76.3	6.5	0.0	0.0	**82.8**
District Tertiary Institutions	24.0	15.7	8.3	0.0	0.0	**24.0**
District Health Training Schools	4.2	2.5	1.8	0.0	0.0	**4.2**
	506.3	**366.4**	**139.9**	**72.3**	**72.3**	**650.9**

Source: MOFPED, MTEF tables

[47] As the MTEF table shows, district spending on primary vs. secondary education is clear but MoES allocations to these two levels of education and lumped together in the line called "Education and Sports (incl. Prim. Education).

Table 3.5. Government Expenditure on Education (based on releases[48])

Education (Ush, billion)	98/99	99/00	00/01	01/02	02/03	03/04	04/05	05/06
Total education expenditure (MTEF number)	**326**	**356**	**415**	**512**	**561**	**587**	**638**	**657**
Recurrent	241	258	286	360	396	436	484	539
Wages	158	153	180	238	274	310	358	396
Non-wages	83	106	106	122	123	127	126	144
Development	85	98	129	152	164	151	154	118
Domestic	36	67	87	96	95	81	88	69
Donor	49	31	42	56	69	70	65	49
Primary (WB estimate based on MTEF)	**205**	**237**	**285**	**349**	**366**	**383**	**407**	**424**
Recurrent	152	172	196	245	259	285	309	334
Wages	99	101	124	162	179	202	248	276
Non-wages	52	70	73	83	80	83	61	58
Development	53	65	88	104	107	98	98	90
Domestic	22	44	59	66	62	53	56	61
Donor	31	21	29	38	45	46	42	29
Secondary (WB estimate based on MTEF)	**72**	**71**	**77**	**98**	**110**	**114**	**126**	**102**
Recurrent	53	51	53	69	78	84	96	93
Wages	35	30	34	46	54	60	71	85
Non-wages	18	21	20	23	24	25	25	9
Development	19	20	24	29	32	29	30	9
Domestic	8	13	16	18	19	16	17	4
Donor	11	6	8	11	14	14	13	5
Other (WB estimate based on MTEF)	**49**	**48**	**53**	**64**	**85**	**91**	**104**	**131**
Recurrent	36	35	37	45	60	67	79	112
Wages	24	21	23	30	41	48	39	35
Non-wages	12	14	14	15	19	20	40	78
Development	13	13	17	19	25	23	25	19
Domestic	5	9	11	12	14	12	14	4
Donor	7	4	5	7	10	11	11	15

Source: World Bank attempt to reclassify total government spending on education into private, secondary and "others"

Trends in inputs: number of teachers, schools and classrooms

3.18 **Teacher numbers in primary education have flattened since 2003/04 after rising rapidly from 1998/9** (table 3.6). The introduction of EMIS has greatly improved the sector's ability to keep track of the number of teachers, schools and classrooms. EMIS data show that the number of primary school teachers on the payroll has been broadly constant since 2003/04 at around 126,000. In addition, there are approximately 2,500 teachers at the primary teachers colleges who are also paid out of the primary school wage bill.

[48] See chapter 1 for the important distinction between releases and actual spending.

Table 3.6. Inputs in Primary Schools, 1995/96 – 2005/06

Year	1995/96	1996/97	1997/98	1998/99	1999/00	2000/01	2001/02	2002/03	2003/04	2004/05	2005/06
Trained	59,747	62,574	80,953	..	86,238	97,722	107,009	..	..	..	..
Untrained	21,817	26,673	18,284	..	24,128	29,316	32,475	..	..	..	..
Total teachers	81,564	89,247	99,237	..	110,366	127,038	139,484	..	..	..	..
Teachers on payroll	..	..	..	86,348	88,048	101,818	115,453	121,772	124,137	126,227	126,990
Number of government-aided schools	..	..	..	..	8,074	9,313	10,420	11,315	11,072	11,332	..
Total primary schools	..	..	..	..	12,480	13,244	14,284	14,540	15,339	..	..
Number of classrooms in government sc	..	..	..	..	..	..	..	73,104	78,403	83,740	85,043

Source: Education Fact File (available online at http://www.education.go.ug/upe.htm) and ESAPR 2005/06.

3.19 **The introduction of UPE resulted in rapid growth in the construction of new schools since the late 1990s.** There were 1000 new government schools yearly between 1999/00 and 2001/02. However, the number of government primary schools has been broadly unchanged at 11,300 since 2002/03.

3.20 **Nevertheless, many of these schools have added additional classrooms.** Since 2002/03, on average, almost 4,000 new classrooms were constructed per year in government primary schools.

Trends in education outputs

3.21 **Uganda achieved impressive increases in net enrollment rates, reaching 84 percent at the primary level in 2005/06[49]. But net enrollment has been flat for 5 years.** While there was an impressive increase from the late 1990s, net enrollment rates have remained flat at this level for the past five years and possibly even declined in recent years (see Table 3.7). Following the introduction of UPE, the number of students enrolled in government schools rose from 2.4 million in 1995/96 to 4.7 million in 1997/98 and reached 6.6 million in 2005/06 (see figure 3.1). The largest ever matriculating grade one class is still the 1997 class where enrollments rose to 2.1 million from less than 600,000 just two years earlier. Partly, the large incoming class reflected the fact that many older students decided to return to (or enter for the first time) school. The incoming classes in primary one has since shrunk but the problem of over-age students persists (see figure 3.2). In recent years, total enrollment in government schools has hovered around 6.5 million, resulting in flat NERs, according to both Uganda Household Survey (2005/06) and National Service Delivery Survey 2004[50]. In contrast to household survey data, MoES figures show that NER has risen in recent years, rising to 92 percent in 2006. Possibly, these figures reflect the overestimation of enrollments in EMIS, an overestimation which we will return to below.

[49] UNHS 2005/06.

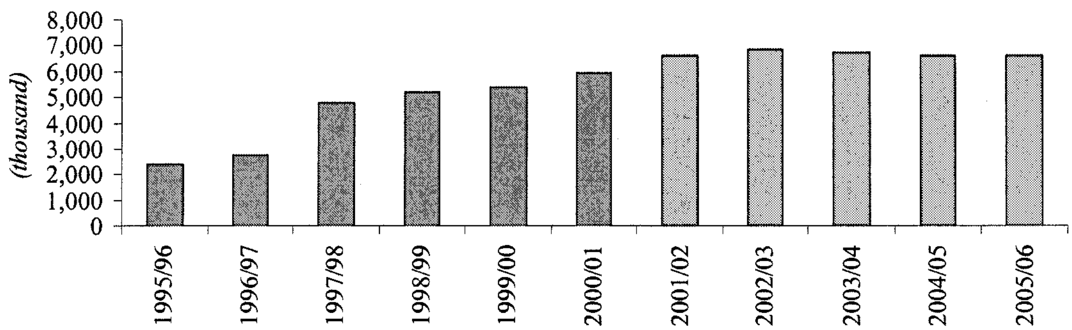
[50] It is worth reflecting on what the dynamics of past enrollment will mean for Uganda's outputs over time. Until the wave of increased UPE enrollment passes through P7 (figure 3 suggests it already has), even a drop in quality will show up in a rise in completers and as a rise in completion as a share of the population cohort. Over time, as enrollment stabilizes or falls further, improvements in outputs can only be achieved if drop-outs and repetition fall. The recent increased growth rate in completion could therefore be a temporary phenomenon.

Table 3.7. Primary School Net Enrollment Rates According to Household Survey Data (in percent)

Survey Year	Male	Female	Total
1999/00	85	84	84
2001 UDES *	87	87	87
2002/03	85	86	86
2005/06	84	85	84

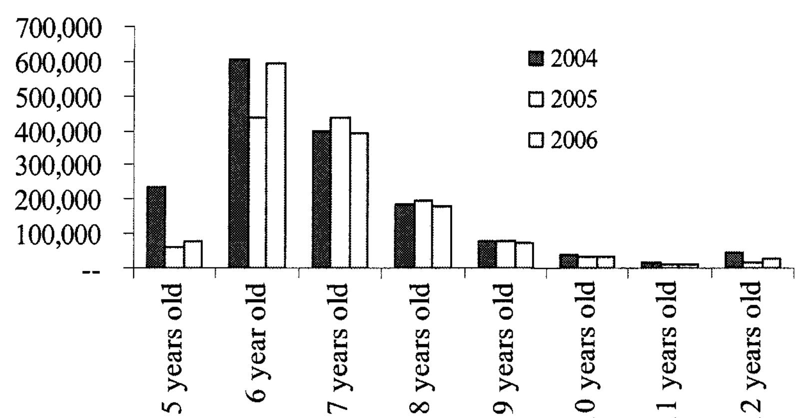
Source: UNHS report 2005/06 and
* UDES = Uganda Ed-data survey

Figure 3.1. Number of Enrolled Students in Government Schools



Source: Annual Education Statistical Abstracts, various issues

Figure 3.2. Students Enrolled in Primary one by Age in 2003/04, 2004/205 and 2005/06

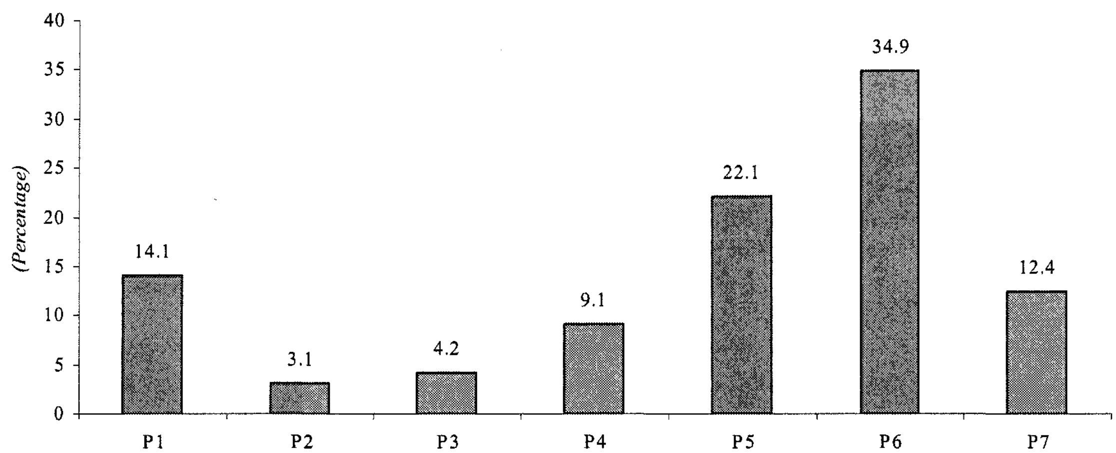


Source: EASPR 2005/06 (table 3.3)

3.22 **The large expansion of the primary education sector meant more challenges.** We have been unable to find historical data on drop-out and repetition rates. Most likely, they would show rising difficulties immediately following the introduction of UPE and a trend improvement since. In particular, larger enrollment numbers meant more graduates in primary education, but it also meant more drop-outs.

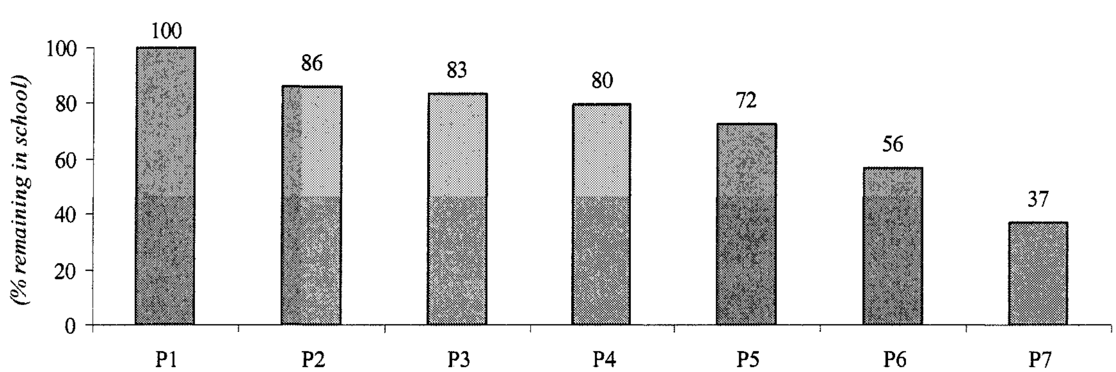
3.23 **While the sector is well on the path to improvements, challenges remain.** For instance, only a third of the students who entered P1 in 1997 graduated P7 in 2003/04. Although the situation has improved somewhat since the first large UPE cohort entered school, the National Service Delivery Survey conducted in 2004 suggest that drop-out rates remain high (see Figure 3.3). The figure shows that drop outs are high in grade 1 and in the higher grades. Figure 3.4 illustrates the implication of the drop-out rates by examining the progression of a hypothetical P1 class of 100 students. The figure shows that high drop-out rates imply that, by grade 7, only 37 students (out of the 100 which entered in P1) would still be in school.

Figure 3.3. Percentage Distribution of Pupils Leaving School by Class



Source: National Service Delivery Survey 2004

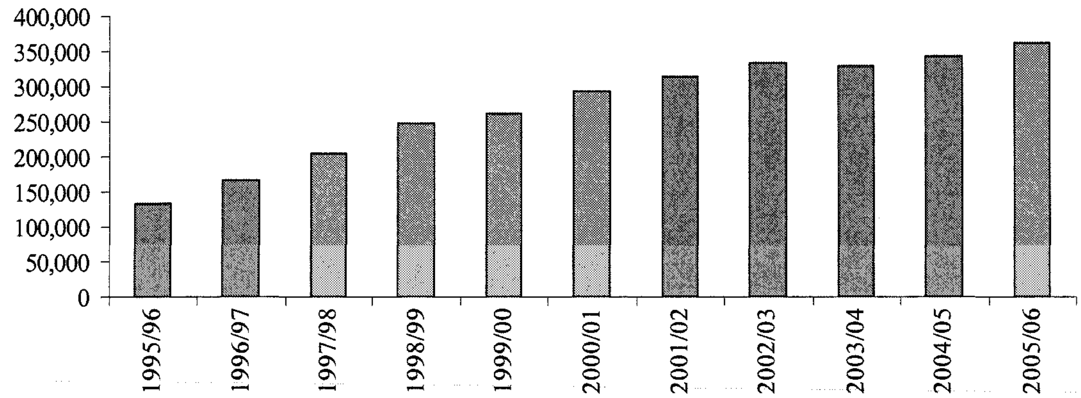
Figure 3.4. Implications of Drop-out Rates on Hypothetical Class Entering with 100 Students[51]



Source: National Service Delivery Survey 2004 and staff calculations

[51] These calculations are not adjusted for repetition rates.

Figure 3.5. Estimate of Number of Graduates from Primary 7 from Government Schools



Source: Annual Education Statistical Abstracts, various issues and World Bank staff estimates

3.24 **Large drop-outs mean the number of graduates from government schools is still low**. In particular, a well performing school system with seven grades to complete, ought to produce approximately an annual number of graduates roughly equal to 1/7 out of total enrollment. In Uganda's case, with 6.6 million enrolled in government schools, there should annually be roughly 950,000 graduates (1/7 x 6.6 million).[52] Instead, the number of graduates is still below 500,000 per year, increasing by only around 20,000 each year in the past 5 years (see figure 3.5).

3.25 **With the sector maturing and enrollment numbers impressively improved, it may be time to shift *even more* focus to "true" output such as graduates and the number of graduates which achieve the desired level of proficiency in numeracy and literacy.** Clearly, the sector has already moved in this direction as is evident by the discussion of quality (e.g. curricula reform) in the Education Sector's Annual Performance Review. We recommend continuing on this path by placing even more focus on the number of graduates and number of graduates acquiring the desired level of competency in literacy and numeracy rather than enrollments.[53]

3.26 **Cross-country evidence also suggests that education sectors with high enrollment rates should focus on giving graduates a quality education.** In particular, country cross-

[52] In an expanding school system due to a growing population, the graduation ratio should be less than 1/7 since there will be disproportionably more students in the younger grades. However, the 1/7-rule is a decent approximation to what Uganda should aspire to, especially since figure 3.1 shows that the years of rapid expansion in enrollment appears to be over.

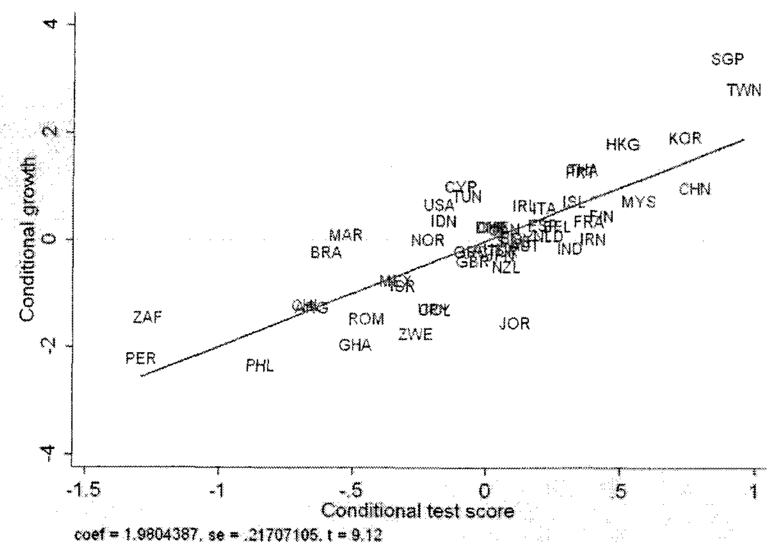
[53] In the most recent Annual Performance Review both quality and access is covered but the emphasis is clearly on access. For instance, the executive summary and the main focus under the pre-primary and primary education sub-sector appear to be net intake rates and gross enrollment numbers. By contrast, the report shows neither the absolute number of graduates from government primary schools, nor the number of graduates who obtained the desired level of proficiency. Moreover, it shows how many of students who sat exams obtained the desired level of proficiency in numeracy and literacy but, the tables do not show how many students were taking the exams. Moreover, the report contains large amounts of information regarding inputs (rather than outputs). For instance, a water pump for UTC Elgon was procured for Ush 3.12, or land and property was bought from Bulemezi community cottages (Block 147, Plot 41) under the presidential pledged construction programme. From an auditor's perspective this information (along with receipts) are necessary but this information is not very helpful in assessing how well the sector is spending its resources, which should be the purpose of a budget strategy paper.

sectional evidence demonstrates that [a] there is a large positive relationship between quality improvements and economic growth and [b] there is almost no relationship between increases in access to education and economic growth.

3.27 In Figure 3.6 below, the economic growth rates of countries are plotted against their scores on international assessments, controlling for other factors which may affect economic growth. As can be seen, the slope of the line fitted to these observations is positive and steep, indicating that, controlling for access, higher test scores contribute significantly to economic growth.

3.28 In Figure 3.7 below, the economic growth rates of countries are plotted against their average years of education of the populace, again controlling for other factors that affect growth. The slope of the resulting line fitted to the observations is almost flat, and the estimated slope of the line is statistically insignificant, indicating that, controlling for test scores, higher years of educational attainment do not contribute significantly to economic growth.[54]

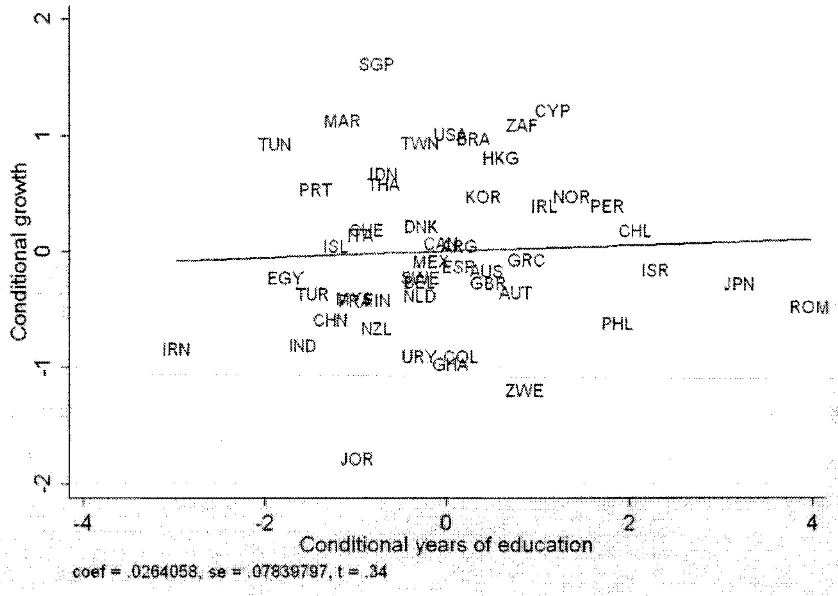
Figure 3.6. Economic Growth and Quality of Education (as measured by test scores)



Source: Hanushek and Woessman (2007)

[54] Hanushek and Woessman (2007). "The role of education quality for economic growth," Policy Research Working Paper Series 4122, The World Bank

Figure 3.7. Economic Growth and Access (Measured by Years of Schooling)



Source: Hanushek and Woessman (2007).

3.29 **The expansion of UPE resulted in a decline in the quality of education but indicators have started to improve.** The biggest output problem is that the majority of the students who reach P6 still don't meet minimum standards in numeracy and literacy. As Table 3.8 shows, only around 30-35 percent of P6 students obtained the minimum desired level of proficiency in numeracy and literacy, an improvement since 2003. Unfortunately, we have been unable to obtain the school-by-school data underlying these averages. As a result, we were unable to analyze which schools (and districts) experienced the largest improvements. In addition, we were unable to decompose the performance of public versus private schools. Presumably, the averages reported in the table below by UNEB is an average across public and private schools but it is not clearly specified.

Table 3.8. % of Pupils with Minimum Competencies in English Literacy and Numeracy

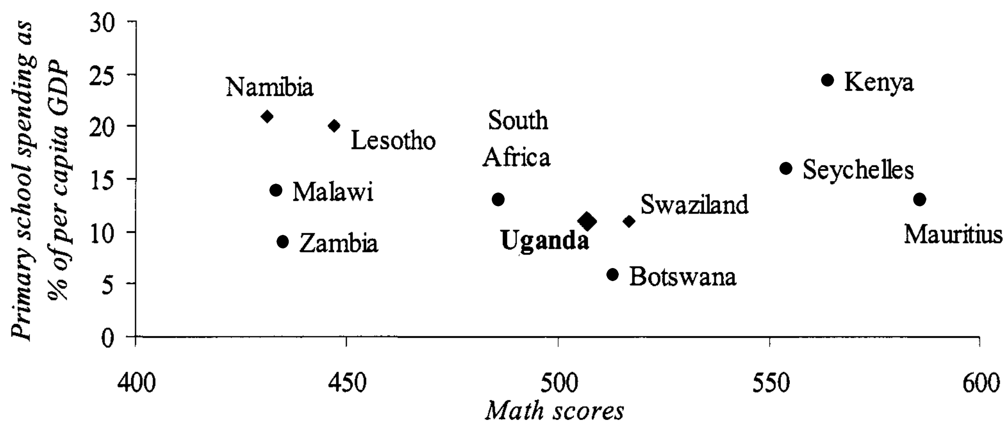
	1999/00	2002/03	2005/06
Literacy Rate: Percentage of pupils reaching defined level of competency in literacy at			
Grade 3	18%	34%	46%
Grade 6	13%	21%	34%
Numeracy Rate: Percentage of pupils reaching defined level of competency in numeracy at			
Grade 3	39%	43%	43%
Grade 6	41%	21%	31%

Source: Uganda National Examinations Board: "National Assessment of Progress in Education"

3.30 **On academic performance, Uganda is in the middle of the African countries participating in Southern and Eastern Africa Consortium for Monitoring Educational Quality (SACMEQ).** As shown in Figure 3.8, Uganda is middle ranking in both academic performance and spending as a share of per capita GDP. Uganda has not participated in larger international tests such as, for instance, Trends in International Mathematics and Science Study (TIMSS). However, South Africa participated in the most recent TIMSS and scored at the bottom of all countries worldwide. The fact that Uganda's performance on the SACMEQ is almost identical to that of South Africa suggests that if Uganda had participated in TIMSS, this would probably have revealed that the quality of Uganda's education, too, lies far below that of international comparators outside of Africa.

3.31 **Uganda could benefit from participating in future TIMSS.** By participating in international standardized tests, Uganda would obtain two pieces of information which could help improve education outcomes. First, the tests results would reveal the strengths and weaknesses of the Uganda education system vis-à-vis other countries. Other countries use such information to help track improvements over the years, and help identify areas most in need of improvements. Second, as part of the international assessments, a common set of possible explanatory factors of quality are collected, both at the student and at the school level. Other countries use analysis of such data to identify underlying factors constraining progress, and to target policy interventions appropriately. A final argument for participating in international assessments is that such tests usually help bring the importance of quality to the attention of policy makers, and stakeholders.

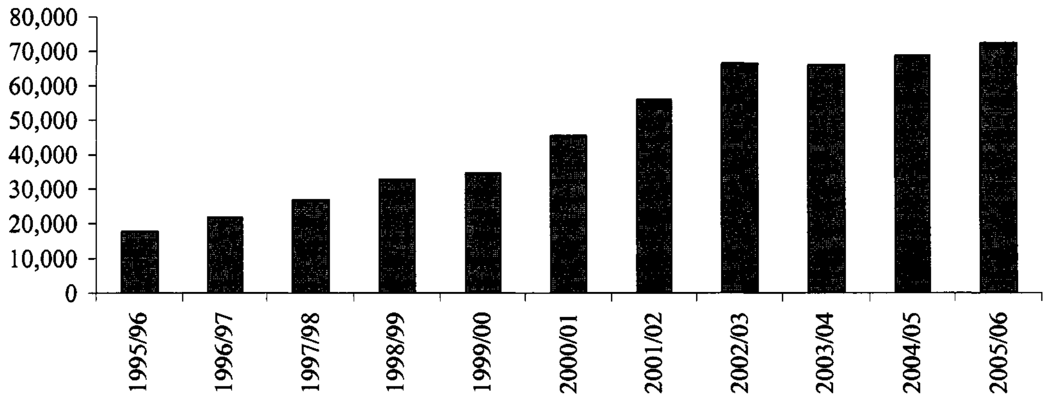
Figure 3.8. Primary School Spending and Maths Test Scores on SACMEQ



Source: Southern and Eastern Africa Consortium for Monitoring Educational Quality data base.

3.32 **The number of graduates from government primary schools in Uganda who achieve minimum required proficiency in literacy seems very low, and seems to have flattened out since 2002/03**. Using the figures in and some additional assumptions (discussed in box 3.1), a rough estimate of the number of graduates with the desired level of proficiency since 1995/96 is estimated. Needless to say, given the assumptions needed to make such calculation such calculations can only give be indicative of this output. However, it is worth highlighting that there is a long way to go. Despite using around 400 billion Ush in 2005/06, less than 80,000 primary graduates with the desired level of proficiency were produced. In other words, each graduate with the desired skill set cost almost Ush 5 million or almost $2,500 (see figure 3.9).

Figure 3.9. Estimate of Number of Graduates from Government Primary Schools who Achieve Minimum Required Proficiency in Literacy



Source: Annual Education Statistical Abstracts, various issues, UNEB scores for P6 students 1999 and 2003, and World Bank staff estimates

Box 3.1 Estimating The Number of Graduates with Desired Level of Proficiency in Numeracy and Literacy

There is limited data available to calculate a time series of graduates obtaining the minimum desired level of proficiency. While we have average test scores for the years 2000-2006, we do not have the data separate for *public* and *private* school students. Moreover, we only have test scores for P3 and P6 students and not for P7 students which is what we would ideally want to measure the standards reached by completers. This is the most problematic data gap regarding the test score data because there are many more P6 students than there are P7 students. Most likely, the higher number of P6 students reflects the fact that the students who qualify for P7 are, on average, more likely to pass the test than the average P6 student.

Another problem is that the test score data are available for numeracy and literacy tests but not the combined performance on the two tests. Thus, only combined with some heroic assumptions can we calculate a time series of the key output of primary education. To estimate how many students have graduated from government schools since 1998/99 with the desired level of competency in English Literacy we assume:

- that P6 scores can be used to assess qualifications of P7 students.
- that the numbers reported in Table 3.8 are roughly the same for government and non-government schools (allowing us to use the figures to estimate competencies of government graduates).

Trends in Unit Costs and their Drivers: Average Unit Costs Based on Government Recurrent Spending Across Time and their Determinants

3.33 **As mentioned in the methodological annex, unit costs are convenient indicators to relate inputs to outputs.** Moreover, as discussed unit costs can be calculated in different ways, depending on which output is examined. Table 3.9 below calculates the three most important unit costs in primary education: the cost of enrolling a student, the cost of graduating a student, and the cost of graduating a student with a desired level of proficiency in literacy and numeracy. In a perfectly efficient system, there would be no drop-outs and no students repeating any classes. Moreover, every student passing through the system would acquire proficiency in literacy and numeracy. Thus, in a school system with seven grades to complete, it would cost exactly 7 times the unit cost of enrolling a student to graduate a student with the desired level of proficiency.

3.34 **In Uganda's case, the unit cost of a graduate is more than twice what it would be in a (hypothetical) perfectly efficient system.** In particular, it costs approximately $27 to enroll a child. Therefore, the aspirational target is that the cost of graduating a child with proficiency in literacy and numeracy should be $189 (7 x $27). In reality (as shown below), the cost is $2,424. Similarly, the cost of graduating a student from primary seven (ignoring the quality aspect) is $497 (i.e. more than twice what it could cost in a more efficient system).

Table 3.9. Main Indicators To Measure Internal Efficiency Of Government Recurrent Spending On Primary Education

		Estimates based on government recurrent expenditure (in 2005/06 constant prices)	
Intermediate outputs/outputs and outcomes	*Measure/Indicator*	*2000/01 (Ush)*	*2005/06 (Ush)*
Enrollment	Cost per primary student enrolled	40,470 ($23)	50,534 ($27)
Graduate	Cost per primary school graduate	817,944 ($464)	923,833 ($497)
Graduate with Minimum Achievement	Cost per primary school graduate demonstrating some minimum achievement level	2,370,853 ($1,345)	4,506,500 ($2,424)

Source: Authors' estimates based on Statistical Education Abstracts (various issues) and fiscal data

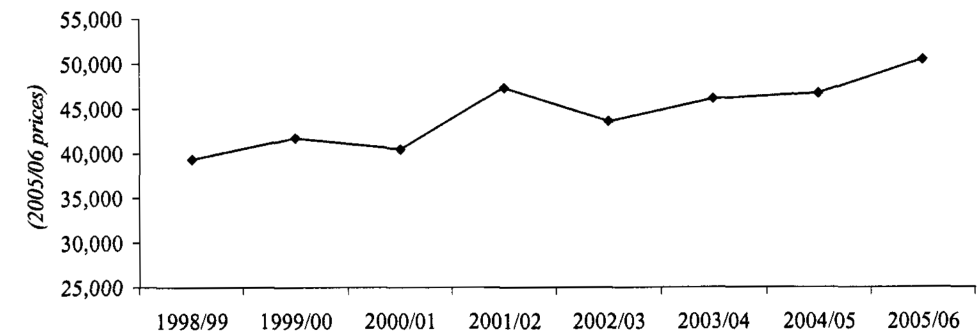
3.35 **Calculating unit costs is not the end point of an efficiency analysis; they merely help summarize the relationship between input costs and achievements in delivering the desired outputs.** The unit cost indicators are also useful in tracking efficiency over time, and comparing schools and districts with one another. It is the task of analysts to help decipher why unit costs have evolved (or are diverging across districts and schools).[55] Below, we demonstrate some uses of unit costs by examining their recent trends.

3.36 **Unit costs in Uganda are rising in real terms, and they have been rising faster per enrolled student than per graduate[56].** The unit cost estimates in Table 3.9 show that the unit cost of an enrolled primary student rose from Ush 40,470 to 50,534 (measured in constant 2005/06 price terms) between 2000/01 and 2005/06, a 25 and 17 percent increase in Ush and US dollar terms, respectively. The increase in the unit cost of a graduate has been more modest, growing by only 13 percent over the same period. The evolution of unit costs per enrolled student and per graduate is shown in figure 3.10 and figure 3.11, respectively. The increase in unit costs implies that improvements in enrollments have not kept up with spending increases.

[55] There appears to be insufficient attention in the EASPR to assessing unit cost trends and their underlying causes. The EASPR includes a discussion on "efficiency". But it does not mention what the current and past unit costs are, and how the proposed interventions will help reduce unit costs in the future. For instance, in the EASPR 2005/6 report, the efficiency sectors includes a list of specific efficiency-related interventions such as performance contracts, or double-shifting but does not mention how much each intervention will cost, save, and are expected to increase outputs. As a result, each proposal's impact on unit costs is unclear.

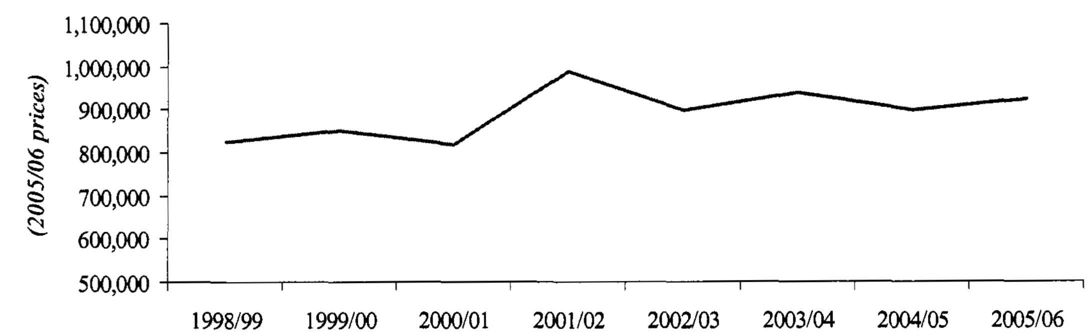
[56] See footnote 9.

Figure 3.10. Government Recurrent Spending per Student Enrolled in Government Primary School



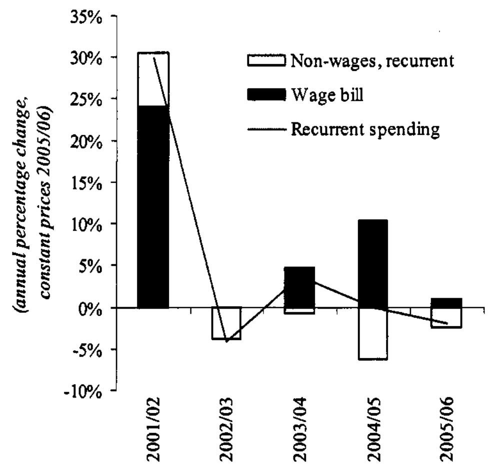
Source: Education Statistical Abstracts, fiscal data from MTEF and World Bank staff calculations

Figure 3.11. Government Recurrent Spending per Graduate from Government Primary School



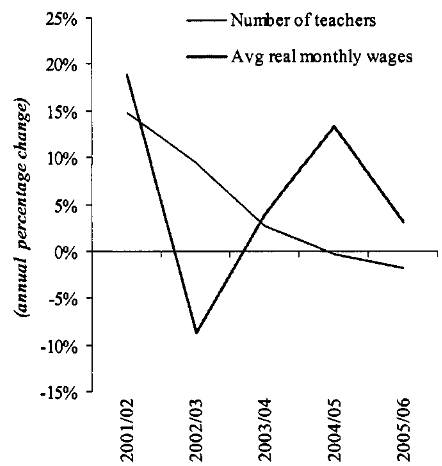
3.37 **We conclude that the most recent increases in the unit cost of an enrolled child stem from wage increases for teachers.** According to our estimates, the MoES raised the average nominal monthly salary of a teacher by 16 percent in 2005/06. With the number of teachers remaining roughly constant between 2004/05 and 2005/06 – although data sources differ on this – the increase in monthly wages was the main factor behind the observed increase in the wage bill. Figure 3.12 shows annual increases in the wage bill and non-wage recurrent spending. The figure shows the wage bill has increased in most years while non-wage spending has not. Figure 3.13 shows the annual increase in the two components explaining the wage bill: the number of teachers on the wage bill and the monthly salary they receive. The graph shows that, except for the year 2002/03, increases in the wage bill were mainly the result of increases in wages, rather than increases in the number of teachers.

Figure 3.12. Primary Education Recurrent Spending, Wage and Non-Wage (constant 2005/06 Prices, Annual Percentage Change)



Source: MTEF and World Bank staff calculations

Figure 3.13. Annual Percentage Change in Number of Primary Teachers and Average Real Monthly Salary of Primary Teachers



Source: EMIS, Ministry of Public Service, MTEF, and World Bank estimates

3.38 **Indeed, teachers' salaries in government primary schools are rising by more than enrollments, and by more than graduates.** Potentially, paying teachers more can result in more motivated and skilled teachers which can help increase total enrollments and graduates by reducing drop-outs and attracting more students to schools. However, whether this happens or not is an empirical question. From 2000/01 to 2005/06, the average monthly salary of a primary school teacher rose by more than 30 percent (measured in constant prices).[57] By comparison (as discussed above), enrollments and graduates from government primary schools rose by only 11 and 21 percent in the same period. It is, of course, possible that wage increases only translate into more motivated teachers with a time lag and that it is too soon to observe a potential positive impact of the wage increases on outputs.

[57] We estimate the average monthly salary was 145,000 (nominal) in 2000/01 and that it was 191,000 in 2005/06 (nominal). Converted into constant prices, the increase between 2000/01 and 2005/06 was 32 percent.

Box 3.2. A Word of Caution on Data Quality

The estimates of unit costs discussed above which are based on administrative data such as fiscal data and the EMIS need to be treated with some caution. There are several reasons why estimates based on these data could be misleading.

First, the fiscal data on primary education used could include more than funding for primary schools. As discussed above, historical fiscal data were not readily available - so an estimate of the break down into primary, secondary and other spending was produced by the authors of this PER ourselves. However, item-by-item analysis of the budget for one year, 2005/06, reveals that spending on primary education includes spending on Primary Teachers' Colleges. For 2005/06, we are able to subtract spending on Primary Teachers' Colleges (worth approximately 6 percent of recurrent spending on what we define as primary education) but data limitations prevent us from doing this for the other years. As a result, we maintained spending on Primary Teachers' Colleges as part of spending on primary education but note that these figures are overestimated by approximately 6 percent for 2005/06 and an unknown amount in 2000/01. To the extent that this share is constant over time, our analysis over time will correctly identify trends but will be incorrect in level terms.

Second, for reasons which are unclear, EMIS does not make an attempt to impute numbers for schools which fail to fill out the annual school census questionnaire. As a result, EMIS data can exhibit year-to-year variation which are purely the result of the coverage of schools varying. There are two possible solutions to this problem: ensure full coverage of schools (via incentives and consequences for non-submission of data), estimate key data such as enrollment and teacher staff numbers in missing schools based on a set of transparent assumptions (e.g. trends observed in the schools which submit data).

Third, administrative data on enrollments and number of graduates can suffer from a number of biases (see, for instance, Ritva and Svensson 1996) and this seems to be the case today. Given that the capitation grant is tied to enrollment numbers, head masters and district municipal officers have an incentive to inflate student numbers to increase funding. Moreover, the EMIS is still a relatively new data management tool. To cement its credibility over the coming years, it needs to regularly demonstrate that its data can be verified when cross-checked against independent surveys, including school surveys.

Finally, there is information which is not collected administratively which we are interested in from the perspective of constructing reliable unit cost figures. These include:

- Information on household contributions in the form of school fees (although abolished on paper, we found that parents still pay in many places), meal fees, etc
- Salary information on non-payroll teachers and non-teaching staff, both of which are not included in the EMIS
- Information on the amount of the capitation grant which actually reaches the schools (more on this later)
- Teacher absenteeism (more on this later, as well)

C. TECHNICAL EFFICIENCY IN PRIMARY EDUCATION'S RECURRENT BUDGET

Leakages from center to school and leakages within school: quantifying waste and the need for behavior change to reduce it

3.39 **This section shows that waste in the form of UPE capitation grant leakages, ghosts, questionable expenditures and teacher absenteeism amounts to 70 billion Ush (21 percent of the recurrent budget in 2005/06) and is, as a result, most likely the largest efficiency problem in primary education.** As discussed in chapter 2, a technical efficiency analysis begins by examining leakages from the center to schools and leakages within schools, and only thereafter examines whether resources that are not wasted are used productively.

3.40 **The good news for Uganda is that there are reasons to believe past efforts have improved the situation considerably since the first half of the 1990s.** Uganda has shown it is possible to make quite dramatic improvements if leadership sends the right message. First, Reinnikka and Svensson's 2002 follow-up PETS survey showed that the government's aggressive response to the first PETS findings (e.g. by publishing monthly inter-governmental transfers of public funds in the main newspapers, requiring details of transfers to primary education to be posted on notice boards in each school and district centre, etc—see Reinnikka and Svensson (2003) for more details)[58]) had already had a significant impact five years later: leakages were "reduced from 80 percent in 1995 to less than 20 percent in 2001" (p. 1). Second, in 2001, with donor support, the government introduced an Educational Management Information System (EMIS) which has enabled the government to monitor enrollment rates and teacher numbers more closely. Third, with all likelihood, the increased media attention on teacher absenteeism and leakages has made it more difficult to get away with corruption and mismanagement.

3.41 **We suggest that a behavior change is needed to further tackle waste in public services in Uganda[59].** On the plus side, there is openness and commitment to uncovering problems in Uganda. But the end-results do not seem to be commensurate with either the willingness of sectors to engage, nor with the donor resources and technical expertise made available as a consequence of this openness. To escape low income status as Asia's 'Tigers' have, a country as poor as Uganda should not tolerate the levels of waste which prevail.

3.42 **The change in behavior is not about renewing commitment to more studies, although the studies do need to be combined to give a better picture of waste.** Uganda has willingly participated in many–often pioneering – studies on tracking leakages, teacher absenteeism and classroom construction costs, exposing policy makers to potentially embarrassing findings. A summary of relevant primary education studies is set out in Annex 2, with some of the results are summarized in tables 3.10, 3.11 and 3.12. Combined, the different surveys show that between 20 and 40 percent[60] of government spending on primary education in the first half of the 1990s did not produce tangible inputs: i.e. it was wasted from an education perspective (see 3.12 for details).[61] Importantly, this money was wasted because of poor governance; that is, it was not wasted because of high unit costs of building classrooms, or employing too expensive or too many teachers, or having too many small schools, or having too small student-teacher ratios, or other commonly looked-at efficiency measures. Rather, weak institutional capacity provided an environment where individuals –teachers, district officials and contractors – could take payments for goods and services which they either did not deliver, or did not deliver to the standards for which they were paid.

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[58] Reinnikka and Svensson (2003): "The power of information: information from a newspaper campaign to reduce capture", World Bank Working Paper No. 3239.
[59] Not just in the education sector.
[60] allowing a margin of error around each survey result.
[61] This is a rough guesstimate, assuming 10-27% ghost teachers (assuming that these include unexcused absent teachers), 50-75% of non-salary funds not reaching schools (as indicated by the 1996 PETS), and assuming that approximately 5-20% of classrooms paid for either don't get built or collapse within a year.

Table 3.10. Share of Capitation Grant Received by Schools, According to PETS 1995 and 2001 Surveys (each observation is a school)

All schools	Mean	Median	St. dev.	Max	Min	Obs
1995	23.9	0	35.1	109.8	0	229
2001	81.8	82.3	24.6	177.5	9.0	217

Regions	Mean (1995)	Mean (2001)
Central	24.3	92.8
North	26.7	102.4
Northwest	11.2	90.3
West	24.0	71.6
Southwest	21.1	83.3
East	20.1	62.4
Northeast	36.0	73.4

Source: Reinnika and Svennson (2003)

Table 3.11. Primary School Teacher Absence, 2002-03[62]

	Absence rate (%)
Bangladesh	15
Ecuador	14
India	25
Indonesia	19
Peru	11
Papua New Guinea	15
Uganda	27
Zambia	17

Sources: Chaudhury, Hammer, Kremer, Muralidharan, and Rogers 2004 for most countries; NRI and World Bank 2003 for Papua New Guinea; Habyarimana, Das, Dercon, and Krishnan 2003 for Zambia
Note: Absent staff are fulltime teachers on current shift who were not found anywhere in the school at the time of an unannounced visit.

[62] Note: The averages of two visits masks even larger absentees: In Bangladesh, the average absentee rate was 15% but there were, in fact, 23.5% of the teachers who were absent during one of the two visits

Table 3.12. Estimates of Fiscal Impact of Leakages in Education in First Half of 1990s

	Recurrent allocation for education in fiscal years (1991 prices) USH (million) 1/	Number of teachers 1/	Enrollments (millions) 1/	Amount per student 2/	Capitation grant 3/	Salary component of recurrent expenditure 4/	Number of public classroom constructed	Cost per classroom 6/	Capital expenditure	Total public expenditure on education 7/
1991	19,202	78,259	3	3,143	7,983	11,219	500	11.4	5,700	24,902
1992	30,002	86,821	2	3,143	7,417	22,585	500	11.4	5,700	35,702
1993	24,569	91,905	3	3,143	8,391	16,178	500	11.4	5,700	30,269
1994	32,258	84,043	3	3,143	8,171	24,087	500	11.4	5,700	37,958

1/ From Ablo and Reinnikke (1998)
2/ Weighted average of USH 2500 per child in P1-P4 and USH 4000 per child in P5-P7: 4/7*2500+3/7*4000
3/ Amount per student multiplied by the number of students enrolled
4/ Recurrent expenditures minus capitation grant
5/ Estimate of the number of public classrooms built in 1991-1995 (author's guesstimate)
6/ Assume $6000, or 11.4 million USH
7/ Recurrent plus capital expenditure (ignoring other capital expenditure than building classrooms)

	Worst case scenario				Best case scenario			
	% ghost teacher and teacher absentee	PETS leakages of non-salary funds	faul rate in construction	Wasted money 8/	% ghost teacher and teacher absentee	PETS leakages of non-salary funds	faul rate in construction	Wasted money 8/
1991	27%	75%	20%	41%	10%	50%	5%	22%
1992	27%	75%	20%	36%	10%	50%	5%	18%
1993	27%	75%	20%	39%	10%	50%	5%	20%
1994	27%	75%	20%	36%	10%	50%	5%	18%

8/ (Share of ghost teachers*salary expenditure + PETS leakage*capitation amount + faul rate * capital expenditure)/(public expenditure)

3.43 **Instead what is needed is a strong commitment to eliminate waste.** There are two aspects to this. First Uganda's political leadership should lead by example and signal that waste and leakage will not be tolerated. Second, the Ministry of Education should commence implementing pilot programs to improve teacher absenteeism. Third, the annual performance reviews should focus sufficiently on the core problem of absenteeism. The EASPR 2005/06 does not mention teacher absenteeism nor governance weaknesses at all.

3.44 The next section discusses how to quantify waste, track progress over time, and introduce policy measures to reduce it. First we start by discussing the current extent of the problem.

Current status on leakages in education (recurrent expenditure only)

3.45 **Capitation grant leakages.** The results from the recently completed PETS show a small improvement in reducing UPE capitation grant leakages (USAID 2006). The survey shows that approximately 16 percent of the UPE capitation grant in 2005/06 was diverted for other purposes. Similarly, showing significant improvements compared with the results from 2001, it took an average of 2 months (down from more than 5 months in 2001) from the time UPE capitation grants were released from the central government, to the time they reached schools.[63]

3.46 **Teacher absenteeism.** Given the previous magnitude of the problem of teacher absenteeism, unannounced school visits were conducted to 160 government and non-government schools during November 2006 as part foof the education sector wide efficiency study (see

[63] USAID (2006): "UPE capitation grant tracking study, FY 2005-06", p. 47

Sondergaard and Winkler, 2007). The schools were randomly selected across three regions (Western, Eastern and Central) and six districts (see annex 2 and Habyarimana 2007 for a further discussion of the sampling strategy). The survey also highlighted that many of the teacher who are present are not teaching. In particular, nearly *one third* of teachers were "*outside of the classroom*" when they were found resulting in around 20 percent of the classrooms taking place without a teacher and 10-25 percent of the classes having already gone home by the time enumerators visited the school (see Figure 3.15).[64]

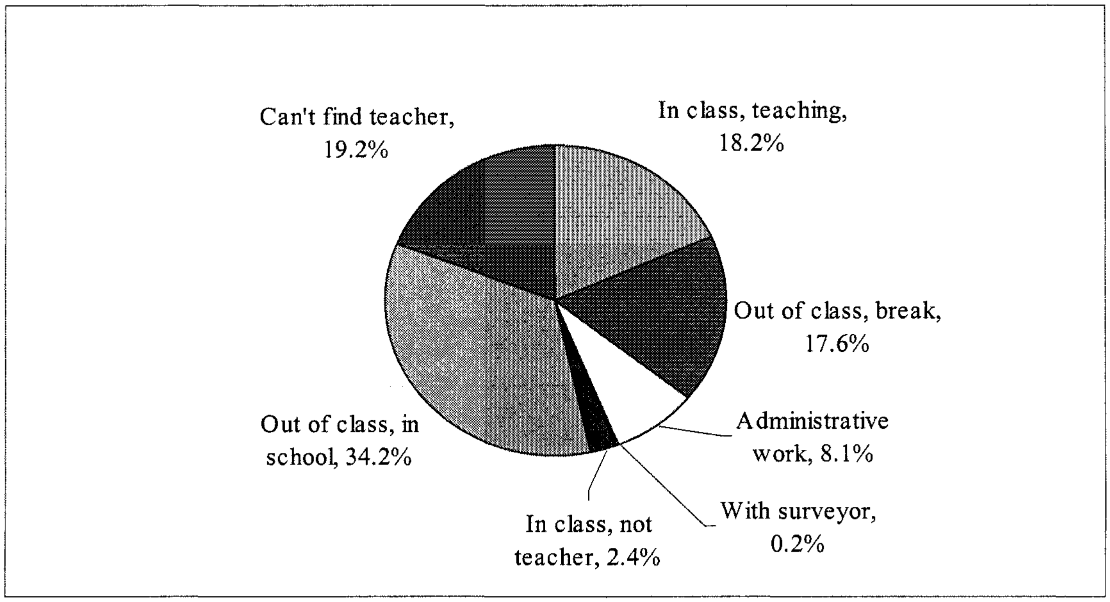
- **Measuring teacher absenteeism is tricky** for the following reasons:
  - Only unannounced visits to randomly selected schools are likely to produce a reliable measure of a typical day at a typical school.
  - Head teachers have incentives to misrepresent absence, since absenteeism could reflect poorly on their capacity to manage the school.
  - Enumerators need to distinguish between teachers who are supposed to be at work, and teachers who are absent because they are on different shifts.
- **We applied a standard methodology and questionnaire**. This was used by the multi-country study carried out by Chaudhury et al (2004). The details of the approach are summarized in Box 5.

- **Unannounced school visits confirmed teacher absenteeism to be an endemic problem in Ugandan primary schools although the problem has been reduced since 2002/03**. The survey revealed that nearly 20 percent of all teachers could not be found in the school at the time of enumerator visits (see Figure 3.14). There is some indication that average absenteeism might have declined from the 2002/03 survey where average absenteeism stood at 27 percent. To confirm this decline, a follow-up survey will be conducted in June 2007. However, we do not expect the results to be too different from those presented here which are based on a sample of almost 2,000 teachers in the 160 schools visited.

- **The survey also showed that head teachers are more likely to be absent than regular teachers**. As table 3.13 shows, 27 percent of head teachers were absent while absenteeism rates ranged from 13-18 percent for senior, regular and other types of teachers. To the extent that head masters are meant to supervise junior teachers and set an example, their absence is particularly worrisome.

[64] The range corresponds to lower grades versus higher grades. This fraction is larger for the lowest grades since they typically only study half day.

Figure 3.14. Where is Teacher When/If You Find Him/Her?



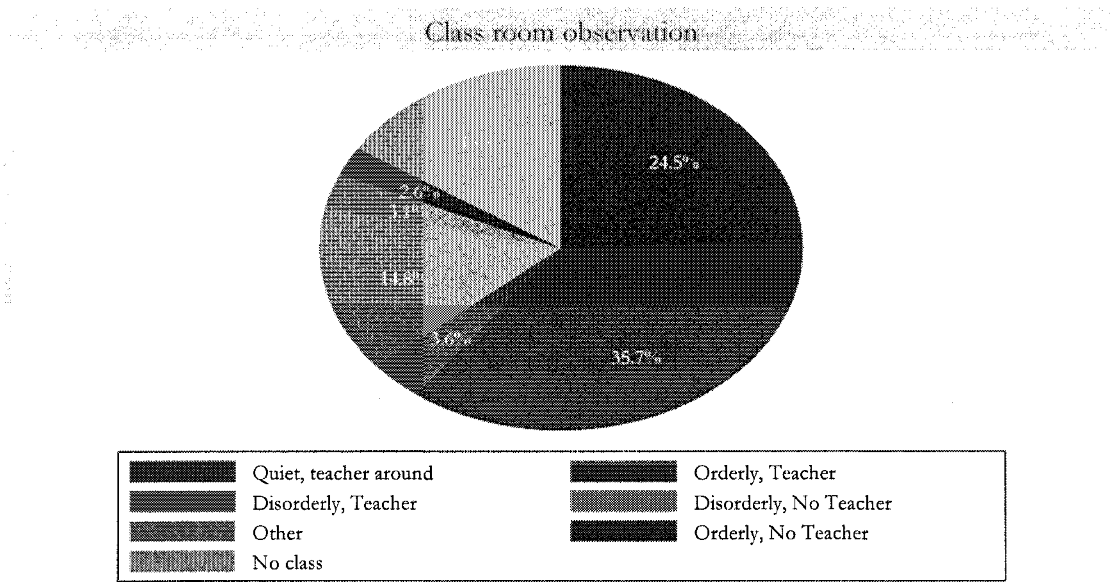
Source: Primary School Unit Cost Survey November 2006

Table 3.13. Teacher Absence Rates by Teacher Rank

Teacher Rank	Absence Rates
Head Teacher	0.27
	(0.04)
Senior Teachers	0.16
	(0.02)
Regular Teachers	0.18
	(0.01)
Others	0.13
	(0.04)
Overall	0.19
	(0.01)

Source: Primary School Unit Cost Survey November 2006

Figure 3.15. Classroom Observation



Source: Uganda Unit Cost Study, 2006

Source: Primary School Unit Cost Survey November 2006

3.47 **Ghost teachers.** Ghost teachers are teachers who appear inappropriately on the payroll. Weak payroll information systems and fraud are two possible reasons for the appearance of ghosts but there are others. Ghosts are normally identified during "payroll clean-up exercises" in which either every teacher on the payroll is examined and verified or a random sample examined. As mentioned above, a 1993 payroll clean-up exercise revealed that 20 percent of primary teachers were ghosts. In 2005/06, another payroll clean-up exercise found that ghost teachers had been drastically reduced but still amounted to 4 percent of primary teachers.

Box 3.3. Measuring Teacher Absenteeism (from Habyarimana, 2007)

Teacher absence is measured in three ways. Firstly we ask the head teacher or the primary respondent (typically a deputy head teacher or senior teacher) about the attendance of up to 20 teachers per school. We call this the *head teacher spot* absence report. Secondly we ask the teacher about the duration of absence for all teachers over the last 30 days. We call this the *head teacher absence duration* report. Finally, we use direct observation to assign absence status; the enumerator marks a teacher as present, if the enumerator can find that teacher during his/her visit during working hours. A teacher is absent if the enumerator cannot find this teacher within the school boundaries. All measures rely on the fact that each of the school visits is unannounced.

The head teacher reports of absence are unreliable for a number of reasons:

1. The head teacher/primary respondent has incentives to misrepresent the absence status of teachers since this reflects on his/her capacity to effectively manage the school.
2. The head teacher is unlikely to be aware of the absence status of all teachers at all times.
3. Recall bias is likely to be significant given a 30-day recall period and the fact that the head teacher must report a duration for every teacher on his roster.

Both of the head teacher based measures could be improved if there was an attendance register in which teacher attendance was accurately recorded. In general, it is not clear how accurately and/or if teacher attendance registers are used. Evidence from this survey indicates that head teachers in 2 schools reported that no such register existed. Furthermore, 55 teachers reported as present by the head teacher did not sign the register, while 50 teachers reported as absent signed the register. It is only coincidental that these two quantities offset each other. Evidence from the previous survey (Habyarimana, 2004), suggests that the bias associated with a reliance on attendance registers is on the order of 5 percentage points.

If absence durations over at 30-day recall period were reported with small/no biases, one could use the estimate as a measure of the probability of a teacher being absent. An unbiased measure of absence duration also allows us to determine the concentration of absence by teacher. In the absence of any recall bias, the probability that a teacher was absent is equivalent to the ratio of monthly absence duration to the total number of days in the month. As the result below demonstrates, absence durations from head teacher recall predict a much lower level of spot absence. According to this data and assuming a teaching month of 22 days, the spot absence rate suggested by head teacher recall is on the order of 10%, about half the size of the corresponding spot absence measure.

Given the shortcomings of both the head teacher reports of absence, we rely instead on the measure of absence that comes from direct observation of the teacher's status. This measure has a number of advantages over other measures that have been used in the literature. Firstly, it does not suffer from biases resulting either from recall or from mis-reporting. Secondly, it provides a measure that is of direct policy interest. Since the enumerator's arrival in the school is random, the estimate of directly-observed absence tells us the fraction of teachers one can expect to find in school during working hours. In addition, this strategy allows us to characterize the allocation of teacher effort for teachers that are present.

3.48 **Ghost schools.** A ghost school is a school with no students which nevertheless has teachers on the payroll and/or receives quarterly capitation grants. Again, weak reporting systems and/or fraud could imply that the closing of a primary school does not get recorded. To our knowledge, there has been no systematic study of this potential problem in Uganda. However, there are two reasons to suspect that the problem does exist. First, the magnitude of the other governance problems discussed above make it plausible. Second, although not specifically mentioning primary schools, the Auditor General's Report to Parliament 2005/06 reports discovering a closed down learning center which, nevertheless, continued to pay out salaries to its instructors. In our school survey, we cannot with certainty confirm the existence of ghost schools. However, 5 out 141 government schools which we selected from the list of schools in the EMIS (4 percent of our sample) could not be visited because they no longer existed, or had been upgraded to a secondary school. From a "ghost" perspective, the issue is whether any of these five schools have continued to receive funding as primary schools. We would need to conduct an additional "tracking survey" of the funds flowing to these schools around the time of their change of status to establish whether this is the case.

3.49 **Questionable expenditure.** Every year, the Auditor General's Office prepares a financial audit to parliament (Auditor General's Report to Parliament on the Financial Accounts). The report is published within nine months of the completion of the fiscal year, usually at the end of March. In the report, the expenditure of every central government vote-holder (i.e. the ministries (votes 01-18), and the agencies, universities and hospitals (votes 100-120) are examined from an auditor's perspective. The reports grade the audits in three categories, ranging from good to very problematic: "an unqualified opinion", "a qualified opinion, except for...", and "a qualified opinion, disclaimer". Given that the audits are prepared vote-by-vote, a full picture of sectors which span across several votes (such as, for instance, the education sector) can only be assessed by compiling different audits. In the case of education, however, the bulk of spending takes place via districts and urban areas (i.e. the votes 500-800+). Unfortunately, the audits of these votes are not included in the report and not consolidated anywhere. Therefore, we focus only on the audit of the Ministry of Education and Sports when examining the Auditor General's audits.

3.50 **The problems identified in the Auditor General's audits reports on MoES in 2004/05 span a wide range of severity and size but their total amounted to Ush 4.3 billion** (see table 3.14. One problem with the Auditor General's report is that it does not rank order the problems in terms of severity (i.e. nature of problem or size); it simply lists the problems. Another problem is that the report does not attempt to summarize the information to allow for comparisons over time and across votes. In this PER, we will not attempt to classify the severity of the problems identified but we will propose a way to summarize the information. Specifically, we propose simply adding up the problems and referring to the total as "questionable expenditures." This amount can be tracked over time and can be reported both as a total and as a share of total expenditure incurred by the particular vote. In this case, table 3.14 reports that questionable expenditures for vote 013 was Ush 4.3 billion in 2004/05 or 5.3 percent of total expenditure of vote 013 in that year (Ush 82 billion). It is important to note that the Auditor General's list of questionable expenditure which makes it into the published report is only a subset of the issues identified initially by the Auditor General's auditors. This initial list is presented for comments to the management of the vote-holder being reviewed. In many cases, this interaction between the auditor and the vote-holder results in additional information being supplied, and the identified potential problem being removed from the Auditor General's report. Thus, the items listed in table 3.14 and included in the published report are issues which could *not* be resolved after weeks of interactions between the auditors and management of the vote holder.

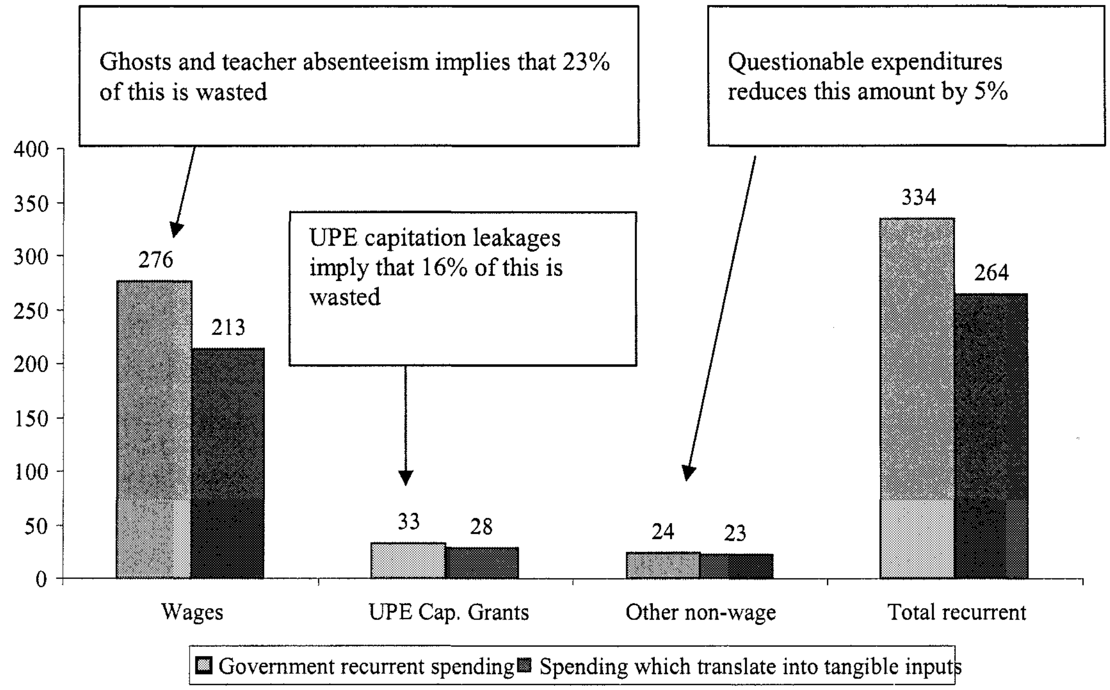
Table 3.14. Questionable Expenditures Identified in 2004/05 Auditor General Report on MoES

Non acknowledgement of receipt of taxes. Withholding tax remitted to a collecting authority (URA) but URA could not verify receipt of funds	1,510,839,225
Withholding taxes were not withheld from suppliers' payments for goods and services	77,479,906
Advances were not accounted for	1,661,052,135
No supporting documentation for funds spend on National Consultative Group workshops	100,333,730
Workshop expenses not accounted for	202,218,000
Funds withdrawn for closed down learning centers	43,200,000
Unaccounted for advances	238,378,000
Non teaching staff on the payroll of a closed down Community Polytechnics Instructors College	7,144,176
Irregular payment of salaries at Paramedical school	39,600,000
Funds not accounted for	229,863,840
Doubtful electricity bill checks	210,974,934
Sum of questionable expenditure	**4,321,083,946**
Total expenditure of MoES (vote 13) in 2004/05	**82,040,000,000**
Questionable expenditure as a share of total expenditure	*5.3 percent*

Source: Auditor General's Report to Parliament for FY 2004/05.

3.51 **Combining the facts on waste with a set of assumptions discussed in annex 2B, we estimate that 21 percent of primary education recurrent expenditure was wasted in 2005/06.** This amounts to Ush 70 billion out of Ush 334 billion. Teacher absenteeism and – to a lesser extent – ghost workers account for the bulk of the waste. The fact that on average 24,510 (19 percent of 129,000) of the teachers on the payroll are not showing up for work on a given day results in almost Ush 53 billion being wasted per year (19 percent of the wage bill of Ush 276 billion). By comparison, leakages in the capitation grant account for only 5 billion of the loss. Waste resulting from the various sources is summarized in figure 3.16 and table 3.15.

Figure 3.16. Summarizing Waste in the Primary Education Sub-Sector (Recurrent Expenditure only)



Source: World Bank estimates

Table 3.15. Details of Waste Calculations (2005/06 data)

	Wages	UPE capitation grant	Other non-wage recurrent	Total recurrent	Total waste	% waste
Government releases	276	33	24	333		
Total waste	63	5	1		70	*21%*
of which: Ghost teachers	11					4%
of which: Teacher absenteeism	52					19%
of which: Questionable expenditure			1			5%
of which: UPE capitation grant leakages		5				16%
Government spending, excl. waste	213	28	23		263	

1/ Implied waste out of total recurrent (i.e. 70/333)
Source: World Bank estimates

3.52 Having established that 21 percent of government resources are not being translated into tangible inputs which can help increase outputs, in the next section we turn to a more standard efficiency analysis and ask the question.

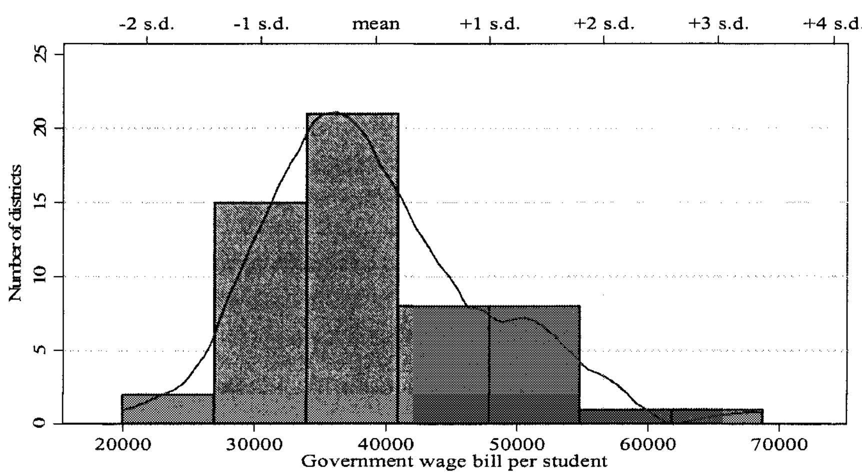
Are resources that are not wasted used productively?

3.53 **We provide four examples of resource allocations which do not seem to be efficient: (i) teacher deployment across districts; (ii) teacher deployment within schools; (iii) wage vs. non-wage spending at schools; and (iv) the mix of teacher and classrooms across schools.** As discussed in methodological discussion in chapter 2, ideally, this step in the efficiency analysis should be an analysis of how inputs can best be combined to achieve the maximum amounts of outputs. However, several data constraints (discussed in chapter 2) made this third step impossible to conduct for this PER. Instead, we relied on international evidence suggesting how resources are best allocated (e.g. on class sizes in lower versus higher grades) and use this evidence to document potential problem areas of resource allocation in the Ugandan school system.

Teacher deployment across districts

3.54 **Teachers are not deployed efficiently across Districts**. Government primary education spending per pupil often varies across districts or schools due to differences in cost structure (especially, due to higher costs of delivering schooling in remote, sparsely populated regions) and differences in poverty levels with education spending being higher to compensate for the effects of poverty. Since the Ugandan EMIS does not capture all spending at the school level and since salaries represent approximately sixty-five percent of all central government funding of primary education, this analysis uses the personnel wage bill as a proxy for total government funding. Nationally, the mean wage bill per student is 39,259 UShs, but there is considerable variation across districts and schools (see figure 3.17 below). For example, the top spending quintile of schools has a mean wage bill per student of 50,526 UShs, almost double the figure (26,585 UShs) for the bottom quintile of schools in the country[65]. It is appropriate to ask whether this variation in wage bill per student is the result of explicit MoES policy—perhaps to compensate for high costs in sparsely populated, rural areas—or is the result of policies and practices having nothing to do with the effective delivery of education to children.

Figure 3.17. Government Wage Bill Spending per Enrolled Student across Districts



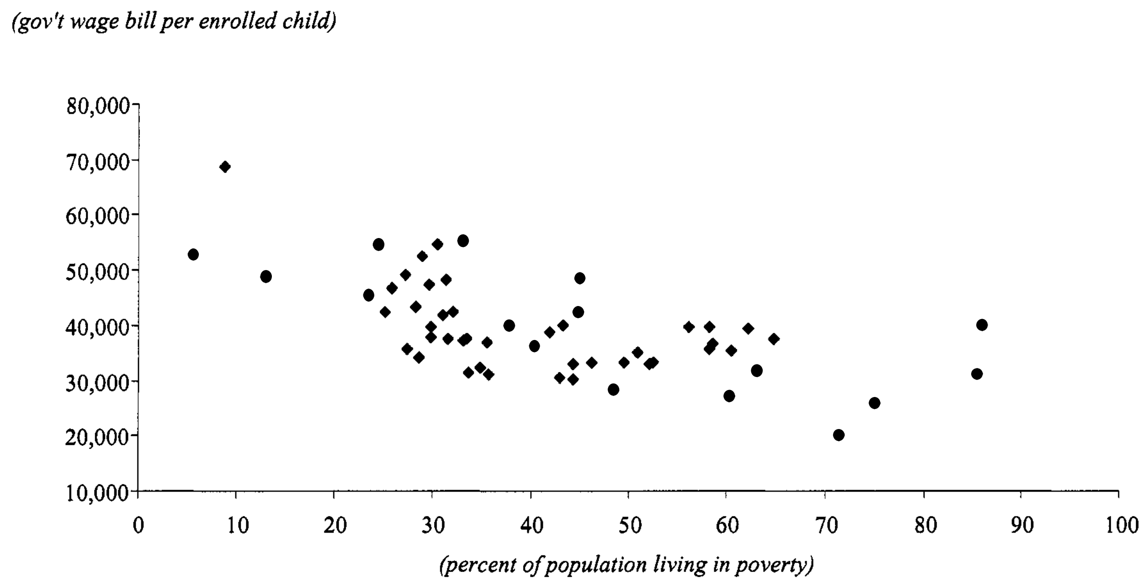
Source: EMIS (downloaded in October 2006)

[65] Calculations based on the 2004-05 EMIS.

3.55 **What could explain the wide variation in wage bill per student across districts?** Most education systems allocate teachers to schools based on the number of students enrolled in a school. If that were the case in Uganda, the distribution in Figure 3.17 would be much more tightly distributed around the mean. Perhaps teachers are allocated according to need as measured by poverty levels of districts. To investigate this possibility, the district wage bill per student is plotted against district poverty rates in Figure 3.18.

3.56 **The wage bill per student across districts in Uganda is in fact inversely, not directly, related to poverty**. Furthermore, regressing the wage bill per student against district poverty rates and the percent of district population that is rural (another common determinant of expenditures internationally) yields the finding that the percent rural population is not related to the wage bill, while the poverty rate is strongly negatively related to the wage bill. In short, primary education expenditures (as proxied by the teacher wage bill) in Uganda appear not to be distributed according to the types of variables often found in other countries.[66] Naturally, teacher deployment may be done on the basis of sound criteria not immediately evident to the analyst. If so, one would expect a positive relationship between the wage bill per student and measures of educational outcomes. However, the absence of school and district-based output indicators (such as test scores) made it impossible for us to examine this in details. Our preliminary analysis using very crude measure of educational outcomes such as the ratio of grade 5 students to grade 1 students (a proxy for survival rates) show no relationship between teacher deployment and educational outcomes (see figure 3.19 below).

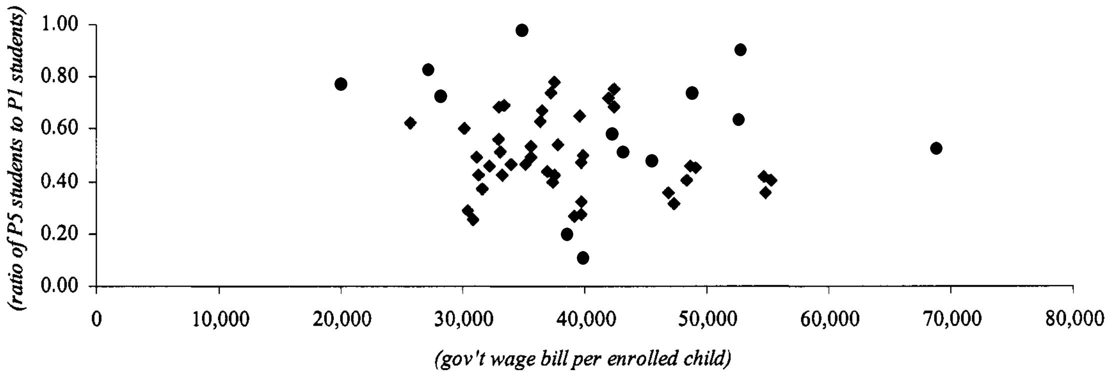
Figure 3.18. Government Spending on Wages per Student and Poverty Share by District



Source: EMIS (downloaded October 2006) and poverty rates from UNHS 2002/03

[66] For example, in Chile a local government's education expenditures are largely based on a capitation grant from the central government's education ministry. Children from poor households or children in rural schools are weighted more heavily than other students such that average expenditures per pupil are somewhat higher in local governments having high percentages of children from poor households and high percentages of rural population.

Figure 3.19. Ratio of P5 Students to P1 Students
(a high number is suggestive of a high survival rate from grade 1 to 5)



Source: EMIS (downloaded October 2006) and poverty rates from UNHS 2002/03

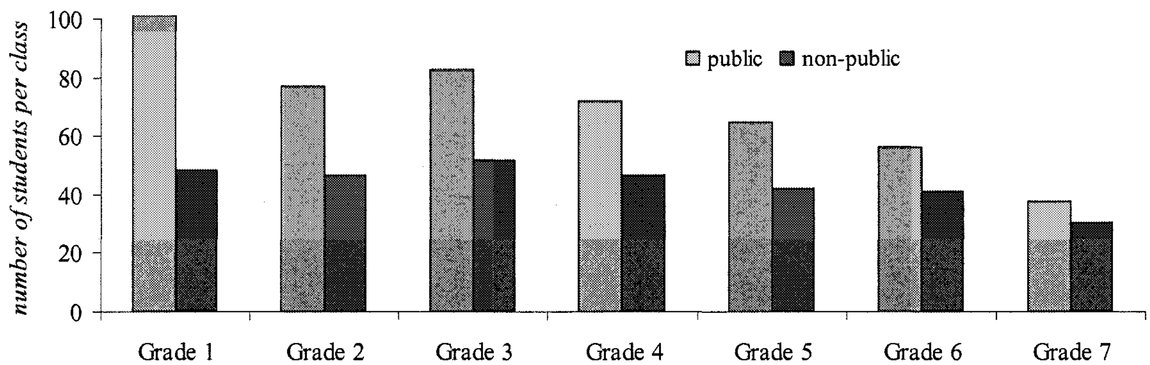
3.57 **The Teachers Service Commission (TSC) Guidelines for teacher deployment across districts are problemmatic.** Since the Central Government's Teachers Service Commission directly deploys teachers in the country, the distribution of teachers is the direct result of its actions and subsequent teacher transfers. The guidelines may be pulling against efficiency in two key ways:

1. First, the guidelines do not explicitly take into account how many classrooms are available at a particular school.
2. Second, the guidelines specifically allocate fewer teachers to the younger classes (1 teacher per 80 pupils in P1 and P2) than to bigger classes (55 pupils per teacher in P3-P7).

Teacher deployment within schools

3.58 **Within primary schools, resources are not being utilized efficiently to maximize learning outcomes**. Government schools allocate fewer resources (measured on a per student basis) to the early grades of primary school relative to the resources allocated to the later grades (see figure 3.20). Most educators would argue for the reverse: the lower grades should have smaller class sizes, and higher unit costs, than the higher grades within primary schools. Large class sizes in the early grades are more likely to contribute to higher repetition and dropout, especially among children from lower income homes. For private schools the differences across grades are relatively minor.

Figure 3.20. Average Class Size in Government and Non-Government Schools



Source: Primary School Unit Cost Survey November 2006

C.3.3 Wage vs. non-wage spending at schools

3.59 **Too few resources are allocated for non-wage expenditures in government schools.** As part of this PER, we undertook a survey to provide a detailed picture of school's expenditures and revenues. Such data are not collected in EMIS and, therefore, our survey provide the first detailed picture (in 15 years, as far as we could tell from our literature review) of how different primary schools spend their resources. Such information is critical for an efficiency analysis and the lack of regularly collecting such data represents an important weakness in Uganda. Box 3.4 provides a summary of some of the broader uses of the data and how it relates to administrative data already collected. Among other things, the survey showed that government schools spend half of what private schools do (measured as a share of total per student expenditure) on non-wages expenditure, including scholastic materials (see table 3.16 below).[67]

Table 3.16. Composition of spending in government and non-government schools

	Total recurrent expenditure	Total wages	Total non-wage recurrent	No. school surveyed
Government	100%	83%	17%	130
Private	100%	65%	35%	19
Community	100%	69%	31%	11

Source: Primary School Unit Cost Survey, November 2006

3.60 **The survey also showed that government schools' non-wage budget is lower because public teachers are paid significantly more than private teachers.** In particular, teachers in government schools earn about 60 percent more on average than a teacher in a private school according to our school survey of 160 private and government schools (see table 3.17). On

[67] To allow for comparisons between government and private schools, our estimate of a government's non-wage expenditure include an estimated value of the resources received from the central government (e.g. textbooks). Similarly, to allow for comparisons of the wage bill, the wages associated with payroll teachers (who are not physically paid at the school) are included in the school's wage cost.

average, a teacher in a government school makes Ush 209,534 per month. By contrast, the average teacher in a private school makes Ush 131,400 per month.[68]

Table 3.17. Average monthly salary of primary teachers, by type of schools

	Avg. monthly salary
Government	209,534
Private	131,400
Government aided (private and NGO)	215,698
Total	200,294

Source: School survey of 160 schools, conducted in Nov 2006

3.61 **As a result of the lower teacher salaries they pay, private schools can afford to hire more teachers and have lower student teacher ratios than government-run schools.** In fact, in our school survey (discussed below), we find that private schools, on average, have 25 students per teacher compared to 50 in government schools. Given that their monthly salaries are 60 percent lower than government salaries, private schools manage to have twice as many teachers per student as a government school but their wage bill per student is only 25 percent higher than in government schools.

3.62 **This finding raises some questions which are worth reflecting on from an efficiency perspective**:

- Why are teachers on the government payroll being paid almost 60 percent more per month than what their counterparts in the private sector earn?
- Which of the two options – which cost the same – is likely to have the biggest impact on learning outcomes:
  a. Deploy ten more teachers to a particular school at private sector costs and qualifications; or
  b. Deploy six more teachers to a particular school at public sector costs and qualifications.

3.63 **The large difference in monthly salary between a public and private teacher suggests that Uganda should evaluate whether learning outcomes would be better served by expanding the teacher force in public or private schools.** This report cannot provide answers to this question; we merely note that large salary differences suggest these questions should be further explored. One way to explore them would be for MoES to conduct an impact evaluation in a randomly selected district and examine whether one strategy produces more education outputs for the same amount of money.

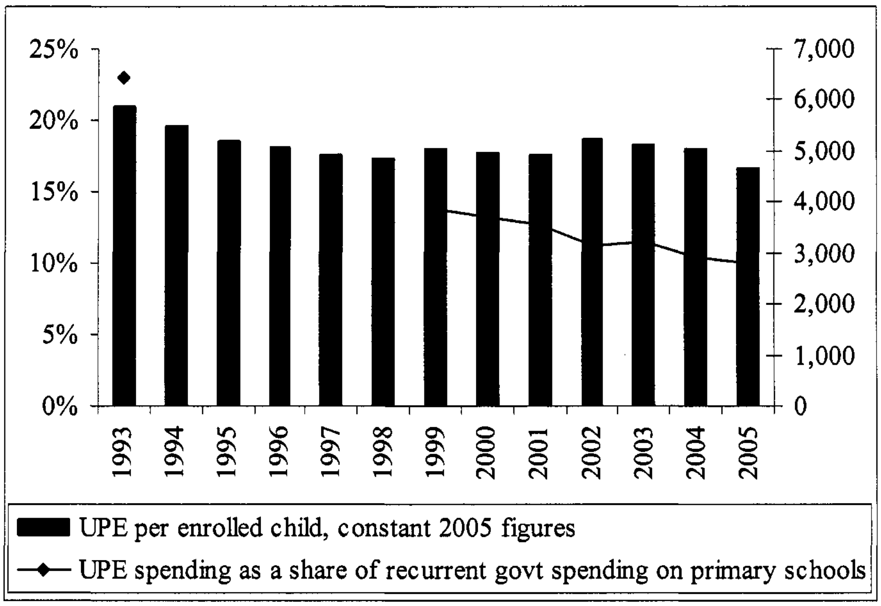
3.64 **The difference in non-wage spending by government and private schools raises an important question from an efficiency perspective**:

[68] One caveat to this result is that the private schools sampled were predominantly rural. Teachers' salaries in urban private schools may more closely resemble (or exceed) the average government salary.

- Are private schools' spending patterns more appropriate to achieve better learning outcomes than government-run schools? That is, private schools seem to be hiring cheaper teachers and spending more on learning materials. In the process, they may be hiring less qualified teachers (although we don't know) but they are achieving lower student-teacher ratios and equipping students with more learning materials than government schools.

3.65 **The most important source of funds to purchase scholastic materials for government school has been declining in importance**. Government schools receive a monthly transfer (UPE capitation grant) from the central government (via districts). It consists of a flat amount per school (100,000 Ush per month) plus a per student amount which is currently around 3,000 but is yearly adjusted at the MoES to reflect available resources. This transfer provides the bulk of the resources schools use to purchase scholastic materials, and conduct small repairs. As figure 3.21 below shows, the grant has been steadily declining in real terms and as a share of total recurrent spending since the early 1990s.

Figure 3.21. UPE Capitation Grant in Constant (2005) Prices And as a Share of Government Recurrent Spending on Primary Schools



Source: Ritva and Svensson for 1993-1995, UPE capitation grant study (USAID for 2005 figure and the the amounts between 1995 and 2005 have been linearly interpolated in nominal terms and converted into constant prices using CPI inflation). Budget figures are from MTEF and own calculations.

3.66 This raises a number of questions worth reflecting on:

- Given that the UPE capitation grant only account for less than 10 percent of total primary education expenditure, does it make sense to still devote so much energy on "UPE tracking studies" (last year, alone, two such studies were conducted);
- With the UPE capitation grant declining, do schools have enough resources available to purchase materials for students and teachers? As mentioned, we found that the government schools we visited spent significantly less on such materials compared to private schools.

- Should the government spent less money on hiring and training teachers (for the next 2-3 years) and, instead, increase the capitation grant?

Table 3.18. Did Schools Receive the Amount of UPE Capitation Grant Which They Expected?

	Luwero	Mayuge	Mukono	Ntungamo	Tororo	Total
Received more than they expected	50%	59%	56%	67%	60%	53%
Received less than they expected	45%	36%	44%	33%	40%	45%
Received exactly what they expected	5%	5%	0%	0%	0%	0%

Source: Unit Cost Survey (Nov 2006)

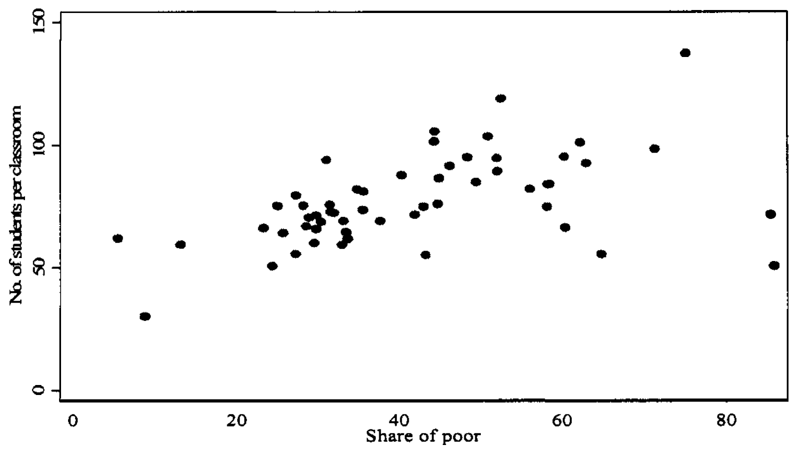
3.67 **Our school survey also revealed a lot of confusion surrounding what exactly a school is entitled to.** Only three out of the 130 government schools visited received exactly what they expected (see table 3.18). More than half of the schools received more than they expected. However, this was mostly the result of schools having very low expectations to the central government: 40 percent of the 130 schools expected to receive 1000 or less per pupil, despite government regulations specifying that schools are entitled to around 5,000 per pupil. Given that Reinikka and Svensson showed that understanding about entitlements led to lower leakages of UPE in the past, two key questions for policy makers to reflect on here are:

- Has awareness of entitlements declined, and if so, how come?
- How can stakeholders (MoES, districts, parents and schools) get a clearer understanding on what entitlements are?

The mix of teacher and classrooms across schools

3.68 **Lower cost public schools use their classrooms more intensively.** There are approximately 11,300 government-run schools in Uganda and the availability of classrooms varies a lot across these schools and the districts they are located in. As figure 3.23 shows, the wealthier districts appear to have more classroom space than poorer districts. Approximately half of Uganda's primary schools (5,628 schools) appear to use their scarce resources very well. In these schools, the number of streams created (averaging 9.5) exceeds both the average number of classrooms (5.1) and the number of teachers available (8.9). The fact that there are more streams created than classrooms available suggests that some classes are taught outside and/or these schools are employing double shifts. The fact that there are more streams taught than teachers available strongly suggests that double shifts are being used, with some teachers teaching more than one stream. Of all the schools in Uganda, these schools do the most with their scarce resources: the number of students per stream is "only" 57 compared to more than 70 for the rest of the schools.

Figure 3.22. No of Students per Classroom and Share of Poor, by District



Source: EMIS (downloaded in October 2006)

3.69 **Higher cost public schools have more classrooms and more teachers than streams.** There is a smaller group of schools (1,530 schools) where teacher and classroom utilization is very poor in comparison. In these schools, the number of classrooms (averaging 11.1 per school) vastly exceeds the number of streams (8.8). That is, at any given time, 2 classrooms stand idle in these schools. Equally troubling, on average, this group of schools has more than 14 teachers assigned to the school. With only around 9 classes in session at any given time, this possibly implies that 5 teachers are sitting idle. Alternatively, the additional teachers could be co-teaching (as a team) but with existing data we have been unable to check this. It is worth investigating why more streams are not created in the schools where both teachers and classrooms are amply available. With almost 90 students per stream in this group of schools, there is clearly a need to create more streams. By region, there appear to be more schools utilizing their resources poorly in the Western and Eastern regions, the regions.

Figure 3.23. Schools that Utilize Resources Well

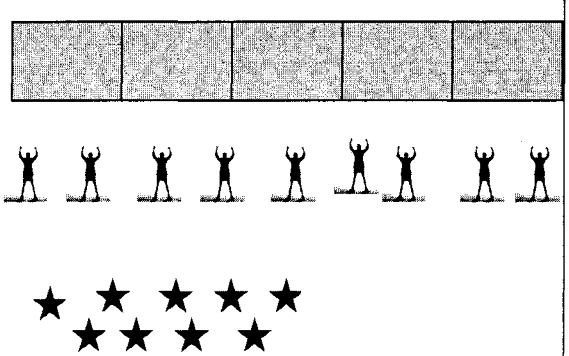
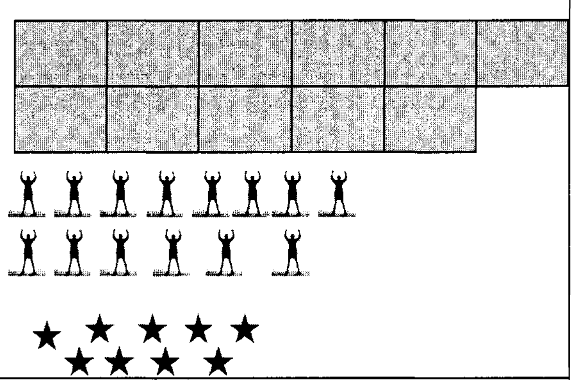


Figure 3.24. Schools that Do Not Utilize Resources Well



= classroom, = teacher, = stream

Source: EMIS (downloaded October 2006) and World Bank staff calculations

3.70 **What does this imply for policy? Uganda should focus more on streams, and should consider double shifting.** The analysis of teachers, classrooms and streams shows that schools with the fewest classrooms and teachers achieve the best result in terms of how many students there are in each stream. Second, it shows that looking at the classic budgeting norms of student-teacher ratios or student-classroom ratios can give a very misleading impression of efficiency in the primary sub-sector. The important variable to focus on (to gauge how many students are actually being taught by a teacher in a class setting) is the student-stream ratio. On average, there are 65 students in each stream in public schools in Uganda. Third, the classroom-teacher-stream analysis suggests it is worth reflecting further before deciding whether classroom construction or hiring more teachers are priority constraints. Large cost savings could apparently be generated by introducing double shifts in schools which lack classrooms. The trade off is between shifts on the one hand and hiring more teachers and building more classes on the other.

Box 3.4. How Reliable Are Enrollment And Teacher Numbers in the EMIS?

A school survey can be used to examine how reliable EMIS data is. Unfortunately, the school survey in this PER can only partly answer this question because our initial school survey was conducted in Nov 2006 data and EMIS data for 2005 only was available at the time of this analysis. This EMIS 2005 data was probably obtained from surveys filled out by headmasters back in the fall of 2004, resulting in a two year gap between the survey data and the EMIS.

Notwithstanding the difference in time when the data was collected, EMIS' enrollment figures appear to be overestimating actual enrollments. This is an important observation because UPE grants are tied to enrollments (i.e. a school's UPE entitlement is based on a flat amount (900,000) and a "per capita" amount (4,480 in 2006/07). In the sample of government schools visited, enrollments were 5 percent lower (in November 2006) than what was recorded in EMIS (based on fall 2004 figures). This implies that these schools could submit higher claims for UPE grants than they would otherwise be entitled for.

The survey also reveals that student-teacher ratios are much healthier than suggested by EMIS data. In government schools, the average student-teacher ratio was 50.4 compared to the EMIS estimate at almost 63. This lower ratio is the result of the lower enrollments found in schools, as well as teacher numbers which were almost 20 percent higher than what was recorded in EMIS. There are other indications that the EMIS data base is not good at accurately capturing the number of teachers employed. Cross-checking the total number of teachers recorded as being on "government payroll" in EMIS (113,967 in the 2005 EMIS) with the actual number of teachers on the payroll from the Ministry of Public Service (122,686 in 2004/05), reveals that the EMIS data base' number underestimates payrolls by at least 8 percent. It is unclear what is behind this underestimating.

Comparing enrollment and no. of teachers in EMIS to information collected in school survey

	Enrollment		No. teachers		No. of	Student-teacher ratio	
	Survey	EMIS	Survey	EMIS	school	Survey	EMIS
Government	77,897	81,024	1,547	1,291	130	50.4	62.8
Private	5,051	5,750	200	213	19	25.3	27.0
Government aided (private and NGO)	4,080	5,404	105	81	7	38.9	66.7
Total	**87,028**	**92,178**	**1,852**	**1,585**	**156**	**47.0**	**58.2**

Source: Survey was conducted in Nov 2006. EMIS data is the 2004/05 data. Thus, the EMIS information may stem from the annual census filled out by headmasters back in the fall of 2004.

Although differences are not small, as a starting point and for regular monitoring, the EMIS appear to be in reasonably good shape in Uganda. This strength, however, does not eliminate the need to conduct regular school surveys on costs and expenditure. Such surveys provide invaluable information which complements the EMIS and allow EMIS' data to be cross-checked.

Source: EMIS (downloaded October 2006) and Unit Cost Survey Nov. 2006

Box 3.5. Double-Shifting

Double-shifting is controversial. Rather than recommending Uganda to proceed one way or the other, we merely provide cost estimates of the trade-offs involved. Specifically, we calculate the savings generated from proceeding along one possible route of introducing double shifts.

Ultimately, considering the merits of expanding double-shifting rests on more than just economics. Uganda's education sector has scarce resources to continuing improving access and further improving quality. These scarce resources are more than just the annual budget allocation; they include the fact that past investments have trained teachers and built classrooms. Deciding to expand double-shifting is a decision to fully utilize these past investments to ensure that more students can achieve a high quality education.

For simplicity of the analysis, suppose every school in Uganda was converted into two schools: a lower primary school for P1-P3-graders starting at 7:30am and ending at noon (4 1/2 hours of teaching time), and an "upper primary school" for P4-P7 graders could start at noon and end at 5:30pm (5 ½ hours of teaching time).

What would be achieved by doing this?

- Total enrollment in the each of the schools ("lower primary" versus "upper primary" would drop to 3.7 and 2.9 million from a total of 6.6 million in the "combined" (single-shift) primary system. Thus, overnight, classroom provision would need to be tailored for a maximum of 3.7 million students (as opposed to 6.6 million).
- With schools on average having 7 classrooms available, a total of 7 streams (three P1 streams and two P2 to P3 streams) could be created in an average school compared to the average of 1.2 streams. Thus, on average, seven out of the ten teachers available in a school would need to work double shifts.
- The number of students per stream would instantaneously drop from 65.5 to 36.

Thus, with all likelihood, the learning environment will be improved which is likely to result in more students being able to graduate with the desired level of proficiency in numeracy and literacy. Moreover, with a closer interaction with teachers, both drop-out and repetition rates can be expected to drop.

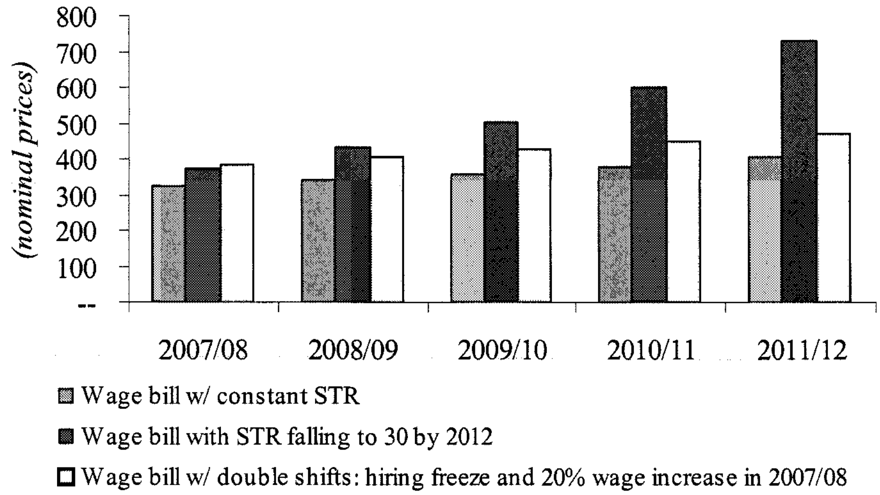
How much would this cost? Let's first look at the savings involved in expanding double shifts.

- All planned classroom construction for primary could be postponed for, say, five years (say, 10,000 classrooms over the next five years at a cost of around 100-150 bn Ush). The annual development budget could be cut in half and focus on rehabilitation, building latrines and procuring desks and chairs.
- No new teachers would be needed to be hired for the next five years. You could expand the teaching work force by gradually convincing more and more teachers to teach double shift. The saving involved in freezing new hires depends on how ambitions teacher hiring plans the Ministry has. To merely keep STR constant at around 53 would involve hiring almost 5,500 new teachers over the next five years (at a cost of 16 bn). However, to reduce STR to 30 over a five year horizon (where learning can start being effective) would involve hiring 100,000 additional teachers between now and 2012, raising the total number of teachers on the payroll to 235,000. This would increase the teacher wage bill from its current level at around 280 bn per year to 406 bn Ush. Thus, to match the student-teacher ratios achieve under double shifts would probably cost an additional 126 bn.

Introducing double shifts would entail costs as well:

- To teach two shifts, the existing teachers on the payroll would need to be compensated. They are currently well-paid from an international perspective at around 3 times GDP per capita in Uganda. Still, they would need additional compensation to accept the longer work-hours.
- Figure 3.25 shows one simulation of costs where a teacher agreeing to work two shifts receives a wage increase of 20 percent (approx. $25 extra per month). It is assumed that all teachers accept the pay increase and work two shifts. The figure shows that already by the second year, the wage bill associated with double shifts would be lower than trying to hire additional teachers to bring down student teacher ratios to 30 by 2012.

Figure 3.25. Comparing the Wage Costs of Introducing Double Shifts



It is difficult to quantify the quality and access improvements which could instantaneously be afforded by introducing double shifting but they are likely to be big. In the example above, double shifting would overnight reduce the amount of students in the classroom, allowing every student a closer interactions with teachers.

D. CONCLUSIONS AND RECOMMENDATIONS

3.71 The primary sector has achieved impressive gains in enrollments in the past decade. But, as discussed above, challenges remain in terms of ensuring that students stay in school and graduate with competencies in numeracy and literacy. The sector has already shifted more of its focus to quality and number of graduates. This is an encouraging sign. This chapter was written to provide analytical underpinning of possible policy measures to support this move.

3.72 How do you place *even more* focus on efficiency and final outputs? It is easy to get lost in the many possible performance indicators one can list in primary education (e.g. survival rates, PLE pass rates, performance of particular districts, etc). However, some indicators have primary importance while others are merely indicators to help explain why the most important indicators has (or has not) performed as desired. The single most important indicator to focus on is the cost of delivering "the (absolute) number of graduates from primary 7 who have obtained the desired level of proficiency". Most other indicators (e.g. survival rates, specific district's performance, enrollment rates, etc) serve the purpose of explaining *why* this indicator has (or has not) performed as desired.

3.73 The Education Sector's Annual Performance Review (and Budget Framework Paper) could be strengthened by placing more emphasis on a discussion of the unit costs of graduates obtaining the desired level of proficiency, and discussing trends in absolutely numbers. Such discussion should be phrased in terms of what was achieved in terms of desired outputs (i.e. graduates) and how inputs (and their costs) helped (or didn't help) achieve those outputs. Presenting recent and projected data alongside their historical trends is a useful way to instill accountability of past performance.

3.74 Overall, the sector appears to be utilizing its scarce resources better than in the past and is on a declining trend in terms of waste. Our literary review showed that the last in-depth efficiency study was conducted 15 years ago which make it somewhat difficult to assess our findings against past performance. Overall, though, we find many examples of improvements in terms of efficiency and waste. Most noticeable, all our waste indicators (from ghosts to teacher absenteeism) are on a declining trend.

3.75 The challenge for the sector is to accelerate the positive steps already taken so that its ambitious goals can be made faster and for less (or the same) costs. This analysis presents the facts: which area is most need of policy interventions. In particular, we find the following problem areas (from an efficiency and waste perspective):

- Teacher absenteeism remains high: on a given day, almost 25,000 teachers paid by the Government of Uganda (or 19 percent of the teaching workforce) are likely to be absent from government primary schools.
- Absenteeism among head teachers is particularly high (27 percent)
- Less than 20 percent of paid teachers who *are* present at school are in a class room teaching.
- As a result, a large number of classes are either taught without a teacher (approximately 20 percent) or gets cancelled.
- Waste amounted to Ush 70 billion, or 21 percent of recurrent spending on primary education in 2005/06.
- Waste as a share of spending has been roughly constant for the past decade, suggesting Government needs to review the incentives and accountability framework governing primary education.

3.76 In terms of its size and severity, waste is the most serious problem facing the sector from an efficiency perspective. However, this chapter documents other problems related to efficiency in the sector.

- Whereas enrollment has risen rapidly with UPE, completion rates remain low, with drop-outs and repetition high. The sector is already shifting more focus to quality and completion rates (rather than access and enrollment rates). This move is encouraging and should be further strengthened. We suggest focusing even more on "final" (as opposed to "intermediate") outputs (i.e. graduates and graduates which have achieved a desired level of proficiency in numeracy and literacy as opposed to enrolled students).
- Unit costs have been rising, driven mostly by wages increases for teachers which have outstripped nominal GDP (ie wages rose by more than inflation *plus* economic growth).
- As a result of high drop-out and repetition rates, the unit cost of a graduate is more than twice what they should be. Poor learning in classrooms imply even higher unit costs of a graduate with acquired level of proficiency in numeracy and literacy.
- Higher monthly salaries to teachers have not translated into more outputs (i.e. more primary graduates).
- Teacher salaries are 60 percent higher in government schools compared to private schools. In turn, government schools can afford fewer teachers per student compared to private schools.
- The deployment of teachers *within* a school is such that primary one and two on average have more than 100 students while primary seven have less than 40 students. Most educators would suggest the resource deployment, with more teachers allocated in early grades.

3.77 **In terms of the severity of the problems identified, waste is clearly the number one problem facing the sector from an efficiency perspective.** Successfully reducing waste is not a guarantee to achieving more output but it is a guarantee to mobilize more resources which could be used to improve outputs. Getting more teachers to show up to school, and having more of those that do turn up teach, is likely to increase output by raising the number of tangible inputs.

3.78 **Combating waste will require behavior change.** The problems highlighted in this chapter are not new. As discussed, some of them have haunted the sector for more than a decade. The new behavior is one which starts by recognizing that, whatever was tried in the past decade, it did not remove the problem. Second, the new behavior is one which does not tolerate that 21 percent of scarce resources are being wasted year after year. Eradicating waste will be impossible and should not be the goal (e.g. some teachers will always be sick on a given day) but significantly reducing it should. The potential benefits of even small progress are huge. A twenty percent reduction in waste would imply saving Ush 14 billion, or the equivalent of hiring almost 6,000 additional teachers.

3.79 **In this PER, we have intentionally focused on presenting the problem areas as facts and resisted the temptation to enter the realm of speculation: *why* are the problems there? We did so because we believe it is important to first agree on the magnitudes of the problems so that the most problematic are can be identified. Only thereafter does it make sense to invest additional resources in exploring solutions.** Obviously, designing policies to reduce the key problem which has now been quantified (i.e. teacher absenteeism) will involve further analysis and some speculation. It is important to resist the temptation to jump to conclusions without supporting analysis. For instance, the lack of teacher houses is often mentioned anecdotally as the reason why teachers are absent. Further analysis may very well prove this to be the case. In designing interventions to combat teacher absenteeism, however, it may not be the most cost effective solution. That is, a teacher house costs more than $5,000 to build while a cheaper intervention may have a similar impact on absenteeism but cost a fraction.

3.80 Still, Box 3.6 reviews some preliminary evidence on the causes of absenteeism in Ugandan school and discusses international experience in addressing the problem. Our preliminary findings from the analysis of the November 2006 school visits are presented in a box below. Clearly, further analysis is needed to pin down underlying causes of absenteeism. To this end, a second school visit is currently on-going. This time, we extended the sample to include more private schools (to be able to better compare absenteeism in government versus private schools), we will be able to address seasonality (by visited the same school as we did in Nov 2006), and by also visiting the schools from the 2002/03 survey, we will be able to examine more closely the schools which have demonstrated the largest decline in absenteeism in the past 5 years. In addition to examining the correlates of absenteeism in Uganda, we discuss experience in combating absenteeism from other countries in Box 3.6.

Box 3.6. Teacher Absence in Uganda: A Summary of Results From a 2006 Survey and Possible Interventions

This box presents some preliminary findings from our analysis of teacher absenteeism, and presents international experience in addressing absenteeism.

Teacher absence is very heterogenous and reflects a wide array of factors from weak incentives, difficult working conditions and a high disease burden. The correlates of absence that we identify at the individual level are

- Age (older teachers more absent)
- residential proximity (less likely to be absent if live very close to the school)
- working in the same district of birth/origin.

The latter raises an important question about decentralization. One of the major trends in teacher hiring, and indeed in other sectors, is that an increasing fraction of new hires are indigenous to the district. Our results suggest that an unintended consequence of the prevailing decentralization policy: higher absence results because individuals with stronger social ties are more likely to be pulled away from work either for personal benefit or to help relatives and friends.

School level correlates include

- greater competition generated by neighboring private schools,
- increased participation and involvement of parents and
- teachers in schools that practice multi-grade teaching are more likely to be absent.

We do not find any strong evidence that district level factors affect teacher attendance.

What can be done?

This section briefly reviews four studies that have attempted to address teacher performance either directly or indirectly. The borders of this dichotomy are defined by where the incentives (to improve teacher performance) are located. In the studies outlined below, the incentives are located at three main levels:

1. Teacher level incentives such as prizes, remuneration bonuses, promotion and individual recognition. Outcomes such as average test scores are measured at the teacher level. This set of incentives can also involve negative interventions such as naming and shaming or even firing poorly performing teachers.
2. School level: high performing schools (measured at the level of the school, rather than teacher incentives as above) are given collective rewards. These can include private rewards to all teachers (sharing a collective pot of money or other prizes), or a local public good at the school such as a staff room, library etc,
3. Household level: Incentives located at the household typically aim at motivating students to increase the effort dedicated to learning. Prizes are announced at the pupil level and recipients typically receive scholarships. Other forms of household level incentives include empowering of parents/local communities to monitor teachers or manage schools directly.

Teacher level incentives work by directly raising the returns to teacher effort (see Jacobson (1989) for an early intervention). For these to work, teachers must believe that the costs of increased effort are less than the expected gains in remuneration (or whatever the form of the performance-based reward). The example below provides a brief summary of two such interventions.

Teacher Incentives in Western Kenya

This study conducted by Glewwe, Illias and Kremer (2003) attempted to measure the impact of a randomized intervention in which high performing teachers would be rewarded with prizes at the end of the school year. Prizes were substantial and included bicycles, mattresses and other household durables. Performance was determined by the rank of the school in the district and the degree of improvement relative to the previous year's exam results. The authors measured a variety of inputs before and after the intervention. While the authors find improvements in test scores in 'treatment' schools, this improvement is

Box 3.6. Teacher Absence in Uganda: A Summary of Results From a 2006 Survey and Possible Interventions (cont'd)

driven primarily by teachers using different teaching strategies – particularly teaching to the test. There is no significant different across treatment and comparison schools in teacher absenteeism (teacher absence rates are around 20% in these schools).

Another teacher incentives paper evaluates the impact of monitoring and performance-based remuneration on teacher absence.

Monitor-less monitors in India
Duflo and Hanna (2006) evaluate a randomized intervention in community schools in which an NGO provides cameras to teachers and institutes attendance-dependent remuneration. Teachers are expected to take pictures every morning, and teachers are paid depending on the number of "full" days attended. Teachers in treatment schools received a base pay as well as performance-based (measured by 'full days attended') component. The results of this intervention were surprisingly large. Teacher absence fell by about half from a high of 36% in comparison schools to 18% in treatment schools. In addition, test scores are higher in treatment schools. The authors discuss the political-economy of this intervention and conclude that it is not a realistic option for national scale up.

School-based incentives rely on mobilizing peer monitoring/head teacher effort to support higher teacher effort levels. The example below outlines an innovative study that evaluates group (or school-level incentives) vs teacher incentives.

Group vs Individual Incentives in India
This paper seeks to determine the impact of two broad interventions are on rural primary schools in India: 1) a set of smart inputs (an additional volunteer teacher and cash block grants vs 2) group or individual teacher incentives (performance-based pay). These four interventions are evaluated in a randomized-control design in Andra Pradesh, India. The four treatment groups are

1. individual-incentives schools
2. group-based incentives schools,
3. schools that get an additional teacher
4. schools that receive block grants (equivalent expenditure).

Karthik and Sundararaman (2006) find evidence in support of the incentives treatment arms. Learning outcomes are higher in incentives schools, but the authors do not find any differences in treatment impact between group vs individual incentives. They conclude that it is likely that the small size of the schools (three teachers on average) makes it easy for peer monitoring to have the same effects as individual incentives. However, like the teacher incentives study in Kenya, teacher attendance is not affected. Instead, teachers choose to increase "cheap effort" – assigning more homework and practice tests rather than show up.

Finally incentives located at the household level typically work through their effects on student effort or mobilizing parental involvement in monitoring or the management of schools. Improvements in student effort can have a profound effect on the learning environment and greatly boost the satisfaction that teachers get from teaching. As such, teacher attendance and preparation can improve dramatically under the 'collective mission' to win merit-based scholarships. An example of such an intervention is described below.

Girls Scholarships in Western Kenya
Miguel, Kremer and Thornton (2005) evaluate a randomized intervention in which a series of pupil scholarship possibilities are announced at the beginning of the school year. About 200 girls are eligible for scholarships in 60 schools in two districts (about 15% of the eligible enrollment). The scholarships pay for tuition for the next two years and parents receive an unconditional cash transfer of $12 per recipient. The results of this intervention were quite dramatic. Performance of all pupils, including non-eligible scholarship recipients (boys), in treatment schools perform much better than control schools. In addition,

Box 3.6. Teacher Absence in Uganda: A Summary of Results From a 2006 Survey and Possible Interventions (cont'd)

teacher absenteeism falls by 6.5 percentage points in treatment schools. The authors attribute this to an improvement in working conditions engendered by increases in pupil effort.

The other form of incentives-at-the-household-level includes empowering parents to monitor teachers. This ranges from establishing school management committees to outright hiring, firing and remuneration policy control. A number of studies have looked at the effects of greater community control in Latin America and show large effects on school performance (see Jimenez and Sawada (1998) and King and Ozler (2001)). When control was transferred to the community, so that parents could hire and fire teachers, teacher attendance and test scores went up (see Lewis (2005) for a recent review). While the associations here are very strong, it is difficult to interpret these results as causal. An interesting analogy is the community monitoring study in Uganda that focuses on health care centers. Using report cards as the means to provide information to the community on the performance of health centers, communities that were randomized to receive this information have much better performing health care centers. Attendance of health center staff and quality of service were higher in report card communities.

Source: Habyarimana (2007)

3.81 **As Box 3.6 shows, further analysis can help narrow down reasons behind the waste problem but we are unlike ever to find a blue print for reducing waste in the primary sector in Uganda.** History has shown that reducing leakages in one place can result in new leakages emerging elsewhere. However, we believe that progress cannot be achieved without a time-bound action plan, some mechanism to monitor progress, and annual process of reviewing what works and what didn't work. Moreover, MOFPED can provide incentives to continue focusing on deliver progress by tying additional resources to measurable results. In summary, the following ingredients seem essential:

3.82 Therefore, we propose the following steps to combat the waste problems identified:

- Sector worker group develops time-bound action plan to achieve progress in reducing teacher absenteeism, ghost workers, UPE capitation grant leakages, and "questionable expenditures":
  - Such plan would include commitment to regularly monitor problem areas (including repeating random school visits to measure teacher absenteeism, using internationally recognized methodology to measure absenteeism).
  - The proposed interventions to combat the problem areas (e.g. teacher absenteeism) should be tested first using randomized control group experiments to measure their impact.
- Governance weaknesses be discussed in Education sector's annual performance report (ESAPR), including:
  - Discussion of progress or lack thereof in reducing teacher absenteeism, ghost workers, UPE capitation grant leakages, and "questionable expenditures"
- Given the documented history of governance problems, we propose that the burden of proof be shifted: we propose that the assumption in the sector working group (and vis-à-vis MOFPED and donors) be that the magnitude of the problems documented in this

chapter on ghosts, UPE capitation grant leakages, absenteeism and questionable expenditure is unchanged until carefully shown otherwise..

- Ministry of Finance, Planning and Economic Development link increases in funding to measurable improvements in reducing teacher absenteeism, ghost workers, UPE capitation grant leakages, and "questionable expenditures".

3.83 To ensure more focus on better utilization of resources in the future, future EASPRs should:[69]

- Focus *even more* on graduates and graduates with a desired level of proficiency and less on enrollments.
- Focus more on what it costs to produce outputs (i.e. unit costs) and why these unit costs have evolved as they have.

3.84 **Our conclusions should not detract from efforts to improve the efficiency of input use in schools.** But it would seem that the first priority for Uganda should be to reduce teacher absenteeism, and such efforts should certainly precede hiring new teachers to further expand the primary education system.

[69] The Education Sector Annual Performance Review already includes a discussion on quality so the point made here is one of relative importance. As discussed below, the review appears to be placing relatively more emphasis on enrollment as opposed to the number of graduates acquiring the desired level of proficiency in numeracy and literacy.

4. FISCAL SUSTAINABILITY ANALYSIS

4.1 Section 1 of this chapter summarizes the main conclusions. Section 2 reviews recent trends in debt, deficits and revenue and expenditure, documents the data used for the base year, and sets out trend growth rates in expenditure used in the baseline ("business-as-usual") projection. Section 3 uses a fiscal forecasting tool to present a baseline ("business-as-usual") forecast, highlighting the likely evolution of expenditures given the PEAP fiscal strategy, recent trends, and recent policy announcements. We end the section by subjecting our business-as-usual forecast to a sensitivity analysis of macro shocks. Section 4 presents a growth-oriented fiscal policy in which the composition of expenditure is more favorable to future economic growth. Section 5 concludes with implications of the analysis for the 2008 and future budget discussions.

A. INTRODUCTION AND SUMMARY

4.2 **Following the Multilateral Debt Relief Initiative and HIPC relief, Uganda is at a low risk of debt distress**. Several rounds of debt relief have removed the bulk of Uganda's public sector debt. As a result, under most likely scenarios, the government's ability to meet its debt obligation when they arise over, at least, the next twenty years is not at risk. Following the latest round of debt forgiveness – the Multilateral Debt Relief Initiative (MDRI) – which forgave debt owed to IDA, other multilateral donors, and the IMF –the central government's foreign debt stood at approximately \$1.7bn (or 17 percent of GDP) in early 2007. While the stock of domestic T-bills and –bonds has been growing in recent years to help the Bank of Uganda (BOU) sterilize donor inflows, the outstanding stock of these instruments (worth around 9 percent of GDP and held mostly by the private sector) is to a large extent offset by large central government deposits held at the BoU. In early 2007, total gross foreign debt plus net domestic debt (i.e. the stock of government securities minus the cash deposit at the BOU) of the government stood at around than 20 percent of GDP, a sharp reduction from the levels reached in the late 1990s when public debt stood at more than 50 percent of GDP (see Table 4.1 and discussion below).

4.3 **Although debt levels are currently low, this will most likely be a temporary phenomenon**. The public sector accumulated debt in the past because of persistently large fiscal deficits which are typical of a low-income country with a small tax base and large development needs. Irrespective of the debt relief, Uganda remains a poor country with demands on the government vastly exceeding its ability to finance domestically through taxation and domestic debt. Even under extremely optimistic economic growth scenarios, this situation will persist for, at least, the next 20 years. Partly, foreign grants (which are not debt-creating) finance the shortfall in revenues but, on average over the past decade, foreign loans have financed a third of the fiscal deficit (see Table 4.2) and this is likely to continue.

4.4 **Nevertheless, if new borrowing is on concessional terms, Uganda should not face a fiscal debt service problem**. Barring reckless fiscal management or cataclysmic events, new borrowing on concessional terms is unlikely to pose any fiscal sustainability risks for at least the next 20 years. Most of Uganda's loans are and most likely will continue to be given on concessional terms, implying long grace periods and interest rates less than 2 percent. As a result, for most of the next two decades, the government will be able to finance large fiscal deficits with loans, without experiencing the stress of having to put aside funds for annual interest payments and loan repayment.

4.5 **The fiscal risk facing Uganda in future is not one of liquidity or fiscal debt sustainability, but of long-term solvency through economic growth.** Although with prudent new borrowing the fiscal sustainability risks (i.e. the risk that the government will be unable to meet its obligations) are negligible, this chapter argues that the main fiscal risk Uganda faces in the coming two decades is rather a risk that the new loans will increasingly be used to finance recurrent expenditure such as, for instance, higher salaries and more civil servants and not invested in addressing infrastructure bottlenecks which currently represent the binding constraint to placing Uganda's economy on a faster growth trajectory (see CEM, World Bank 2007).

4.6 **A shift in fiscal composition is needed for growth.** This chapter argues that unless Uganda takes a number of steps to slow the growth in recurrent expenditure and to shift the composition of spending towards productive infrastructure investments, private investment will slow and eventually economic growth will slow too.

4.7 **We illustrate the likely growth impact of two different fiscal policy scenarios.** In our "business as usual" (or "empirical") business-as-usual we assume[70]:

- Government maintains overall spending at levels consistent with macro economic stability.
- Recent expenditure trends and the announced introduction of Universal Secondary Education (USE) mean recurrent expenditure increasingly squeezes out infrastructure spending,
- With public infrastructure spending postponed or crowded out, private investment growth does not accelerate beyond its average over the past five years (at around 6.5 percent in real terms).
- With private investment growth not accelerating, faster export growth is unlikely.
- Hence growth gradually slows and increasingly becomes aid-financed consumption growth.
- The prudent fiscal stance means debt levels remain comfortable, but in 20 years Uganda remains a country with more than 10 percent of its population in poverty, and in no better position than today to repay debt.

4.8 **We contrast this gloomy business-as-usual scenario with "fiscal policy for growth scenario".** Here the government uses public resources more efficiently to achieve the same outputs, while simultaneously freeing resources to be invested in addressing infrastructure bottlenecks. We use our model to show that:

- If helped and motivated to use its resources more efficiently, the education sector would be able to achieve its ambitious goals (in terms of number of primary and secondary graduates) by spending the same amount of resources (measured as a share of GDP) as it does today. Is this realistic? The analysis in the previous chapter on

[70] A word of caution on the terminology which might differ from other fiscal sustainability analysis: we use the term "baseline" to describe a scenario in which past fiscal trends and current policies continue into the future. We contrast this scenario with a "growth scenario" in which fiscal policy is geared towards mobilizing resources for infrastructure spending. In this sense, our fiscal sustainability analysis also differs from a standard fiscal sustainability analysis (FSA). Our motivation is to contrast the growth impacts of two different fiscal scenarios whereas standard FSA are examining the likelihood that government can service its debt if and when it falls due.

education shows that it is. In particular, chapter 3 showed that waste in the primary education sub-sector alone amounted to 70 billion, or 0.4 percent of GDP. Extrapolating this finding to the rest of the education sector and helping the sector use the resources which are not wasted more productively would imply that much more could be achieved within the existing ceiling.

- Given the importance of employee costs in overall expenditure, fiscal restraint must include lower growth in the wage bill. More analysis will be needed to provide concrete suggestions on how this can be accomplished, and determine how realistic such cuts are. However, any recommendation would likely include proposals to slow down the rate at which new civil servants are being hired and/or lower growth in real wages.
- As a result of the improved public capital stock (e.g. reliable and cheaper provision of electricity, lowered cost of transportation, etc), private investment would likely accelerate compared to the growth rates observed in the business-as-usual, and Uganda could feasibly resume a new process of structural transformation.
- With more private investment, there is every reason to believe that the vibrant entrepreneurial spirit observed in the current search for new exports would accelerate.
- As a result, growth accelerates with exports and investments being more important drivers of growth
- The economy still requires larger amounts of aid (to finance still-large fiscal and balance of payments deficits) but the economy's capacity to repay the loans clearly improves over the forecast horizon.

4.9 The purpose of the analysis presented in this chapter is to assess the long-term macro economic impacts of two distinctly different fiscal strategies, both within an overall framework of fiscal consolidation: a strategy based on a continuation of past trends versus a strategy that explicitly seeks to mobilize resources for infrastructure. For a number of reasons, therefore, the results presented in this chapter differ from recently completed debt sustainability analysis undertaken cooperatively between the government, the World Bank and the IMF. To begin with, the results in this chapter are being generated from a new fiscal and macro model created as part of this PER (and discussed below). A key innovative element in this model is that the composition of spending has growth impacts. Thus, in our model, the same revenue and spending targets (measured as a share of GDP) will have different growth impacts depending on *how* public resources are spent.

B. FISCAL PERFORMANCE IN THE PAST DECADE - LESSONS FOR MODELING FISCAL EXPENDITURE

The Size and Recent Evolution of Public Sector Debt

4.10 **MDRI sharply reduced central government debt from 50 percent at the end of June 2006 to around 20 percent of GDP in early 2007**. This sharp reduction is visible, irrespective of which definition of public debt is used. However, definitions matter in terms of the level of debt before and after the reduction: depending on the definition of public debt chosen, Uganda's current debt is either 20 percent or 37 percent of GDP (down from either 50 or 67 percent of GDP). The definition used in this chapter takes a narrow perspective of the public sector by focusing only on central government debt and assets. Moreover, arrears are included as an expenditure demand rather than include as part of debt (see further discussion below). To define public debt in more detail, first, external debt is measured in gross terms (i.e. the sum of all liabilities, ignoring the fact that the government – broadly defined – is holding assets in the form of international reserves at the BoU). Second, domestic debt is measured in net terms: from total liabilities of the central government (consisting mostly of government securities held by the private sector and a large overdraft at the BoU) we subtract government cash deposits held at the BoU (consisting mainly of the cash proceeds of the securities sold to the private sector). Table 4.1 contains all of the central government's deposits and liabilities and shows public sector debt according to different definitions[71].

4.11 **In addition to the stock of debt Uganda has an unknown and still increasing stock of arrears outstanding**. According to latest IMF estimates, the stock could be as high as 500-600 bn (3.0-3.5 percent of GDP) and new arrears are still being accumulated at a rate of 30-50 billion per year in 2005/06 (and reportedly a smaller but unknown amount the previous year). Moreover, old arrears are still being investigated and added to the outstanding stock. Finally, the government faces contingent liabilities stemming from a number of sources, including implicit or explicit government guarantees in public-private partnerships. Chapter 5 discusses mechanisms for managing contingent liabilities in PPPs.

4.12 **In our analysis, we treat arrears as an expenditure demand rather than part of the outstanding stock of debt for a number of reasons.**[72] First, their exact size is unknown. Second, and more important, arrears in Uganda are neither indexed to inflation nor do they have an interest rate attached to them. Third, the government seems to be able, politically, to postpone honoring the commitments which the arrears represent. Therefore, arrears seem to represent a demand on the budget which is more similar to other policy commitments such as, say, ensuring quality primary education than honoring the debt service payment on a loan. In reality, of course, there is a difference: the government's promise to deliver higher quality in primary education is a political promise; an arrear represents a legal claim on the government from a private entity. In theory therefore, arrears should be included as part of debt.

[71] Contingent liabilities are excluded. These are neither well estimated nor numerically significant at present. The primary liability comes from a public pensions scheme in need of reform.

[72] Our intention is not to suggest that arrears should not be considered as part of the government's outstanding debt. Obviously, it should (as we have also done in the table on "estimates of gross and net public debt"). For the reasons outlined above, though, we found it more convenient from a **modeling perspective** to treat arrears as a demand on the budget, rather than an outstanding stock of debt.

Table 4.1: Estimates of Gross and Net Public Debt

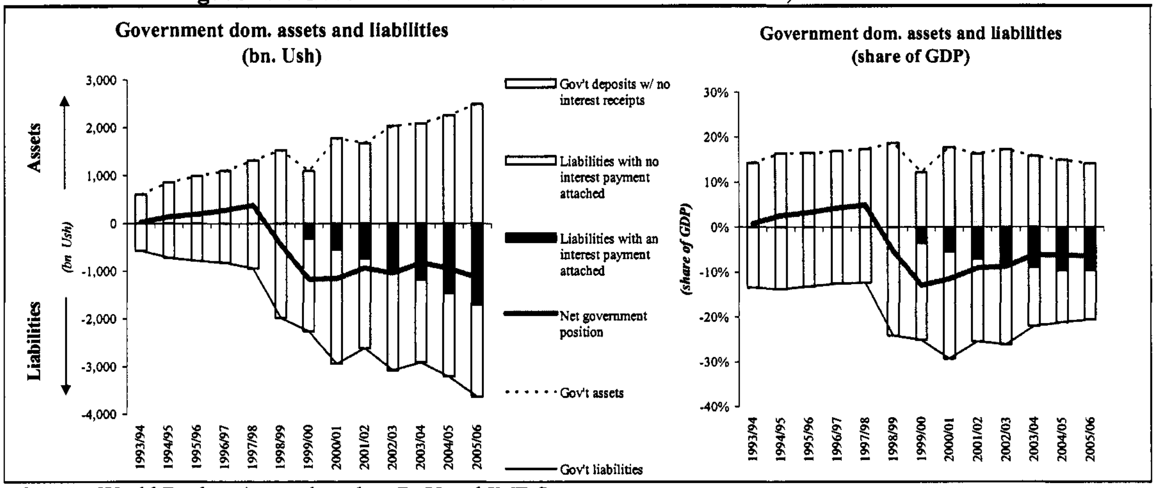
(Billion Ush)	2002/03	2003/04	2004/05	2005/06	2006/07e
Gross domestic debt (most comprehensive)	3,065	2,908	3,206	3,654	3,711
Gross domestic debt (excl. arrears)	**2,540**	**2,394**	**2,530**	**3,057**	**3,111**
Liabilities with an interest payment	1,007	1,173	1,459	1,696	1,750
Treasury bills	997	1,028	919	961	900
Treasury bonds	0	135	530	735	850
Promissory notes	10	10	10	0	
Liabilities with no interest payment	2,058	1,735	1,747	1,958	1,961
Overdraft at BoU	1,533	1,221	1,071	1,361	1,361
Domestic payables (i.e arrears)	525	514	676	598	600
Accounts payables	147	149	278	231	
Pension arrears	313	313	340	309	
Interest payables	64	52	58	58	
Domestic assets					
Assets which earn an interest					
None					
Assets which does not earn interest					
Government deposits in BoU	2,438	2,274	2,363	2,579	2,579
Net domestic debt	627	634	842	1,075	1,132
Net domestic debt, excl. arrears	102	120	166	478	532
Gross external debt (million US$) 1/	4,211	4,465	4,422	4,464	1,770
International reserve (million US$) 1/	720	928	1,219	1,380	1,464
Gross external debt (bn Ush)	8,435	7,987	7,695	8,233	3,317
Total public sector debt, gross	**10,974**	**10,381**	**10,224**	**11,290**	**6,428**
Total public sector debt, domestic net and external gross	**8,537**	**8,106**	**7,861**	**8,711**	**3,849**
% of GDP	72%	61%	52%	49%	19.9%

1/ Incl. IMF debt held at BoU
Source: IMF, BoU and own calculations

4.13 Public debt quickly expanded in the past decades as a result of persistently large fiscal deficits. Excluding grants, Uganda's government has been running deficit in the magnitude of 6-12 percent of GDP for more than two decades. Moreover, these deficits were entirely financed with the help from other countries in the form of grants or concessional loans – loans which usually have grace periods of 10 years and very low an annual interest rates. Table 4.2 provides estimates of the magnitudes of deficits in the period 1994/05 to 2004/05. It shows that almost 70 percent of the 9 trillion cumulative deficit (measured in constant 2004/05 prices) was financed by grants while external borrowing – mostly concessional – more than made up for the residual. In fact, in recent years, assuming borrowing is a last-resort, Government has been using external borrowing in excess of the deficits needed to be financed to build up a large Uganda-shilling-denominated cash deposits at the Bank of Uganda.[73] In 2005/06, the government paid 200 bn in interest on a net position at the BoU worth bn 478 (see Table 4.1), an implied interest expense of more than 40 percent.

[73] The cash deposits are the proceeds from the selling of government securities; not international reserve holdings. International reserved are part of the BoU's (not the central government's) assets.

Figure 4.1. Government Domestic Assets and Liabilities, 1994/5-2005/06[74]



Source: World Bank estimates based on BoU and IMF figures

Table 4.2. Total Deficits Before Grants from 1994/95 – 2004/05 in Constant 2004/05 (Billion Ush)

	1994/05-2004/05	Share of deficit, excl. grants
Total deficit, excl. grant	9,065	100%
Total external grants	6,263	69%
Total external financing, net	2,943	32%
Other means of financing deficits	-140	-2%

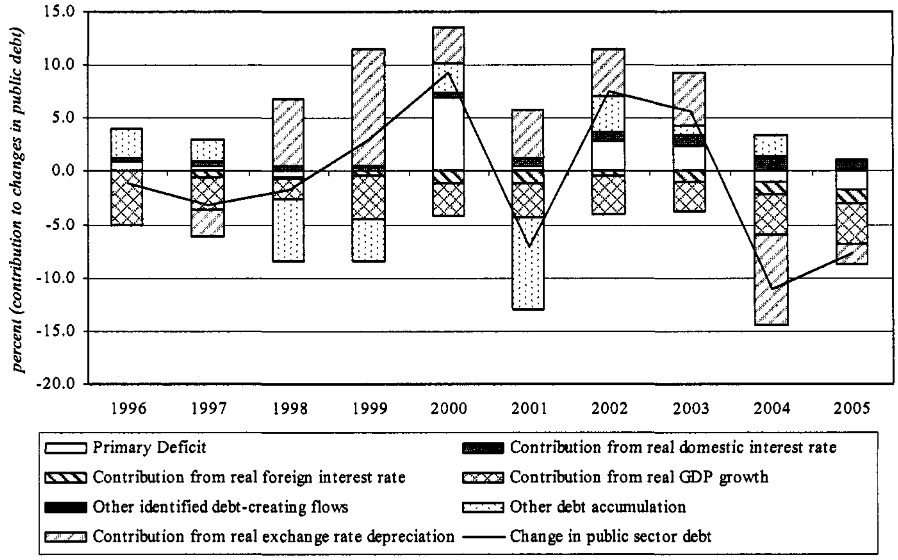
Source: IMF

4.14 **Fiscal deficits are not the only factor that causes changes in debt to GDP ratios.** Other factors include the real growth rate and the exchange rate which changes the value of foreign current-denominated debt. Moreover, debt can change from one-off factors such as debt relief, or the recognition of new debt (such as, for instance, arrears or a contingent liability being realized). Finally, to gauge the impact that changes in real interest rates has on debt dynamics it is convenient to separate the impact that fiscal deficits has before paying interests (i.e. the "primary balance) versus the impact that changes in interest rates has. Figure 4.2 shows a decomposition of changes in public debt from 1996 to 2005 based on the factors mentioned above.[75] As can be seen, the residual factor "other debt accumulation" has played a non-negligible factor in the past decade. Below, we discuss the possible factors included in this "residual".

[74] "Liabilities with an interest payment attached" are government T-bill and –bonds.

[75] When public sector debt is defined more broadly to include the central bank, seignorage should also be included as a factor contributing to changes in debt. At the same time, however, expenditure of the central bank should be included as part of non-interest expenditure (in the primary balance calculation). We choose to focus on the central government's revenue and expenditure and have, therefore, not included seignorage (and the assets of the BoU) in our calculations.

Figure 4.2. Uganda: Public Sector Debt Dynamics, 1996-2005



Source: MOFPED and BOU and World Bank staff calculations

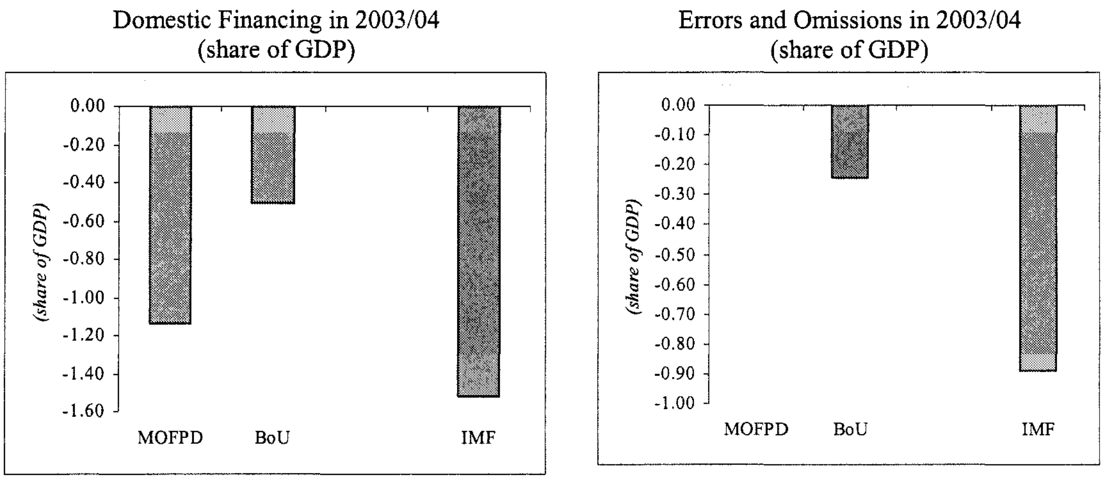
4.15 **For Uganda, "other debt accumulation" captures at least three possible factors.** First, domestic debt issuance in recent years were driven by monetary purposes rather than deficit financing resulting in gross debt expanding (as a result of the issuance of government securities to the private sector) but net debt remaining unchanged (since the cash proceeds of the sales were deposited with the BoU). In principal, then, changes in net debt should correspond to the annual financing flows but, as Figure 4.2 shows, it is not the case. We speculate that a possible reason for the discrepancy is related to the large central government overdraft at the BoU whose evolution in recent years remains as difficult to explain as its origin (see Table 4.1 for the magnitudes of both the deposit account and the overdraft at the BoU). In short, domestic net debt is blurred because of both large assets and liabilities, and its evolution is explicitly not motivated by deficit financing. Second, changes in external debt are difficult to reconcile with annual financing flows due to numerous rounds of debt relief. Finally, although the quality of fiscal and debt data have significantly improved in the past decade, the quality remains insufficient and is, most likely, an important factor explaining the observed discrepancies between flows and changes in stocks.

4.16 **Regarding the quality of fiscal data, in a given year, government spending or deficit financing could be off by as much as one percentage point of GDP.** That is, fiscal tables in print are likely to be off by a magnitude of one percent of GDP. Naturally, data gets revised as more information becomes available. Thus, in most countries, it is not unusual that data will be revised up or down – especially if they are expressed as a percent of GDP. In Uganda, however, the problem is two-fold: first, even fiscal data on spending or revenue which is not expressed as a percent of GDP is likely to be off by as much as bn 100 Ush (roughly 1 percent of GDP). Second, there is no agency which is responsible for maintaining and making publicly available a historical data set containing spending, revenue, financing and debt flows which means that the only reason we, as authors of this year's PER, can see that numbers must have been revised is the inconsistency we can observe between a) different tables in print, b) deficits and financing flows, c) actual expenditure and treasury releases (see discussion below).

4.17 Having weak fiscal data can be costly. Below are some examples of the indirect costs which stem from the absence of a reliable and consistent time series of fiscal data.

- The lack of a consistent historical time series on fiscal data reduces the quality of the work delivered by the many annual consultants examining the fiscal accounts at the sector level. As discussed in chapter 3, the lack of detailed fiscal data on education resulted in additional work and *ad hoc* assumptions which are normally not needed in countries where Ministries of Finance maintain high quality data.
- The credibility of government policy is put at risk. Potential investors in Uganda may equate weak data with non-transparency, and postpone or cancel investments.
- The government's ability for self-evaluation is limited.

Figure 4.3. Comparing Fiscal Numbers Between Different Data Bases



Sources: Ministry of Finance, Planning and Economic Development "Draft Estimates of Revenue and Expenditure", FY 2005/06 (dated June 8, 2005), Bank of Uganda (Quarterly Report), IMF Uganda country economist electronic files.

4.18 **There are many examples of the problems with data quality**. One example is that the exact mixture of foreign and domestic financing of the large fiscal deficits is, not clear. The problem is that the composition between foreign and domestic financing varies widely between different data sources. Moreover, it is not unusual that a particular data source (e.g. IMF or BoU) is unable to determine how the observed deficit is financed, resulting in large "errors and omissions". Figure 4.3 provides an example of the magnitude of the problem: the left graph shows a variation of more than 1 percentage point of GDP between the three fiscal data sources (BoU, MOFPED and the IMF) in the amount which was financed domestically. Equally problematic, the right graph shows that both the BoU and the IMF were unable to determine how a significant portion of the deficit was financed. That is, neither external nor domestic financing were large enough to account for the observed deficit. The unexplained portion ranged from 1.3 percent of GDP in 2001/02 to 0.5 percent of GDP in 2005/06.[76]

[76] It is not entirely clear the reasons behind the large discrepancies: one possible explanation could be difficulties determining net external financing in an environment of complicated and changing rules governing debt relief.

Table 4.3. Budgets, Releases and Actual Expenditures in 2003/04 and 2004/05 (bn Ush)

2004/05	Approved estimates	Actual releases	Actual expenditures	Under/over expenditure
Ministries	1,692.2	1,633.0	1,685.7	6.6
Agencies	316.6	307.6	348.5	-31.9
Referral hospitals	61.5	52.7	54.6	7.0
Embassies and missions	25.1	25.1	28.7	-3.6
District councils	790.2	762.2	762.2	27.9
Urban/municipal councils	40.8	38.4	38.4	2.4
Total	**2,926.5**	**2,819.1**	**2,918.2**	**8.3**
2003/04	Approved estimates	Actual releases	Actual expenditures	Under/over expenditure
Ministries	1,633.8	1,495.4	1,558.1	75.8
Agencies	271.4	260.7	284.4	-13.0
Referral hospitals	51.3	42.6	44.9	6.4
Embassies and missions	26.3	26.3	30.8	-4.6
District councils	725.6	695.5	695.5	30.1
Urban/municipal councils	37.3	32.4	32.4	4.9
Total	**2,745.7**	**2,552.9**	**2,646.1**	**99.7**

Source: Auditor General's report to Parliament 2004 and 2005

Table 4.4. How Were the Deficits Financed in the Years 2000/01-2004/05? (cumulative nominal amounts, according to three different data sources)

	2000/01 - 2004/05		
	BoU	MOFPED	IMF
		(bn Ush)	
Total deficit to be financed	1,603.3	1,577.8	1,578.6
Total financing provided	1,603.8	1,577.7	1,430.8
External	1,739.8	1,907.7	1,788.3
Domestic	-136.0	-330.0	-357.4
Unexplained	-0.5	0.1	147.8
External borrowing in excess of deficit to be financed (bn Ush)	-136.5	-329.9	-209.6
in million US$	-76	-183	-116
		(Shares of GDP)	
Total deficit to be financed	3.0%	2.9%	2.9%
Total financing provided	3.0%	2.9%	2.6%
External	3.2%	3.5%	3.3%
Domestic	-0.3%	-0.6%	-0.7%
Unexplained	0.0%	0.0%	0.3%
	Shares of total deficit to be financed		
Total deficit to be financed	100%	100%	100%
Total financing provided	100%	100%	91%
External	109%	121%	113%
Domestic	-8%	-21%	-23%
Unexplained	0%	0%	9%
GDP (sum of 2000/01 - 2004/05)	54,218.1	54,218.1	54,218.1

Sources: Ministry of Finance, Planning and Economic Development "Draft Estimates of Revenue and Expenditure", FY 2005/06 (dated June 8, 2005), Bank of Uganda (Quarterly Report), IMF Uganda country economist electronic files.

4.19 **Another problem with the fiscal data is that expenditure data reported by MOFPED and BoU is cash releases from Treasury's account, not actual expenditures.** The exact magnitude of the divergence between releases and actual expenditures is still not known but Auditor General's report on the Public Accounts to Parliament hints at potentially serious problems: the report shows that there are ministries that consistently overspent both their budgets *and* their released amounts, while there are ministries that consistently spent less than what is released to them. Since it is actual expenditures (and not releases) which matter for debt accumulation and for growth and poverty reduction, it is conceivable that the primary balance calculated using actual expenditure differ from the primary balance calculated using releases. Although the Auditor General's numbers are not entirely compatible with MOFPED numbers (e.g. donor expenditures are excluded), Table 4.3 shows that actual expenditures may have exceeded releases by up to 100 billion Ush (around 1 percent of GDP) in both 2003/04 and 2004/05. The Auditor General's report also shows that there are ministries that spend more than what is released and budgeted to them, raising concerns about the integrity of the budgetary process, and the commitment control system.

4.20 Yet another data problem is the weak systems set in place to track donor project expenditure. A substantial number of donor-funded projects still receive part (or all) of their funding directly from donors without going through the Uganda Consolidated Fund Account (i.e. the treasury account at the Bank of Uganda). Despite the drive towards donor-financing through budget support approximately 30 percent of total government expenditure is financed outside of the Treasury system. By not going through the Government budget and accounting system these resources are not captured in the government accounts although they are appropriated by Parliament and controlled by the Ministry.[77] As a result, government estimates of "donor financed development expenditure" are often subject to large revisions.

4.21 Additional data problems include:

- **Availability of GFS data**: First, Uganda switched to a new GFS compatible Chart of Accounts in 200/033/04 hence GFS data are available only for the 4three fiscal years from 200/033/04 to 2005/06.
- **Availability of District-level data**: Second, the shift to the new chart of accounts has coincided with a gradual roll out of an IFMS accounting package which now covers all central Minitries, no agencies, and only a handful of Districts. Actual expenditures incurred are available for the IFMS, but not for the other budget centers. In fact, the financial statements of Government presented by the Accountant General in the Auditor General's report show District level spending to be identical to cash releases from the Ministry of Finance, Planning and Economic Development – something that various Auditor General's Reports acknowledge is not the case. For non-IFMS budget centers, Government uses the checks printed by the UCS under the payment system. This gives more accurate actual spending data for Ministries and Uganda-based Agencies, but again not for the non-IFMS local Governments who raise checks through Stanbic Bank. The upshot is that there are significant gaps in fiscal data for the Districts which are the main point of service delivery for most PAF-related spending.
- **For infrastructure spending, we were unable to get audited accounts for extra budgetary funds.** These include the Rural Electrification Fund, Credit Support Facility, Tariff Stabilization Fund, and the Rural Communications Fund. (A Roads Fund is to be added to this list of extra-budgetary funds, and will be managed by RAFU).

The data weaknesses described above has the following implications for the analysis represented below:

- We cannot be certain that the functional expenditure trends described below are accurate since all available numbers are based on "releases" as opposed to actual expenditures
- We cannot be certain that the *level* of debt at the starting point of our analysis (2005/06 and our estimate of 2006/07) is correct. Given the discrepancies in all fiscal data, the level of debt could well be off by up to 5 percent of GDP (i.e. stand close to 25 percent rather than 20 percent estimated).

[77] See Auditor General (March 2004): "The Report and Opinion of the Auditor General to Parliament on the Public Accounts of the Republic of Uganda for the year ended 30th June 2003"

- We might be underestimating recent growth in expenditure and, as a result, underestimate the underlying pressures facing the budget. The continued accumulation of new arrears suggests that reported spending is less actual incurred costs.

4.22 We addressed these challenges by subjecting our results to a battery of sensitivity analysis, testing our results to different starting points in the level of debt, and different trend growth rates in expenditure.

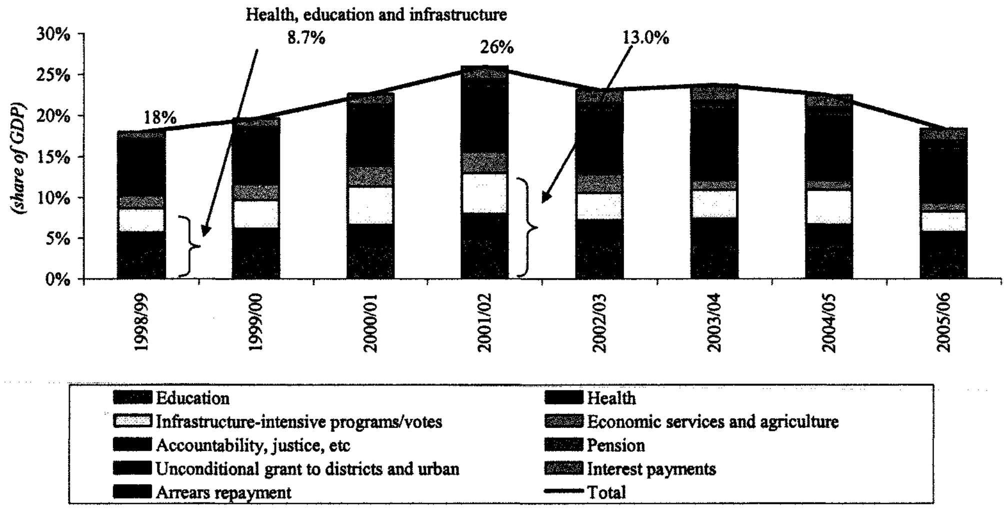
C. TRENDS IN THE LEVEL AND COMPOSITION OF SPENDING (BASED ON RELEASES), 1998/09-2005/06

4.23 **Better fiscal planning and increased poverty focus allowed the government to increase expenditure substantially since the mid 1990s.** Fiscal planning was significantly improved when the Government introduced the Medium Term Expenditure Framework (MTEF) in 1997/98, marking a significant shift from the *ad hoc* budget planning of the early 1990s. The introduction of the MTEF coincided with an increased focus on poverty reducing spending. As a result, donor grant inflows increased substantially and allowed expenditure to increase from around 17 percent of GDP in 1997/98 to almost 26 percent at the peak in 2001/02. Since its peak, expenditure has gradually come down to hovering around 20 percent of GDP. Expenditure appears to have dipped to only 18 percent of GDP in 2005/06 due to lower-than-expected donor inflows.

4.24 **The aid surge resulted in a significant increase in spending on education, health and "infrastructure-related programs and votes"** (see Box 4.1 for a discussion of the functional re-grouping of the MTEF used in this PER). Between 1998/99 and the spending peak in 2001/02, these three groups' combined spending rose from 8.7 to 13.0 percent of GDP (see Figure 4.4).

4.25 **However, in recent years, with fiscal consolidation and arrears clearance, spending on education, health and infrastructure has been outpaced by spending on "Accountability, security, justice, defense, public administration and economic and social services (excl. energy)".** While the education, health and infrastructure spending has fallen sharply as a share of GDP to 8.3 percent in 2005/06, "accountability, etc" grew steadily from 1998/99 until 2004/05 from 5.9 to 7.8 percent and has only falling somewhat in 2005/06 to 6.4 percent.

Figure 4.4. Composition of Spending by Functional Classification, excl. Arrears Repayment



Source: MOFPED Annual Budget Performance Report (various issues)

Box 4.1. Re-Grouping The MTEF For Analytical Purposes

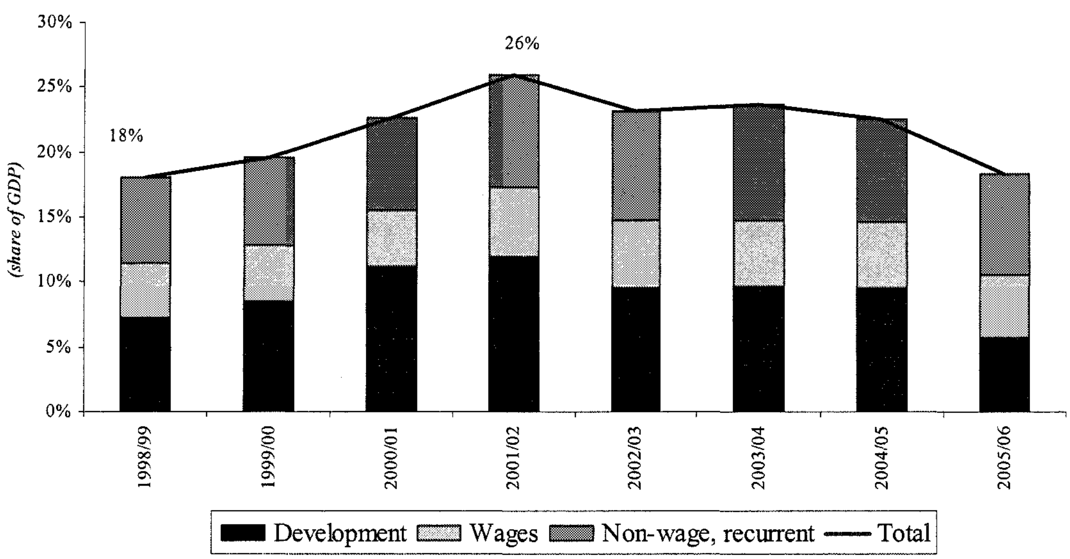
For analytical convenience, we re-arrange the familiar 12 MTEF groupings into 9 groups which will be easier to model in section 3 below. The two main differences are the creation of a large "infrastructure-intensive programs and votes" which contains "roads", "water", "energy" and the part of the Local Government Development Program which focuses on capital improvements, and the grouping together of public administration, security, defense and justice (and large parts of "economic services"). The purpose of re-grouping the MTEF is mainly because it allows for easier modeling. In particular, when forecasting expenditure, it is always the case that some functional groupings are easier to model than others. For instance, education expenditure is tightly related to the number of students enrolled whereas security, defense and public administration is more difficult. Given that we will not be able to model neither security nor defense very convincingly separately, we choose to group them together. Moreover, we have pulled out some sub-components of "public administration" such as pension and "unconditional grants to local governments" into separate categories because, in our forecasts, we will be able to entertain more detailed scenarios of these expenditure items.

MTEF GROUPING	GROUPING FOR ANALYTICAL PURPOSES
EDUCATION	**Education**
HEALTH	Primary
WATER	Secondary
ROADS AND WORKS	Other
AGRICULTURE	**Health**
	Infrastructure-intensive programs/votes
	Roads, water and energy
ECONOMIC FUNCTIONS AND SOCIAL SERVICES	Local Government Development Programme (LGDP)
SECURITY	
JUSTICE/LAW AND ORDER	**Economic services and agriculture**
ACCOUNTABILITY	
	Accountability, Justice, Defense and Public administration, excl. pension, unconditional grant
PUBLIC ADMINISTRATION	**Pension**
	Unconditional grant to districts and urban
INTEREST PAYMENTS DUE	**Interest payments**
Total	Foreign
	Domestic
	Total

4.26 **Most of the recent spending pressure has come from "non-wage recurrent" spending**. Using an economic classification, the increase in expenditure from 1998/99 to 2000/01 was shared between development and recurrent spending. However, following the peak, expenditure settled at a lower level by reducing development expenditure while recurrent expenditure remained roughly constant at the peak's high level (see Figure 4.5). Within recurrent expenditure, wages' share has become more important, having increased by one percentage point of GDP since 1998/99. Even within education and health, growth in wages has outpaced non-wage recurrent and development expenditure in recent years.

4.27 However, it is the large, opaque "non-wage recurrent" expenditure (accounting for almost 8 percent of GDP in 2005/6 and more than 40 percent of total spending) which explains the bulk of the increase in the recurrent budget since 1998/99. Non-wage recurrent expenditure rose from 6.7 percent to 8.6 of GDP between 1998/99 and 2001/02 and has been declining only gradually since.

Figure 4.5. Total Spending (Measured by "Releases" and Expressed as a Share of GDP) by Recurrent and Development (incl. Donor) Expenditure, Exclude Arrears Repayment



Source: MOFPED Annual Budget Performance Report (various issues)

4.28 **A large share of the non-wage recurrent part is in fact "employee costs".** This includes allowances, pensions, and transfers to agencies which, in turn, use the funds to pay salaries. We estimate that as much as bn 520 out of the 1.4 trillion (37 percent) classified as non-wage recurrent expenditure was used to remunerate employees in 2005/06. We base this estimate on a detailed item-by-item analysis of expenditure and budgets. From this analysis, we know that at least 12 percent of non-wage recurrent expenditure is employee costs (see Figure 4.6). These costs include pension expenditure and the large non-wage but nevertheless employee costs spent under defense, security, accountability and public administration (e.g. allowances, welfare and entertainment, training, etc). Moreover, oddly, "statutory salaries (item 211104) are included in the MTEF's "non-wage" expenditure group and not under "wages". Our analysis also reveals that 20 percent of "non-wage recurrent" spending gets classified as "other current grants to agencies". Based on who the recipients of these funds are, we estimate that at least 75 percent of these transfers are, in fact, employee expenditure. The biggest recipients of these funds are the

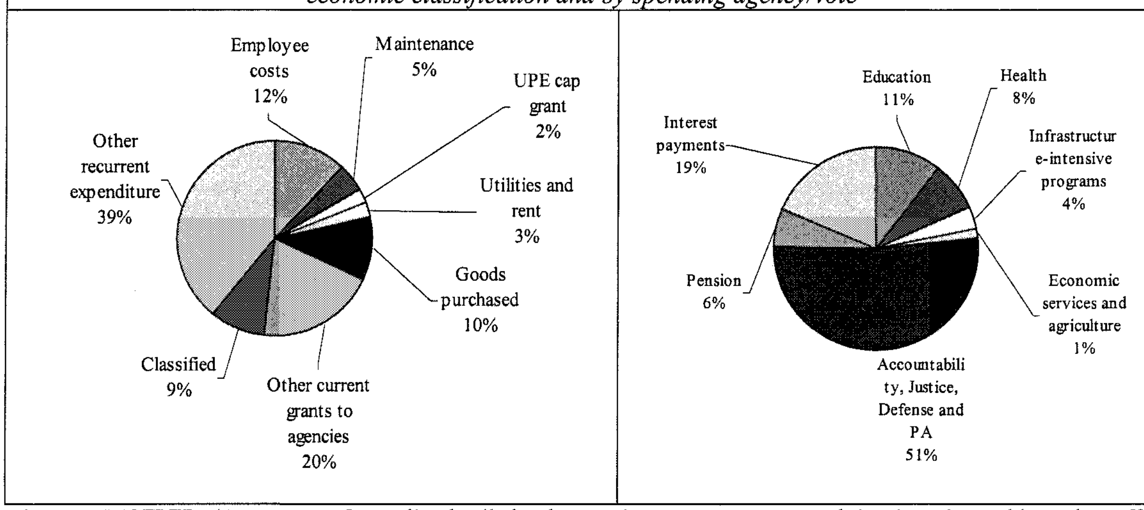
Electoral and the Parliamentary Commissions followed by the police, internal affairs and various law and justice agencies (see Table 4.5). Following the same logic, a large portion of "other recurrent expenditure" (also part of "non-wage recurrent expenditure) includes items which may, in fact, end up as payments to employees.[78]

Table 4.5. Spending Classified as "Other Current Grant to Agencies" by Largest Spenders (bn Ush, Expenditure Measured by "Releases" in 2005/06)

Spending agency/vote	Bn. Ush
Electoral Commission	67
Parliamentary Commission	52
Police, prisons and justice	39
Makarere University	35
Foreign affairs	9
State House and Office of the President and Mass Mobilisation	5
Others	63
Total spending classified as "other current grants to agencies" in 2005/06	269

Source: MOFPED (Accountant General's detailed releases, by vote, program and item) and World Bank staff calculations

Figure 4.6. What is included in "Non-Wage Recurrent Expenditure" and Who Spends it?
2005/06 total releases (excl. donor-financed development expenditure) broken down by more detailed economic classification and by spending agency/vote



Source: MOFPED (Accountant General's detailed releases, by vote, program and item) and World Bank staff calculations

4.29 **Rising employee costs are driving a lot of the recent recurrent spending pressure in Uganda**. In short, our item-by-item analysis of the budget suggests that employee costs may have played a more prominent role in the evolution of spending than what is suggested by only looking at the MTEF's "wage" expenditure. Unfortunately, detailed historical data that would allow us to trace total employee costs (as shown for 2005/06 above) over time is lacking.

[78] Examples include conditional transfers to Agricultural extension, district hospitals, and transfers to URA.

However, as Figure 4.6 (right graph) shows "non-wage" recurrent costs are mainly showing up in "Accountability etc". Since it is "Accountability etc" which has stayed stubbornly high since 2001/02, it is highly likely that the shift from "capital expenditure" to "recurrent expenditure" would be more prominently driven by a rise in employee costs than we are able to show in Figure 4.5.

4.30 **Employee costs may account for a significant portion of overall spending.** In the total domestically-financed spending budget (i.e. incl. domestically-financed development but not donor-development expenditure), employee costs could account for more than 50 percent of spending (see Figure 4.7 and Figure 4.8). This figure is significantly higher than spending on "wages" reported in the MTEF's economic classification (1.4 trillion vs. 900 bn in 2005/06) and underscores the need to develop a more detailed economic classification of spending in Uganda. To maintain consistency with the MTEF (and since we did not have historical data for employee costs) we focus on modeling wage expenditure and not employee costs in our forecast exercise in sections 3 and 4. However, similar projection exercises in the future should seek to relate the number of civil servants and their average wages to employee costs, not only "wages".

4.31 Understanding the drivers of employee costs is more complicated than understanding "wage" growth. With employee costs accounting for more than half of total government spending, it is essential that the recent evolution in employee costs gets documented, analyzed and understood. The "wage bill" (as defined in the MTEF) can be well understood by having access to the number of civil servants and their monthly salary over time. However, the trend growth in employee costs and the factor responsible for such growth is less clear, given the non-transparent way such costs are recorded in the budget (e.g. "transfers to other agencies"). Without such analysis, it will also be difficult to formulate policy recommendations for wider civil service reforms.

4.32 Notwithstanding the data problem regarding employee costs, controlling growth in employee costs can undoubtedly be curtailed by controlling certain functional spending categories. Given that the bulk of non-wage employee costs appear in public administration, justice, accountability, the police and the defense, any attempts to reduce growth in employee costs would need to curtail spending in these functional groupings.

4.33 Looking ahead, controlling growth in employee costs – not just wage growth – is the key to controlling growth in recurrent expenditure. Given that more than half of recurrent expenditure is spent on employee costs, it is difficult to envision fiscal restraint without controlling these costs. Again, the first step in controlling growth will be to understand in more details what the drivers of employee costs have been in recent years.

Figure 4.7. MTEF's Economic Classification (excl. Donor-Financed Development Expenditure)[79]

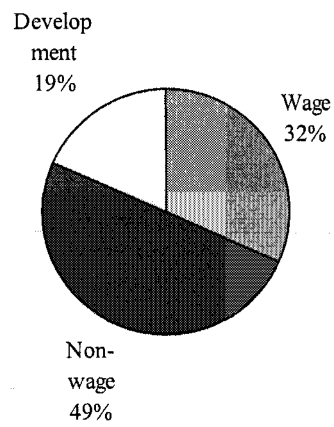
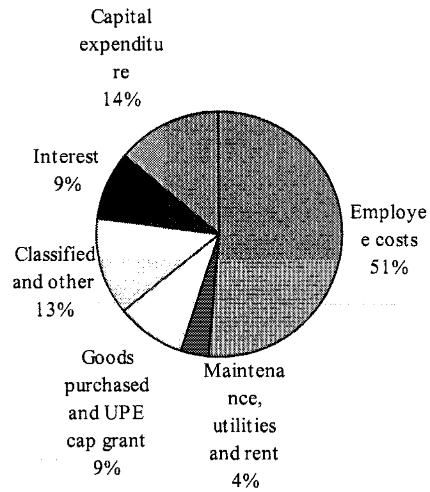


Figure 4.8. More Detailed Economic Classification of Expenditure excl. Donor-Financed Development Expenditure



Source: MOFPED Annual Budget Performance Report (various issues) and MOFPED (Accountant General's detailed releases, by vote, program and item) and World Bank staff calculations

4.34 **Establishing a recent "trend" in spending is not straightforward**. Past trends are usually a good starting point for projecting future trends in expenditure. In Uganda's case total government expenditure since 1998/99 is clearly characterized by two distinct trends which makes it somewhat difficult to determine the appropriate trend to use: there was a rapid built-up in expenditure which was clearly linked to an increase in aid from 1998/99 to 2001/02 and a period of consolidation thereafter. We choose to treat the spike in 2001/02 as an outlier and, when doing so, total expenditure looks reasonably stable at around 22 percent of GDP. Moreover, as our business-as-usual indicator of trend growth in expenditure items, we use the geometric average real growth from 1998/99 to 2005/06. The growth rates are summarized in Table 4.6.

[79] We have been unable to obtain donor releases broken down by items.

Table 4.6. Trends in Government Expenditure 1998/99 - 2005/06 (Average Real Percentage Growth from 1998/99)

	Avg share of GDP (2001/02)	Avg. real growth rate 1998/99-2005/06: Total spending	Recurrent spending	Development spending	No. civil servants	Real wages 1/	Contribution to spending growth 2/
Education	4.3%	4.2%	6.0%	-1.7%	4.0%	6.1%	0.8%
Primary	2.8%	4.3%	5.8%	-0.3%	4.8%	5.7%	0.6%
Secondary	0.8%	0.0%	3.3%	-15.4%	0.4%	13.8%	0.0%
Other	0.7%	8.7%	9.9%	4.7%	-3.3%	3.7%	0.3%
Health	2.4%	7.7%	13.3%	2.7%	9.9%	6.2%	0.8%
Infrastructure-intensive programs/votes	3.6%	2.3%	4.2%	2.1%	7.9%	6.2%	0.4%
Economic services and agriculture	1.7%	-1.5%	14.8%	-3.0%	6.3%	6.2%	-0.1%
Accountability, etc	7.2%	6.6%	7.1%	4.5%	-4.1%	6.2%	2.2%
Pension	0.4%	19.5%	19.5%	..	..		0.3%
Unconditional grant to districts and urban	0.7%	5.2%	5.2%	..	4.6%	6.2%	0.2%
Total 2/	21.6%	5.5%	8.0%	1.1%	2.4%	6.2%	5.5%

Source: MOFPED Annual Budget Performance Report (various issues) and World Bank staff calculations
1/ Rough estimates given limited data and based on growth from 2000/01-2005/06
2/ The total includes pension and interest payments whose growth rates are not shown in this table. Therefore, the sum of the contribution to spending growth will not add to 5.5 percent.

D. TRENDS IN REVENUE

4.35 Over the 10 years to 2005, the rise in revenues from 15 to 20 percent of GDP has been driven mainly by an increase in foreign grants. Foreign grants have risen from around 4 percent of GDP in the mid 1990s to around 8 percent of GDP in 2005.[80] In the same period, growth in domestic revenue has been more modest, rising from 10.5 percent in 1996 to more than 12 percent in 2005, still well below the average of 18 percent in the Sub-Saharan region. For our analysis we combine the projections for revenue set out in with standard macro modeling assumption which link revenues to GDP growth via elasticities.

E. BASELINE: BUSINESS AS USUAL SCENARIO

4.36 This section shows that past trends in expenditure and current policy initiatives will place significant strains on the budget. We show that the most significant pressures over the next five years will come from the education sector, and wage pressures from non-education civil servants. Given the government's commitment to reduce (or at least maintain total expenditure at a fixed level of GDP), these pressures are likely to be met by reducing government investment.

4.37 This modeling exercise illustrate that unless government actively seeks to reduce the growth of recurrent expenditure – where, given its importance in size, controlling employee costs

[80] The government does not maintain and make available a historical fiscal data set. Thus, differences observed between IMF, BOU and MOFPED recorded amounts of grant inflows are most likely due to the fact that they were recorded at different points in time. Moreover, the IMF data imposes consistency between the fiscal accounts and balance of payments inflows. For macro analysis this is desirable but it imposes a consistencies which may not exist in reality due to fluctuations in exchange rates.

will be essential – and introduce measures to reduce the cost of rolling out Universal Secondary Education:

- the education budget will rapidly expand (or, at unchanged expenditure and utilization of teachers and classrooms, quality is likely to continue to suffer as little learning is likely to take place in classrooms with more than 50 students in each classroom),
- the composition of spending will increasingly consist of recurrent spending with infrastructure spending dwindling away,
- with infrastructure bottlenecks remaining unaddressed, private investment growth will remain robust, but will not take off (because of lack of adequate electricity and infrastructure),
- without faster private investment growth, export growth will also not take off, and as a result,
- growth is likely to gradually slow down[81],
- growth will largely be private consumption-driven (as opposed to investment-driven). Moreover,
- growth will continue to require large aid inflows, and
- debt levels will remain at fiscally sustainable levels, rising from 20 percent of GDP in January 2007 to 24 percent of GDP in 2026/27 but with annual debt service payments not exceeding 60 percent of annual exports and eventually declining to around 30 percent as the fiscal deficit narrows.

4.38 In short, our **business-as-usual** case shows that the fiscal sustainability from the perspective **of liquidity is not the main concern in Uganda at this stage**. Rather, existing spending pressures suggest a future composition of spending which will not help lay the foundation for a faster-growing economy.

F. ASSUMPTIONS USED IN BASELINE SCENARIO[82]

4.39 **Our business-as-usual expenditure forecast is based mainly on the historical trends.** These were described in the previous section with three noticeable modifications: first, total spending is fixed at 22 percent of GDP, reflecting the average of the past 8 years and the government's stated commitment for fiscal consolidation (see discussion in chapter 1 of this PER). Second, education spending on primary and secondary education is modeled separately to account for the introduction of Universal Secondary Education in February 2007 and the ongoing efforts to improve the quality of primary education which will require additional funding *unless* more focus in placed on efficiency of spending (see chapter 3). The model used to forecast education expenditure is based on the education sector's own model (Education Sector's Strategic Planning) but augmented to allow modeling of the recent introduction of USE. Third, development expenditure (excl. classroom construction) is modeled as a residual. That is, we forecast recurrent expenditure and, with total spending fixed at 22 percent of GDP, development expenditure is set at whatever there is room for in the budget. Global experience set out in chapter

[81] This is consistent with global experience of growth spurts (see R.Hausmann, L.Pritchett and D.Rodrik, 'Growth Accelerations', JFK School of Government, Harvard University (2004)).
[82] The model has been handed over to MOFPED officials to run alternative scenarios. All are illustrative.

1 suggests the latter is a reasonable reflection of how most countries approach capital expenditure, and, as discussed above, seem consistent with past trend in expenditures in Uganda (i.e. during the period of fiscal retrenchment since 2001/02, it was the development budget which bore the brunt of the adjustment).

Box 4.2. Assumptions Used in Business-as-Usual Scenario

Macro

- Total factor productivity (TFP) is growing exogenously at trend growth rate (1999-2005) of 0.7 percent per year.
- Real GDP is modeled as a production function with constant returns to scale with human capital, physical capital and (TFP).
- Private investment is growing at trend (6.9 percent in real terms) but will accelerated (or decelerate) depending on the growth rate of public infrastructure per worker (which, in turn, is determined by public spending and population growth.

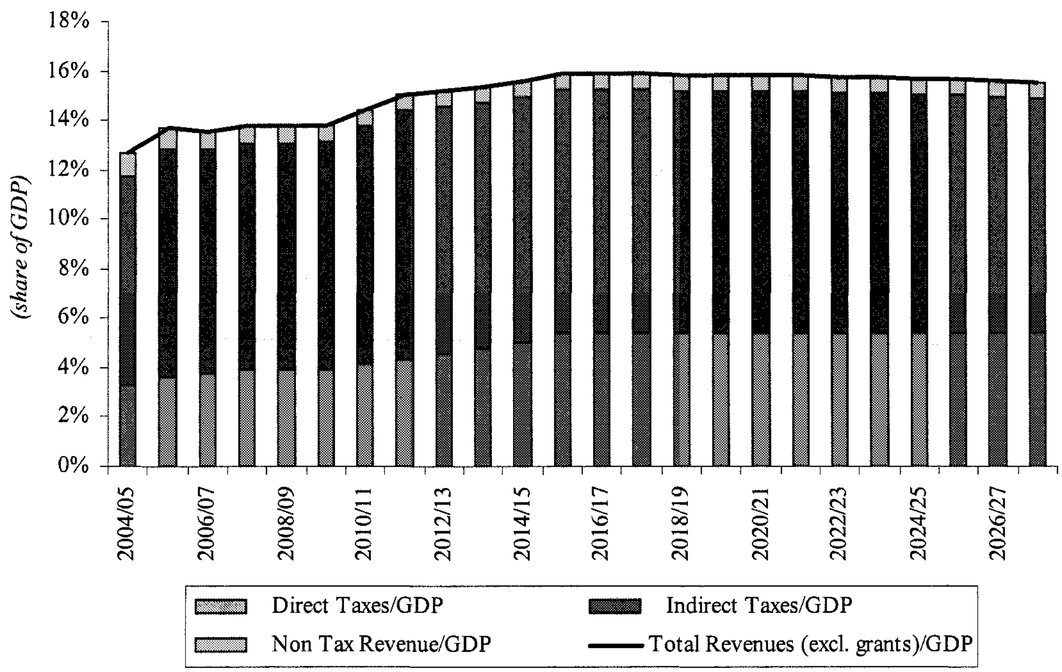
Fiscal

- Fiscal accounts are modeled in as much details as possible: For instance, (i) civil servant (by vote-holder or functional affiliation) and based on past trends; (ii) monthly wages of civil servants (past trends); (iii) maintenance costs (modeled as a fixed proportion of the capital stock); and all education expenditure are detailed modeled based on student numbers and assumptions regarding student-teacher, student-classroom ratios and parameters determining of efficiently these inputs are used.
- Overall fiscal spending is assumed fixed at 22 percent of GDP (reflecting the average since 1997/98 and the government's policy of fiscal consolidation)
- Revenue is partly endogenous by being linked to trade flows (i.e. trade taxes) and GDP (via an elasticity). In our business-as-usual, revenue as a share of GDP gradually climbs to 16 percent of GDP but falters slightly towards the end of the forecast horizon as a result of dwindling trade revenue and lower GDP growth.
- The fiscal deficit is assumed to be financed entirely by aid: we assume that half of the deficit is financed by grants while the other is financed by concessional lending. We assume that donors based financing (both aid and grants) on last year's deficit. As a result, domestic financing occurs when, in the current year, aid falls short of the deficit in need of financing.

4.40 **The revenue forecast is closely linked to the macro assumptions but also incorporates policy changes and increased efforts to raise revenue from existing policies[83].** We assume that increased efforts to expand compliance results in a gradual increase in income tax revenue (measured as of share of GDP) from around 3.6 percent in 2005/06 to 5.4 percent of GDP by the end of the forecast horizon. The amount of trade revenue is determined by the trade forecast (discussed below) but we assume that the average rate is such that trade taxes measured as a share of imports remains at its current level of 25 percent. Finally, non-tax revenue and "other indirect taxes" are assumed to remain constant as a share of GDP. Overall, our business-as-usual revenue forecast assumes that revenues increase from its current level at around 13.5 percent of GDP to around 16 percent of GDP by the end of our forecast horizon.

[83] See chapter 6 for details.

Figure 4.9. Revenue Forecast (Before Grants and Expressed as Share of GDP)



Source: World Bank staff's business-as-usual forecast

4.41 **These fiscal forecasts are combined with assumptions regarding the macro environment** (e.g. overall growth in economy, exchange rates, inflation, and export and import growth). To ensure macroeconomic consistency in the forecast (e.g. fiscal numbers such a government investment and consumption have a macroeconomic impact, or current account deficits need to be financed somehow), the World Bank's long-term consistency framework (the Revised Minimum Standard Model-Extended, RMSM-X) was used. [84]

4.42 We incorporate the government's stated fiscal consolidation strategy in our **business-as-usual scenario**. With total expenditure fixed as a share of GDP and revenue (before grants) gradually improving, our business-as-usual assumption is built around the government's stated desire to reduce the fiscal deficit before grants. In our business-as-usual, the deficit (before grants) is reduced from more than 8 percent of GDP in 2005/06 to close to 5 percent of GDP in 2026/27.

4.43 **Still, our business-as-usual scenario assumes that there will be large fiscal deficit to be financed by either grants, foreign or domestic loans**. With MDRI completed, it seems safe to assume that the question facing Uganda in the next 20 years will not be whether financing will be available to cover fiscal deficits but at what interest rate. In our business-as-usual scenario, we assume that donors will continue to support Uganda to the same extent as in the past 20 years, i.e. by offering to finance the entire fiscal deficit. Moreover, we assume that half of the deficit will be covered by grants and the remaining by concessional loans. As a sensitive analysis, we re-examine our conclusions with different interest rates on debt (e.g. should Uganda have to, or

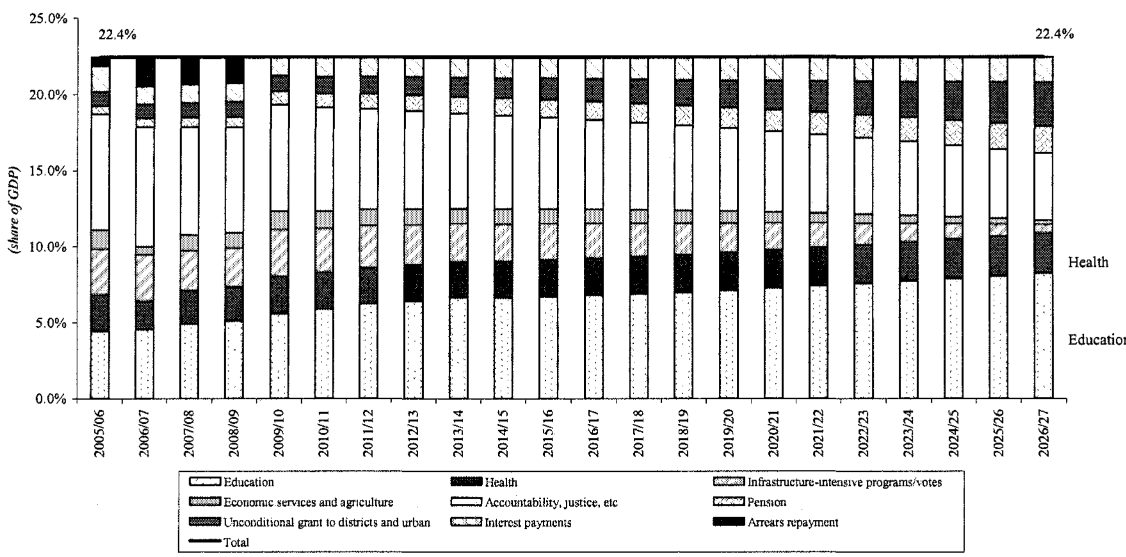
[84] See World Bank working paper 231, May 1989 for brief description of the standard RMSM model.

decide to borrow commercially) and with grants accounting for a smaller portion of deficit-financing.

Results from baseline (business-as-usual) scenario

4.44 **The education sector represents the biggest pressure to the budget from a functional perspective.** At unchanged utilization rate of teachers and classrooms, the sector will require substantially more resources – we estimate an increase from 4.5 to 7 percent of GDP – to fulfill its promises of delivering both Universal Secondary Education and higher quality primary education. Not surprisingly, the higher costs stem mainly from the costs associated with hiring more secondary teachers and building classrooms to accommodate secondary students. However, to deliver on promises to improve the quality of primary education, we assume that average class sizes in primary schools will have to be reduced substantially from their current levels of 65. A reduction to, say, 55 by 2014 will require 60,000 more teachers (in addition to the 129,000 currently on the payroll) and 25,000 more classrooms (in addition to the 85,000 currently standing) to be built. Figure 4.10 summarizes the expenditure forecast from a functional perspective.

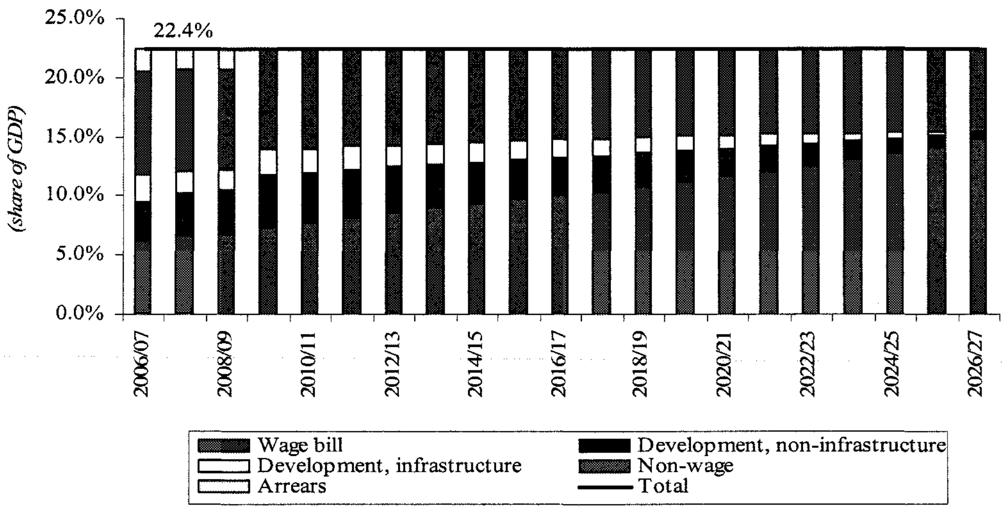
Figure 4.10. Baseline Forecast of Composition of Expenditure, Functional Perspective



Source: World Bank staff's business-as-usual forecast

4.45 **From an economic perspective, the biggest risk to the budget stems from wage bill pressures**, both in terms of pressures coming from hiring new civil servants (i.e. mostly teachers) and assuming that monthly wages continue to grow at their historical rate. Figure 4.11 summarizes the expenditure forecast from an economic perspective. The figure shows that, with unchanged total spending, development expenditure will gradually be squeezed over the next twenty years.

Figure 4.11. Baseline Forecast of Composition of Expenditure, Economic Perspective



Source: World Bank staff's business-as-usual forecast

4.46 **Focusing on the next five years, Figure 4.12 and Figure 4.13 underscore that education spending and wage costs pose the biggest risks to the budget.** However, the figures also highlight that the planned rapid repayment of arrears also require additional funds which could be used elsewhere. In the figures, a budgetary pressure is defined as an expenditure category which results in an increase in its share of GDP beyond the base year (2005/06 level). Thus, secondary education represents a budgetary pressure because the current implementation policies designed to roll out USE will result in spending on secondary education requiring spending to increase as a share of GDP. Unless, that is, class sizes will be allowed to bulge, classes taught under trees, and the quality of secondary education allowed to decline. In the pie chart, the total is the present value of budgetary pressures (as defined above) from 2007/08 to 2017/18.

Figure 4.12. Break-Down of Important Budgetary Pressures from Functional Perspective (2007/08-2017/18)

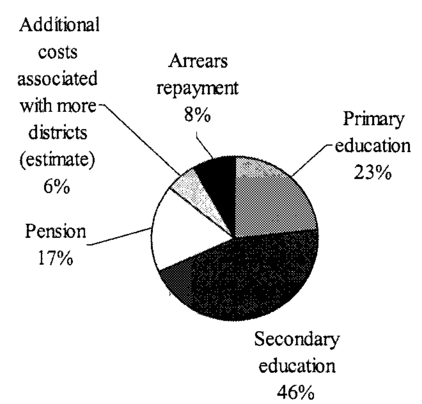
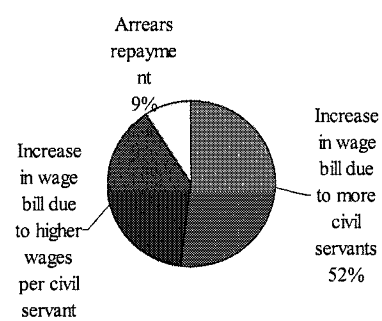


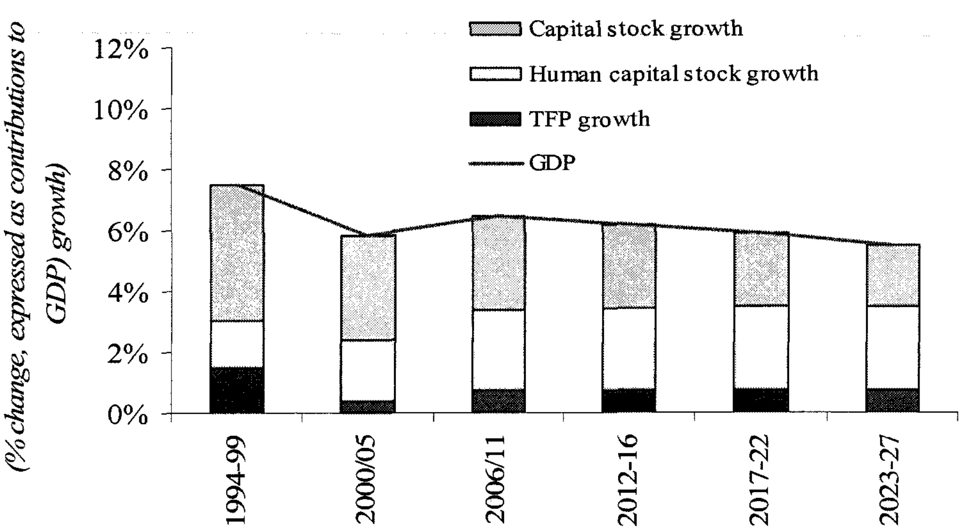
Figure 4.13. Break-Down of Important Budgetary Economic Perspective (2007/08-2017/18)



Source: World Bank staff's business-as-usual forecast

4.47 **The business-as-usual scenario also shows that overall growth is likely to gradually slow down, decelerating to 5.2 percent by 2026/27.**[85] The slowdown would be even more pronounced if we relax our assumption of constant TFP growth over the period (e.g. by assuming that TFP growth also declines as a result of less public infrastructure spending) (see Figure 4.14) or assume a faster rate of depreciation of the public capital stock. The slowdown comes as a result of the forecast decline in infrastructure spending which is being squeezed in the budget. In our model, this has consequences for growth by directly lowering the rate at which private investment is likely to grow.

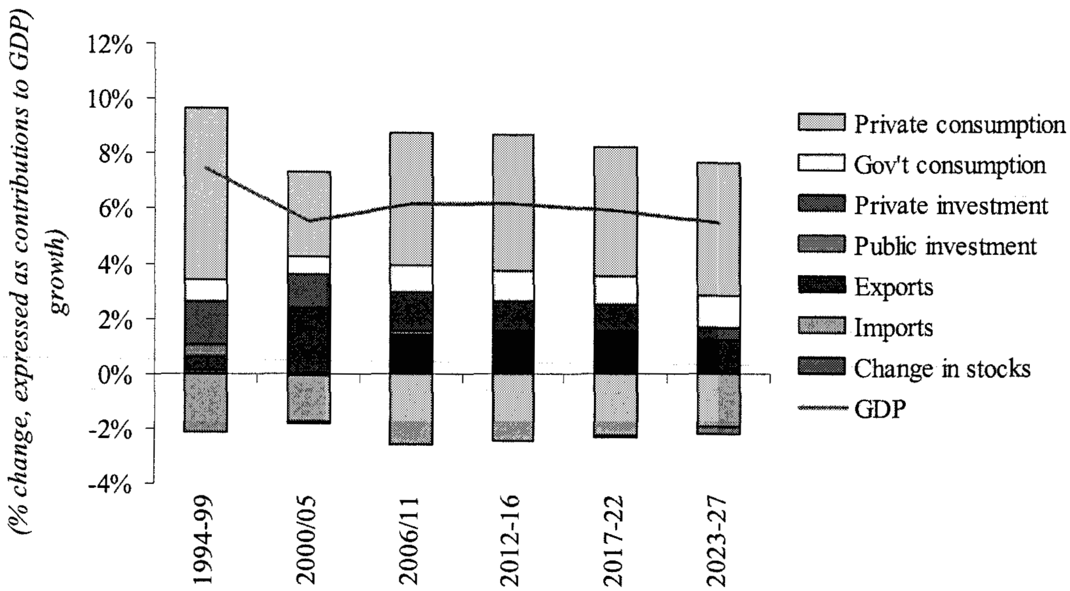
Figure 4.14. Annual Real GDP Growth Decomposed in Contribution from Growth Accounting Perspective



4.48 **The business-as-usual forecast also shows that growth will largely be private consumption-driven** (see Figure 4.15). With public investment shrinking and restraining private investment, export growth remains subdued (although still growing at a brisk pace), private consumption is assumed to be the main driver of growth. To the extent that private consumption growth outpaces population growth, it will help reduce poverty but it will not help lay the foundations for a more dynamic economy.

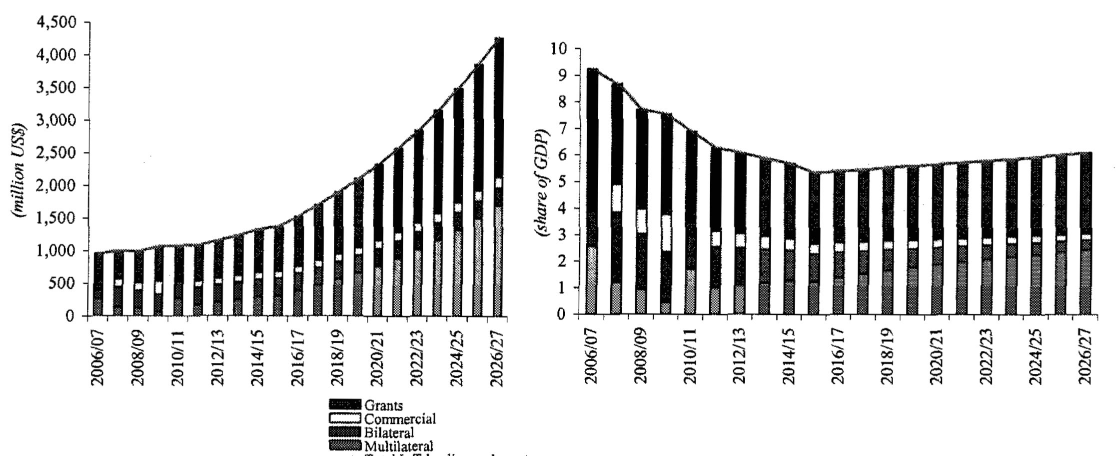
[85] The initial growth acceleration in the initial period is a result of the public capital stock growing as the education sector is assumed to add 25,000 additional classrooms to accommodate USE students and improve student classroom ratios in primary schools.

Figure 4.15. Annual Real GDP Growth Decomposed in Contribution from Expenditure Side



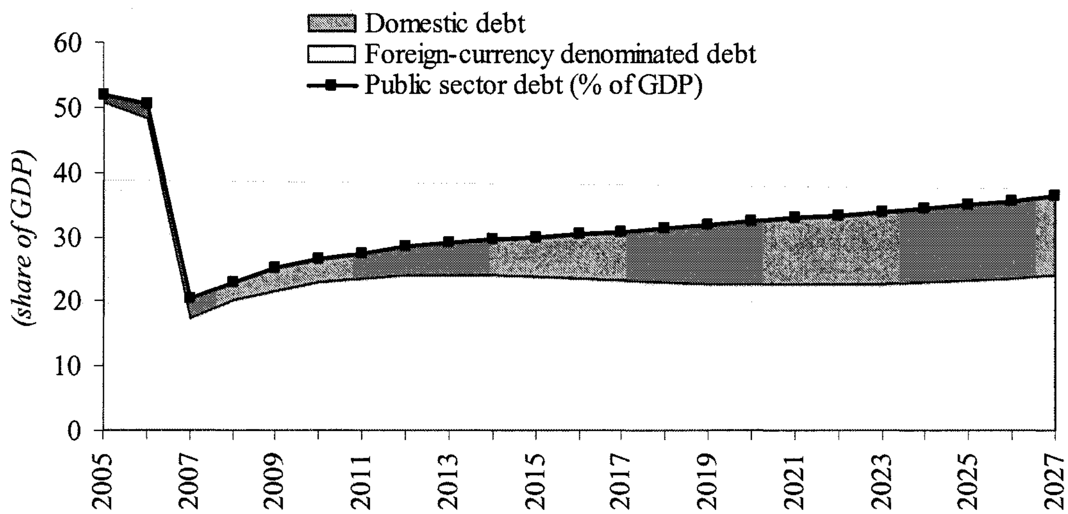
4.49 **With lackluster baseline export growth and overall GDP growth slowing, the economy will continue to require large aid inflows.** In fact, although the required aid as a share of GDP will be smaller than its current level (in line with the smaller fiscal deficit in need of financing), the annual US dollar amount needed quickly rise to more than 4 billion a year by 2027.

Figure 4.16. Assumed Grants and Long-term Lending in Million US$ and as a Share of GDP (2007-2027)



4.50 Debt levels will remain at fiscally sustainable levels, albeit on a rising path from 20 **percent of GDP in January 2007 to 36 percent of GDP in 2026/27** (see Figure 4.17). We judge that the level of debt is fiscally sustainable in the sense that neither the burden of paying interests nor the burden of mobilizing resources – both domestic funds and foreign exchange – to meet debt repayment obligations is at risk at any point. Specifically, interest payments will remain below 2 percent of GDP and 10 percent of the budget (see figure 4.18) and annual debt service requirements on the foreign-currency denominated debt will only temporarily exceed 60 percent of annual export revenue (in 2017-2021) before declining rapidly to less than 40 percent of exports in 2027. The rate of increase is the fastest in the early years of the forecast (where USE is rolled out).

Figure 4.17. Public Sector Debt (Share of GDP)

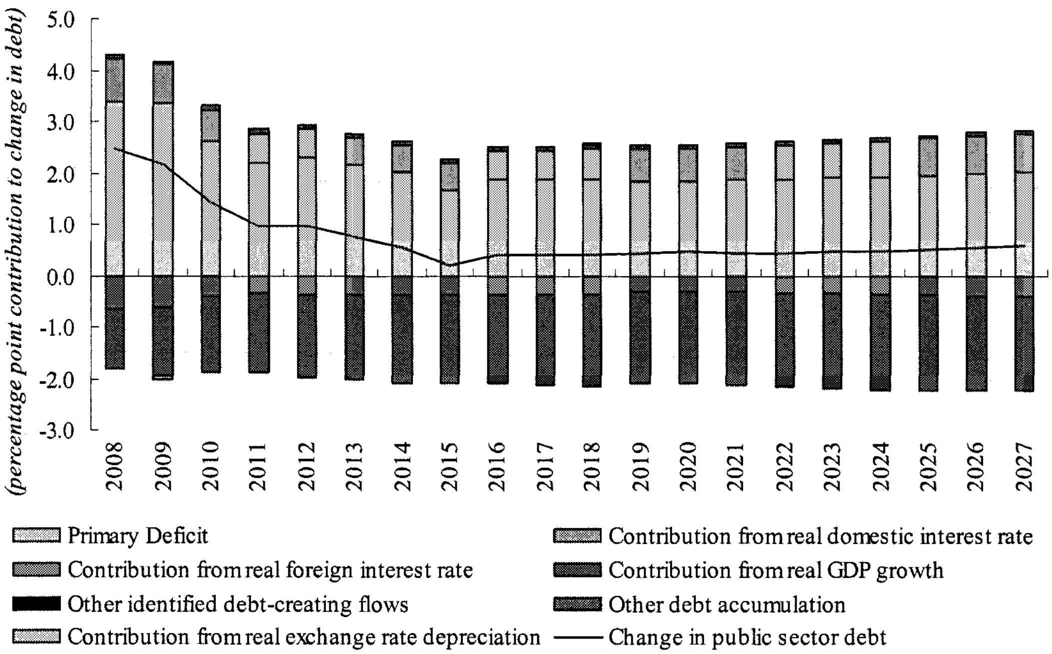


Source: World Bank business-as-usual scenario

4.51 **The result regarding debt is not surprising given the broad assumptions we are using in the baseline**. Specifically, with assumptions which have revenues increasing, set expenditure at a constant level of GDP, with the majority of debt being financing at concessional interest rates and real GDP growing at almost 6 percent for the entire period, it is very difficult to see how public debt would not be sustainable. The fact that debt is on a gradually increasing path is mainly a result of the fact that the government's primary deficit will continue to exceed real GDP growth.

4.52 Indeed, as Figure 4.18 shows, it is the real interest rates on the external debt and the shrinking primary deficit which are the primary drivers behind the benign forecast of public debt. The figure shows that public debt is still expanding at a rapid pace primary due to large fiscal deficit until 2014/15 and increases at a slower rate thereafter as real foreign interest rates and GDP growth gradually offset the upward pressures on debt stemming from primary deficits and real domestic interest rates. Towards the end of the forecast horizon, the primary deficit widens slightly as a result of trade revenues declining (stemming from low demand from imports as a result of dwindling investments).

Figure 4.18. Uganda Public Sector Debt Dynamics, 2007/08-2026/07



4.53 **Naturally, this raises the question as to whether our business-as-usual assumptions are realistic**. On the revenue and expenditure side we think they are. The government has a proven record of at least 8 years showing that it can maintain expenditure under control, even in an environment where aid has varied. Moreover, the government has gradually been increasing revenues and has announced its intentions to increase revenues well beyond our conservative estimate. By contrast, we are less certain about our growth forecasts and linkages between public and private investments, and import and export prices, and the extent to which donors are willing to finance the entire deficit. The latter will ultimately boil down to uncertainty regarding the interest rate which will be attached to the stock of debt.

G. SENSITIVITY ANALYSIS TO BASELINE SCENARIO

4.54 **The sensitivity analysis shows that – from a fiscal sustainability angle – the results of the business-as-usual are broadly robust to a range of shocks**. That is, although the level of debt and associated interest rates will vary depending on the shock considered, the debt burden can be comfortably managed over the next two decades.

4.55 The analysis considered a range of different shocks to test the robustness of our baseline, including a shock to terms of trade, growth, interest rates and share of aid consisting of loans versus grants. Below, a growth slowdown is presented as an example of the shocks examined while the discussion of the other shocks is relegated to an annex.

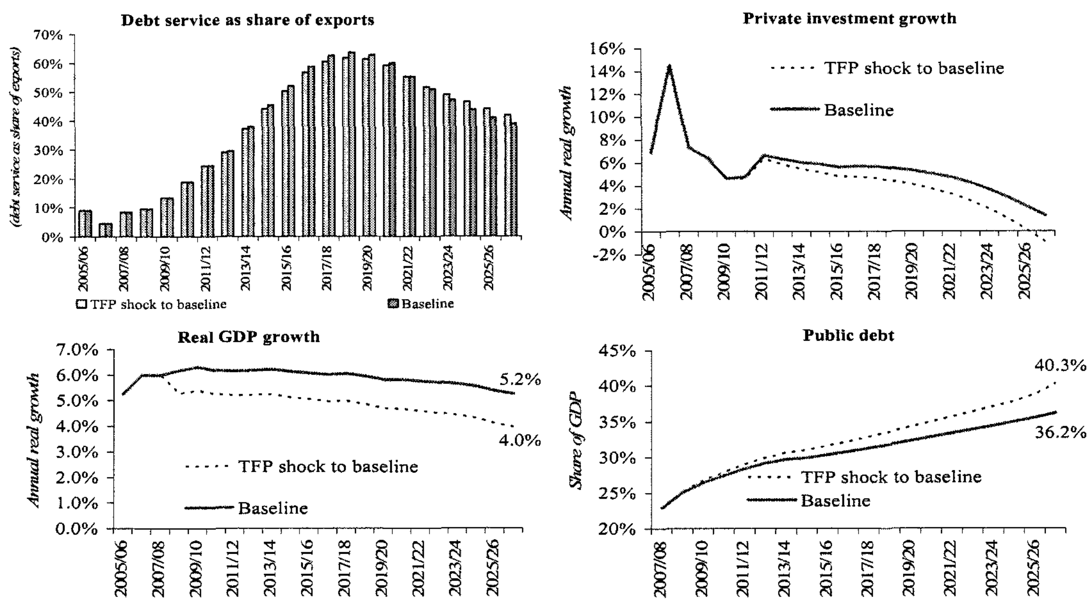
4.56 To test the robustness of our business-as-usual results to a growth slowdown, we re-examined all our results under the assumption that TFP growth slow down significantly. Specifically, instead of the 0.7 percent annual growth assumed in the baseline (i.e. trend growth), we assume that the TFP growth collapses to 0 percent for the duration of the forecast. While zero percent growth is not unusual for a given year, the assumption of 20 years of flat TFP growth is clearly testing the limits of what is plausible and we, therefore, consider this scenario a worst-case scenario from a TFP growth perspective.

4.57 Other things equal, the slowdown in TFP growth causes real GDP to slow down. Such slowdown, mechanically, raises public debt to GDP ratio (by lowering the denominator). Moreover, with total spending set at 22 percent of GDP and recurrent spending pressures unchanged, the slowdown also results in even fewer resources being available for infrastructure spending (compared to the baseline). As a result, private investment growth decelerates further, triggering a further slowdown in real GDP growth and export growth (compared to the baseline). These impacts are shown in Figure 4.19.

4.58 Although the growth forecast looks gloomier, debt levels remain at sustainable levels, increasing only marginally beyond the business-as-usual scenario.

4.59 As a further robustness test, we also examined the impact of a combined TFP growth slowdown and lower "underlying" private investment growth (as explained in the annex, private investment growth is assumed to growth at trend growth plus whatever additional impact public investment growth provides, see annex for details). In particular, we assume that underlying private investment growth is halved for the entire duration of the forecast. Again, the growth forecast looks even gloomier but the impact on the level of debt is only a marginal increase.

Figure 4.19. Comparison of Business-as-Usual and TFP Shock, Selected Indicators



H. GROWTH SCENARIO: A FISCAL POLICY FOR GROWTH

4.60 **How could Uganda deliver a growth-oriented Budget?** The business-as-usual scenario revealed that the main fiscal risk is not a sustainability risk but rather a risk that public funds will increasingly be used to finance consumption rather than investment. The issue, then, is how to change that. Chapter 2 dealt with possible changes to rules and fiscal strategy. Here we consider spending composition. One possibility is to spend more and have the additional expenditure explicitly earmarked for investments. To spend more, more revenue, more grants, or more borrowing would need to be mobilized.

4.61 **An alternative strategy is use public resources more efficiently.** That is, getting more output and outcomes for the same amount of money. As chapter 2 describes, improving efficiency in public spending requires detailed analytical work and careful thinking. Chapter 3 on education provides an example of the type of analytical needed to identify areas where efficiency can be improved. In this section, we assume that overall spending remains the same level of GDP as in the baseline (i.e. at 22 percent of GDP) but we assume that the education sector expands within existing ceilings, an assumption we think is plausible given the identified inefficiencies documented in chapter 3. In addition, as the detailed assumptions below shows, we assume that similar efficiency saving measures are taken in other sectors.

4.62 **The keys are expenditure efficiency and control over employee costs**. The growth scenario shows that a tighter wage policy combined with a policy that actively seeks to improve outputs and outcomes for the same amount of public resources can free up substantial amounts of resources for growth-enhancing investments. Our growth scenario shows that:

- If helped and motivated to use its resources better, the education sector would be able to achieve the same goals (in terms of number of primary and secondary graduates) by spending the same amount of resources (measured as a share of GDP) as it does today.
- Combined with public service reform and a wage policy[86] that actively seeks to lower the annual growth in real wages and new civil servants, significant additional resources would be available for investments, compared to the business-as-usual scenario.
- As a result of the improved public capital stock (e.g. reliable and cheaper provision of electricity, lowered cost of transportation, etc), private investment would accelerate compared to the growth rates observed in the baseline,
- With more private investment, there is every reason to believe that the vibrant entrepreneurial spirit observed in the current search for new exports would accelerate.
- As a result, growth accelerates and becomes more broad-based with exports and investments being more important drivers of growth
- The economy will still require larger amounts of aid (to finance still-large fiscal deficits) but the economy's capacity to repay the loans is clearly improving over the forecast horizon.
- As in the baseline, debt levels will remain at fiscally sustainable levels and will be on a downward trajectory towards the end of the forecast horizon.

I. ASSUMPTIONS USED IN GROWTH SCENARIO

4.63 In this PER we provide examples of the analytical work required to identify efficiency savings within a sector. In this growth scenario, we assume that these recommendations get implemented. In addition, we also assume that:

- The growth rate of the real wages to all civil servants is reduced to ½ of annual real GDP. That is, if real GDP is growing by 6 percent, real wages would grow by 3. Still, with 5 percent annual inflation, nominal wages would still be increasing by 9 percent. However, even at this rate, this would be a significant reduction from the past 8 years where real

[86] The wage policy set before Cabinet last year does the opposite!

wages, on average grew by 6.2 percent (and slightly faster than real GDP) (see Table 4.6).

- The growth rate in the number of civil servants is assumed to slow by 2 percentage points from trend growth, except in education where they will grow as forecasted in education module.
- Pension expenditure is reduced to 9 percent of the wage bill compared to 12 percent in the baseline. In 2005/06, pension expenditure as a share of the wage bill was 9 percent but, at this low rate, the government reportedly still accumulated pension arrears. Therefore, in the baseline we assumed that pension expenditure had to increase to 12 percent to avoid accumulating further arrears.

4.64 The education savings measures which we include in this scenario include:

- Reduce unit costs of constructing fully furnished classrooms at acceptable quality by 15 percent.
- Expand double shifting in both primary and secondary schools.
- Reduce the number of core courses taught in secondary schools to 10.

Box 4.3. Assumptions Used in Growth Scenario

Macro

- Same as in baseline but, given that private investment and growth (and, as result, revenues) are tied to the composition of fiscal spending, key macro variables end up changing compared to the business-as-usual forecast (see summary graphs below)

Fiscal

- All variables are modeled the same was as in the baseline forecast but we assume the following changes:
- The growth rate of the real wages to all civil servants is reduced to ½ of annual real GDP. That is, if real GDP is growing by 6 percent, real wages would grow by 3. Still, with 5 percent annual inflation, nominal wages would still be increasing by 9 percent. However, even at this rate, this would be a significant reduction from the past 8 years where real wages, on average grew by 6.2 percent (and slightly faster than real GDP) (see Table 4.6).
- The growth rate in the number of civil servants is assumed to slow by 2 percentage points from trend growth, except in education where they will grow as forecasted in education module.
- Pension expenditure is reduced to 9 percent of the wage bill compared to 12 percent in the baseline. In 2005/06, pension expenditure as a share of the wage bill was 9 percent but, at this low rate, the government reportedly still accumulated pension arrears. Therefore, in the baseline we assumed that pension expenditure had to increase to 12 percent to avoid accumulating further arrears.

The education savings measures which we include in this scenario include:

- Reduce unit costs of constructing fully furnished classrooms at acceptable quality by 15 percent.
- Expand double shifting in both primary and secondary schools.
- Reduce the number of core courses taught in secondary schools to 10.

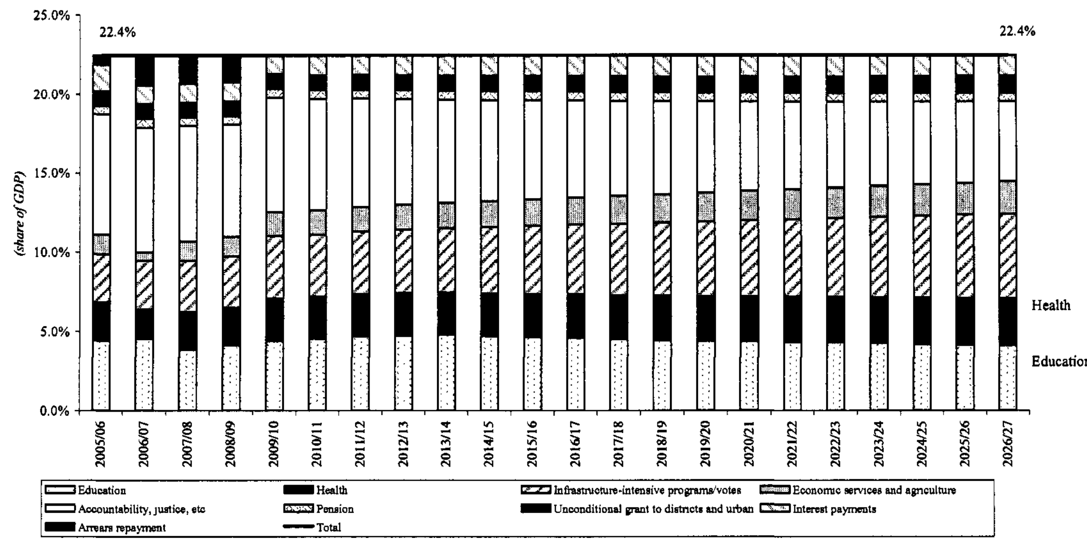
4.65 Our chapter on education efficiency suggests that our assumptions are not entirely unrealistic. In the primary education sub-sector's recurrent expenditure, alone, we document that 21 percent (or 0.4 percent of GDP) is outright wasted. Waste in other non-education sectors could be even larger but further detailed analysis would be needed.

4.66 Still, the assumptions are not meant to be a blueprint for needed policy changes. Rather, the scenario and the assumptions that go with it are meant to illustrate the extent to which "belt-tightening" is needed to mobilize additional resources for infrastructure. Political choices will ultimately determine the extent to which the savings above can be realized.

J. RESULTS FROM GROWTH SCENARIO

4.67 If helped and motivated to use its resources smarter, the education sector would be able to achieve the same goals (in terms of number of primary and secondary graduates) by spending the same amount of resources (measured as a share of GDP) as it does today. Although rising slightly above 5 percent in the coming years to accommodate additional expenditures associated with USE, thereafter, the education sector is forecast to be able to meet the same goals as in the baseline and not exceeding 5 percent of GDP.

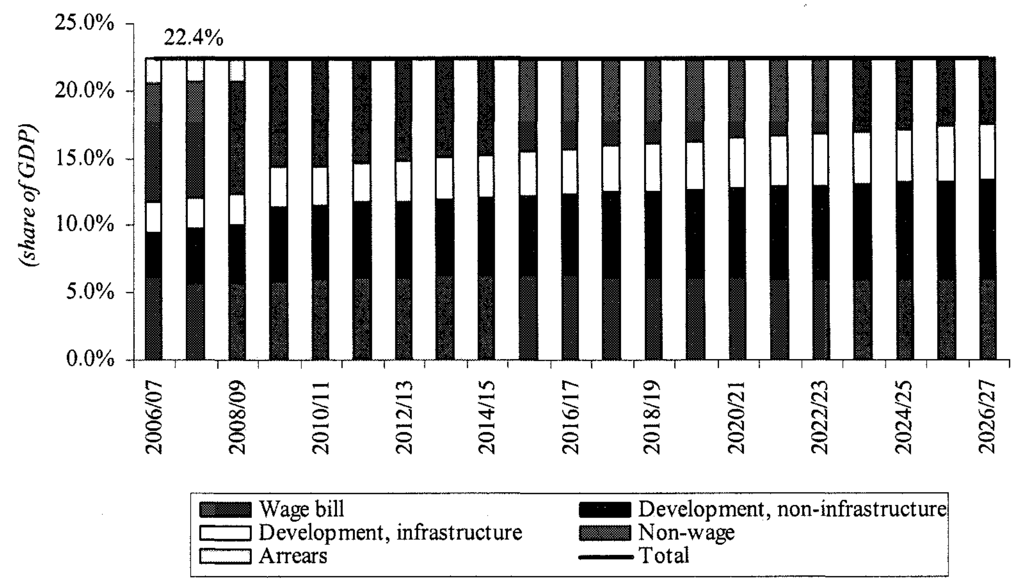
Figure 4.20. Growth Scenario Forecast of Composition of Expenditure, Functional Perspective



Source: World Bank staff's growth scenario forecast

4.68 Combined with a wage policy that actively seeks to lower the annual growth in real wages and new civil servants, significant additional resources would be available for investments, compared to the business-as-usual scenario. From a functional perspective, this wage restraint helps curtail spending in "public administration, defense, justice, etc" category as well as in education spending (where the majority of civil servants are employed).

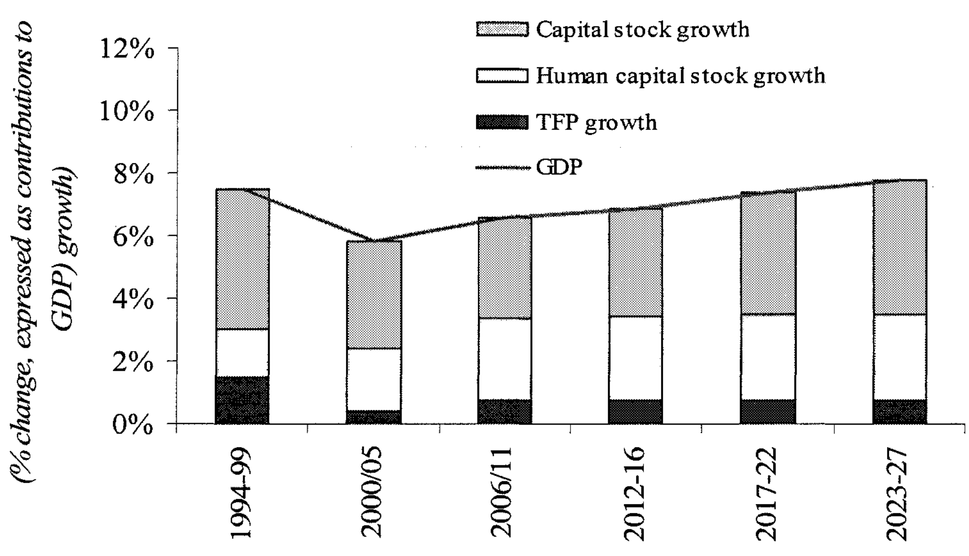
Figure 4.21. Growth Scenario Forecast of Composition of Expenditure, Economic Perspective



Source: World Bank staff's growth scenario forecast

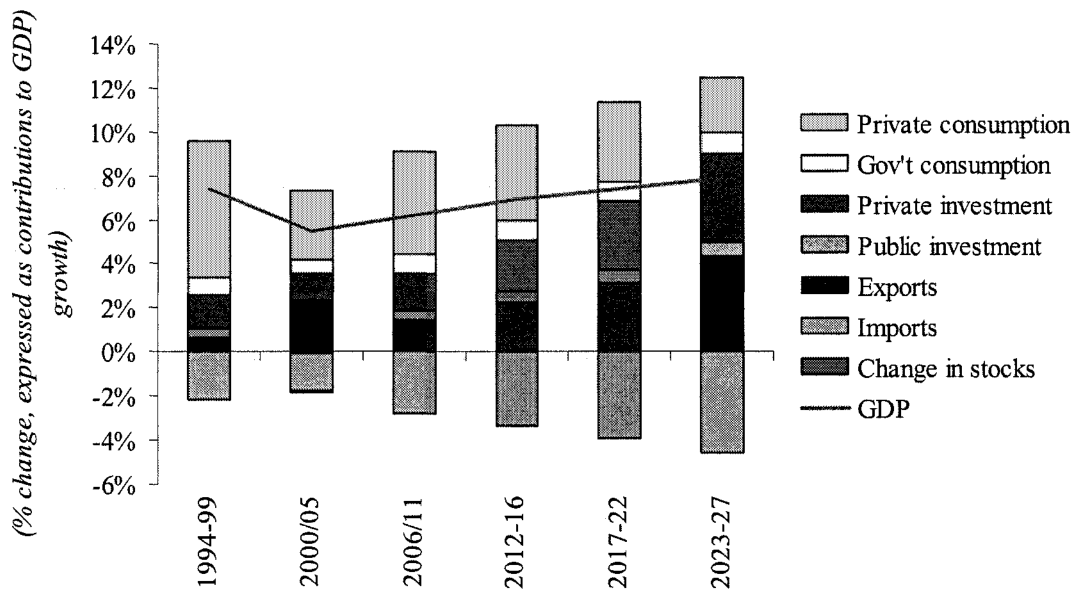
4.69 As a result of the improved public capital stock (e.g. reliable and cheaper provision of electricity, lowered cost of transportation, etc), private investment would accelerate above the growth rates observed in the baseline. And with more private investment, there is every reason to believe that the vibrant entrepreneurial spirit observed in the current search for new exports would accelerate. As a result, growth accelerates and becomes more broad-based with exports and investments being more important drivers of growth (see figure 4.23).

Figure 4.22. Growth Scenario: Annual Real GDP Growth Decomposed in Contribution From Growth Accounting Perspective



Source: World Bank staff's growth scenario forecast

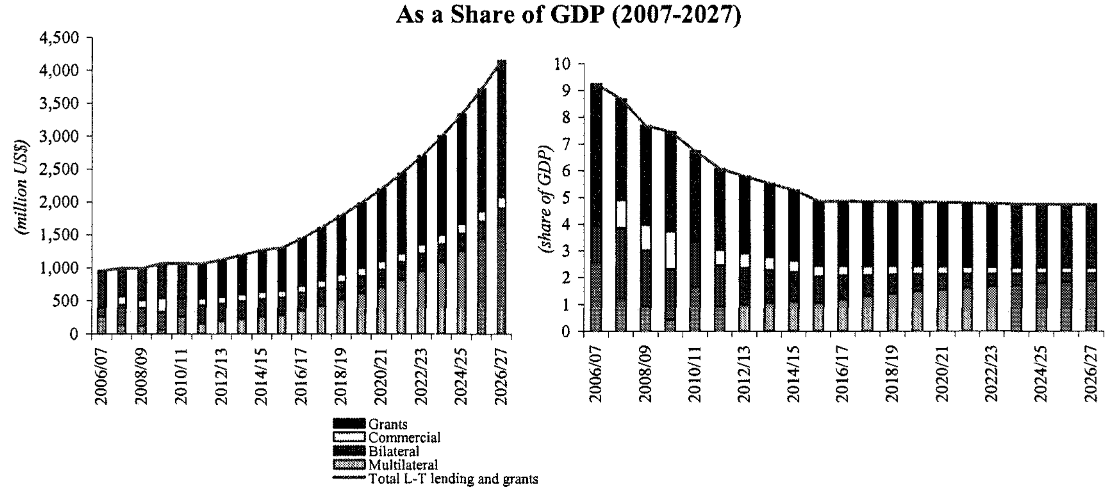
Figure 4.23. Growth Scenario: Annual real GDP Growth Decomposed in Contribution From Expenditure Side



Source: World Bank staff's growth scenario forecast

4.70 The economy will still require larger amounts of aid (to finance still-large fiscal deficits) but the economy's capacity to repay the loans is clearly improving over the forecast horizon.

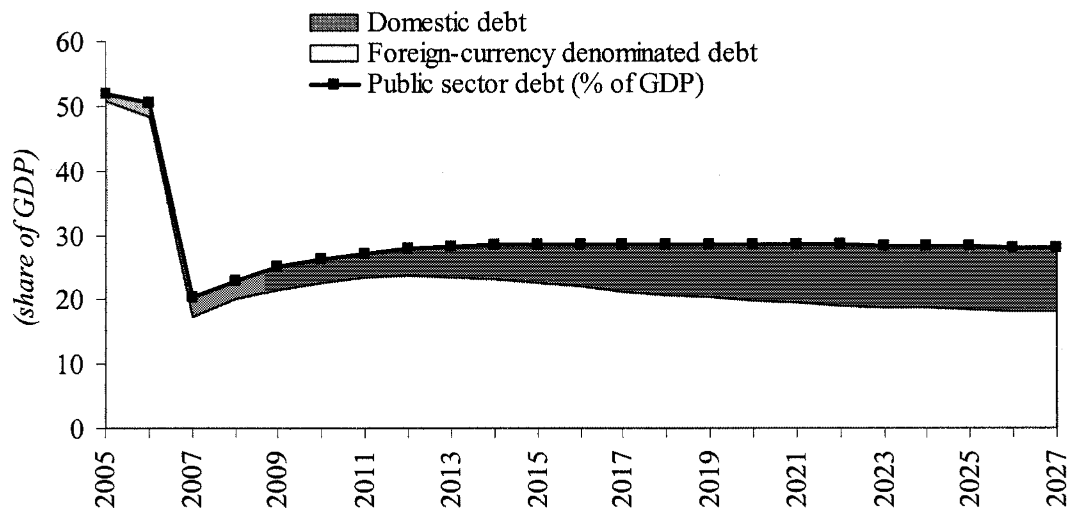
Figure 4.24. Growth Scenario: Assumed Grants and Long-Term Lending in Million US$ and As a Share of GDP (2007-2027)



Source: World Bank staff's growth scenario forecast

4.71 As in the baseline, debt levels will remain at fiscally sustainable levels but, unlike in the baseline case, debt lowers remain broadly unchanged from 2014 onwards.

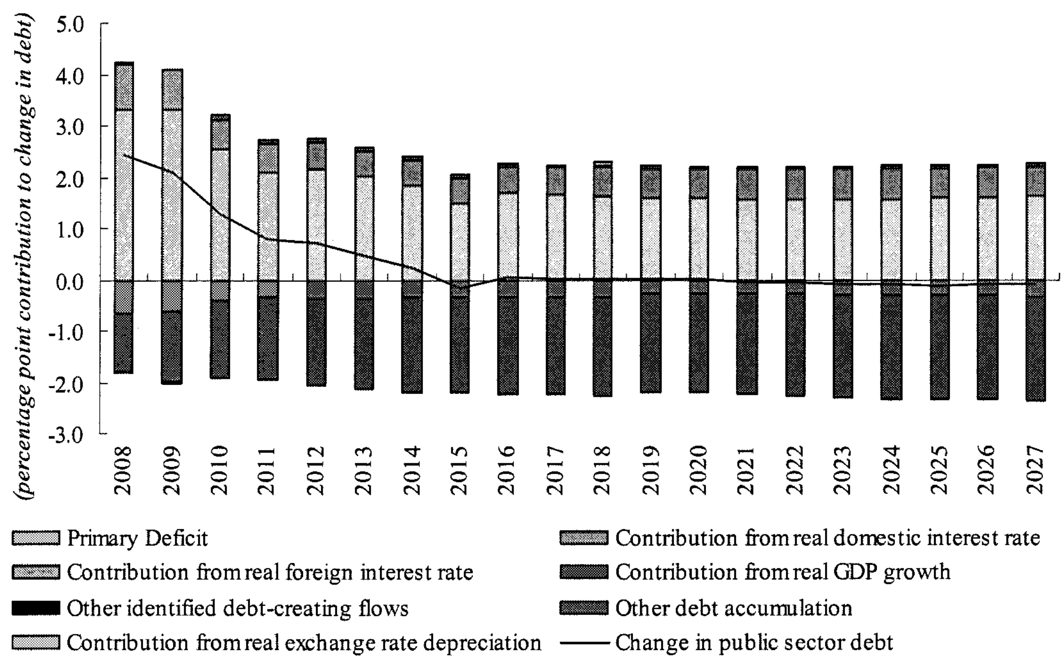
Figure 4.25. Public Sector Debt (share of GDP)



Source: World Bank staff's growth scenario forecast

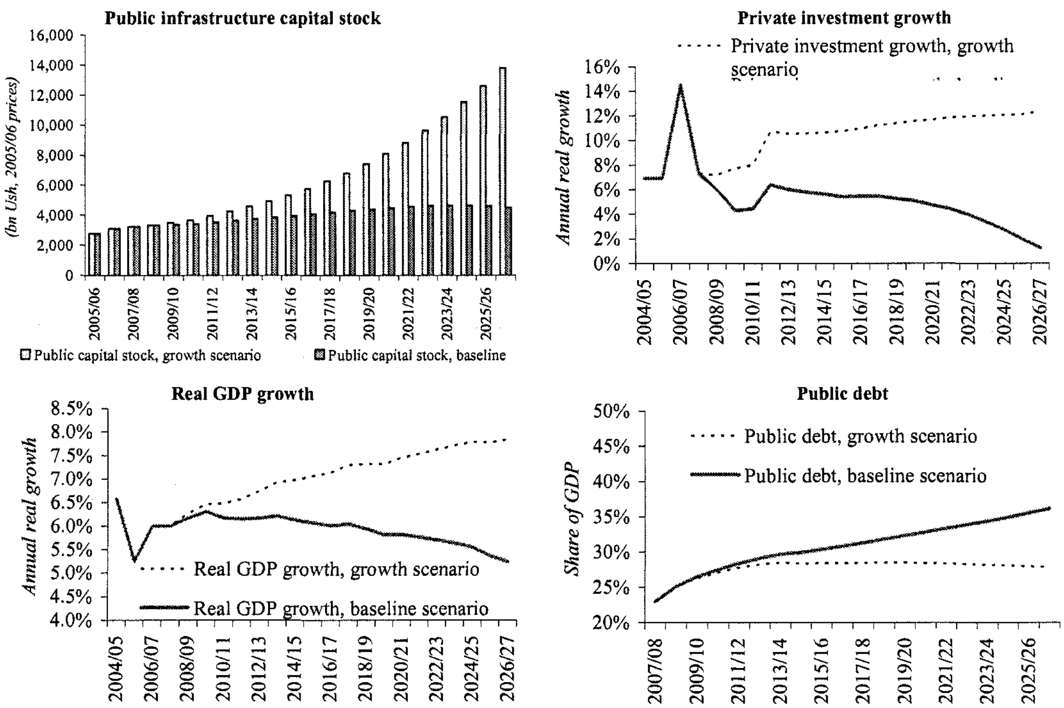
4.72 Similar to the business-as-usual case, it is the real interest rates on the external debt and the shrinking primary deficit which are the primary drivers behind the benign forecast of public debt (see figure 4.26). Figure 4.27 summarizes some key indicators from the baseline and growth scenario.

Figure 4.26. Uganda: Public Sector Debt Dynamics, 2007/08-2026/07



Source: World Bank staff's growth scenario forecast

Figure 4.27. Comparison of Business-as-Usual and Growth Scenario, Selected Indicators



K. Conclusion and Policy Implications for Budget Planning and Discussion

4.73 This chapter used a new macro and fiscal model to demonstrate that:

- Given MDRI which virtually eradicated Uganda's public debt, Uganda is unlikely to be facing a fiscal debt service problem for the coming twenty years, provided that new borrowing continues to be on concessional terms.
- The fiscal risk facing Uganda in future is not one of liquidity or fiscal debt sustainability, but of long-term solvency through economic growth.
- A shift in fiscal composition is needed for growth otherwise growth in recurrent expenditure is likely to squeeze out infrastructure spending, with a likely slowdown in growth to follow.

4.74 The analysis in the chapter and results above give rise to a number of policy implications.

- First, fiscal data is in need of improvement. We propose that MOFPED set up a working group to identify a time-bound action plan to deliver fiscal data which address the data weaknesses identified in this chapter. As a start, there appears a clear need for the MOFPED to start maintaining and make publicly available a detailed functional and economic data base of historical fiscal data. Such historical data is not readily available on the MOFPED's website.

- Second, curtailing recurrent spending pressures cannot be done without addressing growth in employee costs which currently consume more than half of the total budget. This chapter does not provide detailed recommendations on how to reduce such costs. However, this chapter documents – for the first time – an economic classification of the budget which shows the magnitude of employee costs. By June 2006, we had only obtained the detailed fiscal data to investigate employee costs for 2 years (2004/05 and 2005/06). A longer time series of detailed fiscal data – and the help of civil service experts – will be needed to fully understand the drivers of costs in employee costs (as opposed to only wage costs).

- Third, given that infrastructure is the binding constraint to growth, freeing up budgetary resources to spend on adding such infrastructure is likely to help accelerate growth in Uganda. We have purposely downplayed the size of this potential growth impact since such figure is only as good as our underlying assumptions. However, the contrast between growth in the two scenarios is clear: growth will either falter if infrastructure bottlenecks are not addressed or it could accelerate.

- Although the size of contingent liabilities currently appears low, fiscal risks stemming from these should not be ignored. First steps in managing such risks include strengthening transparency, disclosure and accountability requirements.

5. INFRASTRUCTURE SECTORS[87]

5.1 **This chapter suggests how best to *use* the room that could be created to remove infrastructure bottlenecks to growth.** Previous chapters show the pressures restricting spending room in the budget, and how Uganda can create room for more spending on growth-enhancing infrastructure. Here we aim to guide decision makers towards infrastructure spending priorities for growth.

5.2 **The focus on public infrastructure *spending* priorities provides an incomplete picture of priorities to remove binding constraints to growth in Uganda.** It should not be read in isolation from the CEM chapter on binding constraints. Nor should the public spending => growth link be exaggerated. One complication is that in Uganda's case many infrastructure priorities now lie outside of the budget framework; either because the growth constraint is not one that can be fixed by providing more of a traditional public good or service (eg coordination gaps, regulation), or because the State has been withdrawn from direct public subsidies (eg in the energy sector).

5.3 **In addition, there is no guarantee that additional spending on public infrastructure will generate growth.** Since public infrastructure is an intermediary input in private transportation, electricity and water provision, assessing the growth and competitiveness impact of public infrastructure spending requires more than an assessment of public spending trends. It requires accurate data on the prices paid by businesses for transport and energy, and the quality of services that the market is providing. Outturns in price and quality for transport, water and energy services then need to be attributed somehow to public infrastructure. This complex link is not attempted under this study because it would require a detailed impact evaluation of sector specific programs.

- In section A we **benchmark Uganda's infrastructure indicators against neighbors** and peers to seek any obvious areas of under-performance. We ask whether Uganda is allocating enough to infrastructure.
- In section B we assesses whether there is **scope to improve the composition of public expenditure on infrastructure** to better address Uganda's infrastructure constraints[88]. We ask whether Uganda is making adequate use of private and donor finance, and whether allocations across and within infrastructure sub-sectors are appropriate, including whether maintenance is adequate. We end by identifying some institutional issues which Uganda needs to address to improve performance in infrastructure provision.
- We conclude that **Uganda is under-investing in infrastructure and is under-maintaining the existing public infrastructure stock.** We suggest a number of priority actions to reverse these trends. The most significant, to increase absorptive capacity, would be to give parastatals more responsibility for investment decisions. This would need to be coupled with better performance monitoring of parastatals and public private partnerships, including by improving project selection and reporting systems.

[87] This note summarizes work presented in the Volume 2 of the Uganda Country Economic Memorandum, World Bank (2007), and in a summary background paper for this report.
[88] Using the terminology in chapter 2, this is not a chapter on "efficiency". Rather this chapter identifies priorities for resource shifts to make infrastructure spending more "productive".

A. BENCHMARKING UGANDA'S INFRASTRUCTURE

5.4 **Uganda is relatively infrastructure constrained**. Chapter 6 of Volume 2 of the recent World Bank Country Economic Memorandum discussed Uganda's infrastructure and its relationship to growth in some detail. Key findings relative to similar countries are set out in figures 4.1 to 4.3 below:

- **Infrastructure access and quality rates in Uganda are low** by the standards not just of sub-Saharan Africa, but of low-income countries.
- **The electricity sector in particular is in crisis**. There is inadequate generation, the transmission network serves only a relatively narrow area and does not reach areas with potential power generation, and distribution losses are high. Due to a lack of investments and poor distribution and collections, the national grid cannot meet the growing demand for power, which has risen annually on average by 11.5%.
- Almost **half of firms report electricity to be significant constraint to their business** expansion, and Ugandan firms report more power outages than average LIC. The largest 1/3 firms generate their own electricity (which for them is 2-3 times more expensive than the grid). The export intensity of firms in Uganda, which is low anyway, is lower still for firms suffering outages.
- A greater share of firms in Uganda than elsewhere cite **transport as a severe or major obstacle to growth**.
- There are obvious **reasons for high transport costs in Uganda**:
  - Uganda is landlocked – imported materials come mostly from Mombassa, 1300 km away.
  - Only 7% of Uganda's road network is paved.
  - Only 25% of national roads are paved.
- **Uganda has spent a lot of money upgrading and maintaining roads**:
  - National roads in good condition increased from 6% to 75% between 1988-2003.
  - District roads in good condition increased from 15% to 67% between 1988-2003.
  - 67% households reported accessible all year round – outperforming other LICs.

Table 5.1. Standardized Table Comparing Infrastructure Access with Peers and International Benchmarks

Standardized table comparing infrastructure access with peers and international benchmarks
(Latest Observations)

		Year	Uganda	LIC*	% Deviation	Relative Performance	Sub-Saharan Africa*	% Deviation	Relative Performance
Energy	Households Reporting Access to Electricity (% of households)	2001	**8.6**	34.7	-75%	*Underperform*	27.2	-68%	*Underperform*
Energy	Households Reporting Access to Modern Cooking Fuels (% of households)	2001	**3.2**	11.7	-73%	*Underperform*	13.0	-75%	*Underperform*
Water Supply	Improved water source (% of population with access)	2002	**56**	63.8	-12%	*Underperform*	64.1	-13%	*Underperform*
Water Supply	Households using piped water as major source of drinking water (% households)2	2001	**10.8**	30.7	-65%	*Underperform*	31.9	-66%	*Underperform*
Sanitation	Improved sanitation facilities (% of population with access)	2002	**41**	37.5	9%	On Avg.	36.5	12%	*Outperform*
Sanitation	Households Reporting Access to a Flush Toilet (% of Population)	2001	**1.7**	8.6	-80%	*Underperform*	8.7	-80%	*Underperform*
Telecomm	Teledensity (Cell+Fixed Subscribers per 1000 inhabitants)	2005	**56.4**	82.0	-31%	*Underperform*	144.5	-61%	*Underperform*
Telecomm	Internet users (Subscribers per 1,000 people)	2005	**8.7**	19.0	-54%	*Underperform*	29.4	-70%	*Underperform*
Telecomm	Households with Own Telephone (% of households)	2001	**2.7**	8.8	-69%	*Underperform*	5.7	-53%	*Underperform*
Roads	National network in good and fair condition (% national network)	2004	**75**	..	..	..	..	..	..
Roads	Road density, arable land (road-m/sq-km)	2004	**0.2**	..	..	..	0.80	-75%	*Underperform*
Other Transport	Average time to ship 20ft TEU container from port to final destination (days)		**N.A.**	..	..	..	10.4	..	..
Other Transport	Rural Access (Percentage of rural people who live within 2 km of an all-season passable road as a proportion of the 'total rural population')	2004	**67**	34.7	93%	*Outperform*	26.9	149%	*Outperform*
Irrigation	Surface and groundwater resources per capita (cubic meters)	2001	**1,543**	15,129.0	-90%	*Underperform*	12,781.1	-88%	*Underperform*
Irrigation	Agricultural Area Equipped for Irrigation (sahre of arable land)								

Source: World Bank Africa Development Indicators; WB Infrastructure Research Database; Ctry Sectoral Studies; AFTPI Processed
* Avg. of latest obs. 2000-2005
Performance Key:
Outperform - Deviation greater than +10% from benchmark
Underperform - Deviation less than -10% from benchmark
Average - Deviation between -10% and +10% from benchmark

Table 5.2. Standardized Table Providing Private Sector Assessment of Productive Infrastructure (based on Latest Observations)

Standarized Table providing private sector assesment of productive infrastructure
(Based on Latest Observations)

		Uganda	LIC*	% Deviation	Relative Performance	Sub-Saharan Africa*	% Deviation	Relative Performance
Electricity	% of firms identifying electricity as a major or very severe obstacle to business operation and growth.	**44.5**	34.8	28%	*Above Avg.*	50.0	-11%	*Below Avg.*
Electricity	% of sales lost to outages/surges, average.	**17.8**	23.1	-23%	*Below Avg.*	19.2	-7%	On Avg.
Electricity	Power Outages per year, average for firms (median)	**70.8 (30)**	60.4	17%	*Above Avg.*	82.1	-14%	*Below Avg.*
Electricity	% firms that have own generator	**36**	38.3	-6%	On Avg.	49.2	-27%	*Below Avg.*
Electricity	Days for electricty connection, average for firms (median)	**38.7 (14)**	48.2	-20%	*Below Avg.*	71.6	-46%	*Below Avg.*
Telecomm	% of firms identifying telecom as a major or very severe obstacle to business operation and growth.	**5.2**	15.2	-66%	*Below Avg.*	24.6	-79%	*Below Avg.*
Telecomm	Days for a mainline telephone connection, average for firms (median)	**33.4 (7)**	67.2	-50%	*Below Avg.*	86.7	-61%	*Below Avg.*
Telecomm	Phone service outages per year, average for firms (median)	**17.8 (1)**	24.2	-26%	*Below Avg.*	17.3	3%	On Avg.
Telecomm	% of firms using e-mail	**38.7**	48.2	-20%	*Below Avg.*	57.9	-33%	*Below Avg.*
Transport	% of firms identifying tranport as a major or very severe obstacle to business operation and growth.	**22.9**	16.8	36%	*Above Avg.*	24.7	-7%	On Avg.
Transport	% of average cargo lost in transit, average for firms.	**2.9**	1.4	107%	*Above Avg.*	1.9	53%	*Above Avg.*

Source: World Bank Business Surveys, Investment Climate Surveys (2003 data for Uganda), AFTPI Processed
* Avg. of latest obs. 2000-2004
Performance Key:
Outperform - Deviation greater than +10% from benchmark
Underperform - Deviation less than -10% from benchmark
Average - Deviation between -10% and +10% from benchmark

- **Transport is hindered by constraints beyond the roads sector.** There is a trade imbalance through the airport and on the railway. Traffic counts on many rural roads are low. Fuel comprises 50 percent of vehicle operating costs which is high by international comparisons, suggesting high costs of fuel and fuel inefficiency. High fuel costs have been found to be related to high petroleum margins, high taxes on fuel, and the lack of a petroleum pipeline and rail link for imported fuel, which is driven from Kisumu in Kenya.

- **Urbanization will accelerate rapidly in the coming decades** because of Uganda's demographics and labor market trends. After Burundi, Uganda has second lowest urbanization rate in Africa. Just over half of all Ugandans are under 15 years old, making Uganda the world's most youthful country. On average mothers have 6.7 children, the world's third fastest fertility rate. High fertility and youthfulness mean Uganda's population will explode in two generations. Growth in the urban population is already rising faster than the national population, but from a base ¼ of the size of the rural population. As agriculture continues to release labor and plot sizes dwindle, more and more Ugandan workers will seek jobs in the towns and cities.
- **Tele-density in Uganda is lower than in peer countries,** but grew at 55% from 2000-04.
- Uganda also needs to **expand internet access**, which is much lower than peers.
- **Uganda has raised access to infrastructure by more than its peers** over the past decade, except in the electricity sector where improvements have been limited.
- AfDB and WB (2006) states that $200 million per year is needed for water and sanitation alone for Uganda to reach the MDGs.

Table 5.3. Standardized Table Indicating Access to Basic Services, Urban Rural Split

		Uganda	Kenya	Tanzania	LIC	Sub-Saharan Africa 1/
Electricity 2/	Urban	43.9	50.2	38.9	65.1	60.2
	Rural	2.4	4.6	1.6	17.3	10.2
Water 3/	Urban	87.0	89.0	92.0	82.7	82.5
	Rural	52.0	46.0	62.0	54.9	53.9
Sanitation 4/	Urban	53.0	56.0	54.0	58.3	54.3
	Rural	39.0	43.0	41.0	27.8	27.9
Telephone 5/	Urban	14.5	32.7	31.4	20.9	13.8
	Rural	0.6	6.2	3.0	3.0	1.0

Source: Household Surveys, various years; JMP/WHO. AFTPI processed
Notes:
1/ Average of latest Obs. Available 2000-2005
2/ DHS and HHS – Households reporting access to electricity: Uganda (2001); Kenya (2003); Tanzania (2004)
3/ JMP/WHO – Percent of population with access to an improved water source (2002 data)
4/ JMP/WHO – Percent of population with access to an improved sanitation facility (2002 data)
5/ DHS/HHS – Households reporting access to a telephone: Uganda (2001), Jebta (2003); Tanzania (2004)

Table 5.4. Standarized Table Providing International Benchmarking of Infrastructure Spending Level

	latest year available	Infrastructure Spending 1/ 2/ (US$-person)	(% GDP)	Income Per Capita (Current US$)
Uganda	2004	19.26	7.18%	250.00
Tanzania	2004	30.9	9.2	297.3
Kenya	2003	43.86	10.50	430.00
Indonesia	2004	93.2	7.9	1166.5
Turkey	2004	240.1	5.7	4209.7
Chile	1998	317.3	6.4	4923.5

Source: Fiscal Cost Baseline, WB PPI Database

Note:(1) Only Includes Public Spending

1/ Infrastructure includes energy (oil, gas and electricity), water and sanitation, irrigation, transportation (air, land, port, roads), and telecommunications.

2/ Overall spending includes operations and maintenance, investment for both public and private sector; unless otherwise indicated

B. COMPOSITION AND MANAGEMENT OF INFRASTRUCTURE SPENDING

B1. Does Uganda Allocate Enough To Infrastructure?

5.5 **Uganda allocates less to infrastructure as share of GDP than other Low Income Countries** analyzed under the Bank's recent fiscal cost baseline study, and lower than its neighbors Kenya and Tanzania. Compared to upper middle income countries Turkey and Chile, Uganda devotes a greater share of its GDP to infrastructure, but as can be seen, in per capita terms these countries invest many times Uganda's amount. In 2004 (the most recent year for which data are available for all countries, Uganda spent 7.2% of GDP and about $US 19.3 per capita on infrastructure (box 4.4). Despite recent privatization and utilities reform, about half of this amount is still on-budget. Parastatals account for around 33 percent of infrastructure spending. Local Governments provide just under 5%.

B2. Is Uganda Making Adequate Use of Private Funding?

5.6 **Uganda has been more open to private sector participation in infrastructure than the rest of Africa**. In part this was to try and improve the efficiency of infrastructure service delivery. Over 2001-2005, the private sector committed infrastructure financing to Uganda's telecom and energy sectors worth an average of 1% of Uganda's GDP per year. This was just a little better than other LIC and SSA countries. Yet this amount has grown over 150% from the average over 1996-2000, and Uganda had eight operational private sector projects as of 2005. Public private partnerships (PPPs) through concessions have been used in electricity generation and distribution, and more recently in the railway. In addition management contracts have been used in the water sector.

5.7 **Nevertheless, the country still receives only 14% of total infrastructure from private financing.** The average size of the projects was also smaller compared to peers. The average investment per project was US$58 million, compared to over US$150 million in other LICs and

US$110 million in SSA. This however, may just reflect Uganda's willingness to involve the private sector in more areas than its peers. The picture will change dramatically with the onset of the Bujagali Dam project expected this year, which will provide a second major private power generation concession. This fraction in rail transportation can also be expected to grow in the coming years.

Table 5.5. Standardized Table Showing Level and Main Actors in Infrastructure Provision

Standardized table showing level and main actors in infrastructure provision

Nominal GDP Shares (%)	2003-2004	2004-2005
INFRASTRUCTURE SPENDING	**6.53%**	**7.18%**
Total Gross Investment	**3.68%**	**4.62%**
Total O&M	**2.85%**	**2.57%**
On-Budget 1/	2.67%	3.44%
Gross Investment	2.04%	2.88%
O&M	0.63%	0.56%
Local Governments 2/	0.35%	0.35%
Gross Investment	0.19%	0.20%
O&M	0.16%	0.15%
Parastatals/SOE	2.34%	2.36%
Gross Investment	0.28%	0.50%
O&M	2.06%	1.86%
Special Funds		
Gross Investment		
O&M		
Private Sector		
Gross Investment 3/	1.17%	1.04%

Source: Fiscal Cost Baseline, WB PPI Database

Notes:

1/ Excluding transfers to Local Governments

2/ Explain whether includes expenses funded by transfers from the CG and/or own revenues

3/ Proxy by Committed Investment with Private Participation

5.8 **The majority of Uganda's private infrastructure investment is in the telecom sector.** This is typical of most SSA and LIC countries, but Uganda has fared particularly well in telecoms. Uganda's three mobile players had a combined subscriber base of around 1.75 million as of the end of 2005, with MTN leading the way followed by "Mango." Uganda also sold a 51% majority stake in UTL to a private consortium which included Deutsche Telecom in 1999 in a competitive bid for US$33.5 million. A year earlier, Uganda took the important step of introducing competition not only in their mobile sector, but also in fixed line services by awarding a second national operator's license (SNO) to MTN and creating an independent regulator for the sector.

5.9 **The energy sector has ambitious plans for private investment.** In 2003, Uganda became one of only a handful of SSA countries to open its energy sector up to private investment when it signed a 20-year generation concession with Eskom Enterprises of South Africa to take over the operations and maintenance of the 120-MW Kiira and 180-MW Nalubaale power stations at Owen Falls in Jinja from Uganda Electricity Generation Company Limited (UEGCL). At the time of the concession, the two power stations generated 95% of Uganda's electricity supply. In 2005, the major functions of The Uganda Electricity Distribution Company Ltd. (UEDCL) were also privatized under a 20 year concession contract to Umeme Limited for a distribution network that covered roughly 40% of Uganda and provided service to approximately 250,000 customers. Umeme Limited was jointly owned by Eskom Enterprises (44%) and Globaleq of the UK (56%). The Bujagali Dam project is expected to commence this year, at a total cost of close to $800 million.

5.10 **A drawback of electricity sector concessions has been the high degree of government support offered to the private operators**. In generation, revenue is guaranteed during the first seven years of the concession, meaning that the government bears the risk of changes in generation capacity. In distribution, the concession included US$11 million worth of government investment and guarantees against government default. Such government support may be necessary to initially attract private participation into previously state dominated sectors, but Uganda also needs to ensure that private operators assume all ordinary market risks to their investments. This theme is picked up in more detail in section B6.

5.11 **Uganda is one of only eight other SSA countries to experiment with private participation in the water sector**. In two successive management contracts beginning in 1998, portions of the state-owned utility National Water and Sewerage Corporation (NWSC) were brought under private sector management. Since both contracts were for management only, neither required the private firms to commit any of their own resources to the network and they were based primarily on a fixed monthly fee paid by the government.

5.12 **Private investment is also being sought in the railways sector**. In October 2005, the Kenyan and Ugandan governments awarded a 25 year concession to Rift Valley Railways Consortium (led by South African investors) to operate and manage the national railways systems of both Kenya and Uganda. The private operator bears most of the commercial, performance and financial risks under the contract while the government has provided a partial political risk guarantee through MIGA.

B3. Is Uganda Making Adequate Use of Donor Funding?

5.13 **Uganda receives significantly more money from donors towards its infrastructure sector than its peers**. This amount grew over time, from an annual average of 1.7% of Uganda's GDP over 1995-1999 to 2.6% over 2000-2004 which is more than double both the average for SSA and LICs. Uganda received 60% of this funding from multilateral sources and 40% from bilateral sources over the decade 1995-2004. Among the top donors to Uganda's infrastructure sector over the last decade were IDA (US$550 million), Germany (US$120 million), Denmark (US$108 million), and the EC (US$103 million). Major recent World Bank (IDA) projects include the three phase Road Development Program and the Energy for Rural Transformation project.

5.14 **Donors played a highly significant role in determining infrastructure spending allocations.** In Uganda's 2004-2005 budget, donor resources accounted for about 60% of total on-budget spending and about one-third of total public spending (i.e. including expenditures by parastatals). The largest sustained increase in infrastructure spending since 1997 was driven by donors, specifically responding to the inception of the Poverty Action Plan Fund (PAF). The PAF earmarks funds for "priority sectors" which the program defined largely as rural (feeder) roads and water supply for consumption.

Table 5.6. Standardized Table Comparing Official Development Assistance (ODA) Per Infrastructure Sector with Peers and International Benchmarks

Standardized table comparing Official Development Assistance (ODA) per infrastructure sector with peers and international benchmarks

		Uganda	LIC	% Deviation	Relative Performance	Sub-Saharan Africa	% Deviation	Relative Performance
TOTAL INFRASTRUCTURE	Annual Average ODA (95-99)	1.73%	1.68%	3%	On Avg.	0.77%	125%	*Outperform*
	Annual Average ODA (00-04)	2.59%	1.20%	116%	*Outperform*	0.73%	255%	*Outperform*
Energy	Annual Average ODA (95-99)	0.36%	0.59%	-39%	*Underperform*	0.15%	140%	*Outperform*
	Annual Average ODA (00-04)	1.20%	0.30%	300%	*Outperform*	0.15%	700%	*Outperform*
Water Supply and Sanitation	Annual Average ODA (95-99)	0.63%	0.29%	117%	*Outperform*	0.21%	200%	*Outperform*
	Annual Average ODA (00-04)	0.44%	0.23%	91%	*Outperform*	0.20%	120%	*Outperform*
Telecomm	Annual Average ODA (95-99)	0.01%	0.06%	-83%	*Underperform*	0.04%	-75%	*Underperform*
	Annual Average ODA (00-04)	0.03%	0.03%	0%	On Avg.	0.02%	50%	*Outperform*
Transport	Annual Average ODA (95-99)	0.73%	0.74%	-1%	On Avg.	0.37%	97%	*Outperform*
	Annual Average ODA (00-04)	0.91%	0.64%	42%	*Outperform*	0.33%	176%	*Outperform*
Irrigation	Annual Average ODA (95-99)	N.D.	N.D.			N.D.		
	Annual Average ODA (00-04)	N.D.	N.D.			N.D.		

Source: OECD DAC Statistics. AFTPI Processed.

Performance Key:
Outperform - Deviation greater than +10% from benchmark
Underperform - Deviation less than -10% from benchmark
Average - Deviation between -10% and +10% from benchmark

5.15 **There are problems with the allocation of this donor money**. One has been the potentially excessive focus on investment in extending the feeder roads rather than on maintenance for and paving of existing urban roads which facilitate a greater share of commerce. Second, the emphasis on MDGs such as water consumption has also come at the expense of other sectors with arguably higher returns in terms of economic growth, such as electricity and irrigation. The largest sustained increase has come in the PAF. Third, the conditions on donor funds which limit their use to investment rather than O&M also distorts planning, programming and project appraisal since as a result, many of the funds get misclassified as investment rather than O&M in order to meet these conditions. In 2003, for example, it was estimated that 25% of donor earmarked "investment" funds were O&M misclassified. Fourth, as noted in the forthcoming debt strategy, implementation rates on externally financed infrastructure projects have been too slow. Since donor commitments will remain an important *and necessary* source of Uganda's infrastructure financing for the foreseeable future, donors must coordinate efforts and correct the above mentioned problems. Above all, donors should look to allocate resources with a broad-based aim to improving efficiency in service delivery, thus curing bottlenecks and spurring economic growth.

Table 5.7. Standardized Table Quantifying Donors Participation in Spending by Infrastructure Sector (based on most recent year available: 2004-2005 Fiscal Year)

% of GDP	(A) Public Resources 1/		(B) Donor Funded		Total Public Channelled Spending (A+B)
	O&M (including wages)	Investment	O&M (including wages)	Investment	
Electricty	1.05%	0.50%	0.07%	0.24%	1.85%
o/w Rural					
Other Energy	0.01%	0.01%	0.01%	0.00%	0.03%
Road Transport	0.35%	0.78%	0.09%	1.28%	2.50%
o/w Rural					
Rail Transport	0.28%	0.03%	0.00%	0.04%	0.35%
Air Transport	0.26%	0.05%			0.31%
Water Transport	0.00%	0.03%			0.03%
Communications	0.00%	0.00%			0.00%
o/w Rural					
Water Supply	0.33%	0.40%	0.11%	0.20%	1.03%
Sanitation	0.00%	0.00%			0.00%
Irrigation	0.02%	0.02%	0.00%	0.00%	0.04%
TOTAL PUBLIC INFRASTRUCTURE	2.30%	1.81%	0.28%	1.77%	6.16%

Source: Fiscal Cost Baseline

Notes:

1/ Includes budget, parastatals, special funds and local governments

2004-2005 Fiscal Year, actual expenditures.

Is the Institutional Framework for Infrastructure Service Delivery Appropriate?[89]

5.16 **Uganda's institutional framework for public infrastructure financing is similar to its regional peers.** Government Ministries are responsible for the on-budget allocations and parastatals are responsible for a significant share of spending off-budget. In 2004-2005, on-budget allocations made up just under half of the total infrastructure spending while parastatals accounted for about one-third and local governments were responsible for just 5% of the total spending. Decentralization in Uganda is a relatively new and on-going process and local implementation capacity is still lacking, due in part to unclear and overlapping directives from the central ministries[90].

5.17 **Weak budget monitoring and evaluation exposes Uganda to the risk of waste and corruption.** During the budget process, the Budget Policy and Evaluation Unit in MoFPED is responsible for vetting sector investment plans. However, it currently has limited capacity to perform this function. Ex post monitoring of budget implementation is the responsibility of the sector line ministries, which produce quarterly budget implementation reports. In theory, MoFPED is only supposed to release funds once expenditure obligations have been met. In practice, however, control is limited. While PAF expenditures undergo relatively stricter monitoring to ensure that money is spent on intended purposes, oversight in non-PAF sectors is weak. A case in point is the electricity transmission company UETCL which has been allocated a sizeable investment budget, which to date has not been implemented. Furthermore, MoFPED currently does not conduct ex post value for money analysis which might help reduce inefficiencies in sectors with large cost overruns.

[89] See especially annex to Volume II, chapter 6, of World Bank (2007).

[90] See World Bank (2005); "Uganda Transport Sector Review", a background paper to the CEM for a description and suggestions to improve the institutional framework in the road sector.

5.18 **Non-transparent extra-budgetary funds pose a fiscal risk in a country with weak governance and capacity.** Uganda has five "special funds" for which accounts are unavailable. A sixth (the road fund) was recently established (see Box 4.1). Global experience has shown that extra-budgetary funds create opportunities for waste and corruption in countries with weak governance structures. The Tariff Stabilization Fund which was designed to smooth tariffs until the Bujagali hydropower project comes on stream, is already being utilized to subsidize higher tariffs from thermal power generation. This Fund is also being used to fund selective rural electrification projects despite the existence of a separate Rural Electrification Fund. Fiscal liabilities and contingencies created through extra-budgetary funds are not accounted for in the Government's budget.

Table 5.8. Overview of Extra-Budgetary Funds in Uganda's Infrastructure Sectors

Fund	Administrating Authority	Funding sources
Rural Electrification Fund	Rural Electrification Agency	5% levy on bulk transmission, government budget, donors including World Bank, excess fees from regulatory authority
Credit Support Facility	Rural Electrification Agency	World Bank, Government
Tariff Stabilization Fund	UETCL	Tariff levy on cost of generation
Rural Communications Fund	Uganda Communications Commission	1% levy on gross revenue from telecom and postal services providers; World Bank
Road Fund (under preparation)	RAFU	Vehicle registration fee, license fees, drivers' permits; exact structure under preparation

5.19 **Off-budget management of State-Owned Enterprises (SOEs) understates the fiscal costs of infrastructure.** All SOEs are managed outside the budget with an arms' length relationship. Consequently, the fiscal cost of outstanding arrears, such as taxes, debt payments to government or pension arrears, is not captured in the budget. The Government is considering providing significant off-budget subsidies through debt-equity swaps for some SOEs with debt payment arrears to the government, including the Uganda Railway Company and National Water and Sewage Corporation. Furthermore, SOEs are eligible to receive direct donor funding. This can undermine government policy priorities and makes coordination and monitoring complex. The absence of a clear policy regarding government support for SOEs, such as with regard to further government investment in the rail network, increases the risk of ad hoc public spending and reduces the predictability of public spending. Investment strategies for SOEs need to be integrated with the annual budget process.

5.20 **Parastatals appear to be more specialized in O&M spending while on-budget allocations are more dedicated to investment**. Indeed, over 80% of on-budget spending was directed at new investment in 2004-2005 while parastatals committed about 80% of their resources to O&M. Over the last three years for which we have data, investment spending as a share of GDP by parastatals has declined, and this is starting to become a more general pattern across Africa as well.

Table 5.9. Standardized Table Showing Institutional Responsibilities for Infrastructure Delivery

Standardized table showing institutional responsibilities for infrastructure delivery

Shares	Budget	Infrastructure Service (Operator)	Policy Umbrella	Regulatory Functions
Shares on: Total Public Spending: 60% Public Investment: 89% Public O&M: 32%	On-Budget	• National Roads (MoWHC) • Feeder Roads (local authorities financed by PAF) • Railways Construction (MoWHC)	Ministry of Works, Housing and Communications	
		• Energy generation (MoEM)	Ministry of Energy and Minerals	
		• Rural Water (local governments financed by PAF) • Water Resource Management (MoLWE)	Ministry of Lands, Water and Environment	• Ministry of Lands, Water and Environment
Shares on: Total Public Spending: 40% Public Investment: 11% Public O&M: 68%	Off-Budget	• Railways Operations: Uganda Railways Corporation (URC) → In the process of concessioning • Airport Operations: Civil Aviation Authority (CAA) • Rural Communications (Rural Communications Fund-Administered by UCC)	Ministry of Works, Housing and Communications	• Airport Operations: Civil Aviation Authority (CAA) • Uganda Communications Commission (UCC)
		• Transmission Company: Uganda Electricity Transmission Company Limited (UETL) • Administration of Assets: Uganda Electricity Board (UEB) • Electricity Generation - Concession to Private Sector : Uganda Electricity Generation Company Limited (UEGCL) owns assets and ESKOM operates and maintains • Electricity Distribution - Concession to Private Sector: Uganda Electricity Distribution Company Limited (UEDCL) owns assets and UMEME and WENRCO operate, maintain and expand • Rural Electrification (Extra Budgetary Fund) • Tariff Stabilization Fund	Ministry of Energy and Minerals	• Electricity Regulatory Agency (ERA)
		• Urban Water Supply: National Water and Sewerage Corporation (NWSC)	Ministry of Lands, Water and Environment	• Ministry of Lands, Water and Environment

5.21 **Parastatals should be given more responsibility for investment**. In order to improve efficiency, Government should consider allowing parastatals to assume more risk and responsibility for investment decisions made within their sector. If the central government can commit to keeping subsidies for parastatals low, and prices competitive, forcing them to take more investment spending decisions could achieve greater allocative efficiency: firms would have to internalize more of the costs of their decisions, and would have an incentive to curb losses.

5.22 **Since parastatals will be the key instruments to develop infrastructure in the future, it is essential that they be efficient producers**. This will necessarily entail taking more responsibility for infrastructure spending off-budget, and this includes giving greater responsibility to local governments as well, once the institutional capacity for implementation of these responsibilities exists at that level.

Table 5.10. Standardized Table Showing Level and Main Actors in Infrastructure Provision

Nominal GDP Shares (%)	2003-2004	2004-2005
INFRASTRUCTURE SPENDING	**6.53%**	**7.18%**
Total Gross Investment	**3.68%**	**4.62%**
Total O&M	**2.85%**	**2.57%**
On-Budget 1/	2.67%	3.44%
Gross Investment	2.04%	2.88%
O&M	0.63%	0.56%
Local Governments 2/	0.35%	0.35%
Gross Investment	0.19%	0.20%
O&M	0.16%	0.15%
Parastatals/SOE	2.34%	2.36%
Gross Investment	0.28%	0.50%
O&M	2.06%	1.86%
Special Funds		
Gross Investment		
O&M		
Private Sector		
Gross Investment 3/	1.17%	1.04%

Source: Fiscal Cost Baseline, WB PPI Database

Notes:

1/ Excluding transfers to Local Governments

2/ Explain whether includes expenses funded by transfers from the CG and/or own revenues

3/ Proxy by Committed Investment with Private Participation

Table 5.11. Standardized Table Characterizing Special Funds

Fund	Administrating Authority	Funding Sources	Objectives
Rural Electrification Fund	Rural Electrification Agency	5% levy of transmission bulk purchases by generation companies; World Bank; Government	Capital subsidies to private rural generation companies; tariff subsidies to rural distribution companies
Credit Support Facility		World Bank, Government	Refinancing facility for long-term private lenders to rural electrification projects; partial risk guarantees
Tarrif Stabilization Fund	UETCL	Tariff levy on cost of generation	Smoothing of electricity tariff increase until Bujagali comes on stream
Rural Communications Fund	Uganda Communications Commission	1% levy on gross revenue from telecom and postal services providers (Universal Service Levy)	Support for rural communication. Provide at least one public telephone per 5,000 people at sub-county level, and to ensure internet access at every District headquarters
Road Fund (to be set up)	RAFU	Vehicle registration fee, license fees, drivers' permits; Exact structure tbd	Road maintenance

Is the composition of infrastructure spending appropriate?

5.23 **Inter-sectorally, the answer is broadly "yes":** The roads sector has received the largest share of total spending over the last two years, including on and off budget allocations. In 2004, roads received about 40% of total public expenditure allocated to infrastructure, of which over 80% was directed at new investment. In 2003, the sector received less relative investment (31% of total public expenditure) but O&M received a higher share of that spending (26%). In both rail and air transport, the majority of funds were committed to O&M. The fact that hardly any of this public money was spent on communications is probably an efficient outcome since the private sector can now largely be relied upon to provide resources for network expansions and O&M. Air transport, water transport, and irrigation were relatively minor in terms of total public spending. As a result of the power crisis, the electricity sector now also receives a substantial share of the budget again, after under-investing for many years.

5.24 **Intra-sectorally, the answer is "no": urban infrastructure and trunk roads are relatively neglected, project prioritization evaluation and prioritization is weak, and is maintenance is under-funded[91].** The bias towards rural roads was discussed above in the discussion of donors. Below we argue that better composition could be achieved partly by strengthening prioritization processes. For maintenance we estimate - based on standardized unit costs – Uganda's allocation is lacking for roads and electricity. Maintenance in roads was just 21% of the normative benchmark in 2004, whereas for electricity maintenance was 37% of the benchmark. Maintenance in the water sector through NWSC is significantly higher than the normative level, perhaps due to a significant backlog. While the lack of precise estimates of maintenance needs makes it impossible to say with certainty whether or not Uganda is spending enough on O&M, these figures are meant to provide us with a snapshot of what infrastructure maintenance spending looks like today and raises the need for more detailed analysis. Nonetheless, trends in the quality of infrastructure services from the earlier discussion serve to provide some support for our estimations.

5.25 **Project evaluation and prioritization capacity in infrastructure line ministries needs to be strengthened**. So too do the economic decision criteria used for infrastructure budget allocations by MoFPED. At the project level, guidelines for project appraisal exist while implementation capacity varies by sector. The budget policy and evaluations unit in MoFPED provides line ministries with project preparation and evaluation guidelines. PEAP requires that "expenditure within sectors be prioritized based on cost-benefit or cost-effectiveness analysis as well as econometric or participatory impact analysis". But this practice is not widely observable. The road agency, RAFU, has developed capacity in best practice economic cost-benefit analysis.[92] However, budget allocation and execution in the road sector to differ considerably from economic priorities.

5.26 **Investment planning needs to include an evaluation of the fiscal costs and benefits of private sector participation**. Government needs to analyze public and private options for financing infrastructure projects. Analytical tools include a Public Sector Comparator which establishes the costs of public versus private procurement for an identical project. As project risks should be allocated to the party which can manage them at least cost, PPPs are most

[91] World Bank (2005), "Uganda Transport Sector Review".

[92] RAFU is using the HDM-4 model for road project evaluation which is also used by the World Bank. Projects are eligible for investment if their economic rate of return is above 12%.

beneficial if they involve a significant transfer of economic and financial risks to the private sector, at limited cost to the government. As Box 4.2 shows, infrastructure PPPs in Uganda have left a number of risks in the public domain. These fiscal costs of PPPs need to be accounted for and monitored.[93] Financial risk modeling provides a useful tool in identifying costs to government of different risk sharing arrangements. In annex 4.2 we present an example of use of this tool for the UMEME concession.

Table 5.12. Risk Sharing Arrangements in Uganda's Infrastructure PPPs

PPP / Risk	UMEME	Eskom	Rail Concession
Political risk[1]	Public/Private	Private	Public/Private
Regulatory risk[2]	Public	Public	Public/Private
Generation risk[3]	Private	Public	Not applicable
Commercial risk[4]	Public/Private	Public	Private
Performance risk[5]	Private	Private	Private
Financial risks[6]	Private	Private	Public/Private

[1]Risk of expropriation, non-payment by government etc.
[2]Risk of changes in regulation, lack of tariff adjustments etc.
[3]Risk of insufficient power generation
[4]Risk of insufficient revenue generation
[5]Risk of delayed availability of infrastructure service
[6]Risk of changes in interest rate, exchange rate etc.

5.27 **There seems to be serious backlog for road maintenance**. Uganda's road network consists of national, district and urban roads for which on-budget spending is responsible. These are paved roads, dirt, gravel or bitumen, and the unit maintenance costs for each differ across type. Given the makeup of Uganda's road network, we calculated that, on average, normative unit maintenance costs should be around US$2,200 per kilometer of road. Yet Uganda only allocated about one-fifth of that estimated standard to road maintenance over the last two years, providing a potential source of the inefficiency reported in that vital sector.

5.28 **The electricity sector is under-maintained.** In electricity sector operations and maintenance spending was estimated at under one-half normative maintenance estimates in both 2003 and 2004 given the generating capacity of Uganda. Total recorded O&M spending by UEDCL, the distribution company, was around US$1 million dollars over 2003 and 2004, a figure which is only about 2% of international benchmarks. Unless existing distribution assets are adequately maintained, UEDCL will continue to face significant system loss problems.

[93] For example, the contingent liabilities created through government guarantees for the UMEME concession and the railway concession need to be assessed and monitored. In line with best practice standards, the expected costs of guarantees provided under PPPs which are likely to be called in the next fiscal year need to be appropriated for, either in form of a general contingency appropriation or a separate appropriation for each guarantee. Different valuation techniques can be employed, such as expected loss calculations or option pricing methods. Looking at the example of Chile, such calculations showed that the contingent liabilities associated with minimum revenue guarantees provided to concessionaires amount to about 0.25% of GDP in expected value terms. Chile, like Colombia, also sets aside funds for the costs of called guarantees.

Table 5.13. Standardized Table Assessing Actual Unit Maintenance Expenditures

Standardized Table Assessing Actual Unit Maintenance Expenditures.

	Total O&M Spending (US$millions) 1/	Installed Capacity	Normative Unit Maintenance Costs (US$)	Actual Unit Maintenance Expenditures (US$)	Ratio Actual/Normative
Electricty					
2004	43.79	322,000 kw	$168/kw	63	*37%*
2003	24.71	322,000 kw	$168/kw	77	*46%*
Roads					
2004	19.29	40,890 km	$2,265/km	472	*21%*
2003	18.80	40,890 km	$2,265/km	460	*21%*
Railways					
2004	26.06	1,244 km	$25,000/km	20946	*84%*
2003	22.93	1,244 km	$25,000/km	18434	*74%*
Water					
2004	2.64	007 connections	$12/conn.	43	*358%*
2003	2.71	611 connections	$12/conn.	52	*433%*

1/ Includes budget, parastatals, special funds and local governments

2004-2005 Fiscal Year, actual expenditures.

5.29 **Maintenance in water and sanitation is less clear-cut, but there are signs of a backlog.** Unit maintenance expenditures were significantly higher than benchmarks in the water sector. However, this may have been due to several factors, such as NWSC addressing backlogs in maintenance, duplications of functions, and inefficiencies in spending rather than over-investment. To detect a history of under-maintenance, we used a standardized chart of efficiency measures. With an unaccounted for water rate of around 45%, NWSC, falls into the middle range for maintenance backlogs. However, with over 300 pipe breaks recorded per 100 kms of network in 2004, NWSC falls into the significant maintenance backlog category. Despite these problems, the parastatal still managed to provide water to customers an average of 21 hours per day over 2002-2004, a figure associated with a minimum maintenance backlog.

5.30 **The railway sector was closest to matching its actual maintenance expenditures with normative rates in 2003 and 2004.** This may have been due in part to efforts aimed at improving the condition of the assets prior to the concession agreement reached in 2005.

Are State Owned Enterprises and Utilities Using Resources Efficiently?

5.31 **Electricity and water sector parastatals have been surprisingly good at recovering operating costs but only UETCL was doing so before subsidies, and is no longer.** Of the seven major infrastructure parastatals for which we have data, only UETCL had been fully covering its costs on average over the past three years, after the exclusion of indirect government subsidies from their bottom lines. The figures on cost recovery beg questions about the efficiency of public-sector spending performance and demonstrate that there is room for improvement.

Table 5.14. Indicative Values for Proxy Variables on Asset Condition

	Green : Limited maintenance backlog	*Amber* : Medium maintenance backlog	*Red* : High maintenance backlog
Non Revenue Water	<30	40-50	>50
Hours of Service	>20	10-20-	<10
Pipe Breaks (100 km/year)	<30	30-50	>50

Source: Yepes (2004); Inception Report for WSS.

5.32 **All parastatals still rely on implicit subsidies**. While Uganda officially has a zero subsidy policy, the government makes regular "transfers" to parastatals which are apparent in their annual audited statements. Together these amount to just less than 1% of GDP. The two parastatals receiving the largest shares of implicit subsidies are the Uganda Railways Corp. (URC) and the NWSC, which received indirect government support worth 0.21% and 0.18% of GDP respectively. Uganda, with its relatively experimental stance towards privatization, also gave tax credits to private providers which were small relative to GDP. Furthermore, in the electricity sector deductions from concession fees are a common means through which UMEME receives payment from the public sector for unpaid electricity bills. It will remain to be seen if the new private consortium (Rift Valley Railways) recently set up to take control of the rolling stock of URC, will reduce or eliminate the need for further government support to that sector.

5.33 **Some parastatals and private concessionaires also enjoy generous guarantees**. This overall level of government support to parastatals is a lower bound estimate however, because Uganda also supports many of its parastatals through such means such as government guarantees for borrowing, which in effect shifts risks onto taxpayers and reduces overall efficiency. Parastatals in Uganda also receive subsidies through implicit debt swaps as well as tax credits and deferrals. The government could greatly improve efficiency by making all subsidies explicit (on-budget) and committing itself to keeping this level of direct support small.

5.34 **PPPs in the electricity sector involve substantial government support.** In electricity generation, the majority of risks remain in the public sector. As Eskom's revenue level is guaranteed during the first 7 years of the generation concession, the Government – through Uganda Electricity Transmission Company (UETCL) – bears the risk of changes in generation capacity. UETCL therefore has to absorb the full increase in the unit price of generation resulting from the current reduction in hydropower generation. In electricity distribution, the Government's support to the UMEME concession includes (i) Government investment of $11 million financed through borrowing, (ii) a guarantee for a $2.5 million escrow account and defined events of Government default, and (iii) a put option for the concessionaire after 18 months with a right to recover 50% of the $5 million mandatory investment. Additional Government support is provided through tariff subsidies from the Tariff Stabilization Fund. None of these fiscal costs are currently reflected in government accounts.

5.35 **Costs and benefits of the rural electrification scheme should be reviewed.** The Rural Electrification Fund provides subsidies of up to 60% of initial capital costs for rural electricity generation and distribution projects. In addition, a Credit Support Facility provides refinancing and partial risk guarantees to private lenders. This scheme results in de facto very limited risk-taking both by the equity sponsor and private lenders. Given the high costs of attracting private sector participation to a sector which is not commercially viable[94] and the complexity of arranging and monitoring such transactions, public sector procurement may be a more feasible option for rural electrification.

5.36 **By comparison, the railway concession provides an example of appropriate risk sharing between public and private sector.** The structure of the recently awarded Kenya-Uganda railway concession reflects a major transfer of risk to the private sector. The concessionaire bears commercial, performance and most financial risks. Limited Government support is provided in form of a partial risk guarantee to lenders against political risk and specified government payment obligations. Given the small size of the guarantee ($15 million), it will likely cover a fraction of total project costs, limiting Government exposure and fiscal costs.

Box 5.1 International Experience With PPP Units

Emerging evidence regarding PPP units points to the benefits of a central PPP unit located in the Ministry of Finance, Planning and Economic Development.

South Africa's PPP unit is a best practice example. The PPP unit reports to the Budget Office of the National Treasury. Its mandate includes a wide range of tasks, such as formal approval at three different stages of project preparation to ensure compliance with Treasury regulations; in-depth technical assistance to line ministries and other departments throughout the PPP project cycle; assistance in appointing transaction advisors; development of policy, guidelines, and instructions, including the PPP Manual and the Standardized PPP Provisions (contract terms); training courses and workshops; promotion of public awareness of PPPs through the PPP Quarterly publication, website, and conferences; and management of a Project Development Facility that provides funding for government transaction costs related to PPPs. The implementing line departments are responsible for contract management, but variations or amendment to the contract will have to be approved by the PPP Unit. The PPP Manual developed by the unit sets best practice standards in PPP evaluation, including a requirement for ex ante cost-benefit analysis.

Similar PPP units have been established in the UK, Canada, Australia, India, Indonesia and the Philippines.

In the **UK**, responsibilities for policy guidelines, PPP development and procurement and technical assistance are allocated to separate institutions. The unit responsible for PPP development and procurement (Partnerships UK) is owned jointly by government and the private sector. The National Audit Office carries out ex post reviews of PPP and assesses their efficiency in achieving value for money. Value for money analysis is mandatory before entering PPPs.

5.37 **New institutional structures could help reduce fiscal costs of future PPPs in infrastructure.** The Government should consider establishing institutional structures to (i) analyze the costs and benefits of private sector participation as an alternative form of infrastructure procurement, (ii) develop a dedicated PPP unit in the MoFPED (see box 4.1) and (iii) regularly test the availability of the private sector for alternative project procurement, perhaps using the capacity of the Uganda Investment Authority. In addition, PPP transactions should be structured with the help of an experienced advisor.

[94] In one of the first concessions awarded under the scheme, the Western Nile concession, subsidies are estimated in the range of 60-80% of total project costs.

5.38 **Operating efficiency in the electricity and water sectors is a problem.** In 2004, revenue losses at UMEME (formerly the Uganda Electricity Distribution Company (UEDCL)) and NWSC together cost over 1% of Uganda's GDP[95].

5.39 **UMEME suffers electricity losses around 35% and a collection rate which hovers around 80%.** Taken together, these technical and administrative inefficiencies have cost the company an average of US$35 million per year over the last three years. UMEME (formerly UEDCL) has been able to set its tariffs to cover costs for the last two of three years for which we have data. However, 2004 saw an end to that trend as under-pricing cost the firm an amount equivalent to 0.3% of Uganda's GDP.

Table 5.15. Standardized Table Informing on Financial Implicit Subsidies

% of Nominal GDP	2,003	2,004
Total Annual Implicit Subsidies	**0.36%**	**0.69%**
Fiscal Grants, Tax Credits, Deferred Taxes and "Extraordinary" Items		
to Infrastructure Parastatals	0.29%	0.49%
Electricty	0.05%	0.10%
Uganda Electricity Board (UEB)		
Uganda Electricity Generation Company Limited		
Uganda Electricity Transmission Company Limited	0.04%	0.03%
Uganda Electricity Distribution Company Limited (UEDCL)	0.01%	0.07%
Transport		
Uganda Railways Corporation (URC)	0.23%	0.21%
Civil Aviation Authority		0.00%
Communications		
Uganda Communication Commission	0.00%	0.00%
Water Supply		
National Water and Sewerage Corporation (NWSC)	0.00%	0.18%
Tax Credits to Private Providers	0.07%	0.20%
Fuel & energy	0.02%	0.11%
Coal & other solid mineral fuels	0.00%	0.01%
Petroleum & natural gas	0.00%	
Energy for rural transformation - MoEMD (0325)	0.00%	0.01%
Rural electrification (0331)	0.00%	0.02%
Uganda Electricity Distribution Company Limited (UEDCL)	0.01%	0.07%
Transport	0.05%	0.09%

5.40 **NWSC has an unaccounted for water (UFW) rate of around 45%.** This cost them an average of over US$8 million per year over the last three years, while under pricing has cost them about US$5 million per year. NWSC's UFW rate has actually improved over the last several years, from around 60% in 1998 before the first experiment with private management, which also reportedly improved collection rates to around 100%.

[95] The most recent year for which comprehensive data are available.

Table 5.16. Standardized Table Informing on Quasi Fiscal Costs 1/

Standardized table informing on quasi fiscal costs 1/

% of Nominal GDP	2002	2003	2004
TOTAL	0.76%	0.66%	1.05%
Electricty			
Uganda Electricity Distribution Company Limited (UEDCL)			
Unaccounted Losses 2/	0.31%	0.21%	0.37%
Under-Pricing 3/	0.00%	0.00%	0.31%
Collection Inefficienices 4/	0.19%	0.27%	0.19%
TOTAL HIDDEN COST	**0.50%**	**0.48%**	**0.87%**
Water Supply			
National Water and Sewerage Corporation (NWSC)			
Unaccounted Losses 2/	0.12%	0.11%	0.13%
Under-Pricing 3/	0.12%	0.07%	0.06%
Collection Inefficienices 4/	0.01%	0.00%	0.00%
TOTAL HIDDEN COST	0.25%	0.18%	0.18%

Table 5.17. Standardized Table Informing on Capacity to Execute Resources

Ratios (in Percent)	Release/Estimate 2003-2004	Release/Estimate 2004-2005
Electricity	45.2%	164.6%
Road Transport	55.0%	96.4%
Rail Transport	46.2%	88.0%
Air Transport		
Communications	100.0%	45.0%
Water Supply	67.1%	60.9%
Sanitation		
Irrigation	83.7%	87.8%
TOTAL ON-BUDGET INFRASTRUCTURE	56.23%	90.50%

Source: Fiscal Cost Baseline

Does Uganda face absorptive capacity constraints in infrastructure sectors?

5.41 **Yes: in last two budget years, every major infrastructure sector in Uganda experienced significant problems executing funds.** In 2003-2004, sub-sectors were only able to spend just over half of the resources that were initially promised to them in the budget. This situation improved in 2004-2005 however, when they ended up spending about 90% of the resources initially budgeted to them, but this was largely because much higher actual expenditures were realized in the electricity sector than were budgeted, from emergency spending triggered by the drought.

5.42 **Studies in the roads sector have reported a slew of implementation problems with road construction contracts, which increase costs** and slow down donor disbursements[96]. Yet

[96] Ref: study of road maintenance and construction costs in Uganda, Policy research Corporation and Royal Haskoning, for the Ministry of Finance, Planning and Economic Development, December 2004.

recent road sector reforms have focused more on institutional set-ups rather than on improving processes and performance. Problems identified for national road works projects include:

- Procurement: incorrect pre-qualification of contractors, cumbersome and slow procurement process, inadequate engineering designs and cost estimates, inadequate contract sizes;
- Inadequate contract advance;
- No budget for land acquisition;
- Late payments;
- Late refund of non-VAT taxes;
- Poor supervision of works;
- Insufficient delegation of authority within agencies to act promptly.

5.43 **A lengthy budgeting procedure slows down disbursement rates for District roads and some urban roads**. Districts submit a work program to MoWHC. MoWHC then suggests District allocations to MOFPED based on a formula for equalization which combines population, human development, economic potential, accessibility, road network length and its condition. The process takes months and tends to complete only after the fiscal year has started. It makes it hard for Districts to plan ahead and to discuss with priorities openly with communities.

Summary and Conclusions

5.44 **Additional spending on infrastructure alone – however it gets financed - will not solve the bottlenecks holding back Uganda's overall economic growth**. Unless proper institutional structures are in place to identify growth priorities, allocate infrastructure resources efficiently, to maintain the capital stock, and to address absorptive capacity, additional spending will not necessarily generate commensurate benefits.

5.45 **To continue to make gains in allocative and operational efficiency within its infrastructure sectors, Uganda should:**

- **Increase access and quality levels of existing road and energy infrastructure**. There are well known economies of scale in utilities. Low access is significantly driving up unit costs and therefore the prices paid by firms. Poor quality and expensive energy and expensive transport are becoming binding constraints to growth and structural transformation. This will become an increasingly important issue as Uganda continues to urbanize.
- **More effectively utilize donor funding**. Donor money is strongly skewed towards social projects aimed at rural access with little focus given to areas of infrastructure more closely linked to economic growth.
- **Strengthen infrastructure planning and project evaluation**. Project evaluation and prioritization capacity in infrastructure line ministries. Investment planning needs to include an evaluation of the fiscal costs and benefits of private sector participation.
- **Ensure that private participants are bearing their fair share of business risks**. The advantages of utilizing private resources may be diminished if the private sector is not taking appropriate risks. MOFPED should consider establishing a PPP unit to appraise private participation in infrastructure.

- **Carefully assess costs and benefits of PPPs before entering into new ones.** Balanced risk sharing with the private sector is key to limiting the on-going fiscal costs for Government.
- **Give parastatals a relatively greater responsibility for investment decisions** and make them internalize more of the costs of their business decisions. The central government, in turn, should commit to limiting the level of subsidies it indirectly provides to these firms; subsidies if needed should be up-front.
- **Comprehensively monitor and measure the performance of parastatals** and off-budget accounts. Monitoring of unit costs and physical losses will be key.
- **Reduce losses in electricity and water**. The electricity distribution company, has a loss rate of 35% and a collection rate of just over 80%. The water company, NWSC, has a UFW loss rate of 45% and the network is in obvious need of repair since almost four pipe breaks per kilometer of network was recorded in 2004.
- **Adequately fund maintenance**. Spending is too low, especially in the roads and electricity sectors. Our estimates show that spending on both road and electricity network maintenance is well off international benchmarks. These figures are especially troublesome for landlocked Uganda, where road infrastructure is critical for growth and the country is already struggling to expand electricity supply amidst a shortage exacerbated by the recent drought.
- **Identify and address the causes of slow disbursement** in roads and energy sectors.

6. RAISING ADDITIONAL TAX REVENUE

6.1 **In this chapter we assess Uganda's ability to raise additional revenues to finance higher infrastructure spending, based upon URA's Corporate Plan and data received from URA[97].** Our analysis of Uganda's revenue performance relative to its peers concludes that to get buoyancy in the tax system in the longer term, Uganda will need to expand the tax base gradually over time. Exemptions and concessions have further narrowed an already narrow tax base, and weak administration greatly hinders compliance. Short-term efforts must focus on improving compliance and collections in line with URA's modernization and reform program[98], which is showing initial promise. However, the emphasis in the short-term must stay on these modernization efforts and on controlling tax incentives. Uganda should not seek to raise revenue by increasing tax rates on the existing narrow base of tax payers. This would prove ineffective and inefficient for growth, especially if it results in ad hoc taxes on important sectors for competitiveness and growth, such as transportation and telecommunications as occurred in 2005/06. URA's revenue growth target of around 16 percent of GDP over the next 5–6 years will be very challenging. It is only feasible if policy measures and government actions are supportive, and most crucially, if the recent promising reform momentum is maintained. The annexes to this chapter contain data and descriptive assessments of Uganda's tax structure and administration. Below the main arguments and conclusions are set out, starting with overall conclusions.

A. SUMMARY AND CONCLUSIONS

6.2 **The tax system needs to be further simplified, and the base needs to be broadened over time by rationalizing exemptions and ineffective fiscal incentives.** The establishment of the East African Customs Union, and the deepening process of globalization require that domestic revenue mobilization be enhanced to compensate for the expected loss of trade taxes. However, for developing countries - and Uganda is not an exception – global experience shows that there is little scope for raising statutory rates on capital income because of efficiency problems (Harberger, 1982) nor on consumption-based taxes because of equity and evasion concerns (Zee, 2004). One lesson learned from worldwide tax reforms is that in countries where revenue administration and its governance are weak, simplifying the tax regime by maintaining competitive rates and broadening the base by rationalizing exemptions and ineffective fiscal incentives are the only feasible approach to attain high elasticity, not just buoyancy on the basis of frequent, ad-hoc policy changes. Success stories (e.g., Romania, Slovakia) further indicate that simplification of tax regime should be an integral part of a comprehensive anti-corruption strategy in revenue administration (World Bank and IFC, 2006).

[97] The historic data presented in the annexes to this chapter was gratefully received from URA. The elasticities we use in the analysis which follows are weighted averages from URA's Corporate Plan. These elasticities result in a considerably higher weighted average than those derived from the past (see Ayoki, Milton; Marios Obwona; and Moses Ogwapus (2005), "Tax Reform and Domestic Revenue Mobilization", EPRC. The baseline weighted average elasticities in URA's Corporate Plan suggest that the tax to GDP ratio would reach 14.4 percent of GDP by 2011 applying LTEF assumptions for growth, somewhat lower than the revenue collection target of 16 percent.

[98] Developed with technical assistance from the IMF's Fiscal Affairs Department and financial support from DFID.

6.3 **Some of Uganda's medium to long-term tax revenue collection problems are structural**. The assessment of the tax system in Uganda, and empirical studies of tax buoyancy globally, tend to indicate that the current low level of development and other unfavorable fiscal architecture (high population growth rate and age dependency ratio, low literacy rate, agriculture-dominated economy, high perceived corruption) are constraints to Uganda's taxable capacity. Over the long term, with the benefits of gradual transformation of key fiscal architectures Uganda should improve the revenue intake by consistently broadening the tax base, rationalizing fiscal incentives, exploring additional sources of non-tax revenues, and greatly enhancing the effectiveness of revenue administration. However if new industries are founded upon generous tax holidays and exemptions, the revenue buoyancy of future growth might not rise with structural transformation as it should.

6.4 **Quick fix increases in tax rates could hamper growth**. Raising the tax burden further on Uganda's existing narrow pool of taxpayers could stifle innovation and growth. Further ad hoc taxes on easy-to-tax yet important sectors for competitiveness, such as transport and telecommunications, may impede growth.

6.5 **Instead, Uganda would achieve much more by simplification, by improving administration and compliance, and by rationalizing tax expenditures**. Recent tax policy reforms have moved in opposite directions; with higher tax rates on the one hand, and extended incentives on the other (including discretionary tax aid and extension of VAT exemptions and zero-rates). Revenue administration capacity remains weak - according to the Uganda Revenue Authority (URA), the overall compliance rate of domestic taxes is exceptionally low, at just 38 percent. Uganda's strategic revenue reform program[99] aimed at enhancing revenue effort through administration reform rather than ad-hoc policy changes is therefore sensible, and must be resolutely followed. In addition, existing systems of exemptions, zero-rating in VAT and other indirect taxes as well as discretionary incentives in direct income taxes should be rationalized. In the short run, Uganda should focus on simplifying the tax regimes and tax administration procedures to lower tax-induced barriers to firms' entry and operation.

6.6 **Uganda should initiate a program to keep track on an ever-increasing number of tax expenditures**. Uganda's tax system is eroded by many "tax expenditures". The aggregate total of tax expenditures in Uganda is already quite large, yet more are being considered. Tax expenditures should be treated and scrutinized as government spending channeled through the tax system. Tax expenditures which are granted by administrative discretion should be made more transparent. The size of tax expenditures should be reported to the public through Parliament.

- All tax expenditures should be governed by the tax law. A list of tax benchmarks for each type of tax should be established, with tax expenditures defined and listed as deviations from the provisions in the benchmarks.
- Tax databases and models should be developed to estimate and document the cost of tax expenditures in the Budget.
- Integrate tax expenditures into revenue policy analysis, into the budget appropriation process, and into the various Budget laws and regulations designed to increase transparency.

[99] This is monitored in IMF (2006), "Uganda: Building On Revenue Administration Reform Achievements:"

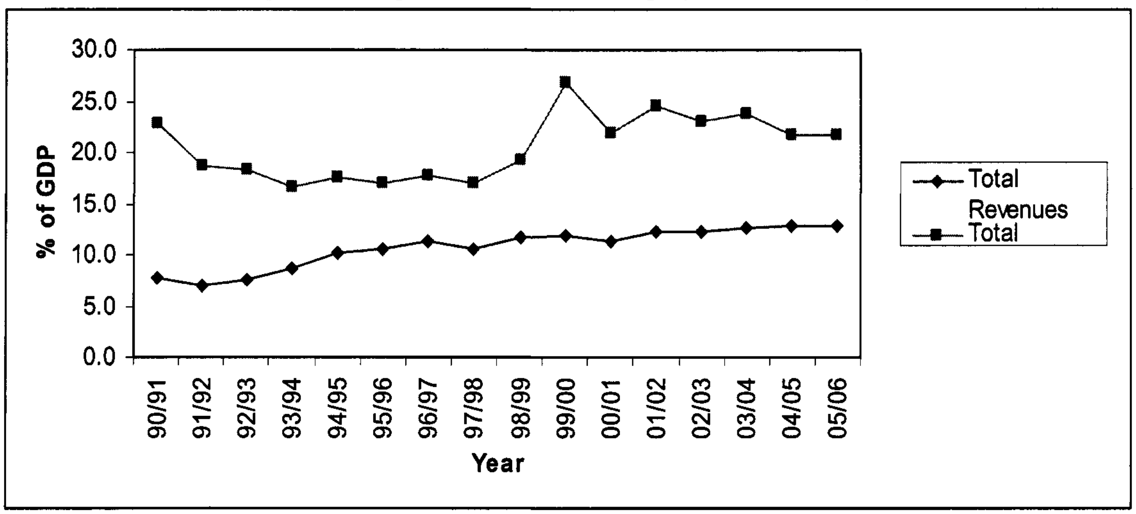
6.7 **There are downside risks to not reforming the system of incentives and concessions.** Some macro-based revenue estimation shows that Uganda can easily run into different revenue paths, which directly impact both the level and structure of growth financing. The tax-GDP ratio is projected to range from approximately 14 percent to more than 18 percent by the end of 2015/16, depending on the design and success of the revenue reforms. Given the government's projected public expenditures, the resultant fiscal deficits (excluding grants), even in the best case scenario in estimation of the revenue intake, can still be as high as 6.5 percent. It is worth noting that the URA revenue target of 16 percent by 2011 appears to reflect the upper bound of projected achievability and can only be reached if a sound tax policy regime is established, and administration reforms are successfully implemented.

B. ANALYSIS OF REVENUE PERFORMANCE AT THE MACRO LEVEL

6.8 **Uganda's revenue performance in the 1990s was strong.** Improvements in economic growth and a favorable shift in the structure of the economy combined with aggressive tax policy and administration reforms in the 1990s led a surge in revenue collection during the 1990s. Revenue rose steadily from a low level of 5 percent of GDP in 1988/89 to the peak of 13 percent in 1998/99, but has since gone down and leveled off at about 12 percent in recent years (Annex 1).

6.9 **However revenues now cover a smaller share of total spending.** In 1997/8, domestic revenues were sufficient to finance almost 80 percent of the total public expenditures, but in 2001/2 they could cover just 60 percent (MFPED, 2004). Pressure for raising revenues has intensified. For long term sustainable growth, the country needs to focus on enhancing the effectiveness and efficiency of the revenue system, which has become a priority target of the PEAP policy action.

Figure 6.1. Expenditure-Revenue Gaps (1990/91-2004/05)



* *Source:* IMF database, and author's estimates.

6.10 **Uganda's revenue collection relies on the five main types of taxes, with heavy reliance on trade taxes.** Corporate income tax (CIT), personal income tax (PIT), VAT, excise, and trade (annex 1). Combined direct income taxes (CIT and PIT) account for almost 24 percent of the total revenues. VAT collects almost 34 percent, excise—11 percent, and trade tax (duty on petroleum imports and other import duties)--24 percent. Trade taxes are significant. The collection on petroleum imports and other tariff duties, alone, are equivalent to the total collection of the CIT and PIT, while the total taxes collected on international trade (including the VAT and excise on imports) account for more than half of the total revenue. As the country follows the global trend of trade liberalization, a major challenge is how to increase the collection of domestic revenues for compensation of the expected loss in trade taxes.

6.11 **The tax-GDP ratio is among the lowest in Sub-Saharan Africa and low compared with low income countries in general.** Annex 2 summarizes the revenue performance in selected African countries, with reference to the averages for the region, low-income African countries, and the low-income countries worldwide for the period 2000-04. Uganda's collections as a share of GDP are less than 61 percent of Kenya's and significantly lower than the averages for Sub-Saharan Africa (SSA), low-income countries in Africa, and low income countries worldwide.

6.12 **Low revenue mobilization in Uganda is constrained by a relatively low taxable capacity**. A closer look at the comparative fiscal structures in selected SSA countries reveals that revenue collection in Uganda has been hindered by structural factors. Annex 3 shows that the low level of collection in Uganda is attributed to the country's low level of development, high share of agriculture (a hard-to-tax sector), high population growth, low share of working age population, low literacy rate, and high prevalence of corruption—which in turn could limit the aggregate potential tax base and compliance. These features are similar for Tanzania which collects almost the same level of taxes as a share of GDP[100]. The policy implication of this is that as the economy industrializes and transforms, new emergent sectors should not be exempt from paying taxes if buoyancy is to be maintained.

6.13 **However, even relative to its peers, Uganda's main taxes have low productivity**. Compared with some of its peers in Sub-Saharan Africa (ie Kenya, Tanzania, Namibia, Rwanda, and Zambia) we estimate that foregone corporate income tax collections amount to 0.8 percent of GDP. Uganda's foregone VAT collection due to low productivity (using the SSA average as the benchmark) is estimated at 0.6 percent of GDP. The revenue losses from these two major taxes would be substantially higher by a factor of 2.4 and 4 respectively, if Kenya's levels of tax productivity were used as the benchmarks. In addition, we conclude that certain taxes (particularly, excessively high excise on petroleum and mobile phone) are largely collected at high efficiency costs, and may impede growth.

6.14 **Revenue collection in Uganda is not only relatively ineffective (as signaled by the low collection level), but it is relatively inefficient, being highly dependent on trade tax collection.** Annex 2 presents the structure of the revenue collection in six Sub-Saharan African countries for the period of 2000-04. Uganda's revenues are not only sharply different from Kenya in level, but also in composition. While Kenya has more balance in the collection of direct and indirect taxes (34 and 66 percent respectively), collections in Uganda are still largely dependent on indirect taxes (80 percent of the total tax collection). In particular, Uganda as a land-locked country remains highly dependent on distortionary trade taxes, which account for a third of the total collections, compared with just 13 percent in Kenya or 24 percent in the average SSA.

[100] The empirical estimation is based on a worldwide comparison, controlling for countries' income level (GDP per capita), agriculture value added, trade openness, population growth, and corruption (Le, Moreno-Dodson, and Rojchaichaninthorn, 2007).

6.15 **Not only is the overall structure problematic, but the evolution of tax policy has been relatively inefficient**. Chen, Matovu, and Reinikka (2001) find that while the existing aggregate level of revenue in Uganda is low, its rapid increase with ad-hoc policy measures in the 1990s may have been a constraint to private investment at the firm level. With the major reforms of the tax system during the post-recovery period, the government put an explicit revenue raising target of 1 percentage point of GDP per year. However, the tax policy is burdened with numerous ad hoc, inconsistent and highly distorting tax policy features, particularly, raising tax rates (especially fuel taxes) combined with multiple exemptions and other fiscal incentives. High rates and complexity in the tax regime combined with weak administration, led to the prevalence of tax evasion and avoidance during the 1990s tax reforms. The FIAS (2003) survey reports that investors perceived tax rates in Uganda as the top obstacle to investment – with a total of more than 75 percent of respondents rating this factor as moderate, major or very severe obstacle.

6.16 **A narrow group of taxpayers bear the brunt of revenue collections**. Chen et al. (2001) demonstrate that compared with Kenya (which has higher revenue collection), foreign firms regard Uganda as a more highly taxed country. Furthermore, Gauthier and Reinikka (2006) confirm empirically the undesirable inverted U-shaped relation between tax burden and firm size for the case of Uganda: selective tax exemptions benefit large businesses to a disproportionate degree, while evasion is common among small businesses, leaving medium sized firms to bear a disproportionately high tax burden. This may imply the Ugandan tax system is a driving force for informality.

6.17 **In sum, Uganda's revenue system features contrasts**. First, while collection appears to be robust compared with the estimated taxable capacity, it remains far below the level needed to meet the increasing demand for public expenditures.[101] Second, whereas the aggregate level of taxation is low by regional standards, it is collected at relatively high economic costs, as the system is heavily dependent on trade taxes, is levied on a narrow group of taxpayers, and is regarded as a significant impediment to investment.

C. LONG-TERM PERSPECTIVES FOR REVENUE COLLECTION

6.18 **We project revenue collection in Uganda to reach 16 percent of GDP by 2015/16, a much lower forecast than in the Government's LTEF**. To gauge the scope for additional revenue space over the next decade, we applied the macro-based approach for revenue estimation. The approach is based on the concept of tax elasticity set out in box 6.1. Table 3 summarizes the results of revenue projections in the three case scenarios and juxtaposes them with the government projected expenditures for the estimated fiscal deficits (excluding grants). Annexes

[101] It is not surprising that the estimated taxable capacity for Uganda is significantly lower than that of Kenya. Applying an OLS specification with fixed effect for the worldwide coverage, controlling for GDP per capita, trade openness, share of agricultural value added in GDP, and corruption, Le, Moreno-Dodson, and Rojchaichaninthorn (2007) find that the average taxable capacity for Uganda and Kenya was approximately 11 and 13 percent of GDP during 1994-2003, respectively. The result or at least the ranking of the taxable capacity in these two countries appears consistent to the one by Stotsky and WoldeMariam (1997). They apply an OLS with random effect for a sample of 30 SSA countries during the six year period, 1990-95, controlling for a number of tax handles, including agriculture, manufacturing and mining and find that respectively, the taxable capacity for Uganda and Kenya, was about 10 and 16 percent of GDP in 1995. The empirical approaches and the samples used in the two mentioned studies may be different and there may be some legitimate reservation about their absolute results regarding the empirical specification and quality of the data. But it is quite clear that the taxable capacity in Uganda is relatively low, which could be explained by the current low level of development and by a number of structural and institutional deficiencies.

7-9 show the detailed projection of revenue collection by major tax. In 2015/16, the revenue collection is projected to reach 16 percent of GDP in the base case scenario, implying the fiscal deficits of almost 9 percent.

6.19 **High elasticity assumptions still only yield a tax to GDP forecast of 18.1 percent of GDP**. In the lower and higher revenue elasticity case scenarios respectively, the collection is expected to be 13.6 percent and 18.1 percent, with associated fiscal deficits of approximately 11 percent and 6.5 percent. Note that in all three case scenarios, the tax system is assumed to be elastic. Especially, in the higher elasticity case scenario, revenue collection is assumed to respond at a rate of 50 percent higher the rate of economic growth, in real terms. These are non-trivial assumptions in practice and hence the results of the revenue projection in the higher elasticity case scenario should reasonably be interpreted as the upper limit of the expected annual revenue collection over the next ten years.

Box 6.1. Forecasting Revenue Potential Through Tax Elasticity

First we derive tax elasticity from the equation:

$$Ln(T_{it}) = \beta_0 + \beta_{1i} Ln(Y_t) + \varepsilon_t$$

Where, the revenue series T_{it} are adjusted for annual discretionary changes due to change in tax regime (tax base and/or rate); and Y_t is the appropriate base (e.g., GDP or private consumption). Both the T_{it} and Y_t are in real terms. The coefficient β_{1i} is interpreted as the elasticity of the respective tax i.

It follows that the forecast tax collection (nominal) in year t+1 is defined as:

$$T_{t+1}^n = T_t^n * \left(1 + \beta * g_y^r + \pi_{t+1} + \beta * g_y^r * \pi_{t+1}\right)$$

Where, $\beta * g_y^r$ expresses the estimated growth rate of tax in real terms.

The revenue series of the year 2004/5, the latest available data on actual collections from URA are used as the base for revenue forecasting. The projection is based primarily on the elasticity of major taxes as estimated by the URA: the elasticity of direct income tax to GDP is 1.8, and the elasticity of all major indirect taxes to private consumption is 1.1. This implies that the weighted elasticity of all taxes is approximately 1.3.[1] Parameters used for expected annual inflation and the growth rate of GDP follow Government's Long-Term Expenditure Framework (LTEF). Also, note that the URA assumes that the real growth rate of consumption per capita is 3 percent per annum. Given the expected growth rate of population as assumed in the LTEF, the real growth rate of total private consumption is expressed by:

$$c_t = g_{cc} + g_p + g_{cc} * g_p$$

Where, c_t denotes the growth rate of total private consumption; g_{cc}: real growth rate of per capita consumption; and g_p: population growth rate.

Table 6.1: Projection of Revenue Collection and Expenditure-Revenue Gaps (2005/06-2015/16)

		Forecast of revenues and predicted expenditure-revenue gaps					
		Lower revenue elasticity case		Base case		Higher revenue elasticity case	
Fiscal year	Government Expenditures/ GDP/1	Total Revenues/ GDP/2	Expenditure -Revenue gap/GDP	Total revenues/ GDP/2	Expenditure -Revenue gap/GDP	Total revenues/ GDP/2	Expenditure -Revenue gap/GDP
2005/06	22.8%	12.7%	-10.1%	12.8%	-10.0%	13.0%	-9.8%
2006/07	20.7%	12.8%	-7.9%	13.1%	-7.6%	13.5%	-7.3%
2007/08	19.9%	13.0%	-6.8%	13.5%	-6.4%	14.0%	-5.8%
2008/09	20.3%	13.2%	-7.1%	13.8%	-6.5%	14.5%	-5.8%
2009/10	20.8%	13.3%	-7.5%	14.1%	-6.7%	15.0%	-5.7%
2010/11	21.3%	13.4%	-7.9%	14.4%	-6.9%	15.6%	-5.7%
2011/12	21.8%	13.5%	-8.3%	14.7%	-7.1%	16.1%	-5.7%
2012/13	22.4%	13.6%	-8.8%	15.1%	-7.3%	16.6%	-5.8%
2013/14	23.1%	13.6%	-9.5%	15.4%	-7.7%	17.1%	-6.0%
2014/15	23.8%	13.6%	-10.2%	15.6%	-8.2%	17.5%	-6.3%
2015/16	24.6%	13.6%	-10.9%	16.0%	-8.6%	18.1%	-6.5%

1/ Estimations in LTEF.
2/ Author's estimates.

D. ASSESSING THE PERFORMANCE OF UGANDA'S MAIN TAXES – AND PRIORITIES FOR INCREASING REVENUE

6.20 ***There is scope for Uganda to greatly increase CIT productivity and revenue collections.*** The CIT regime in Uganda is regionally comparable in terms of its rate and base. However certain types of fiscal incentives (e.g., initial capital allowance) need to be transparent with specific criteria for eligibility. Outright tax subsidies (tax aid) are both economically inefficient and inequitable, and should be abolished. Complexity and ambiguity in a tax regime combined with weak administration capacity inevitably open opportunities for corruption, and lead to lower compliance and greater revenue leakage. Some broad estimates of the CIT efficiency or productivity can gauge the extent of the revenue loss. CIT efficiency is defined as the ratio between the share of CIT collection in GDP and the maximum CIT statutory rate. Table 1 indicates that among the six selected SSA countries in the sample, Uganda and Tanzania are ranked at the bottom of the efficiency criteria (both with the CIT efficiency of 0.03), while Kenya is placed at the top (0.1). If Uganda strives to reach the average efficiency of the sample (0.056) or the level of efficiency in Kenya, while retaining the statutory rate of 30 percent, the estimated CIT potential could be substantially higher than the current level of collection by 0.8 and 2.1 percentage points of the GDP, respectively.[102]

[102] It is worth noting that the estimated magnitudes of the CIT revenue foregone (or the VAT as discussed later) compared with other countries are indicative, as the comparison does not control for the cross-country differences in the key fiscal architectures attributing to the size of taxable capacity.

Table 6.2. Estimated CIT Productivity in Selected SSA Countries (2003/2004)

Countries	Max. CIT Statutory Rate (%)	CIT Collection (% of GDP)	CIT Efficiency Rate
Uganda	30	0.9	0.030
Kenya	30	3	0.100
Tanzania	30	0.9	0.030
Namibia	35	3.1	0.089
Rwanda	40	1.8	0.045
Zambia	30	1.3	0.043
		Average	0.056

* *Source*: IMF database; and author's estimates

6.21 **The poor performance of the VAT in Uganda can be attributed to excessive exemptions, multiple zero-rating, and low compliance.** That is, there are policy as well as administration problems. While the standard rate is set at a relatively high level (18 percent), collection is relatively weak (just more than 4 percent of GDP in 2003/04—annex 1). The system of exemptions and zero rates may be designed to support the poor, its unintended fiscal and economic consequences (e.g., high compliance and administrative costs; loopholes for revenue leakage) are substantial, given weak administration capacity. In addition, many studies have shown that the equity impact of VAT exemption and non-export zero-rating is insignificant at best. Exemption may erode the base or even create cascading. Currently, the list of exemptions in the Ugandan VAT is even much broader than the EU's standard list under the Sixth Directive - which is itself considered excessive and non-conforming to international best practice (Cnossen, 2003).[103] Note that the number of exemptions has not reduced but has tended to increase since the early 2000s. Compared to exemption, zero-rating is even more detrimental to revenue objective: ultimately no collection is gained, while additional channels are opened for VAT fraud, and the administration incurs substantial costs for managing VAT refunds. Currently the VAT Act allows for 7 groups of commodities, in addition to exports, to be subject to a zero-rate, of which two were just added in 2001 and 2002.

6.22 **VAT revenue foregone in Uganda is substantial by either of the two measures (GCR or VAT efficiency index).** Uganda's VAT productivity is similar to the one in Madagascar, and Tanzania but is significantly lower than in Kenya and Namibia. Table 2 presents our estimates of the two measures of the VAT productivity for Uganda and four other SSA countries: VAT gross compliance rate (GCR) and VAT efficiency ratio.[104] The depleted base, high evasion and avoidance may explain the situation. At the level of Kenya's GCR, VAT collection in Uganda would have been more than 7.3 percent of GDP. At this GCR, revenue foregone is estimated to be in the magnitude of more than 3 percentage points of GDP. At the level of Kenyan VAT efficiency ratio, the VAT collection should have been almost 6.8 percentage of GDP, and thus

[103]The EU standard exemptions under the Six Directive cover health care, education, social services, cultural services, public radio and television broadcasts, postal services, immovable property, insurance, financial transactions and gambling. Currently, a number of EU member states start bringing in their VAT regime some selected items in the EU standard list Finland and Sweden start taxing postal services; Denmark, Finland, Greece, Italy, the Netherlands, Spain, Sweden and the UK: cultural activities; and Denmark: radio and TV broadcastings. Such services and commodities have become elective in the EU exemption list.

[104] Gross Compliance Ratio (GCR) is defined as the ratio between actual VAT collections to the potential, which in turn is determined as the share of VAT rate in private consumption. VAT efficiency is estimated as the ratio between actual VAT collection in terms of GDP and a VAT standard statutory rate.

VAT leakage is approximately 2.5 percentage points of GDP. Using just average VAT efficiency for SSA (0.27) Uganda's revenue loss amounts to almost 0.6 percentage points of GDP.[105]

Table 6.3: Estimated GCR and VAT Efficiency Ratios in Selected SSA (2003/2004)

Countries	Private Consumption (% of GDP)	VAT Collection (% of GDP)	Standard Statutory VAT Rate (%)	Estimated VAT potential (% of GDP)	Gross Compliance Rate (GCR)	VAT Efficiency Ratio
Uganda	76	4.3	18	13.7	0.31	0.24
Madagascar	81	3.2	15	12.2	0.26	0.21
Kenya	70	6	16	11.2	0.54	0.38
Tanzania	79	4.7	20	15.8	0.30	0.24
Namibia	49	8.3	15	7.35	1.13	0.55

**Source*: WDI 2006; Ministry of Finance (Kenya; Tanzania); Bank of Namibia; and author's estimates.

6.23 **To enhance compliance and, related to it, the VAT efficiency, Uganda should consider phasing out all non-export zero rates and rationalizing the list of exemptions**. On the other hand, the policy on VAT refund should be changed: firms that get refunded after the statutory due date should be entitled to an interest equivalent to the one applied to overdue tax arrears.

6.24 **Uganda should be cautious in raising excise taxes on petroleum and telecoms**. The transport and telecoms sectors are important for growth and competitiveness. To enhance revenue collection petroleum products should be brought into the VAT net, while maintaining a reasonable excise rate in line with the one in Tanzania or Kenya. Also, the rate on mobile phone is already high (with an effective rate of more than 30 percent: 18 percent VAT plus 12 percent excise), which may exert unintended consequence on growth of new technology. Some may argue that a high tax burden on mobile phones is justified as it is applicable to a new dynamic sector. But this argument does not hold, as it is inconsistent with the broad direction for recent amendments of the VAT regime in 2003 (in particular, technology-related commodities such as computer software, printers, computers, photosensitive semiconductor devices have just been added to the list of VAT exemptions).

6.25 **While Uganda has made big strides toward simplifying the duty regime, some critical issue on trade policy needs to be resolved**. Particularly, the high effective rate of protection still exists in certain sectors: For example, textile, used clothing, and sugar imports are all unjustifiably subject to the maximum statutory rate of 15 percent. This high level of protection needs to be reduced for both equity and efficiency objectives.

E. ANALYZING TAX EXPENDITURES

6.26 **Uganda's tax system is eroded by many "tax expenditures"**. As in most countries, the tax system in Uganda is being used not just to raise revenue for spending in an equitable, efficient and simple way, but to promote specific economic and social objectives through tax foregone, or "tax expenditures". Some concessions and exemptions were introduced during the tax reforms in the 1990's, others subsequently. In addition to tax laws, certain tax provisions contained in the Ugandan Constitution (Articles 106 and 128) are tax expenditures. Furthermore, tax provisions which were granted by administrative discretion to favor special economic and social groups are tax expenditures. However, the number of tax expenditures granted by administrative discretion

[105] The average VAT efficiency ratio in the SSA is based on the estimate by the IMF (Ebrill et al., 2001).

in Uganda is unknown. Box 4.X (below) lists the identified tax expenditures under Income Tax Act (ITA) and VAT, as amended through July 2005.

- ***Out of the 51 tax provisions under ITA, some 24 items were initially identified as tax expenditures.*** The most significant categories are: (1) tax subsidies to economic sectors, particularly to the agriculture and energy; (2) tax subsidies to specific regions and locations; (3) tax deductions for start-up corporate expenditures or initial public offerings; (4) tax allowances for industrial buildings; (5) tax deductions for scientific research and training expenditures. These categories are not exhaustive and sometimes overlap.

- ***Out of 32 exempt supplies and zero rated supplies under VAT, 17 are tax expenditures***. VAT exempt supplies include supplies relating to immovable property, specific sector activities (agriculture, energy and computer electronics), as well to designs and technical services. The VAT zero rated supplies relates to educational materials, agriculture sector, and others which have coverage broader than international practices.

- **In the Uganda Constitution**, there are two types of tax expenditures: tax exemptions for (1) salaries, allowance and/or other benefit for certain public servants (Article 106 and Article 128 of the Constitution)[106]; (2) tax exemptions for pension contribution and pension benefit.

- **Regarding administrative discretion**, tax expenditures are granted to foreign investors, local business and individuals in terms and conditions which are not listed in the tax laws and are unknown to the public.

6.27 **The aggregate total of tax expenditures in Uganda is already quite large, yet more are being considered**. Based on publicly available information, we estimate total tax expenditures under corporate income tax and VAT to be in the region of 4 to 5 percent of GDP (see table 4.1). The estimated aggregate total of all tax expenditures (4.7-5.2% of GDP) was approximately 38%-42% of the total revenue received (12.5% of GDP) in 2003/04. These estimates are imprecise and should be regarded as indicative ballpark figures[107]. Nevertheless a figure of this magnitude merits further more detailed analysis. Estimating the incidence of tax expenditures on tax revenues requires detailed analysis of the individual tax measures. This in turn involves detailed information on the tax base, including information relating to exempt persons and activities. However Uganda's tax databases are very incomplete - owing to the limited tax return information. Our estimates exclude customs and excise taxes and tax expenditures granted through administrative discretion, and those tax expenditures which are part of the Constitution. The aggregate total would be significantly higher if we were able to accurately include tax expenditures granted under customs and excise. Along with the structure and informality in the economy, tax expenditures seem a significant reason that tax revenue collections in Uganda remain low, and they may be contributing to over-burdening the capacity of tax administration.

Table 6.4. The Scope and The Size of Tax Expenditures

[106] Refer to the Article 106, subsection 4, Article 128, subsection 7 in the Constitution (1995)

[107] NB the implicit assumption on CIT here is that the difference between estimated profits from the UBI and taxable income by URA is tax expenditure – an assumption which seems unrealistic.

		Amount of Tax exempted (billion USh)	Amount of Tax exempted (% of GDP)
Tax expenditures under the Tax Laws:			
(A) Income Tax			
Resident individual 1/		N/A	N/A
Non Resident individual 1/		N/A	N/A
Corporate income 2/	2000/01	216	2.13%
(B) Value Added Tax 3/			
VAT exemptions VAT zero rated	2003/04	N/A	2.5-3.0 %.
(C) Customs and Excise Duties			
Import duty exemptions and rates reductions		N/A	N/A
Tax expenditures under the Constitution			
(D) Pension 4/		N/A	N/A
Pension arrears for resident	2003/04	16.4	0.14%
Pension arrears for workers in East African Community	2003/04	4.4	0.04%
Civil Servant Pension Fund for both civil servants and for public officers	2003/04	12	0.10%
AFPS		N/A	N/A
NSSF and OPS		N/A	N/A
(E) Personal income tax exemption for civil servants (Article 106 & 128)		N/A	N/A
Estimated total (excluding those from pension arrears)			4.7-5.2 %

Sources: Uganda Bureau of Statistics, Uganda Ministry of Finance, Planning and Economic Development and Uganda Revenue Authority.
1/ For those tax exemptions in the same nature as pension benefits. See Box 2 for detail.
2/ The estimates are based on "Report on the Uganda Business Inquiry, 2000/2001"with the share of companies with 50 million sales to total companies sales calculated from the initial UBI database.
3/ the estimates from the VAT analysis in the "mobilization tax revenues in Uganda."
4/ There are four pension schemes in Uganda : PSPF for civil servants, PSPF for public officers, AFPS for military officers, National social security Fund and voluntary arrangements. The exempted amount of tax expenditures are estimated by using the lowest income tax threshold at 10%.

6.28 **Tax expenditures should be treated and scrutinized as government spending channeled through the tax system**. Since tax expenditures are actually governmental spending which is being channeled through the tax system[108], they should be subject to the same accountability and transparency criteria which are applied to direct expenditures.

6.29 However many tax expenditures in Uganda are neither subject to overall control within sustainable limits, nor judged with regard to effectiveness, equity and efficiency. A brief review of seven selected tax expenditures in Uganda (granted under the Income Tax Act, the Value

[108] Surrey, S.S. and McDaniel, P.R., "Pathways to Tax Reform, the Concept of Tax Expenditures," Harvard University Press, Cambridge, Massachusetts, 1973.

Added Tax Act, the Constitution, and through administrative discretion see box 4.3 [109]) suggests that compared with many public spending priorities, these tax expenditures could be ineffective, inequitable and inefficient. Because, by definition, tax expenditures are automatically funded from the tax bases (prior to the collection of tax revenues), they automatically have a higher budget priority than direct expenditures, including the PAF. Their treatment presently is inconsistent with the principles for prioritizing fiscal allocations defined in the Budget Speech and the PEAP.

6.30 **Tax expenditures which are granted by administrative discretion should be made more transparent**. Whereas Ugandan tax laws are published, tax expenditures which are granted by administrative discretion are sometimes not published in the Gazette and are not otherwise available to the public. Nor are the archives of the Ugandan Gazette available to the public. This significantly impedes public access to information on tax concessions.

6.31 **The size of tax expenditures should be reported to the public through Parliament.** Without transparency, there is no assurance of integrity regarding tax expenditures, and there can be no contestability relative to other fiscal priorities. Various systems of reporting are used in OECD countries to provide information on tax expenditures[110]. A list of the basic steps which we recommend Uganda could follow to establish a system of tax expenditure reporting is as follows:

- **All tax expenditures should be governed by the tax law.**
  - The tax law should be the only law under which tax provisions are enacted.
  - All tax expenditures should be integrated into the tax laws, whether they were granted by the enactment of a law, under the Constitution or by administrative discretion.
  - At the time of the enactment of a new tax expenditure into the tax law, a sunset date for this tax expenditure, where possible, should be stipulated. Upon this sunset date, this tax expenditure may or may not be renewed, depending on whether or not it is still necessary, under the circumstances at that time.

- **A list of tax benchmarks for each type of tax should be established, with tax expenditures defined and listed as deviations from the provisions in the benchmarks.**
  - All tax expenditure provisions should be reviewed carefully; where possible, tax expenditures should be either repealed or replaced with direct budget expenditures.
  - Where necessary, tax expenditures, (as well as the tax provisions in the tax benchmark) should be redesigned to ensure accountability and cost-effectiveness, as well as to remove ambiguities which may provide opportunities for abuse.

- **Tax databases and models should be developed to estimate and document the cost of tax expenditures in the Budget**. Based on the status of the current tax databases in

[109] Customs and Excise Acts are not discussed because they are most straight forward. The number of commodities listed in these two Acts was deemed too large to be reviewed and analyzed in this study give time constraints. The East Africa Community Customs and Excise Act could be a general guideline to establish the relevant tax benchmarks against which tax expenditure analysis is carried out on C&E.

[110] Swift, Z.L. "Managing the Effects of Tax Expenditures on National Budget," Tax Notes International, Washington DC, March 2006.

Uganda[111], we recommend that Uganda take the following steps to establish a complete mechanism for estimating tax expenditures:

- Estimate the impact on tax revenue of tax expenditures which have been granted with respect to the VAT imports, and with respect to the Customs and Excise Duties. Because the Uganda Customs administration has installed ASYCUDA***, tax expenditure data can be generated by extracting the necessary data from this system, with assistance from specialists.112
- Construct a CIT tax analysis database, by drawing a special sample of tax returns in order to analyze tax expenditures on CIT and projections in both the current year and the next 1-3 years.
- Evaluate the capacity in Uganda to estimate tax expenditures under the VAT (other than the VAT imports), evaluate both the capacity and sophistication of national accounts and the complex VAT exemptions/zero ratings.
- Develop methodologies and models for projecting tax expenditures.

- **Integrate tax expenditures into revenue policy analysis, into the budget appropriation process, and into the various Budget laws and regulations designed to increase transparency**.
    - Analyze the impact of tax expenditures on the sufficiency, equity, efficiency and simplicity of the tax system when proposing new measures.
    - Assess the functional (sector) classification of tax expenditures.
    - Develop tax expenditure discussion documents, including effective analysis and accountability analysis.
    - Integrate tax expenditure projections into the MTEF.
    - Developing a plan to apply the IMF's Code of Fiscal Transparency to tax expenditures.

[111] see Appendix 3 of Swift, Z.L. (2007)

[112] Though this study has not review the Customs and Excise duty regarding tax expenditures, but work in this area is straight forward. These special tax laws can be review along with the reviewing the data bases.

Box 6.2. Draft: Tax Expenditures Under Income Tax Act and Value Added Tax Act

The following gives a list of tax expenditures under the Income Tax Act (Cap 340) (ITA) and Value Added Tax (Cap 349) (VAT) based on international practices. This is by no means a complete list of tax expenditures for ITA and VAT.

Income Tax Act 1997, Cap 340 (As Amended To July 2005)

Exempt Income

Article 21/1 g education grants with commissioner's gave ruling
Article 21/1 h alimony or alimony under an judicial order or written agreement of separation
Article 21/1 j value of any property acquired by gift, bequest, devise or inheritance that is not included in business, employment or property income
Article 21/1 k, any capital gain that is not included in business income
Article 21/1 n a pension
Article 21/1 o a lump sum payment made by a resident retirement fund to a member of the fund ;
Article 21/1 o a lump sum payment made by a resident retirement fund to a dependant of a member of the fund;
Article 21/1 p the proceeds of a life insurance policy paid by a person carrying on a life insurance business
Article 21/1 q the official employment income of a person employed in the armed forces of Uganda, the Uganda police force, or the Uganda prisons service, other than a persons employed in a civil capacity
Article 21/1 r the income of an investor compensation fund established under section 81 of the Capital Markets Act (Amendment June 30, 2006)
Article 21/1 t income of a collective investment scheme to the extent of which the income is distributed to participants in the collective investment scheme
Article 21/1 u interest earned by a financial institution on a loan granted to any person for the purpose of farming, forestry, fish farming ...

Deductions

Article 25 deduction on interest payment to a debt obligation
Article 27/1 deduction of tax for the depreciation of the person's depreciable assets, other than specified in article 26.
...Class 1: computers & Data handling equipment, 40%
...Class 2: Automobiles, construction and earth moving equipment, at 35%
...Class 3. Buses. Goods vehicles, tractors, trailers, plant & machinery for farming, manufacturing and mining at 30%
...Class 4. Railroad cars, Locomotives, Vessels, office furniture, fixtures, etc. 20%
Article 28/1a, initial tax allowance (for an eligible assets outside Kampala, Entebbe, Namanve, Jinja and Njeru) for that year of amount equal to 75% of cost base of the property at the time it is placed in service (for the first time).
Article 28/1b, tax allowance (Kampala, Entebbe, Namanve, Jinja and Njeru) of a 50% of cost base of the eligible property at the time it is placed in service (for the first time)
Article 28/4-8, tax allowance of 20% of the cost base of the industrial building (exclude. approved commercial building), at the time it was place in service (for the first time), after 1st July 2000
Article 32. allowed tax deduction of 100% of expenditure incurred during the year of income for scientific research
Article 33. Training expenditures is allowed to deduction for expenditures at 100% during the year of income
Article 36. Mineral exploration expenditures is allowed to be deducted 100% in income
Article 38. Carry forward loss more than 5 years.

Box 6.2. Draft: Tax Expenditures Under Income Tax Act and Value Added Tax Act (Cont'd)

VALUE ADDED TAX ACT, Cap.349 (As Amended To July 2005)

Exempt Supply

Article 19/ 1e, the supply of unimproved land, exempted
Article 19/1f, supply by way of lease or letting of immovable property (except, a lease or letting of :commercial premises, hotel or holiday accommodation, parking or storing cars, service apartments, a lease or letting for periods not exceeding three months) , exempted
Article 19/1o, supply of petroleum fuels subject to excise duty, exempted
Article 19/1q, supply of dental, medical and veterinary equipment, exempted
Article 19/1r, supply of feeds for poultry and livestock, exempted
Article 19/1s, supply of machinery used for the processing of agricultural or dairy products.
Article 19/1t. supply of photosensitive semiconductor devices (including photovoltaic devices both in modules and panels), light emitting diodes; solar water heaters and solar cookers; exempted (solar energy equipments)
Article 19/1u, supply of accommodation in tourist lodges and hotels outside Kampala and Entebbe;, exempted
Article 19/1v, supply of computers, printers, parts and accessories falling under 84.71 and 84.73 categories under the harmonized coding system of the customs law, exempted
Article 19/1w, supply of computer software, exempted
Article 19/1x, supply of lifejackets, life saving gear, head gear and speed governors, exempted
Article 19/1y, supply of mobile toilets and Ekoloo toilets made from polyethylene, exempted
Article 191/z, supply of insecticides, exempted
Article 19/1aa, supply of feasibility studies, engineering designs and consultancy services and civil works related to roads and bridges' construction, exempted.

Zero rated supplies

Article 19/2ae, supply of educational materials other than books, zero rated
Article 19/2af, supply of seeds, fertilizers, pesticides, and hoes
Article 19/2ag, supply of cereals (grown, milled or produced from Uganda), zero rated
Article 19/2ah, supply of machinery, tools and implements suitable for use only in Agriculture, zero rated

Box 6.3. Examples of Trade Offs In the tax System: Ineffectiveness, Inequity, and Inefficiency in Tax Expenditures

1. Both pension contributions and pension benefits are tax exempt (ITA Article 21).

This scheme erodes the tax base, causing revenue loss and making the income tax regressive.

2. Tax exemption for salaries, allowance and other benefits of certain public servants under the Constitution (Article 128)

This creates inequity, in addition to the loss of tax revenue in the tax system.

3. Article 38 of the ITA permits the indefinite carry-forward of losses.

Indefinite loss carry- forwards are an inefficient use of tax revenues. This provides opportunities for abuse by using transfer pricing.

4. Article 28 of the Ugandan ITA provides as follows: "Initial tax allowance for eligible assets (plants and machinery) at cost base of the property at service for the first time during the year of income: 50% such tax allowance for assets invested in the five main cities (Kampala, Entebbe, Namanve, Jinja and Njeru) and 75% for investing in outside the five main cities".

This policy is ineffective, but is a windfall to investors. It is questionable whether this type of accelerated-depreciation tax benefit actually influences investment decisions. Investors in plants and machinery would not close their plants or business if this ***one- year*** accelerated-depreciation tax allowance were not available. Considering that plants and machinery are durable goods and long term investments, there are other factors (such as the existence of roads and infrastructure) which are more important than this one-year accelerated-depreciation tax allowance in determining the location of these investments. Therefore, this tax allowance is ineffective in influencing investment decisions, but provides a windfall to investors.

5. Start-up costs. A person who has incurred expenditures in starting up a business is allowed to deduct an amount equal to twenty-five per cent of the amount of these "start-up costs" in each of the initial four years of the operation of the business (ITA, Article 30).

This policy is subject to abuse. It is a classic case of the problems caused by "company churning". It complicates tax administration and results in large revenue losses.

6. Article 19 of the Ugandan VAT exempts "unimproved land" from VAT.

This causes horizontal inequity.

7. Tax incentives/tax expenditures are granted by administrative discretion to favored business enterprises, creating effective tax rates that vary both between and within sectors.

This results in inequality, ineffectiveness and inefficiency.

All tax incentives/tax expenditures should be in the tax law, and should be available to all enterprises on the same terms and should not be granted by administrative discretion.

ANNEXES

Chapter 1 Annex. Methodology For Calculating Macro Investment Requirements[113]

Total factor productivity trend: Growth in GDP per capita decomposition.

The basis for this growth accounting exercise is a 'benchmark' Cobb-Douglas aggregate production function in physical capital K and effective labor $H.L$. The argument here is that labor force would treat all workers as if they were identical, yet workers characteristics clearly influence marginal productivity. Thus:

$Y = AK^{0.35}(HL)^{0.65}$ or in per-worker terms,

$y = Ak^{0.35}H^{0.65}$ where y = Y/L and k = K/L. The unobserved variable A is an index of total factor productivity; its contribution to growth is calculated as a residual in the usual way. The elasticity of output with respect to labor, at .65, is broadly consistent with international evidence. Thus the growth of output per worker can be decomposed into the contributions of the growth of capital per worker, the growth of labor quality index and the residual measuring the total factor productivity.

Due to data limitations, labor input is measured by the population of labor force age rather than by the actual labor force. Capital stocks are derived by applying the perpetual inventory method to year-by-year investment data, assuming an annual depreciation rate of 0.04. The initial capital stock is assumed to be proportional to GDP.

As in most literature of growth accounting, to calculate the labor quality index H, following Collins and Bosworth (1996) we use the educational attainment to reflect the quality of labor force such that $H = Exp(E * S)$. Where S denotes the return to an additional year of schooling and E the average years of schooling. To estimate the return to education, we use the estimates obtained in the return to education analysis by Mugume and Canagarajah, (2005). The return to an additional year of schooling ranges from 0.46 in 1992 to 0.13 in 1999 and 2002. Since the Uganda was emerging from a long economic decline trough after the civil war and macro-economic distortions in 1992, we consider the 1999 and 2002 estimate as reasonable estimate for the returns to schooling. Moreover, this is comparable to estimates obtained elsewhere (Cohen and Soto, 2001; Collins and Bosworth, 1996). Thus, we assume that each year of educational attainment, other things equal, raises labor's marginal product by 13 percent. However, for the projections, we assume that eventually the return would decline as the stock of education increases, hence we assume that it declines to about 10% as assumed in most African studies for the period 2004-2015.

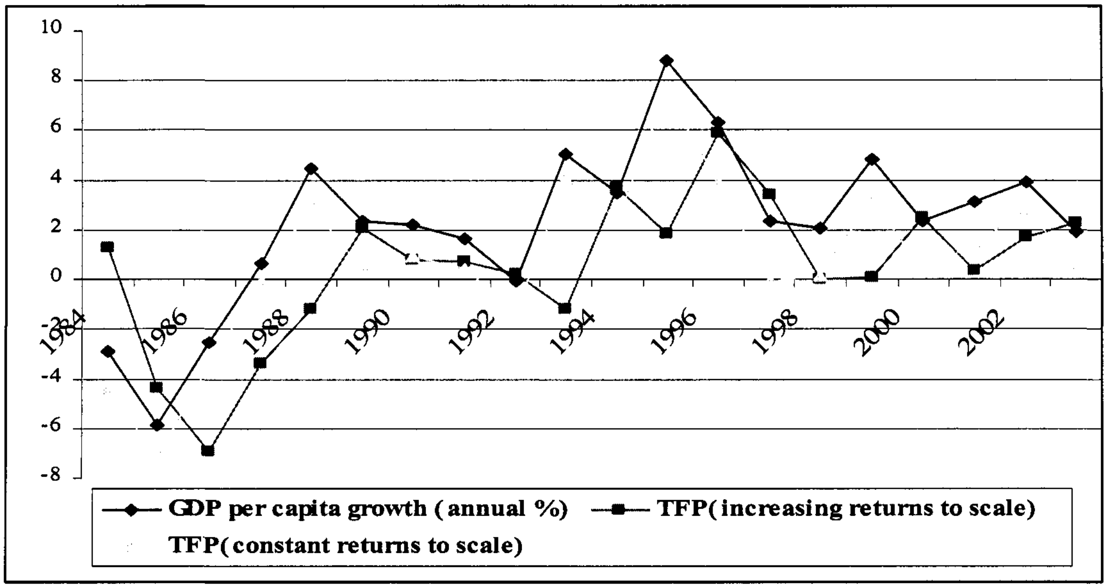
Taking the logarithm and differentiating with respect to time:

$$\frac{\dot{y}}{y} = \frac{\dot{A}}{A} + 0.35\frac{\dot{k}}{k} + 0.65 \times 0.13\dot{E}$$

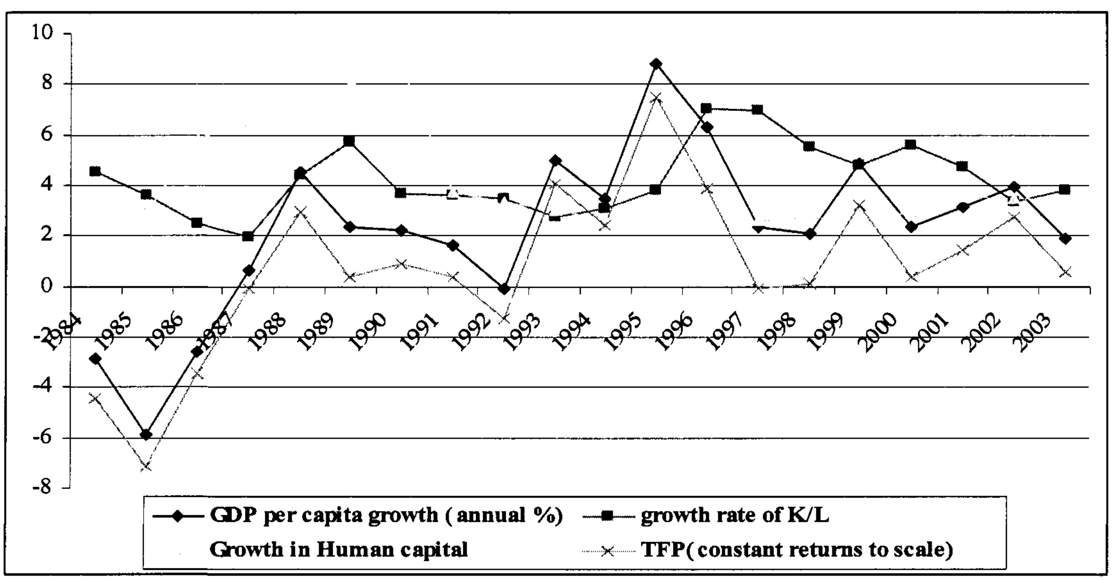
Thus, $g_y = g_A + 0.35g_k + 0.65 \times 0.13\dot{E}$, where g denotes a growth rate.

[113] This note was prepared by Adam Mugume.

For given $g_y, g_A, \dot{E}$, g_k can be derived as $g_y - g_A - 0.65 \times 0.13\dot{E}$ and thus $K_t = K_{t-1}(1+n)(1+(g_y - g_A - 0.65 \times 0.13\dot{E}))$ where n is the labor force growth rate. Aggregate investment at any given time t can also be derived as $K_{t+1} - 0.96K_t$ given the assumption that capital decays at a rate of 4%.



Constant returns to scale TFP seems to closely trend the growth in GDP per capita



The growth rate in capital per worker seems to be driving the total factor productivity and the growth in output per worker.

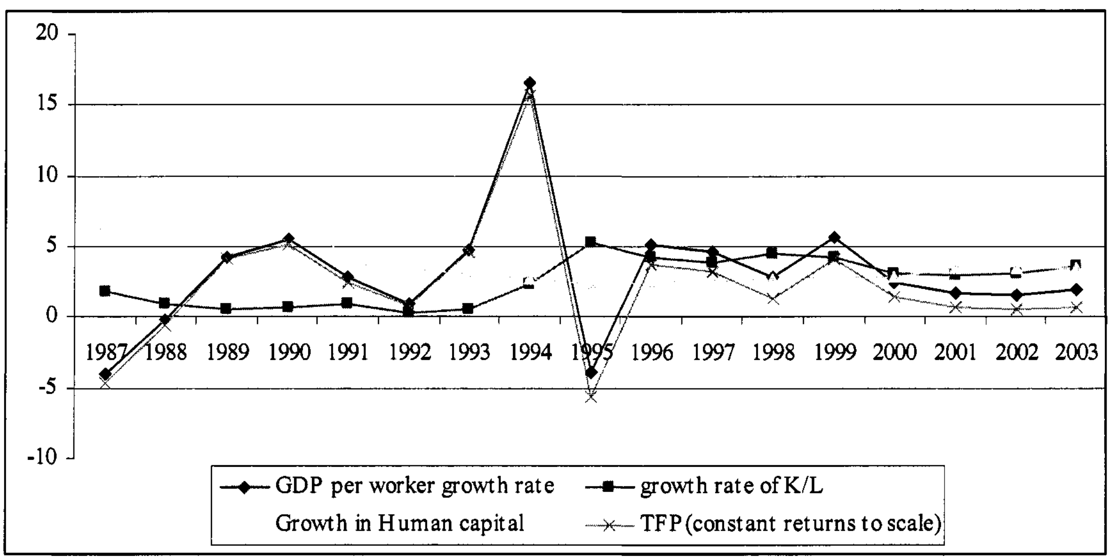
Sources of growth

	Growth of per capita GDP	Growth rate of capital per worker	Human capital growth rate	TFP
1986-92	1.9	3.8	6.2	0.53
1993-98	4.7	4.9	2.6	3.0
1999-03	3.2	4.4	3.2	1.7

On the basis of a simple growth accounting exercise, Dunn identifies three distinct growth episodes for Uganda in the post-conflict period. During the first, from 1986 to 1992, total factor productivity (TFP) contributed just under 0.5% per annum to overall growth. From 1993 to1998 its contribution rose to 3% per annum before falling back to around 1.7% more between 1999 and 2003. This is consistent with what is reported by Bevan, et al., (2003).

Using Penn Data Set

We also use data from Penn web and use the same methodology of decomposing growth into various sources. The figure below shows a slight difference from the one obtained using data from WDI. **Could this be due to the fact that the Penn data set is in Dollars?**



Sources of Growth for Uganda (using return to extra year of schooling of 13%)

	1985-1989	1990-1994	1995-2000	2001-2003
GDP per worker growth	-3.13	6.10	2.81	1.93
Growth in K/L	0.18	0.95	4.20	3.23
Growth in Human capital	6.11	4.31	2.64	3.47
TFP	-1.73	3.20	1.34	0.60

Sources of Growth for Uganda (using return to extra year of schooling of 7%)

	1985-1989	1990-1994	1995-2000	2001-2003
GDP per worker growth	-3.13	6.10	2.81	1.93
Growth in K/L	0.18	0.95	4.20	3.23
Growth in Human capital	4.74	3.77	2.48	3.03
TFP	-1.72	3.08	1.34	0.60

The results seem to give a more prominent role to the productivity residual in explaining Uganda's growth. For instance, for the period 2001-2003, although growth in capital per worker and human capital were respectively about 3.2% and 3% GDP per capita growth was only 1.93. The correlation coefficient between GDP per worker growth, and TFP, capital per worker and human capital indeed seem to suggest that growth Uganda for the period 1985-2003 was predominantly due to TFP.

	19985-92	1993-03
Correlation coefficient between TFP and GDP per capita growth rate	0.997	0.997
Correlation coefficient between K/L growth and GDP per capita growth rate	0.390	-0.43
Correlation coefficient between human capita growth and GDP per capita growth rate	0.658	0.263

In comparison to a sample of 19 SSA countries and a few African countries in Ndulu and O'Connell, (2003), Uganda's sources of growth seem to be markedly different from the rest of the African countries. Uganda seems to have a substantially larger capital per worker and human capital growth rates but their contributions to GDP per capita growth is low even after adjusting the return to extra year of schooling to 7% as assumed by Ndulu and O'Connell (2003).

Sources of Growth for 19 SSA countries

	1985-1989	1990-1994	1995-2000
GDP per worker growth	0.45	-1.74	1.51
Growth in capital	-0.22	-0.08	-0.12
Growth in Human capital	0.34	0.30	0.26
TFP	0.33	-1.95	1.37

Sources of Growth for Some African countries

Ghana	1985-1989	1990-1994	1995-2000
GDP per worker growth	1.52	1.05	1.77
Growth in capital per worker	-1.28	0.05	1.17
Growth in Human capital	0.15	0.15	0.15
TFP	2.65	0.86	0.44
Kenya	1985-1989	1990-1994	1995-2000
GDP per worker growth	2.02	-1.91	-0.94
Growth in capital per worker	-0.79	-0.66	-0.28
Growth in Human capital	0.35	0.36	0.29
TFP	2.46	-1.60	-0.96
Mauritius	1985-1989	1990-1994	1995-2000
GDP per worker growth	4.95	3.37	3.83
Growth in capital per worker	0.63	1.07	0.95
Growth in Human capital	0.32	0.26	0.24
TFP	4.01	2.09	2.4

Tanzania	1985-1989	1990-1994	1995-2000
GDP per worker growth	0.92	-0.59	1.29
Growth in capital per worker	-0.04	0.45	-0.26
Growth in Human capital	0.16	0.10	0.14
TFP	0.80	-1.14	1.41

Projections

Assuming that for the period 2004-2015 TFP was to be constant at the SSA average of about 1.4 (Ndulu and O'Connell and 2003), labor force was to grow at 3.8% per annum as projected by the UN, GDP per worker was to grow at about 2.2 percent derived by assuming a 4 year moving average, and the average years of schooling was to grow from 3.4 in 2003 to about 7 in 2015 reflecting a constant growth rate of about 6 percent per annum, which could be a reasonable growth rate given the emphasis on the UPE and USE, then the derived average growth in capital stock per worker would be about 3.3%. This would require real investment growth average of about 7 percent per annum, and assuming that the share of public investment in total investment remains at 25% as in the last 7 years (1998-2004), this would translate into public investment growth requirement of 7 % per annum on average for the period 2004-2014. If TFP was assumed to be a 4-year moving average of Uganda's recent experience, the public investment growth holding other factors constant would still be 9 %. We take the mid-point.

Chapter 3. Household expenditure on education[114]

Annex 1 of Chapter 3

The objective of this annex is to discuss household spending on primary and secondary education in Uganda in 2002 and 2006. Discussion of levels of expenses on primary and secondary school can be found in section 1. Section 2 describes changes in household expenses on primary and secondary education from 2002 to 2006. Appendices illustrate methodological issues.

Summary conclusions of the analysis:

- 52% increase in per student primary school expenses and 36% increase in per student secondary school expenses from 2002 to 2006;
- 3.3 times difference between urban and rural per student primary school expenses in 2006 (with urban higher than rural);
- 36% of per student primary school expenses and 63% of per student secondary school expenses covered school fees in 2006, while much lower shares of total expenses covered uniform and books and supplies;
- 6.1 times higher per student total primary school expenses on nonpublic than on public schools at the primary level and 6% higher at the secondary level in 2006;

1. LEVELS OF HOUSEHOLD EXPENDITURES ON PUBLIC PRIMARY AND SECONDARY SCHOOLS, 2006

Mean per student total household expenses on secondary public schools were 16 times higher than on primary public schools in 2006, with the former amounting to 391,512USh and the latter to 24,568USh. Disparity between urban and rural expenses on primary schools was high, with urban equaling 66,107USh and rural amounting to 20,326USh or 3.3 times lower than urban. Disparity between urban and rural expenses in secondary schools was much lower: urban expenses amounted to 487,519USh and were only 1.3 times or 34% higher than the rural, the latter equaling 363,633USh.

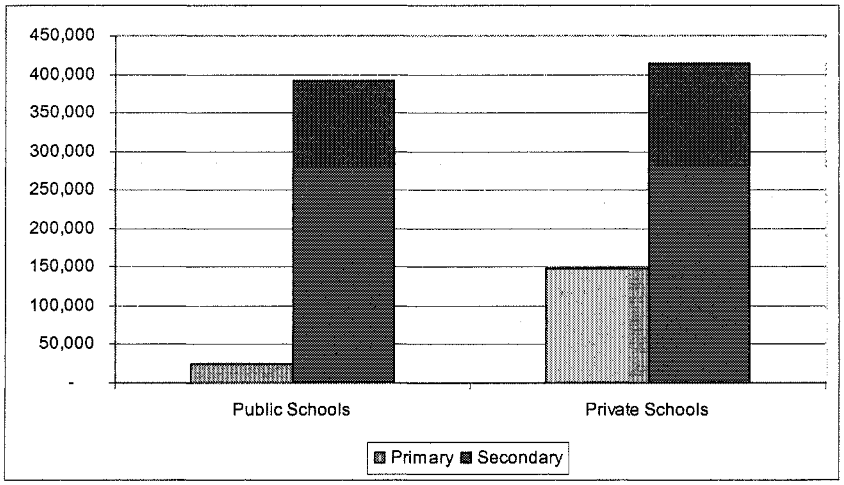
School fees were the biggest expense item in both secondary and primary public schools in 2006: in secondary public schools, they amounted to 270,123USh and comprised 63% of the total expense; in primary public schools, they constituted 9,006USh or 36% of all expenses. In secondary public schools, there was almost no disparity between urban and rural areas in the structure of school expenses by item. School fees constituted 65% of all expenses in urban and 62% in rural areas, books and supplies – 8% across the urban/rural subdivision and uniforms and sports clothes – 4% in the cities and 5% in the rural areas. In primary public schools, the situation was not the same: the structure of expenses of urban households was quite differed from that of the rural ones -- while in the cities school fees constituted 49% of all school expenses, in rural areas they amounted only to 32% of the total. Books and supplies accounted for 15% of total expenses in the cities and for 25% in rural areas. Uniform and sports uniform amounted to 12% and 20% of the total in urban and rural areas correspondingly.

Public schools were substantially, 6.1 times, cheaper than private schools at the primary level and only 6% cheaper at the secondary level. Per student private school total expenses amounted to 149,643 in primary schools and to 413,703 in secondary schools. The ratio of private/public expenses was much lower in urban than in rural areas, equaling to 3.5 and 5.5

[114] This annex has been prepared by Maria Shkaratan.

correspondingly. This ratio was close to 1 in both urban and rural secondary schools. **The structure of expenses on public and private schools differed at the primary level and was very similar at the secondary level.** In private schools, most of expenses (65% at the primary level and 61% at the secondary level) covered school fees, while expenses on school uniform and books were below 10% each.

Figure 1. Total Primary and Secondary School Expenditures in Public and Private Schools. Uganda Household Survey 2006. Ugandan Shillings



2. TRENDS IN HOUSEHOLD EXPENDITURES ON PRIMARY AND SECONDARY SCHOOLS, 2002-2006

Overall, school expenses increased substantially from 2002 to 2006, more so in primary rather than secondary and in private rather than public schools. In primary schools, per student mean total expenses increased by 52% in the observed period from 2002 to 2006[115]. By comparison, total per student mean expenses only grew by 32% in secondary schools in the same period. The increase in primary school expenses was higher in private (62%) than in public schools (40%). Moreover, this gap was even larger in rural areas, with 74% increase in private schools vs. 35% in public ones, than in urban areas, where the corresponding numbers were 43% in private vs. 20% in public schools. The increase of expenses for secondary schools was not as large as for primary schools, but followed the same pattern by being higher in private (38%) than in public (17%) schools. This increase in rural areas was approximately the same in both public and private schools -- correspondingly 30% and 34%, respectively, while urban areas demonstrated a higher increase in private school expenses (45%) and a decrease in public schools expenses (-28%).

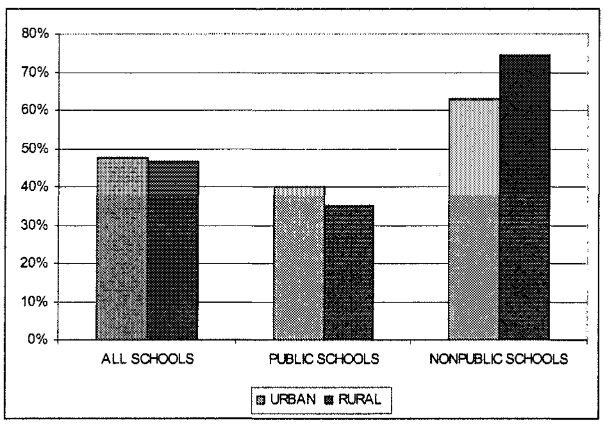
Overall, the structure of primary school expenses did not change noticeably from 2002 to 2006, with the exception of urban schools and private schools – both experienced a slight decrease in the share of school fees in total expenses. The structure of secondary school expenses changed in the same period, with the share of fees decreasing in all types of schools (urban, rural, public and private). In primary schools, payments for the biggest expense item,

[115] All 2006 expenses are expressed in 2002 prices. Therefore, the 52% increase is measured in constant terms.

school fees, increased by 44%, expenses on books and supplies grew by 64% and expenses on uniform – by 19%. The increase in itemized expenses was similar in urban and rural areas. This resulted in a structure of 2006 primary school expenses that was very similar to the 2002 one, with two exceptions: the share of fees in total expenses decreased from 72% to 63% in urban areas and from 77% to 69% in private schools. In secondary schools, school fees, the biggest expense item, increased only by 9%, while expenses on books and supplies grew more than twice and expenses on uniform – by 46%. Urban and rural households experienced different structure of increases by item: for urban households, the biggest increase was in the payments for school uniform (2.2. times) and for books and supplies (2.5 times), while for rural ones the only large increase was in payments for books and supplies (2.0 times). Payments of school fees increased 15% in the cities and 2% in the rural areas. These changes resulted in a structure of 2006 secondary school expenses that differed from that in 2002: the main change was in the share of school fees, which constituted 82%-84% of total school expense for all types of schools (urban, rural, public and private) in 2002 and a lower share ranging from 62% to 72% for the same school types).

Figure 2. Percentage Changes in Total Primary and Secondary School Expenditures from 2002 to 2006, by Urban/Rural. Source of data: Uganda Household Surveys 2002 and 2006

PRIMARY SCHOOLS



SECONDARY SCHOOLS

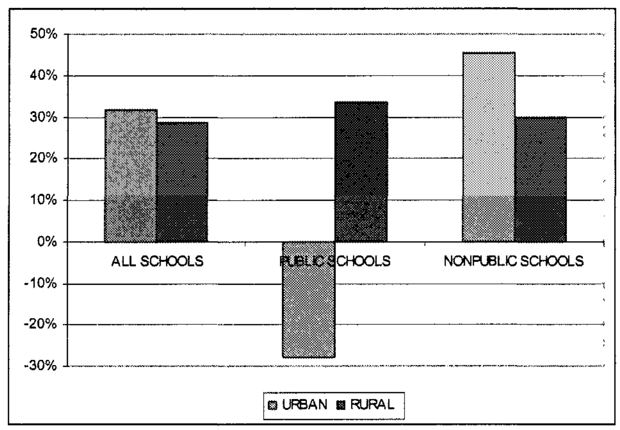
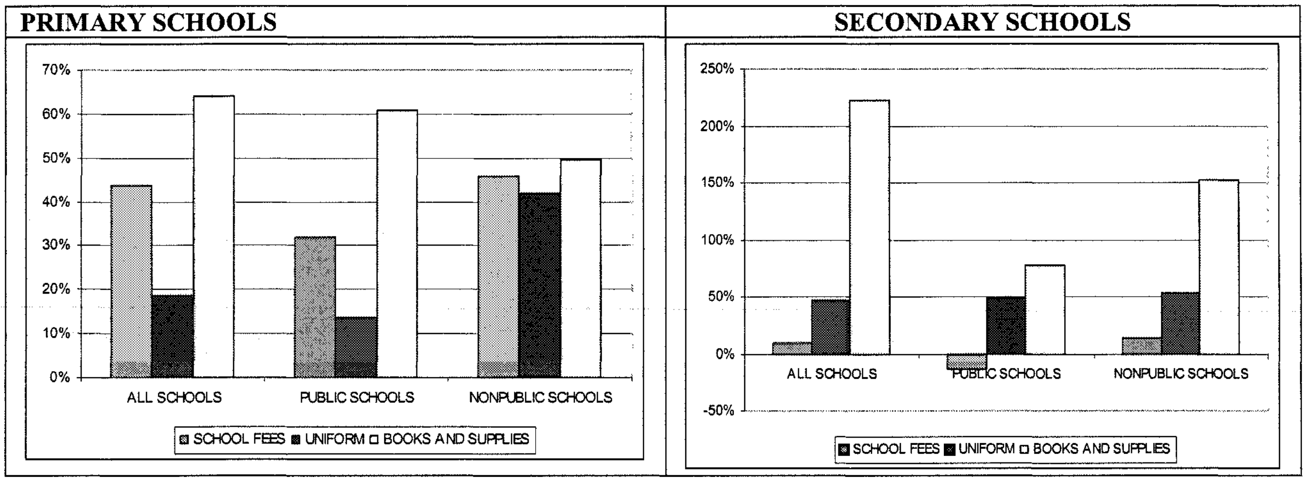


Figure 3. Percentage Changes in Total Primary and Secondary School Expenditures, by Expense Item, 2002-2006. Source of data: Uganda Household Surveys 2002 and 2006



Attachment 1. Methodology

The analysis included estimation of (1) level of expenses on primary and secondary schools in 2006 and (2) trends in primary school expenses from 2002 to 2006. The analysis was based on data from Uganda household surveys of 2002 and 2006. Household school expenses were calculated on a per student basis and analyzed by type of school (public/nonpublic), rural/urban location of the household and by expense item (e.g. school fees, transportation, etc). The 2006 expenses were deflated using CPI and presented in 2002 prices to make comparison with 2002 data meaningful.

Levels of per student household expenditures on education (section 1) were estimated using only 2006 data, as the 2002 survey questionnaire did not include a sufficiently detailed set of questions on education expenditure. In the 2006 survey, a section to collect data on education expenses at individual level was added to the questionnaire, thus making the analysis of expenses on different levels of education possible for the entire sample of households, including those with children at different school levels.

The analysis of ***expenditure trends*** (section 2) was conducted using 2002 and 2006 household level data for the sub-samples of households that only had, correspondingly, primary and secondary school students. Households with students attending different school levels were excluded from the analysis. This methodology was used, because the only data on education expenditures common for 2002 and 2006 household surveys was collected at the household level and thus estimated household expenditures on education of all students in the household. If a household had children at different school levels, it was not possible to know what part of its education expenses went to primary schools and what part went to secondary and tertiary school.

This methodology described in the paragraph above gives correct estimation of expenditure trends, to the extent that subsamples compared between 2002 and 2006 are broadly similar. We confirmed that these sub-samples were similar by comparing basic socioeconomic indicators. However, because the subsamples are not representative of the original sample and thus not representative of the population, such analysis gives somewhat biased estimation of expenditure levels. Therefore, we only use these data to discuss changes in expenditures, not their levels.

Annex 2 of Chapter 3. Review Of Past Studies of Absenteeism and Leakage in Uganda's Primary Education sub-Sector.

Uganda was the first country to have a Public Expenditure Tracking Survey (PETS) conducted in 1995 (Ablo and Reinikka, 1998), and several follow-up surveys have been completed since[116]. Uganda was also amongst eight pioneering countries which participated in the first cross-country analysis of teacher absenteeism. Third, Uganda has undertaken several teacher payroll clean-up exercises. One such exercise in 1993 which found that some 20 percent of teachers on the payroll were "ghosts" has been widely quoted.[117] Finally, the construction of classrooms in the period 2002-2003 has been scrutinized by both the Ministry of Local Government and the Auditor General in two separate reports.[118]

Possibly as a result of pioneering these types of surveys – and, thus, subjecting itself to be scrutinized half a decade (or more) before its peers – Uganda universally come out at the bottom (i.e. performing the worst) when its results are compared to other countries. The PETS conducted in 1996 revealed that "only 24 percent of the total capitation grant from the central government reached the schools. Thus, for every dollar spent on nonwage education items by the central government, roughly 80 cents were captured by local government officials (and politicians). In fact, a majority of schools received nothing." (Reinikka and Svennson, 2003). Countries which have since undertaking PETS universally perform better than this (see, for instance, table 2 in Reinnikki and Svensson (2003)).

Apart from the regular "PETS" Uganda has conducted other studies to track whether financial inputs translate into tangible inputs. In 2002/03, Uganda participated in the first cross-country study of teacher absenteeism. The study showed that 27 percent of Uganda's primary school teachers were absent, the worst performer in Chaudhury et al's (2004) sample of eight countries. Equally problematic, the Auditor General's (2003) Value for money study of classroom construction showed non-negligible leakages –defined here as money spent on purchasing a classroom of a certain standard but either not delivered at all or delivered at sub par quality. Specifically, the report revealed that many of the construction works were not to the standards set in the technical guidelines and could not be expected to remain standing for the 30 years they were designed to last. The Auditor General report notes that "in extreme cases structures collapsed before they were completed" (p. 12). In the worst case – Kampala – 27 percent of the newly finished classrooms visited were judged to be of a "very bad" quality, possibly implying – although not it is not clear from the AG report –that they are at imminent risk of collapsing.[119]

[116] Republic of Uganda (2000). "Tracking the Flow of and Accountability for UPE Funds." Ministry of Education and Sports. Report by International Development Consultants Ltd, Kampala, Uganda. Reinikka and Svensson's 2002 survey revisited the original schools from the 1996 survey plus an additional 170 schools from 9 of the original 18 districts surveyed. Although the two surveys were done in 1996 and 2002, they refer to financial flows in 1995 and 2001, respectively.

[117] Among other places, this finding is quoted in Table 3 in Reinnikka and Svensson (2003): "Survey Techniques to Measure and Explain Corruption" and in Reinnikka and Svensson (2004): "Local Capture: Evidence From a Central Government Transfer Program in Uganda (2004).

[118] "Technical and Value for Money Audits" done for at least 13 districts and dated February 2004, available on the Ministry of Local Government's website, and in the Auditor General's "Value for Money Audit Report on the Universal Primary Education Programme" (undated, but reportedly released in 2003)

[119] The report does not define exactly what is meant by "very bad" but it is the worst category of four.

Table 1. Mean and Median per Student Household Expenses on Education, by Urban/Rural and Expense Item. Uganda Household Survey 2006. In Ugandan Shillings

	Public Schools						Private Schools					
	School Fees	Uniforms and Sports Clothes	Books and Supplies	Boarding Fees	Other	Total School Expenses	School Fees	Uniforms and Sports Clothes	Books and Supplies	Boarding Fees	Other	Total School Expenses
Means												
Primary												
Urban	32,182	7,784	9,603	1,701	14,751	66,107	161,775	13,331	18,046	19,501	27,117	233,601
Rural	6,701	4,102	5,218	993	3,831	20,326	71,223	6,777	8,824	10,445	16,322	112,451
Total	9,006	4,435	5,616	1,055	4,823	24,568	98,066	8,745	11,592	13,075	19,602	149,643
Secondary												
Urban	354,126	20,348	44,812	45,846	82,794	487,519	331,663	22,270	48,281	82,116	57,949	545,547
Rural	247,221	18,043	32,334	42,349	55,818	363,633	224,759	16,961	31,545	42,145	55,840	344,247
Total	270,123	18,522	35,053	43,053	61,821	391,512	259,336	18,705	37,040	55,179	56,530	413,703
Medians												
Primary												
Urban	7,000	5,500	5,200	-	1,000	26,000	114,000	10,000	11,000	-	5,000	155,800
Rural	-	4,000	3,600	-	600	11,500	33,500	5,000	5,000	-	2,100	54,000
Total	-	4,000	3,600	-	700	12,200	49,200	6,000	6,000	-	3,000	79,000
Secondary												
Urban	270,000	15,000	26,000	-	22,000	405,000	230,000	20,000	28,000	-	20,000	362,000
Rural	165,000	15,000	20,000	-	16,000	273,000	150,000	15,000	21,500	-	15,000	227,500
Total	183,000	15,000	20,500	-	20,000	291,000	170,000	15,000	24,000	-	18,000	283,800

Table 2. Trend in Household Primary School Expenditure: Mean and Median per Student Annual Household School Expense by Item*. Uganda Household Surveys 2002 and 2006. In Ugandan Shillings

	2002					2006, in constant 2002 prices				
	School Fees	School Fees and Boarding/Lodging	Uniform	Books and Supplies	Total School Expenses	School Fees	School Fees and Boarding/Lodging	Uniform	Books and Supplies	Total School Expenses
Means										
All Schools										
All Students	8,273	8,720	2,909	2,547	15,658	11,893	12,511	3,447	4,182	23,776
Urban	39,509	41,977	4,632	4,480	54,341	52,254	54,553	6,538	8,011	80,211
Rural	5,134	5,378	2,735	2,353	11,770	7,227	7,650	3,090	3,739	17,251
Public Schools										
All Students	3,712	3,969	2,749	2,299	10,452	4,892	5,068	3,125	3,696	14,604
Urban	21,357	22,492	4,267	3,597	33,942	25,712	26,980	5,063	6,481	47,466
Rural	2,521	2,719	2,647	2,211	8,866	3,236	3,326	2,971	3,475	11,991
Nonpublic Schools										
All Students	34,979	36,466	3,758	3,970	45,576	50,997	53,729	5,339	5,945	73,617
Urban	65,690	68,989	4,876	5,575	81,499	94,064	98,607	8,971	9,728	132,740
Rural	19,653	20,235	3,199	3,169	27,648	32,498	34,453	3,778	4,320	48,221
Medians										
All Schools										
All Students	-	-	2,500	1,500	6,000	-	278	2,920	2,670	10,013
Urban	14,000	15,000	3,750	2,800	26,250	12,516	12,516	4,589	4,172	36,922
Rural	-	-	2,500	1,500	5,800	-	-	2,698	2,503	9,262
Public Schools	-	-	-	-	-	-	-	-	-	-
All Students	-	-	2,500	1,500	5,533	-	-	2,781	2,503	8,511
Urban	-	-	3,333	2,100	12,750	4,172	4,673	4,172	3,504	18,294
Rural	-	-	2,500	1,500	5,400	-	-	2,642	2,503	8,205
Nonpublic Schools	-	-	-	-	-	-	-	-	-	-
All Students	17,500	18,000	3,000	1,800	23,000	25,032	25,032	4,172	3,588	38,215
Urban	55,400	55,400	4,000	3,100	67,000	75,096	75,096	6,675	5,757	99,794
Rural	9,000	9,000	2,500	1,333	15,200	12,516	13,350	2,920	3,087	25,338

Table 3. Trend in Household Secondary School Expenditure: Mean and Median per Student Annual Household School Expense by Item*. Uganda Household Surveys 2002 and 2006. In Ugandan Shillings

	2002					2006, in constant 2002 prices				
	School Fees	School Fees and Boarding/ Lodging	Uniform	Books and Supplies	Total School Expenses	School Fees	School Fees and Boarding/Lodging	Uniform	Books and Supplies	Total School Expenses
Means										
All Schools										
All Students	123,770	128,524	8,430	8,946	150,794	134,699	142,201	12,333	19,869	198,539
Urban	145,034	154,728	7,417	9,609	176,407	166,775	167,421	16,184	23,976	232,387
Rural	113,741	116,165	8,909	8,634	138,714	115,625	127,204	10,043	17,426	178,412
Public Schools										
All Students	121,185	123,458	7,541	7,991	144,776	104,547	118,709	11,204	14,189	169,404
Urban	129,624	130,111	8,122	9,914	153,916	55,055	55,055	13,270	8,070	110,940
Rural	118,872	121,634	7,382	7,464	142,269	121,946	141,086	10,477	16,341	189,956
Nonpublic Schools										
All Students	120,243	121,041	8,998	8,756	142,870	136,107	138,128	13,841	22,067	197,718
Urban	160,352	160,622	7,112	7,397	178,627	187,360	187,360	19,738	29,123	259,665
Rural	94,667	95,802	10,201	9,622	120,069	101,640	105,019	9,875	17,322	156,060
Medians										
All Schools										
All Students	108,000	115,000	6,500	8,000	143,800	111,809	112,644	10,013	14,519	173,137
Urban	106,000	144,000	2,500	7,200	189,000	138,927	138,927	16,688	23,085	222,367
Rural	108,000	114,000	8,000	8,000	131,500	100,128	101,129	8,344	12,516	147,688
Public Schools										
All Students	114,000	115,000	5,000	6,500	129,000	100,128	100,128	10,013	9,011	134,505
Urban	120,000	130,000	8,000	6,000	163,500	-	-	11,682	2,086	54,236
Rural	108,000	115,000	5,000	7,000	129,000	112,644	112,644	8,344	10,013	147,688
Nonpublic Schools										
All Students	90,000	90,000	6,000	8,000	114,000	120,988	125,160	10,847	16,688	174,556
Urban	144,000	144,000	-	6,750	189,000	175,223	175,223	17,940	25,032	249,485
Rural	90,000	90,000	10,000	8,000	101,000	87,612	87,612	6,953	12,516	125,160

Table 4. Trend in Household Primary and Secondary School Expenditure: Mean per Student Total Annual Household School Expense by Groups of Households. Uganda Household Surveys 2002 and 2006. In Ugandan Shillings.

	2002			2006, constant 2002 prices		
	Households with at Least One Member in School of Any Level	Households with Children Only in Primary School	Households with Children Only in Secondary School	Households with at Least One Member in School of Any Level	Households with Children Only in Primary School	Households with Children Only in Secondary School
Means						
All Schools						
All Students	38,686	15,658	150,794	61,443	23,776	198,539
Urban	94,505	54,341	176,407	148,816	80,211	232,387
Rural	29,519	11,770	138,714	45,913	17,251	178,412
Public Schools						
All Students	20,251	10,452	144,776	27,759	14,604	169,404
Urban	55,576	33,942	153,916	72,771	47,466	110,940
Rural	17,449	8,866	142,269	23,685	11,991	189,956
Nonpublic Schools						
All Students	66,779	45,576	142,870	114,172	73,617	197,718
Urban	101,050	81,499	178,627	178,050	132,740	259,665
Rural	46,869	27,648	120,069	80,634	48,221	156,060
Medians						
All Schools						
All Students	9,750	6,000	143,800	16,354	10,013	173,137
Urban	63,000	26,250	189,000	99,877	36,922	222,367
Rural	8,250	5,800	131,500	13,100	9,262	147,688
Public Schools						
All Students	6,333	5,533	129,000	9,700	8,511	134,505
Urban	19,100	12,750	163,500	23,889	18,294	54,236
Rural	6,000	5,400	129,000	9,262	8,205	147,688
Nonpublic Schools						
All Students	36,480	23,000	114,000	62,163	38,215	174,556
Urban	78,500	67,000	189,000	127,329	99,794	249,485
Rural	21,667	15,200	101,000	36,296	25,338	125,160

Table 5. Comparison of Households in 2002 and 2006 Samples by Selected Socioeconomic Characteristics. Uganda Household Surveys 2002 and 2006.

2002	All Schools			Public Schools			Nonpublic Schools		
	Percentage of Urban Households	Level of Household Head Education*		Percentage of Urban Households	Level of Household Head Education*		Percentage of Urban Households	Level of Household Head Education*	
		Mean	Median		Mean	Median		Mean	Median
Households with Members Attending School	15%	2.7	2.0	8%	2.5	2.0	39%	3.1	3.0
Households with Children Only in Primary School	9%	2.4	2.0	6%	2.4	2.0	34%	2.8	2.0
Households with Children Only in Secondary School	34%	3.4	3.0	26%	3.2	3.0	39%	3.4	3.0
All Households	15%	3.9	4.0	8%	2.5	2.0	39%	3.1	3.0

2006	All Schools			Public Schools			Nonpublic Schools		
	Percentage of Urban Households	Level of Household Head Education*		Percentage of Urban Households	Level of Household Head Education*		Percentage of Urban Households	Level of Household Head Education*	
		Mean	Median		Mean	Median		Mean	Median
Households with Members Attending School	16%	3.1	3.0	9%	2.8	2.0	36%	3.5	3.0
Households with Children Only in Primary School	10%	2.7	2.0	7%	2.7	2.0	30%	3.1	3.0
Households with Children Only in Secondary School	40%	3.7	4.0	28%	3.7	3.0	43%	3.6	3.0
All Households	16%	3.1	3.0	9%	2.8	2.0	36%	3.5	3.0

* Ranges from 0 to 6, with 0 indicating no education and 6 reflecting "diploma or above"

Table 6. Percentage of Households in Each Education Category by Region. Uganda Household Surveys 2002 and 2006.

	2002			2006		
	Households with at Least One Member in School of Any Level	Households with Children Only in Primary School	Households with Children Only in Secondary School	Households with at Least One Member in School of Any Level	Households with Children Only in Primary School	Households with Children Only in Secondary School
Central	98.6%	46.2%	2.7%	99.8%	52.0%	3.9%
Eastern	98.5%	68.2%	2.7%	100.0%	65.9%	2.8%
Northern	97.0%	77.3%	2.2%	99.5%	77.4%	1.8%
Western	99.5%	62.6%	2.4%	99.8%	63.8%	2.0%
Total	98.5%	61.9%	2.5%	99.8%	63.4%	2.7%

Annex 3 of Chapter 3. Details of primary school sample (from Habyarimana 2007)

The districts were selected from the eligible universe (excluding Northern districts) with a sampling probability proportional to population share. Population estimates were drawn from estimates provided by the Uganda Bureau of Statistics. Care was taken to ensure that the drawn sample reflected the broad attributes of the population with respect to socio-economic status (see Habyarimana 2007 for details).

Having selected the districts, the list of schools was stratified by ownership into two categories: government-owned schools and non-government owned schools. The list of the schools and their attributes comes from the 2005 Education Management Information System (EMIS) school returns data. The stratification was done in order to generate a sample of non-government owned schools that was large enough to produce reliably precise measures of central tendency for non-government owned schools. Within the each strata, schools were selected with sampling probability proportional to pupil enrollment. 23 government schools and 7 non-government schools were selected for each district. By design, about ¾ of the schools sampled are government owned and the rest non-government owned. Non-government owned schools include community schools, private-owned and schools owned/managed by non-governmental organizations (chiefly religious organizations). The appendix shows the characteristics of schools in selected districts and sampled schools relative to non-sampled schools in the country and selected districts respectively.

Owing to time constraints, only 160 out of the sampled 180 schools could be visited before the end of the third term. Of the 160 schools that were visited 127 were government schools and 33 were non-government schools. Table 1 below presents means of selected characteristics. The average school in the sample had an average of 11.6 teachers and a pupil-teacher ratio of 48.4. Only 5% of the schools in the sample were multi-grade schools, where more than one grade is taught in the same classroom. 20% of the schools were located in peri-urban or urban areas, with nearly two thirds in peri-urban areas. One third of schools is located within 5 kilometers of a main road and nearly two-thirds are located within 5 km of a taxi stage.

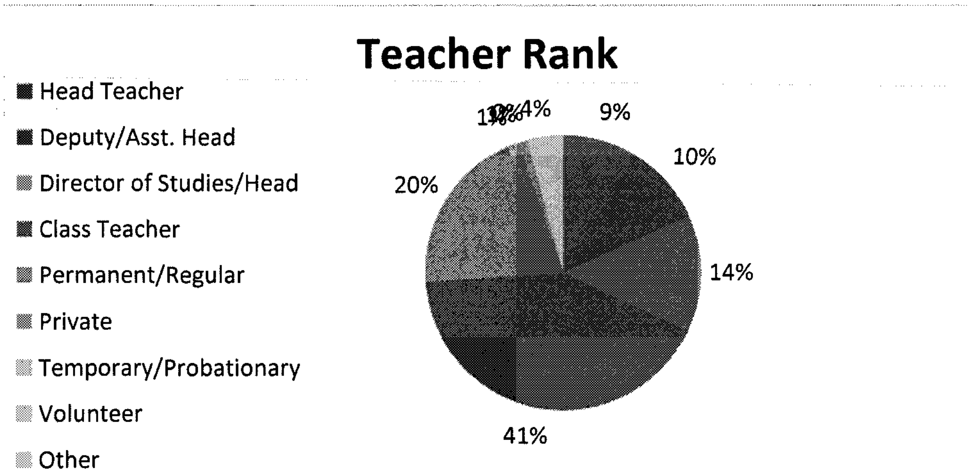
Table 1. School characteristics, selected means by ownership

	Non-government	Government	Total
Number of teachers	10.64	11.90	11.63
	(0.98)	(0.42)	(0.39)
Pupil teacher ratio	38.42	51.09	48.44
	(3.29)	(1.23)	(1.25)
Proportion of mother's literate	0.72	0.60	0.63
	(0.07)	(0.04)	(0.03)
Multi-grade school	0.06	0.05	0.05
	(0.04)	(0.02)	(0.02)
Proportion of schools within 5km of main road	0.39	0.32	0.34
	(0.09)	(0.04)	(0.04)
Proportion of schools within 5km of taxi stand	0.79	0.62	0.65
	(0.07)	(0.04)	(0.04)
Proportion Urban	0.36	0.15	0.20
	(0.09)	(0.03)	(0.03)
Number of observations	127	33	160

Since the primary focus of this chapter is on teachers, it is worth pointing out a number of important teacher characteristics. A total of 1837 teachers were surveyed as part of this exercise. This includes all teachers in schools with less than 20 teachers and a maximum of 20 for schools that had more than 20 teachers. A little more than 10% of schools had 20 or more teachers.

As figure 1 shows, 60% of all teachers are class teachers/permanent teachers; 14% are heads of department and 20% are head-teachers or deputy head teachers. The rest are volunteer, private or other part-time teachers.

Figure 1. Teacher composition



The demographic structure of our sample of teachers is presented in table 2 below. 58% of teachers are male. The average age in the sample was 35.6 and they had been teaching in the school for an average of 4.4 years. Only about a quarter of all teachers had completed A-levels. More than ¾ of teachers are married and fluent in the language spoken in the area around the school. More than half the teachers are teaching in their district of birth. This is in part an outcome of autonomy in hiring by the districts and the tendency for districts boards to hire 'locals'. 63% of all teachers live in the same parish as the school.

In the sections that follow, we restrict our analysis to full time teachers. This involves dropping about 1.7% of observations that correspond to part-time, private, volunteer teachers or teachers that have been transferred but are still on the school roster.

Table 2. Teacher characteristics, selected means

	Mean (Standard Error)
Gender (1=Male)	0.58 (0.01)
Age, years	35.60 (0.32)
Tenure of current posting	4.40 (0.12)
Proportion that have completed A-levels	0.24 (0.01)
Proportion teaching in district of birth	0.54 (0.01)
Proportion that live in the same parish as school	0.63 (0.01)
Proportion fluent in language spoken around school	0.77 (0.01)
Proportion married	0.76 (0.01)
Number of Observations	1843

Annex 4 of Chapter 3. Assumptions used to quantify the amount of waste

We quantified the implied annual cost to the sub-sector making four assumptions.

1. That **the findings are nationally representative**. For instance, we assume that our survey results from 3 regions of Uganda can be extrapolated into a national average absenteeism rate.

2. **That the problems identified are cumulative.** Although some of the problems identified are potentially overlapping problems (e.g. an absent teacher could be a ghost), we assume they are cumulative. That is, with teacher absenteeism of 19 percent and ghost teachers at 4 percent, we assume that, on average, on a given day 23 percent of the teachers who receive a monthly salary from the government are absent or a ghost.

3. **The most recent reports on problems are reasonable indicators of the present magnitude**. We assume recent reports documenting a problem (e.g. ghost teachers) give a reasonable indicator of the current magnitude of the problem. For instance, although the Auditor General's report refers to 2004/05 figures, we assume that the findings – that 5.3 percent of MoES' expenditure are questionable – is a good estimate of questionable expenditure for 2005/06 (until the next report comes out).

4. **All questionable expenditures are "wasted"**: we assume they represent government expenditures which were not translated into the tangible inputs which were their intended use, and that is why they are being questioned by the Auditor General.

The first and second assumptions are reasonable. If sampling was carefully done when surveys of teacher absenteeism or UPE capitation grant leakages were done, the first assumption is non-controversial. Similarly, the survey instrument used to measure absenteeism should be measuring only absent teachers and not ghosts. When enumerators arrive at schools, the head master provides them with a teacher roster. It is this teacher roster which is the basis for identifying who is present and who is not. The only way teacher absenteeism could include ghosts is if this roster includes names of teachers who do not exist. We do not think this is the case because it would suggest that both the head master and the other teachers (who will be asked questions about the missing teacher) were all accomplishes – with no incentives for them to do so – in covering up the ghost.

The third and fourth assumptions require more justification. Assuming that all questionable expenditure is wasted may lead to an overestimation of the governance problem. However, we impose this assumption for three reasons: first, categorizing the severity of each item identified is inevitably an *ad hoc* and subjective judgment call that we wanted to avoid. Second, as explained, the items listed in the published report are the items which could not be reconciled after lengthy discussions between MoES and the auditor. Third, an auditor relies on sampling, and the items listed in ...... are items uncovered in this sub-sample of total expenditure. For instance, the example of "non teaching staff on the payroll of a closed down Community Polytechnics Instructors College" was discovered by sampling some of the Community Colleges. Presumably, if the auditor had the resources to check all facilities and receipts, more questionable (but never fewer) would have been recovered.

In the third assumption, we are assuming that simply because problems get identified, documented and even responded to in terms of new policies, they do not disappear. Specifically, the payroll cleanup exercise in 2006 showed that 4 percent of primary teachers were ghosts. In effect, it cleaned up the payroll and removed those ghosts. Still, given that it is the latest report documenting the size of the problem, we will assume that the problem of ghosts remains at 4 percent until the next payroll exercise is conducted which – hopefully – will show that the problem has been reduced. The reason for this assumption is the long history of governance problems in the sector which shows that problems quickly re-emerge. An alternative assumption would be to assume that once a problem has been documented and a particular policy aimed at reducing the problem has been implemented, the problem has disappeared or been reduced. The problem with this assumption, we find, is that policies are often implemented without a clear sense of how much of an impact they will have.

Annex 5 of Chapter 3. Addressing the problem of conducting efficiency analysis in an environment with large private contributions

Unit costs based only on government spending could be misleading. What if household contributions offset changes in public funding? Suppose household contributions were $10 per enrolled student in 2000/01 but only $5 per student in 2005/06. Total spending per enrolled student would have fallen from $33 ($10 plus $23, as shown in ……) to $32 ($5 plus $27). If, total spending had been falling, this could explain why outputs have not increased further.

We find *no evidence* from household data that additional public spending has offset declining private contributions. To investigate we first used household survey data to estimate how much households spent on schooling in 2002/03 and 2005/06[120]. A detailed analysis of the UNHS data is included in Annex 1. The analysis in the annex shows that household expenditure per student enrolled has in fact increased in recent years. It has *not* declined. Rather, public schools' resources per child have clearly increased in recent years by more than outputs.

[120] The two UNHS years for which household expenditure data on education (where the child went to public school) are available.

Annex 6 of Chapter 3. Scope

This PER focuses narrowly on internal efficiency and we cover only the primary education sub-sector. An analysis of internal efficiency should ideally focus on a particular sub-sector because the analyst needs to assess the costs per unit of a broadly homogenous output or outcome. We attempt to answer the questions: (a) can outputs be increased without raising the current level of resources or funding? (b) Can expenditures be reduced without adversely affecting the current level of educational output? **We narrowed in on the primary sub-sector because**:

1. Spending on primary education consumes more than 60 percent of the education budget (or around 400 bn in 2005/06). Thus, even a 5 percent saving would be worth 20 billion Ush. Savings of similar sizes in absolute terms would be relatively more difficult to find elsewhere in the education budget.
2. The ongoing efficiency study in the MoES is a "sector-wide" study which will report broader efficiency problems from the entire sector.
3. Unit cost data are available for primary schools - we undertook a survey of 160 primary schools, which provided cost information that is unavailable for the other sub-sectors.
4. This provides a pilot technique for use in other sectors. Efficiency studies should not be seen as one-off reports; rather, they need to be part of regular reviews done every year, or every other year. Thus, even more important than the specific findings reported in this chapter are the methodology, indicators, and questions being asked. The entire sector may find the analytical framework useful to undertake similar analysis for the other sub-sectors.

We chose to focus on recurrent expenditures only – that is, the recurring costs of educating a child. To measure the efficiency of Government's capital expenditure on primary education – where the output is for instance a classroom - a different set of indicators is needed. In a country with already high enrollment and low population growth, the education system would be stable in terms of student numbers. Then recurrent and capital expenditure can safely be lumped together when conducting efficiency analysis because, on average, capital expenditure would most likely be a small amount, and exhibit little annual variation. In Uganda's case, where the school system is rapidly expanding, capital expenditure accounts for 10-15 percent of total expenditure. It is therefore important to distinguish the two: recurrent expenditures are the expenditures needed to *run* the existing system; capital expenditures are investments in physical infrastructure to *expand* the system.

Annex 7 of Chapter 3. Efficiency indicators needed to measure efficiency of capital spending

The indicators used to measure the efficiency of capital expenditure are similar to indicators used to measure the efficiency of most types of new construction and are shown in Table 3.1). As can be seen, compiling the indicators involves close cooperation between architects and engineers (to estimate whether constructions meet specified requirements) and economists and auditors to analyze the numbers.

More work is needed to examine the efficiency of classroom construction and maintenance in Uganda. In most forecasts of classroom needs, Uganda will need to increase its amount of classrooms by 35,000 over the next ten years (in addition to its current stock of 85,000). Put differently, around 30-40 percent of the classrooms standing ten years from now will have been built between now and then (with the range depending on how many of the existing classrooms will need to be retired in this period). However, since efficiency analysis of capital and recurrent spending is distinctly different, this PER focuses only on recurrent spending. The sector working group should develop an efficiency study of classroom construction.

Table1. Main indicators to measure efficiency of government capital spending on primary education

Output	Measure/indicator
Contract signed to construct classroom	Cost per accepted contract bid price of classroom
Classroom constructed	Total capital expenditure per constructed classroom
Classroom constructed and verified to have fulfilled requirements in contract	Total capital expenditure per constructed classroom verified to have fulfilled requirement of contract 1/
Classroom standing in good condition for desired number of years	Total capital expenditure plus sum of annual maintenance expenditure per classroom over number of years examined

Source: Authors' framework to measure efficiency of capital spending

1/ If classrooms is not fulfilling requirements, an estimate of what it would cost to fulfill the short-fall in quality should be added to the cost

Chapter 4 Annex. Tools used to model fiscal sustainability

As part of this PER, we developed a tool to help policy makers analyze the sustainability of new or planned policy initiative. The tool consists of six excel spreadsheets which are interlinked. A diagrammatic overview of toolkit is provided in Annex A.

There are two cornerstones in this modeling framework: an augmented version of the World Bank's long-term forecasting tool (the RMSM-X model) used to ensure macroeconomic consistency, and a detailed excel file in which expenditure can be forecast using a range of different assumptions. In our business-as-usual scenario, the assumptions are as described above.

The innovate elements of our model is two-fold:

- first, we have augmented the RMSM-X model in such a way that the composition of expenditure (modeled in a separate file) is linked to overall growth whereas, in the traditional RMSM-X, growth is set exogenously and independently of how the government spends its resources,
- second, the model rigorously links fiscal data on "development spending on infrastructure" and "maintenance spending" to the possible growth impact they might have. Specifically, in the model, we recognize that not all fiscal spending on capital becomes part of the capital stock. Specifically, we assume that only a portion of spending is actual capital spending, the rest gets reclassified as government consumption. There are two ways to interpret this "rescaling" of spending: a) partly, as shown in section 2, part of what is classified as development expenditure is, in fact, consumption; b) spending on public investments may result in shabby roads or be spent on inappropriate projects. Moreover, we recognize that it takes time for capital spending to become part of the useable (and from a growth perspective, relevant) capital stock (i.e. simply initiating work on a road does not mean that the road will help spur growth). Instead, we assume that spending this year only shows up in the capital stock two years later. Finally, we allow maintenance expenditure to offset depreciation, thereby, indirectly helping to reach a desired size of the capital stock. In this PER, we will not delve further different strategies of reaching a particular capital stock. However, we believe we have allowed for enough flexibility in the current model to allow for such analysis in future PERs.

The key macro economic assumptions of our modeling approach are as follows:

- We assume that the entrepreneurial spirit which the private sector has shown in the past decade will largely continue, despite infrastructure weaknesses. We reflect this by assuming an underlying growth in private investment growth at its trend average (of around 6 percent in real terms).
- We also assume that private investment growth around its trend can either be improved upon (or deteriorate) based on how successful the government is in expanding the public capital stock of infrastructure.
- Export volume growth of non-commodities is based on the past performance of real private investment growth: again, we assume that product discovery will continue at a healthy pace but that, say, double-digit growth rates observed will not be realized unless private investment takes off first. Import volume growth is determined by elasticities linking import volumes to the real exchange rate, import prices, investment and growth.
- Export and import prices are determined exogenously and assume – conservatively – that Uganda will face a slight terms-of-trade deterioration over the forecast horizon (i.e. import prices are assumed to grow slightly faster than export prices). Other prices and

real interest rates are determined exogenously and do not have any equilibrium impacts in the model. The nominal exchange rate is assumed to adjust in such a way that the real exchange rate (determined by the nominal exchange rate and differences in domestic and foreign prices, both of which are exogenous) is kept constant.

- Economic growth is determined from a growth accounting framework where we assume a) a Cobb-Douglas production function with constant returns to scale; b) determine human capital accumulation as a function of exogenously determined population growth and "years of schooling" linked to our forecast of the number of graduates from primary and secondary education (from the education model described in more details in chapter XX); c) physical capital is given by total investment growth (where public and private investment are determined as above) minus assumed 7 percent depreciation[121]; d) TFP growth is exogenously given by its trend growth rate (over the past 5 years).
- From a national account's "expenditure" perspective, then, private consumption is the only variable which remains to be determined and, as is commonly done in the RMSM-X, we let it be determined residually (as $C = Y - I - G - X + M$)

Most of the macro assumptions above are standard when using the RMSM-X in projection exercises. However, two assumptions are non-standard:

- the link from the composition of public spending to private investment, and
- exogenously determining GDP growth from a growth accounting framework where TFP is the exogenous variable. In the process, we decoupled the RMSM's standard tight relationship between investment and GDP growth (via an exogenously assumed "Incremental Capital-Output Ratio" (ICOR)).

The motivation for imposing these new linkages comes from the analysis undertaken as part of the CEM. In the CEM, lack of adequate infrastructure was identified as the binding constraint to growth. Using this evidence, we ask the question: is it likely that private investment will reach double-digit growth rates with no electricity, and with an inadequate roads network? Similarly, is it likely that export growth takes off without faster private investment growth?

The extent to which the link between public infrastructure and private investment will materialize will depend, of course, on whether the selected public infrastructure projects have been thoroughly vetted and aimed at reducing identified bottlenecks for the private sector.

We have constructed our model sufficiently flexible to allow policy makers to test the sensitivity of our results to different "elasticities" between the chosen composition of public spending and private investment growth. Similarly, it is possible to "turn off" this link completely and examine the results using a standard "ICOR" assumption linking total investment to GDP growth.

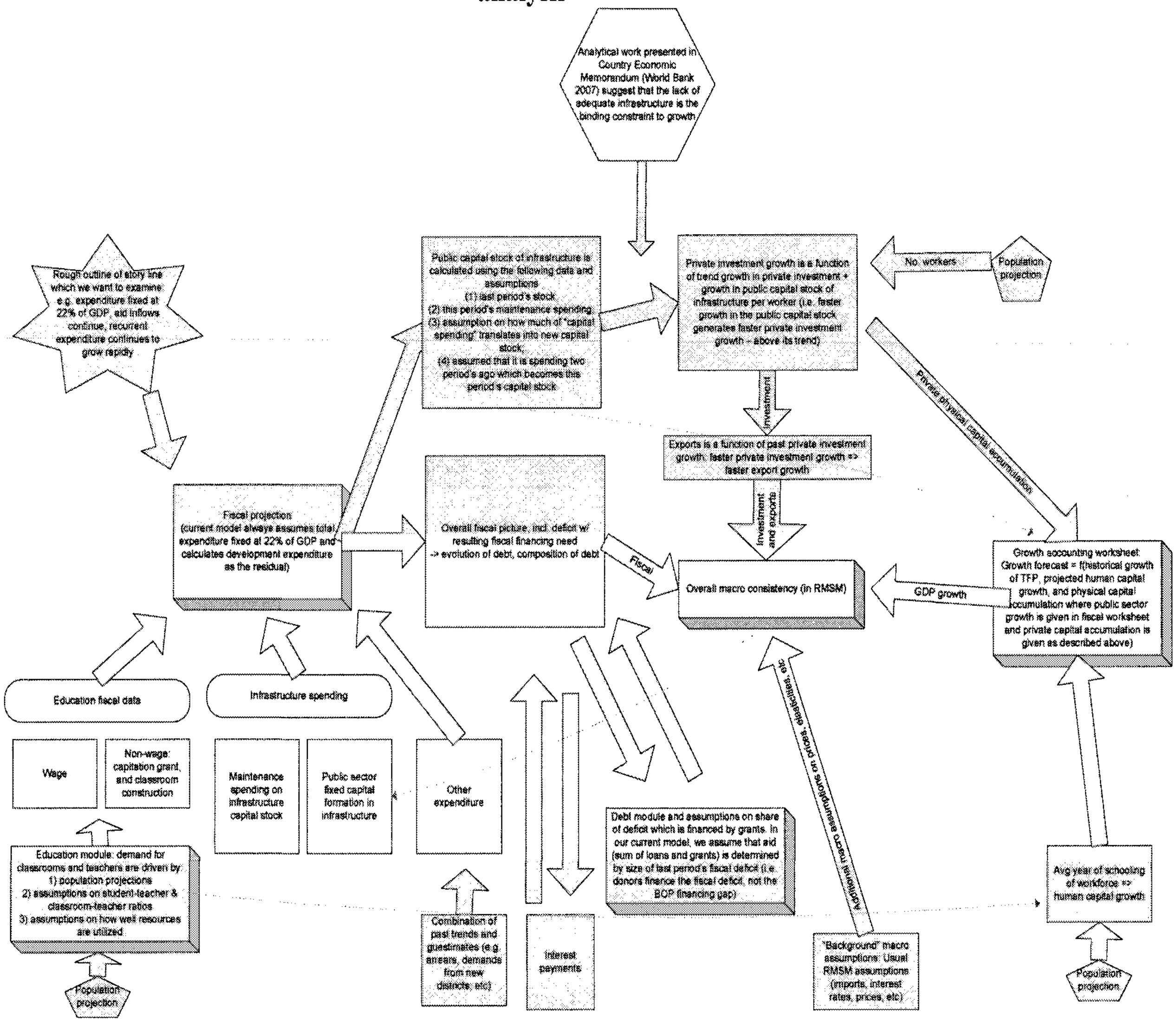
The model is also sufficiently flexible to incorporate different determinants of private investment. For instance, analytical work five to ten years in the future might find that, by then, the binding constraint is the lack of skilled labor. If so, the model can easily be modified to reflect this constraint. Annex B shows a schematic representation of such hypothetical modification. In the

[121] At seven percent, the assumed depreciation may seem on the high side. The reason for the slightly higher depreciation is that we are departing from the usual equation which determines the evolution of the capital stock by including maintenance expenditure. That is, normally, only spending on large renovations (which expand the capital stock) should be included in this equation. We include maintenance (i.e. small repairs) under the assumption that this probably helps reduce the effective rate of depreciation. To offset this inclusion, we raise the rate of depreciation.

diagram, the model is modified to link private investment to the average number of years of schooling in the work force.

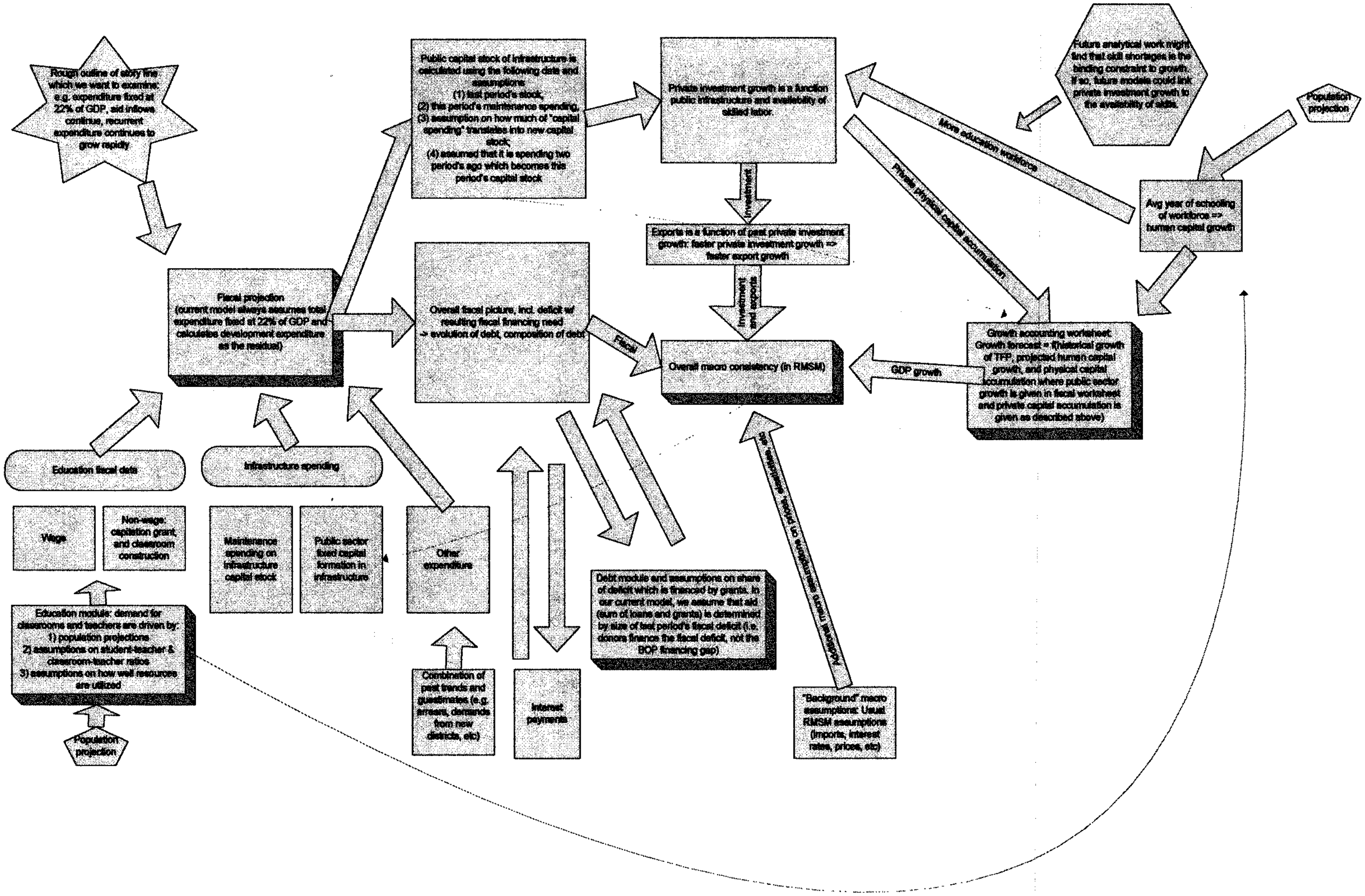
It is important to note that just as the standard RMSM-X model is not a general equilibrium model (where, for instance, wages would rise in response to scarcity of labor and result in firms changing their capital-labor ratio, or domestic interest rates would rise in response to rapidly rising public domestic debt) our modifications do not change this basic limitation of the RMSM-X. At the end of the day, our model remains a consistency framework that conveniently allows policy makers to generate forecasts by exogenously imposing key assumptions. The model does not report whether the forecasts are likely or plausible; users have to make this assessment themselves. It is easy to generate completely implausible forecasts: for instance, nothing prevents the user from eliminating all recurrent expenditure (e.g. by firing all civil servants) and force the government to only build roads. Doing so will generate impressive growth rates in the "model", and result in rapid private investment growth. Similarly, nothing prevents the user from generating very large current account deficits that would require donors to triple their annual aid. Thus, similar to the RMSM results, the user will have to look critically at the results generated, and conduct careful sensitive analysis.

Annex 1 of Chapter 4. Schematic overview of linkages in model used for fiscal sustainability analysis



Source: World Bank staff

Annex 2 of Chapter 4. Example of how future analytical work would result in private investment growth being determined by different factors



Source: World Bank staff

Annex 3 of Chapter 4: Stress Tests and Sensitivity analysis[122]

The baseline scenario showed that with increasing revenues, expenditure set at a ceiling of 22.5 percent of GDP and bulk of debt financed at concessional interest rates, public debt would rise but remain within sustainable levels, at. 36.4 percent of GDP by 2026/27. This annex presents a range of stress tests and sensitivity analysis from a fiscal sustainability point of view. With the exception of the variables under shock, these simulations take the same assumptions as the baseline (see Annex Box 1 below).

Box 1. Key Macroeconomic and Fiscal Assumptions for the baseline scenario

Real GDP growth – is driven by the composition of fiscal expenditures and hence endogenously determined. In the baseline, this turns out to be about 6.0 percent for the first 10 years before gradually declining to 5.2 percent by 2026/27.

Fiscal policy – based on fiscal consolidation strategy adopted by government.

- Domestic revenues are closely linked to the macro evolution, but also assume some benefits from increased effort – income tax to increase from 3.6 percent of GDP to 5.4 percent by 2026/27; Trade revenue, to maintain the same share of 25 percent of imports.
- Total expenditure is fixed at 22 percent of GDP, reflecting an average of 8 years and the government's commitment to fiscal consolidation. The deficit before grants is assumed to decline from more than 8 percent of GDP in 2005/06 to 5 percent by 2026/27.
- With grants tapering off to about 3 percent of GDP over the forecast period, non-interest expenditures (primary expenditures are projected stay around 20 -21 percent of GDP. The resultant primary deficit path envisages a gradual decline from 9 percent in 2004/05 to about 5 percent by 2026/27.

Nominal interest rate on domestic public debt are assumed to stabilize at about 11 percent(about 6 percent in real terms), slightly higher than the average of 8.7 percent over the past 10 years, while that on foreign debt is expected to decline to about 0.6 percent, from the average of 0.8 percent over the 10 years to 2005.

Inflation: Domestic inflation which has remained above the target of below 5 percent for 3 consecutive years is expected to remained above 5.0 percent until 2008 on account of the effect of the energy crisis, before declining to the target of 5 percent for the rest of the forecast period. **Foreign inflation** based on trading partners inflation, is assumed to gradually decline from 5 percent to 3.5 percent by the end of the forecast period.

Nominal Shilling Exchange rate is assumed to change in line with changes in inflation differentials between Uganda and its foreign trading partners, for the real exchange rate to remain constant.

The current account deficit (excluding interest payments) in terms of GDP gradually rises from 2.8 percent in 2004/05 to about 6 percent of GDP over the forecast horizon

While several other alternative scenarios[123] under the standard fiscal sustainability framework are available, here we report on 5 major shocks (interest rate, terms of trade, grants reduction, contingent

[122] Prepared by Rachel Sebudde

[123] Alternative scenarios can include:

Historical Average Scenario: This scenario presents an alternative evolution of the debt ratio under the assumption that all key variables (real growth rate, real interest rates, inflation, real exchange rate, primary balance to GDP ratio, seignorage revenue to GDP ratio are constant and equal to their historical averages over the projection period. The template calculates averages over a ten-year period, and uses that information to project debt dynamics five years ahead.

liabilities and fiscal shocks) under what would be referred to as country specific shocks. The assumptions underlying these are summarized in Box 2 below.

1. Interest Rate Shock

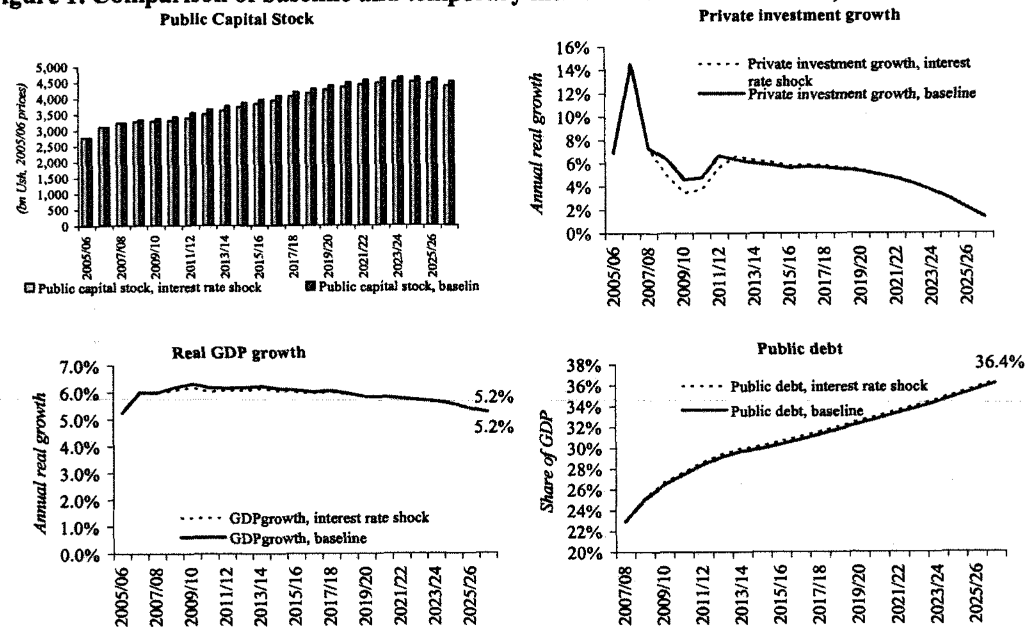
The baseline scenario assumes that real domestic interest rates would remain stable at around 6 percent over the forecast horizon (11 percent in nominal terms). This assumed that the sterilization choices would carefully balance the use of domestic securities and sales of foreign exchange and that private investor appetite for government securities remains unchanged. This may not be the case especially in the event that the foreign exchange market would not absorb the forex sales[124]. In this analysis, we shocked the system with various increases in domestic interest rate of both short term nature (one time increases) and over a long term nature over the forecast horizon.

A temporary real interest rate shock of 10 percentage points on the real interest temporarily reduces private sector investment on account of the crowding out arising from more attractive government securities and higher cost of investment. This trend is however reversed after the shock and both investment and growth return to baseline values. Public debt slightly increases as interest cost and the primary deficit rises.

Country-Specific Alternative Scenario: This scenario is user-defined and models the impact of country-specific shocks that the country is considered most vulnerable to in the medium term. Shocks could consist of fiscal slippages; a slowdown in the growth rate, higher borrowing costs, terms of trade shocks, sudden stops in the availability of external financing, or realization of contingent liabilities. Country teams are at liberty to design this scenario based on the particulars of the country.
No Policy Change Scenarios: This scenario assumes that current policies (as of the base year, defined as the last year for which actual data exist) regarding revenue and expenditure are maintained for the projection period. As negotiations with authorities entail indications to raise revenue and cut expenditure, it would be instructive to compare the baseline projections thus obtained against a no-change scenario. This alternative scenario assumes that the primary balance to GDP ratio remains unchanged with respect to the last year of actual data. Fiscal revenue and primary expenditure remain constant in percent of GDP and interest expenditure in national currency assumes the same nominal interest rate on domestic and foreign debt as in the baseline. All other macroeconomic data are assumed unchanged with respect to baseline.

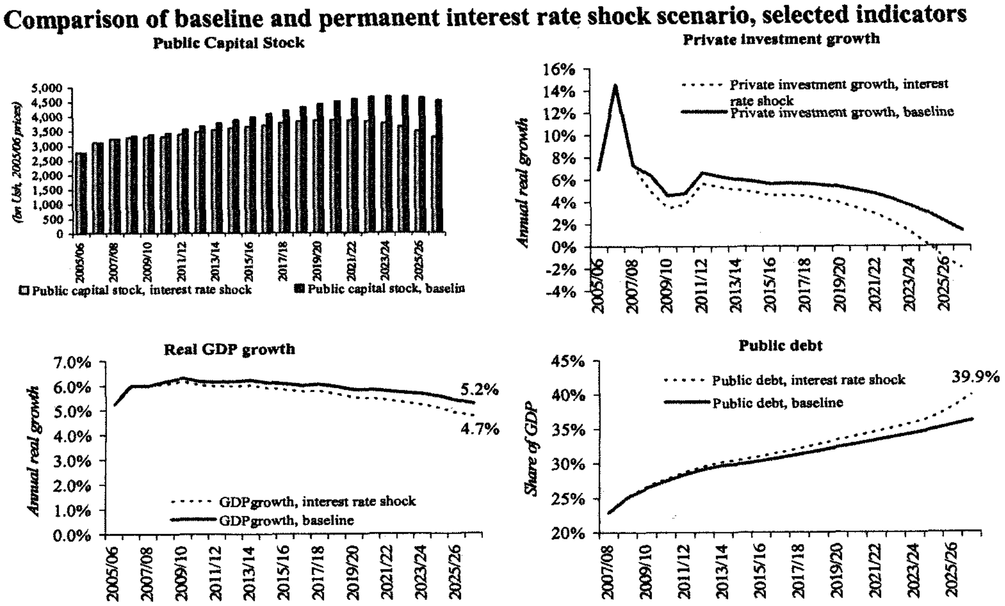
[124] In situation of strong appreciative pressures in the market, monetary authorities have in the past resisted effective sales forex exchange as per program to avoid destabilizing the market and appreciating the currency beyond reasonable levels.

Figure 1. Comparison of baseline and temporary interest rate shock scenario, selected indicators



On the other hand, a permanent rise in the real interest rates by the same magnitude but running through the forecast period has a more severe impact on the economy – private investment is completely crowded out and records a negative growth rate by the end of the forecast period. Government would also not be able to maintain the capital stock on account of increased cost of investment. With an increasing primary deficit, due to increasing interest costs, lower growth, public sector debt explodes as would be expected.

Figure 2. Comparison of baseline and permanent interest rate shock scenario, selected indicators



Box 2. Key assumptions under Specific Country Shocks to the Baseline Scenario

1. **Interest rate on public debt shock:** This shock is assumed to take be both temporary and permanent. Under the temporary shock, we assumed that interest rates on domestic securities increase by 10 percent in year 1, 2 and 3 of the projection as was the case in the early 2000s. The underlying assumption is that there would be a temporary change in the mix of sterilization instruments towards domestic instruments, say on account of appreciative pressures in the foreign exchange market. For the permanent rise, the assumption would be that agents change their expectations and hence raise the expected return which eventually raises interest rates, especially if the central back, has to continue sterilization operations. The likelihood of having a permanent raise in interest rates is however much smaller than the temporary shocks to the system.

2. **Terms of trade shock**: Assumed to be through a decline in export prices by 20 percent per annum for 5 consecutive years. It is also assumed that exports have a price elasticity of 0.2 percent as per historical average. Such a scenario has been observed in the past and likely to be repeated.

3. **Grants reduction**: Assume a reduction in grants to 25 percent of total budget. This scenario is consistent with Uganda's new position after MDRI. Uganda being in the "green light" window implies that it cannot access grants from the World Bank, which is the biggest source of external financing. This implies that actually in reality, the grants financing of the budget will be declining.

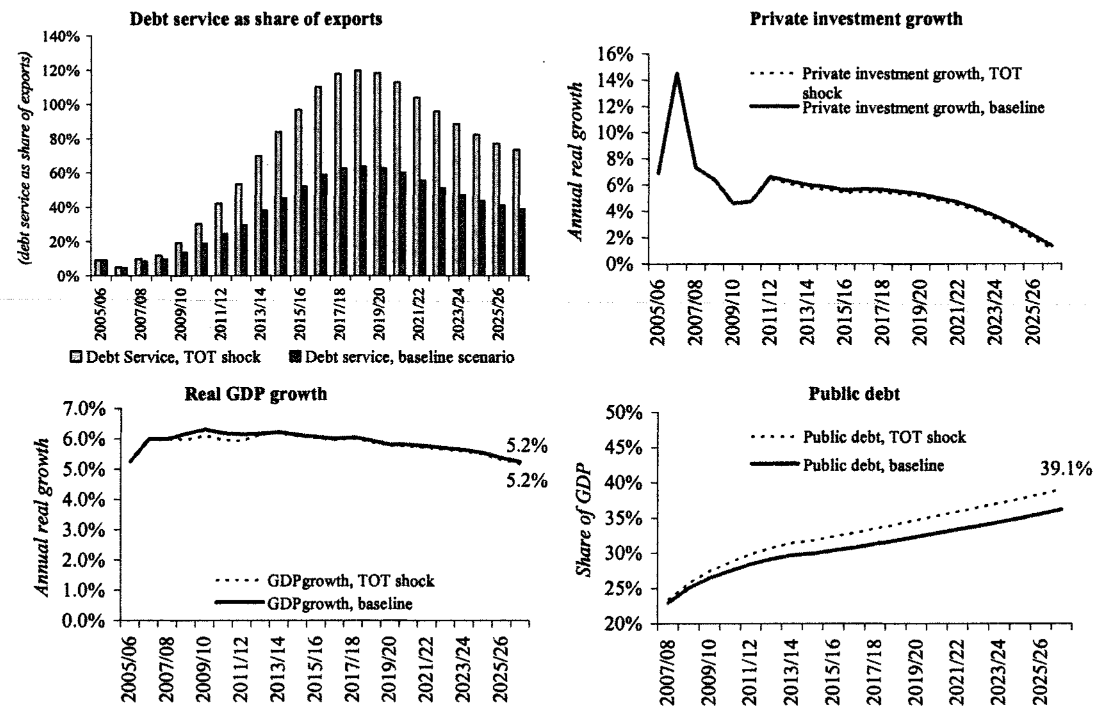
4. **Contingent liabilities shock:** Assume that contingent liabilities double to 5 percent of GDP for 3 years starting 2007 thereby increasing the government transfers to the private sector and reducing the resources available for infrastructure financing. Given the increasing incidence of private-public partnerships, and court awards, there is a high possibility this scenario could materialize.

5. **Fiscal Shock**: Assume that government relaxes the ceiling on expenditures to allow further infrastructure spending. The underlying assumption is that additional expenditure on infrastructure would not jeopardize macroeconomic stability as it increases productivity and the absorptive capacity of the economy. Therefore spending on infrastructure under this scenario rises to 4 percent, compared to 2.5 percent of the baseline scenario. A possible scenario for financing accelerated infrastructure investment

2. Terms of Trade Shock

Another shock of particular relevancy to Uganda is the terms of trade (tot) movements. In the macro stock-take conducted under the CEM, terms of trade shocks were confirmed to be major sources of short-term changes in growth, through its effect on total factor productivity – a one percent shock of ToT would raise the contribution of TFP to growth by up to 0.2 percent over the 1990s. The decline in growth over the period 1998/99-2004/5 was also accounted for by the declining trend of ToT which amounted to over 40 percent between 1998/99 and 2004/05. This was on account of both a decline in export unit prices and an increase in the import unit value largely driven by the oil prices. While in the past Uganda has been able to weather tot shocks arising from oil price increases quite well, this has not been the case with those arising from declining commodity prices, the bulk of Uganda's exports. Even with the on-going diversification, such shocks are bound to affect Uganda's economic performance. Assuming that export have a price elasticity of 0.2, in this scenario, we show that if export prices declined by 20 percent per annum for 5 consecutive years starting 2007, the largest impact on the economy comes through a reduction in export values during the years of the shock. This results into a permanent reduction in the contribution of exports to GDP, and consequently worsens the debt service to exports ratio to a peak of 120 percent, before it declines to 73.4 percent by 2027. With a reduction in GDP growth over the shock period, on account of lower export performance, public debt would increase slightly and hence remain within manageable levels. (see Figure 7).

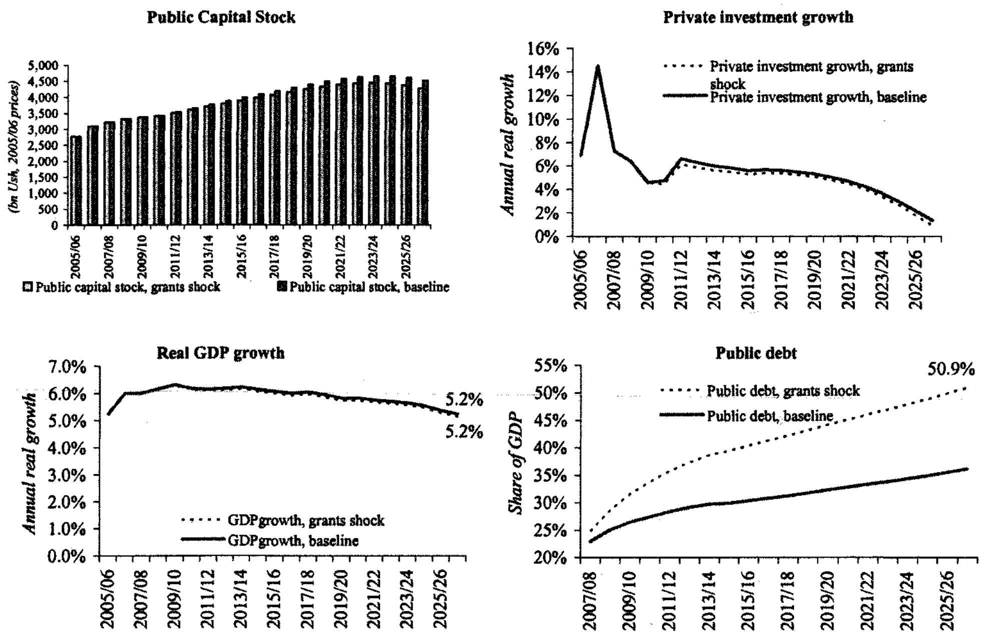
Figure 3: Comparison of baseline and terms of trade shock through export commodity prices, selected indicators



3. Grants Reduction shock

Over the past 2 decades, over 60 percent of the GOU's fiscal deficit was financed by external resources, mainly grants. This scenario tested the system with a reduction in grants from 50 percent of the deficit in the baseline to 25 percent. By having fewer grants, the government has to take on more debt- this explodes to 50.6 percent of GDP, compared to the baseline of 36 percent. Thus, automatically, we see debt increasing as well as interest payments from 1.6 percent of GDP to 1.8 percent of GDP.

Figure 4: Comparison of baseline and Grants reduction shock scenario, selected indicators



4. Contingent liabilities

Because of their implicit nature, little attention in reporting or planning is normally given to contingent liabilities. Yet, achieving and monitoring fiscal sustainability requires monitoring the entire range of risks[125] from contingent to covariant type risks. The first step to controlling and managing contingent fiscal risks requires for government to have an open stance regarding the type of risks it faces, as well as the volume and eventual costs these commitments will represent, and an estimation of the probability that such contingent obligations will be triggered.

In Uganda, the Auditor General's report identifies a number of contingent liabilities recognized during fiscal years 2003 to 2005. The data shows that the biggest source of such liabilities was legal fees. The total amount of these liabilities is no more than 2.6 percent of GDP (see Table below), but as big as the overall deficit large (i.e. after grants).

125 It would be useful to think about such al along the following lines of categorization:

(i) Explicit liabilities, whose repayments are established by law or contract;

(ii) Implicit liabilities , which involve a moral or expected obligation on part of the government, not mandated by law, but by public expectation, political pressure, etc;

(iii) Direct liabilities which occur for certain and thus are predictable, based on determined factors

(iv) Contingent liabilities, which are obligations activated by an event that may occur or not occur;

Table 1: Contingent Liabilities recognized in the Auditor Generals Reports 2002/03-2004/05

	2003	2004	2005
Legal fees	10,000,000	239,377,146,040	342,040,612,081
Guarantees & indemnities	29,944,000	-	26,700,000
Guarantees of bank overdrafts	29,944,000	-	26,700,000
Other contingent liabilities	1,457,192,052	107,000,000,000	4,674,780,860
Total in Shs	1,497,136,052	346,377,146,040	346,742,092,941
Total % of GDP	0.0%	2.6%	2.3%

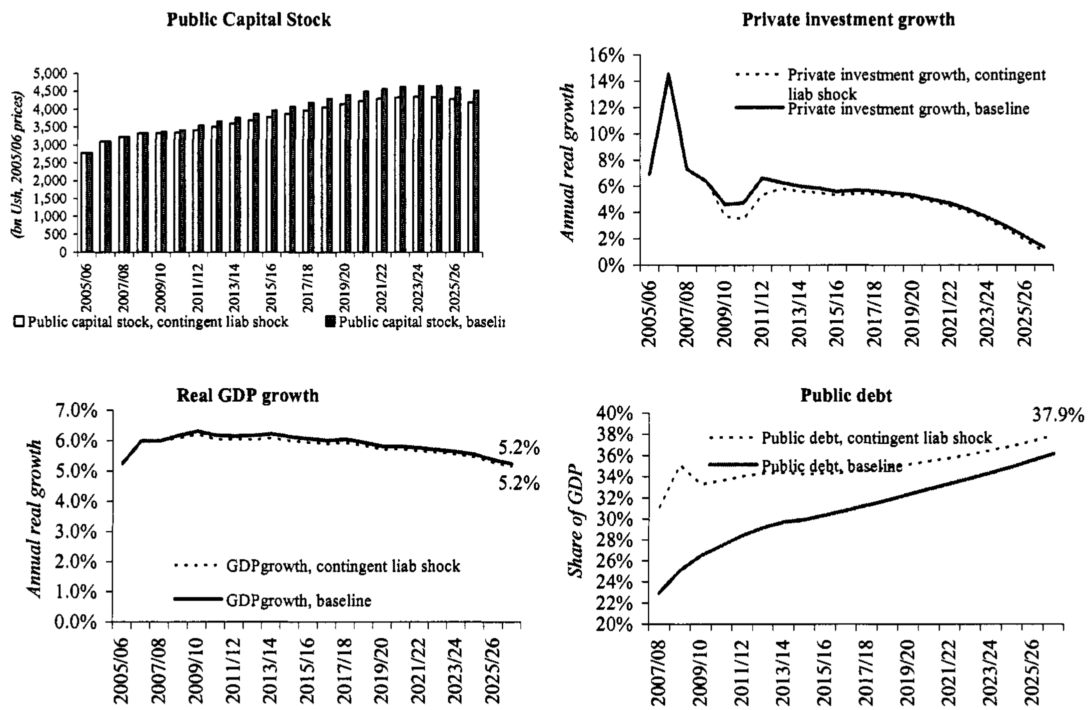
Yet, data availed to us for the FSA work shows that there is no comprehensive framework for properly monitoring these risks. Consequently, they are not fully always incorporated into the fiscal framework. There is, for instance, an increasing interest in use of private-public-partnerships to finance infrastructure projects, especially in the energy sector. Specific examples include Eskom, which was offered a revenue guarantee for the first 7 years of the concession, where the GOU bears the risk of changes in generation capacity. On the other hand, UMEME's concession includes a GOU investment of US $ 11 million financed through borrowing, a guarantee of US $ 2.5 million and defined events of GOU's defaults and a put option for the concessionaire after 18 months with a right to recover US $ 2.5 million and additional support through tariff subsidies.

In the pension sector, apart from arrears and trend of pension liabilities in the public pension sector, the semi-autonomous National Social Security Fund (NSSF) is funded with a mandatory contribution of 15 percent by formal sector workers' wage. Total contributions to the NSSF grew by almost 300 percent between 2000 and 2004 to about Shs 400 billion (or 3.0 percent of GDP) by end 2004. Unfortunately, the contributions have had a negative return and a significant portion of them are absorbed by operational costs and fraud (including valuation of assets issues). The key fiscal risk in this area include the fact that in spite being excessively liquid (due to bad management), any reform that would induce early withdrawals from the fund, may in fact spark off an inequitable outcome at the expense of those who remain in the fund or leave last. Furthermore, in the event that NSSF went under, it would be the responsibility of GOU to ensure that contributors to the funds are appropriately compensated[126].

In this contingent liabilities shock scenario, we show that in the event that any of such liabilities increased to 5% of GDP, for 3 years starting 2007, it would have severe effects on fiscal operations and the economy. We treat this as an additional expenditure item that raises the transfers to the private sector by 5 percentage points of GDP, and accordingly raises the primary deficit over the shock period. Even though government would have allowed expenditure to go beyond the cap of 22 percent of GDP, on account of increased interest cost as borrowing increases to finance a bigger deficit, the resultant re-allocation of resources would see an increase in recurrent expenditures to the detriment of public spending in the infrastructure intensive programs. This one-off shock also decelerates private investment during the years of the shock. Both the rise in the primary deficit and the increased interest payments increase the public debt to close at 37.9 percent, compared to the baseline of 36.4 percent.

[126] GOU may have no formal obligation to preserve and grow the capital of NSSF, but member reserves have been capitalized by NSSF and it would be unlikely for GOU, partial owner (appoints the Board) to default on these obligations, if the cause of any shortfalls lies in poor fund management.

Figure 5: Comparison of baseline and Contingent liabilities shock scenario, selected indicators



5. Fiscal shock

In the baseline, we assume the government remains determined to maintain total spending at 22 percent of GDP, even if this comes at a cost of seeing infrastructure spending dwindle. In this shock scenario, we assume that the government is willing to relax its spending limit to allow infrastructure spending to increase. In particular, we assume that infrastructure's share of GDP is the same as in the growth scenario discussed in the main text of the chapter (i.e. at 2.5 percent of GDP and rising to more than 4 percent by the end of forecast horizon).

This shock impact the evolution of public debt measured as a share of GDP in two opposite directions. On the one hand, the higher spending widens the fiscal deficit, pushing up overall debt. At the same time, however, the additional spending translates into faster private investment and overall GDP growth, raising the denominator in the debt to GDP calculation. As Figure 7 shows, the combined impact of these two factors is that public debt as a share of GDP would increase compared to the baseline case (i.e. faster growth is not enough to offset the impact of larger fiscal deficits).

Figure 6: Composition of spending in fiscal shock

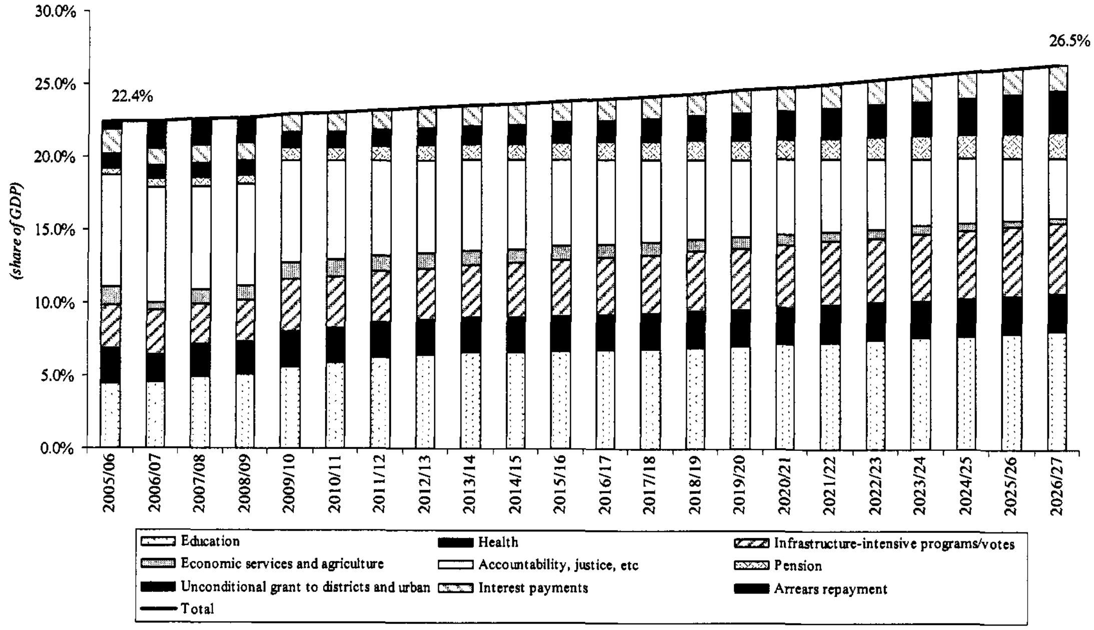
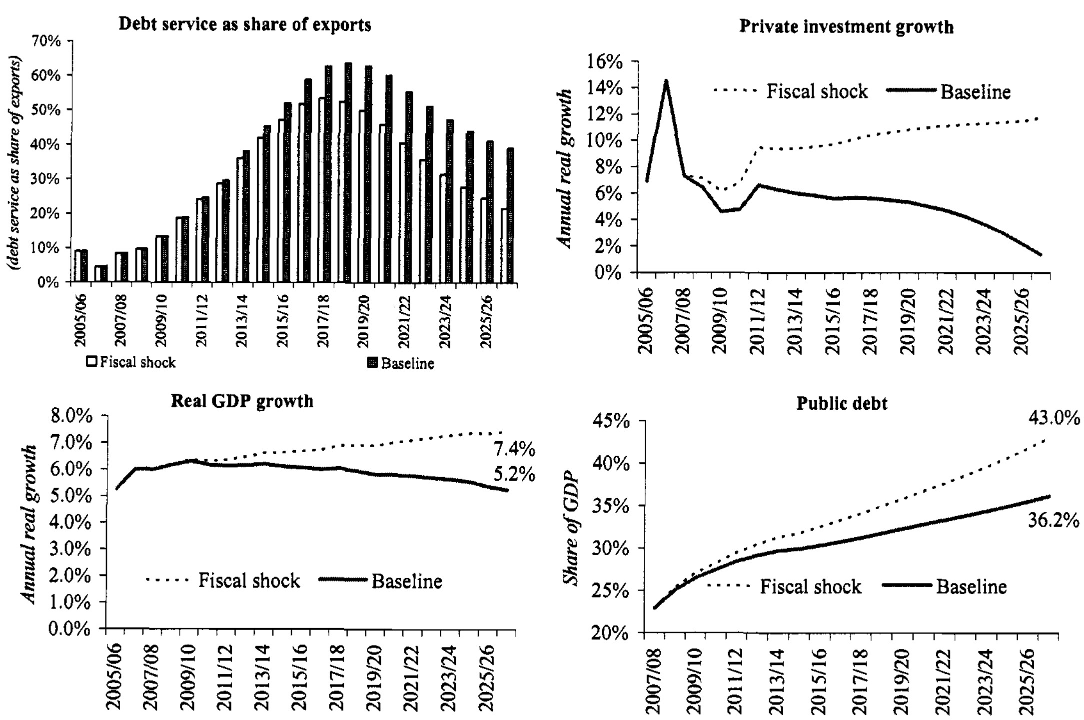


Figure 7: Comparison of baseline to fiscal shock scenario, selected indicators



Chapter 5 Annexes. Quasi Fiscal Deficit Methodology

Annex 1 of Chapter 5. Model -- Calculating Hidden Costs in the Energy and Water Sectors[127]

The model to calculate "hidden costs" in the infrastructure sector is described below. While there are more detailed and complex ways in which a model could be developed to reflect a loss in specific countries and sectors, this model has been formulated specifically to provide an insight into three key components of hidden costs: poor collections, tariffs set below cost-recovery levels and losses above normative levels. The intention in developing this model was to devise a simple-to-use methodology with which to monitor trends and to benchmark across sectors and countries without the need for intensive data collection efforts.

Let H be the "hidden costs" in the (electricity, gas or water) sector, defined as:
$H = R^* - R$ .... (1)

Where,

R^* is the expected revenue in a system operating with tariffs that cover costs, where bills are paid and where losses are within normal levels expected for a system of that age and design.

R is the actual revenue.

If,

Qs, is the volume (electricity, gas or water) supplied to the transmission network.
Qe, is the end-user consumption (of electricity, gas or water)
Lm, are the losses in transmission and distribution (of electricity, gas or water). Lm includes normative losses, Ln where Ln are those losses that are expected in a system of that design and age as well as losses due to system inefficiencies outside norms and theft.

Then,
$Qs = Qe + Lm$ .... (2)

And,
$R^* = (Qs - Ln)\ Tc$ .... (3)

Where Tc defined as the cost-recovery tariff, is the long run cost of operation and maintenance and includes a reasonable allowance for investment and normative losses.

And,
$R = (Qs - Lm)\ Te\ Rct$ .... (4)

Where Te is defined as the weighted average end-user tariff, and Rct is the rate of collection of billed amounts.

Substituting for R^* and R in (1), using (3) and (4), then
$H = (Qs - Ln)\ Tc - (Qs - Lm)\ Te\ Rct$

[127] Quoted from Ebinger, J. (2006), *Measuring Financial Performance in Infrastructure*, World Bank Working Paper

$H = (Qe + Lm – Ln) Tc – Qe\ Te\ Rct$

$H = Qe (Tc – Te) + (Lm – Ln) Tc - Qe\ Te (Rct - 1)$

$H = Qe (Tc – Te) + Tc (Lm – Ln) + Qe\ Te (1 – Rct)$ .... (5)

Total losses, Lm, can also be defined as lm Qs, or the rate of total losses multiplied by the volume (of electricity, gas or water) supplied to the system.

Normative losses, Ln, can in turn be defined as ln Qs, or the rate of normative losses multiplies by the volume (of electricity, gas or water) supplied to the system.

Therefore,
$Lm – Ln = (lm – ln) Qs$ .... (6)

Where Qs is defined in (2) as:
$Qs = Qe + Lm$

Substituting for Lm, we have,
$Qs = Qe + lm\ Qs$

$Qs (1 – lm) = Qe$

$Qs = Qe / (1 – lm)$ .... (7)

Substituting for Qs in (6) using (7) gives:
$Lm – Ln = (lm – ln) Qe / (1 – lm)$ ....(7)

Hidden costs defined in (5) can therefore be expressed as:
hidden costs would be

$H = R^* - R – T$

Where, T is the amount of capital or other transfer.

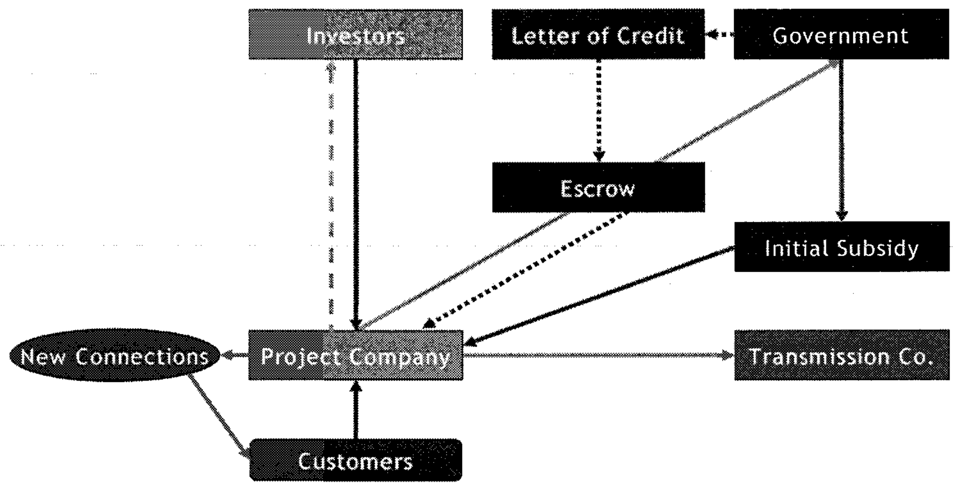
Annex 2 of Chapter 5: Financial risk modeling for PPPs – Case study UMEME

Before entering PPPs, governments need to carefully assess its costs and benefits. Balanced risk sharing with the private sector is key to limit ongoing fiscal costs for government. Financial risk modeling provides a useful tool to analyze the potential costs to government and provide guidance for negotiating PPP transactions. Indicative financial risk modeling has been undertaken using Uganda's electricity distribution concession as a case study. The model uses a cash flow simulation methodology based on Monte Carlo analysis. Simulations are based on 20 years of historical data on key macroeconomic variables including foreign exchange rates, inflation and interest rates, based on a normal probability distribution function.

In simulating the concession structure (see Figure 1), the main assumptions of the model include:

- Private concessionaire agrees to build a set number of new concessions which are paid by (1) subsidies, (2) operating income and (3) equity investments in this order.
- The concessionaire pays the bulk supply price to the transmission company, line rent to the government and its own operation costs. The rent is being paid into an escrow account. In the case of government breach of contract, the rent is reduced and the escrow account and a guaranteed Letter of Credit can be used to pay any shortfall to the concessionaire.
- Tariffs are derived from indexed costs (supply cost + staff costs + operating costs + line rent + min. return on investment) and divided by the power supplied and expected collection rate.
- The government is assumed to break the agreement if tariffs rise by more than 30% in real terms (proxy assumption to trigger an event of government failure in the model).
- Government can terminate the agreement if (i) the concessionaire break the law, (ii) the concessionaire goes bankrupt, (iii) investment in new connections falls below target. If government terminates, the concessionaire receives a lump sum equal to 80% of its investment.
- The concessionaire can terminate the agreement if (i) government interferes with the agreed tariffs, (ii) force majeure occurs. If the concessionaire terminates, it receives a lump sum equal to its investment plus 50% of the min. return on investment.

Figure 1: Concession Structure



Results of scenario analysis show that

- The concessionaire breaks even in the average case but there is a 44% probability of the concessionaire making a loss.
- The concessionaire has a strong incentive to improve collection rates, as company profits are highly correlated with collection rates.
- The costs to government are laid out in the Table below.
- In 5% of the cases, government can be surprised by a termination payout of more than $25 million in one year

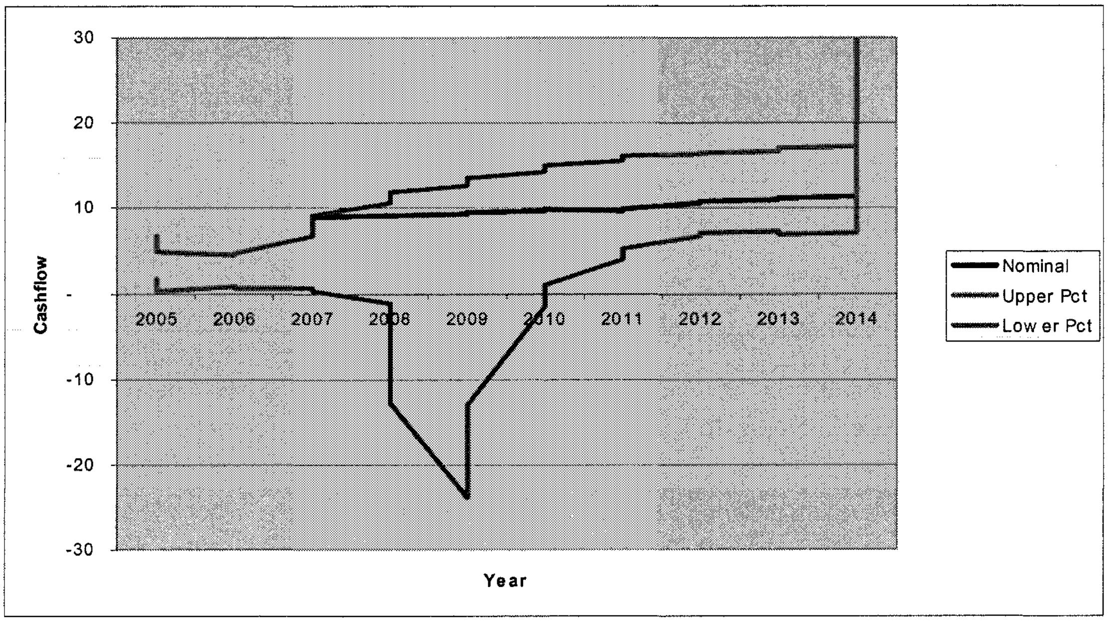
US$ million	Average	Max Added*	Total
Subsidy to rehabilitate existing lines	6		
Subsidy to help build new lines	5		
Cost of stabilizing tariffs	4	8	12
Cost of termination	8	13	21
Total	23	21	44

*5% maximum probable added cost

In conclusion, this case study illustrates the following in terms of risk sharing in the modeled PPP:

- The main project risks are due to changes in operational efficiency and are largely borne by the private operator
- Economic risk is borne by the consumer unless government intervenes
- Regulatory risk is borne by government
- Political risk is borne by government
- PPPs are not for free
  - In the worst case scenario (lower percentile), government would need to budget to absorb a potential payment of $25 million in one year. This happens if termination occurs after the concessionaire has made all its investments.
  - Consequently, government needs to closely track PPP performance.

Figure 2: Government cash flows



Annex 3 of Chapter 5. Road Sector Project Status

	Project Title	Approved	Began	Duration (Yrs)	Physical Progress	Expected Physical Progress to Date At Inception	Source of Finance	Initial value of project	Revised project cost	Budget Released - up to the start of the ongoing financial year	Planned exp. for the ongoing financial year	Actual expend. for the last financial year	Estimated budget to finalise the project - Balance to completion	Estimated Rate of Return at Time of Appraisal
	Name of project	*The year when the primary funding source was approved and project officially entered the government's active portfolio*	*The year when physical implementation began*	*Years from approval to compln.*	*Km complete if still being built*	*Km complete*		*Total value of the project when the project was approved (million US$)*	*Total value of the project if any project revision made (million US$)*	*The aggregated amount released to finance capital expenditures (million US$)*	*The budgted expend. for the present financial year (million US$)*	*The actual expenditures incurred in the last financial year (million US$)*	*The estimated budget required to complete the project after the close of the present financial year (miilion US$).*	*Internal Rate of Return %*
Project														
1	Upgrading to bitumen Nebbi - Arua road (54km)	1999	2001	2	54	54	IDA & GOU	24.3		20.62	0.00	0.00	0	25.7
2	Upgrading to bitumen Pakwach - Nebbi road (76km)	1999	2001	2	76	76	IDA & GOU	13.67		13.51	4.18	0.63	0.16	26.8
3	Upgrading to bitumen Karuma - Olwiyo road (46km)	2001	2003	2	46	46	IDA & GOU	17.4		11.42	4.95	4.33	5.98	14.6
4	Upgrading to bitumen Olwiyo - Pakwach road (54km)	2001	2004	3	37	54	IDA & GOU	14.6		5.79	4.95	3.46	8.81	16.8
5	Upgrading to bitumen Busunju - Kiboga road (69km)	1999	2001	2	47	69	IDA & GOU	20.06		12.33	7.04	2.69	7.73	26.8
6	Upgrading to bitumen Kiboga - Hoima road	1999	2001	2	53	77	IDA & GOU	25.96		11.79	7.04	2.14	14.17	16.1

	(77km)													
7	Rehabilitation Fort Portal - Hima road (55km)	2001	2004	3	33	55	IDA & GOU	13.4		4.92	5.04	2.93	8.48	12.5
8	Rehabilitation Hima - Kasese - Kikorongo - Kilembe road (53km)	2001	2004	3	42	53	IDA & GOU	14.2		7.15	5.04	4.40	7.05	21.3
9	Rehabilitation Kikorongo - Katunguru and Equator roads (54km)	2001	2004	3	54	54	IDA & GOU	15.5		7.25	5.04	3.28	8.25	14.9
10	Upgrading to bitumen Kagamba - Rukungiri road (35km)	2001	2003	2	35		EU & GOU	21.6		17.65	0	6.24	3.95	
11	Construction of Kampala Northern Bypass road (21km)	2003	2004	1	11	35	EU & GOU	57	0	24.17	13.72	7.47	32.83	
12	Rehabilitation of Jinja-Bugiri road (73km)	2001	2004	3	0	73	EU & GOU	53.64	62.4	11.49	0	0	42.15	
13	Upgrading to bitumen Kyotera - Mutukula road (45km)	2000	2001	1	45	45	ADB &GOU	13.6		11.38	0	0	2.22	
14	Upgrading to bitumen Gayaza - Kalagi road (21km)	2001	2002	1	21	21	ADB &GOU	3.5		3.47	0.52	0.57	0.03	
15	Tarmacing Kyegegwa - Kyenjojo road (54km)	2003	2003	0	54	54	GOU	8.84		9.8	0.44	0.35	-0.96	
16	Upgrading to bitumen Ntungamo - Kagamba road (15km)	2001	2001	0	15	15	GOU	6.1		6.1	0	0	0	
Source: Road Agency Formation Unit (RAFU)														

Chapter 6

Annex 1 of Chapter 6. Uganda Revenue Performance During 1988/89-2004/05 (Billions Uganda Shillings)

Year	1988/89	1989/90	1990/91	1991/92	1992/93	1993/94	1994/95	1995/96	1996/97	1997/98	1998/99	1999/00	2000/01	2001/02	2002/03	2003/04
Total revenue	44.60	89.57	133.79	180.46	282.60	373.35	506.99	611.70	735.92	812.29	958.28	1008.00	1110.15	1251.25	1438.42	1694.81
Income tax	4.78	9.47	13.87	23.64	43.57	50.78	64.50	67.68	88.25	107.89	148.43	158.79	200.12	259.49	319.94	402.21
-Payee	0.50	0.71	1.01	3.21	11.19	14.61	20.33	25.04	38.33	48.41	67.65	83.47	103.55	137.31	168.27	200.27
-Corporate Tax	4.28	8.76	12.86	20.43	31.25	24.79	24.49	24.08	25.65	29.25	44.31	40.89	54.27	69.41	84.22	121.58
-others	0.00	0.00	0.00	0.00	1.13	11.38	19.68	18.56	24.27	30.23	36.47	34.43	42.30	52.77	67.45	80.36
Excise duty	4.91	7.25	12.42	15.03	18.78	40.37	48.63	61.36	86.26	98.18	104.64	107.77	101.27	116.25	113.24	128.55
International trade	19.29	49.55	80.58	100.58	165.98	211.22	297.59	356.69	453.44	466.18	543.55	555.45	611.07	633.26	721.75	861.31
-Petroleum	3.11	15.90	36.14	54.50	82.52	92.83	119.66	150.76	197.46	188.27	193.21	197.20	199.30	222.17	240.68	269.81
-Import duty	4.68	9.18	13.44	21.54	38.68	52.56	65.14	75.86	72.27	78.05	96.48	105.09	141.01	117.22	133.07	133.95
-Excise duty	0.00	0.00	0.00	0.00	0.00	1.40	1.98	5.73	17.75	17.60	25.01	23.60	24.30	23.01	34.83	48.84
-VAT(Imports)	5.30	9.78	16.45	20.41	36.76	46.96	67.11	77.70	129.26	144.61	181.24	178.87	199.45	218.37	251.06	332.59
-other duties	6.21	14.69	14.55	4.13	8.02	17.47	43.70	46.64	36.70	37.64	47.61	50.69	47.01	52.48	62.11	76.13
VAT(Domestic)	13.18	20.51	23.16	28.25	47.08	61.92	83.86	110.25	87.83	116.32	140.02	164.26	174.75	213.43	244.41	240.93
Total Import (petroleum + import duty)	7.79	25.08	49.58	76.04	121.20	145.39	184.80	226.62	269.73	266.32	289.69	302.29	340.31	339.39	373.75	403.76
Total VAT (Domestic + Imports)	18.48	30.29	39.61	48.66	83.84	108.88	150.97	187.95	217.09	260.93	321.26	343.13	374.20	431.80	495.47	573.52
Fees and Licenses	1.41	2.8	3.76	3.65	6.45	8.7	12.41	15.73	20.14	23.73	21.65	21.73	22.94	28.84	39.08	61.8
Memo item: GDP at factor costs, current prices	895	1376	1830	2589	3625.94	4069	4922	5565	6048	6868	7381	8984	10033	10305	11884	13242

Tax Revenues as percentage of GDP

	1988/89	1989/90	1990/91	1991/92	1992/93	1993/94	1994/95	1995/96	1996/97	1997/98	1998/99	1999/00	2000/01	2001/02	2002/03	2003/04
Total revenue	5.0%	6.5%	7.3%	7.0%	7.8%	9.2%	10.3%	11.0%	12.2%	11.8%	13.0%	11.2%	11.1%	12.1%	12.1%	12.8%
Income tax	0.5%	0.7%	0.8%	0.9%	1.2%	1.2%	1.3%	1.2%	1.5%	1.6%	2.0%	1.8%	2.0%	2.5%	2.7%	3.0%
-Payee	0.1%	0.1%	0.1%	0.1%	0.3%	0.4%	0.4%	0.4%	0.6%	0.7%	0.9%	0.9%	1.0%	1.3%	1.4%	1.5%
-Corporate Tax	0.5%	0.6%	0.7%	0.8%	0.9%	0.6%	0.5%	0.4%	0.4%	0.4%	0.6%	0.5%	0.5%	0.7%	0.7%	0.9%
-others	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	0.4%	0.3%	0.4%	0.4%	0.5%	0.4%	0.4%	0.5%	0.6%	0.6%
Excise duty	0.5%	0.5%	0.7%	0.6%	0.5%	1.0%	1.0%	1.1%	1.4%	1.4%	1.4%	1.2%	1.0%	1.1%	1.0%	1.0%
International trade	2.2%	3.6%	4.4%	3.9%	4.6%	5.2%	6.0%	6.4%	7.5%	6.8%	7.4%	6.2%	6.1%	6.1%	6.1%	6.5%
-Petroleum	0.3%	1.2%	2.0%	2.1%	2.3%	2.3%	2.4%	2.7%	3.3%	2.7%	2.6%	2.2%	2.0%	2.2%	2.0%	2.0%
-Import duty	0.5%	0.7%	0.7%	0.8%	1.1%	1.3%	1.3%	1.4%	1.2%	1.1%	1.3%	1.2%	1.4%	1.1%	1.1%	1.0%
-Excise duty	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.3%	0.3%	0.3%	0.3%	0.2%	0.2%	0.3%	0.4%
-VAT(Imports)	0.6%	0.7%	0.9%	0.8%	1.0%	1.2%	1.4%	1.4%	2.1%	2.1%	2.5%	2.0%	2.0%	2.1%	2.1%	2.5%

Annex 1 of Chapter 6. Uganda Revenue Performance During 1988/89-2004/05 (Billions Uganda Shillings) (cont'd)

Year	1988/89	1989/90	1990/91	1991/92	1992/93	1993/94	1994/95	1995/96	1996/97	1997/98	1998/99	1999/00	2000/01	2001/02	2002/03	2003/04
-other duties	0.7%	1.1%	0.8%	0.2%	0.2%	0.4%	0.9%	0.8%	0.6%	0.5%	0.6%	0.6%	0.5%	0.5%	0.5%	0.6%
VAT(Domestic)	1.5%	1.5%	1.3%	1.1%	1.3%	1.5%	1.7%	2.0%	1.5%	1.7%	1.9%	1.8%	1.7%	2.1%	2.1%	1.8%
Total Import (petroleum + import duty)	0.9%	1.8%	2.7%	2.9%	3.3%	3.6%	3.8%	4.1%	4.5%	3.9%	3.9%	3.4%	3.4%	3.3%	3.1%	3.0%
Total VAT (Domestic + Imports)	2.1%	2.2%	2.2%	1.9%	2.3%	2.7%	3.1%	3.4%	3.6%	3.8%	4.4%	3.8%	3.7%	4.2%	4.2%	4.3%
Total Excise (Dom.+ imports)	0.5%	0.5%	0.7%	0.6%	0.5%	1.0%	1.0%	1.2%	1.7%	1.7%	1.8%	1.5%	1.3%	1.4%	1.2%	1.3%
Fees and Licenses	0.2%	0.2%	0.2%	0.1%	0.2%	0.2%	0.3%	0.3%	0.3%	0.3%	0.3%	0.2%	0.2%	0.3%	0.3%	0.5%
Tax revenue as percentage of total tax revenues																
	1988/89	1989/90	1990/91	1991/92	1992/93	1993/94	1994/95	1995/96	1996/97	1997/98	1998/99	1999/00	2000/01	2001/02	2002/03	2003/04
Income tax	10.7%	10.6%	10.4%	13.1%	15.4%	13.6%	12.7%	11.1%	12.0%	13.3%	15.5%	15.8%	18.0%	20.7%	22.2%	23.7%
-Payee	1.1%	0.8%	0.8%	1.8%	4.0%	3.9%	4.0%	4.1%	5.2%	6.0%	7.1%	8.3%	9.3%	11.0%	11.7%	11.8%
-Corporate Tax	9.6%	9.8%	9.6%	11.3%	11.1%	6.6%	4.8%	3.9%	3.5%	3.6%	4.6%	4.1%	4.9%	5.5%	5.9%	7.2%
-others	0.0%	0.0%	0.0%	0.0%	0.4%	3.0%	3.9%	3.0%	3.3%	3.7%	3.8%	3.4%	3.8%	4.2%	4.7%	4.7%
Excise duty	11.0%	8.1%	9.3%	8.3%	6.6%	10.8%	9.6%	10.0%	11.7%	12.1%	10.9%	10.7%	9.1%	9.3%	7.9%	7.6%
International trade	43.3%	55.3%	60.2%	55.7%	58.7%	56.6%	58.7%	58.3%	61.6%	57.4%	56.7%	55.1%	55.0%	50.6%	50.2%	50.8%
-Petroleum	7.0%	17.8%	27.0%	30.2%	29.2%	24.9%	23.6%	24.6%	26.8%	23.2%	20.2%	19.6%	18.0%	17.8%	16.7%	15.9%
-Import duty	10.5%	10.2%	10.0%	11.9%	13.7%	14.1%	12.8%	12.4%	9.8%	9.6%	10.1%	10.4%	12.7%	9.4%	9.3%	7.9%
-Excise duty	0.0%	0.0%	0.0%	0.0%	0.0%	0.4%	0.4%	0.9%	2.4%	2.2%	2.6%	2.3%	2.2%	1.8%	2.4%	2.9%
-VAT(Imports)	11.9%	10.9%	12.3%	11.3%	13.0%	12.6%	13.2%	12.7%	17.6%	17.8%	18.9%	17.7%	18.0%	17.5%	17.5%	19.6%
-other duties	13.9%	16.4%	10.9%	2.3%	2.8%	4.7%	8.6%	7.6%	5.0%	4.6%	5.0%	5.0%	4.2%	4.2%	4.3%	4.5%
VAT(Domestic)	29.6%	22.9%	17.3%	15.7%	16.7%	16.6%	16.5%	18.0%	11.9%	14.3%	14.6%	16.3%	15.7%	17.1%	17.0%	14.2%
Total petroleum + import duty	17.5%	28.0%	37.1%	42.1%	42.9%	38.9%	36.5%	37.0%	36.7%	32.8%	30.2%	30.0%	30.7%	27.1%	26.0%	23.8%
Total VAT (Domestic + Imports)	41.4%	33.8%	29.6%	27.0%	29.7%	29.2%	29.8%	30.7%	29.5%	32.1%	33.5%	34.0%	33.7%	34.5%	34.4%	33.8%
Total Excise (Dom.+ Imports)	11.0%	8.1%	9.3%	8.3%	6.6%	11.2%	10.0%	11.0%	14.1%	14.3%	13.5%	13.0%	11.3%	11.1%	10.3%	10.5%
Fees and Licenses	3.2%	3.1%	2.8%	2.0%	2.3%	2.3%	2.4%	2.6%	2.7%	2.9%	2.3%	2.2%	2.1%	2.3%	2.7%	3.6%

* Source: Ayoki et al. (2005).

Annex 2 of Chapter 6: Assessment of Uganda's major tax regimes

Corporate income tax (CIT). Annex 4 summarizes the key features of the corporate income tax regimes in Uganda, Kenya, and Tanzania. The statutory tax base and rates in Uganda are comparable to the ones in the other two countries.

- ***Base.*** The CIT base is determined as follows: Taxable base = gross sales turnover minus costs of goods sold, fiscal depreciation, interest payment, and other eligible deductions. Turnover for small businesses is set at a threshold of Ushs 50 million, which is the same as the one established for the VAT. Businesses below the threshold are subject to a presumptive income tax.[128]
- ***Rate.*** A single standard rate of 30 percent is applicable to both resident and non-resident companies. The treatment of mining companies is exceptional: firms are subject to a rate of 25 percent or 45 percent, depending on the volume of taxable income.
- ***Depreciation.*** The fiscal depreciation does not deviate from the norms of the international practice. It consists of a mixture of declining balance and straight line methods.
- ***Thin capitalization rule.*** Thin capitalization rule is set even more stringent than the one in Kenya (2:1 debt-equity ratio in Uganda, compared with 3:1 in Kenya).
- ***Loss carried forward.*** Losses are allowed to carry forward to offset future profits for an indefinite period. By international standard, this provision is generous, but it is in line with the one in Kenya and Tanzania.
- ***Fiscal incentives.*** Fiscal incentives for investment are reflected in the Investment Code 1991 and are typically granted on the basis of geographical location, type of business, economic activities, and type of capital asset.
    - *Initial investment allowance.* In addition to some types of accelerated depreciation, generous initial investment allowance is available with various rates applied to the actual cost of initial investment. The incentives are discretionary. In many cases, initial investment allowance is not automatic-- approval for specific investment project in certain sectors (e.g., hotels and hospitals) is required on a case basis.

Tax holiday and tax aid. Tax holidays were formally abolished in 1997. This is an encouraging development as this type in fiscal incentives is generally considered as a blunt, costly, and ineffective instrument.[129] However recently, a distorted form of incentive, tax aid or tax subsidy, with similar net revenue effect as tax holiday has crept back into the system, and is available to selected firms or organizations. Basically, tax aid is a non-traditional form of tax expenditures. It specifically refers to the part of the tax liabilities that line ministries, subject to approval by the MFPED, assume from selected firms investing in priority or development projects, mostly hotel

[128] The rate structure of a presumptive taxation is rather complicated with 6 brackets applicable on presumptive gross turnover. Tax liability ranges from Ushs100,000 to Ushs 450,000 or 1percent of the gross presumptive income, whichever lower.
[129] In many cases, when tax sparing treatment does not exist, tax holiday simply leads to a resource transfer from the treasury of a host country to the treasury of the firm's home country with a worldwide income regime (e.g., UK, US). The net result would be that firms do not get any real benefits for their investment, while the host country incurs revenue loss.

development, higher level education institutions, and non-profit organizations. It is worth noting that unlike tax holidays, tax aid is not confined only to CIT, but is made available to cover other major types of indirect taxes such as VAT and trade taxes. While tax aid currently accounts for a small share of the total domestic collection (estimated at 0.7 percent in 2005/06), it is non-transparent, prone to corruption, and is likely to induce unfair competition, drive up transaction costs, and create additional uncertainty in the tax system.[130] Firms from particular sectors (e.g., hotel developers) that typically enjoy other types of preferable fiscal treatment (e.g., initial capital allowance) are the regular recipients of tax aid. From a regional perspective, the IMF has recently proposed that the Eastern Africa Community (EAC) countries establish a Code of Conduct to limit harmful tax competition.

Personal income tax (PIT) Annex 5 presents the structure of the personal income tax applicable to residents according to the Income Tax Act 1997. While the PIT regime is regionally comparable, its collection is relatively low (1.5 percent of GDP or approximately 12 percent of the total revenue collection in 2003/04).

- ***Rate***. There are four brackets with marginal rates of 0, 10, 20, and 30 percent. The highest marginal rate of the PIT is equivalent to the single statutory rate of the CIT (30 percent). This parity helps alleviate the potential problem of arbitrage of income source. The rates are comparable to Kenya's and elsewhere in Sub-Saharan Africa.[131] Like many other developing countries, Uganda does not apply an imputation income tax system; but instead, the tax law allows for a partial relief of double taxation by levying a lower rate of 15 percent on dividend (or half of the highest marginal PIT rate of 30 percent).

- ***Base and threshold***. Currently the PIT still follows the scheduler income tax regime contrary to the worldwide trend of moving toward simpler and more efficient global income taxation. In particular, zero-rate thresholds are different for wage income earners and non-wage income residential individuals: the level applicable to resident individual is at UShs1,560 thousand, almost three times the average annual GDP per capita and 12 times as high as the one for PAYE (Ushs130 thousand). The relative ease of collecting PIT from wage income earners may attribute to the establishment of the differentiated thresholds. However, such discrimination creates unfair burden on wage income earners, who are already typical 'captive' taxpayers. Such equity issue could potentially distort the choice of job and further encourage informality. For both equity and economic efficiency purpose, these thresholds should be adjusted toward uniformity.

Value Added Tax. VAT was enacted in 1996 to replace Sales Tax and Commercial Transaction Levy, and has undergone a number of amendments. The VAT regime follows the common international setting: it is consumption-typed, adheres to destination principle, and is based on credit invoice method for calculation of tax liability.

- ***Rates***. There is a single positive rate, 18 percent. In addition to exports, the VAT Act allows for a number of zero rates which include selected commodities in medical and agricultural sectors and are specified in the Third Schedule.

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[130] MFPED (2006) estimates that tax aid in the amount of approximately UShs19 billion (out of the total expected revenue collection of Ushs2,600 billion in fiscal year 05/06) is to be made to hotel developers, higher level education institutions, NGOs, and selected enterprises.

[131] For example, in Kenya, there are 5 brackets with no zero-rate threshold and with the highest marginal rate of 30 percent.

- ***Base and exemptions***. The base extends to retail level and covers both goods and services. However VAT is riddled with exemptions (listed in the Second Schedule of the VAT Act), which include selected agricultural, postal, financial, and medical sectors, selected supply to the Bank of Uganda, social welfare and education services. The supply of petroleum fuels, which is subject to a high excise, is also exempt. Businesses, especially transporters, using petroleum as inputs have to pay a cascading tax—first, import duty and then, excise at the pump--and are ineligible to claim for any refund. These costs which are ultimately borne by both suppliers (transporters) and consumers are economically inefficient, especially for Uganda, a land-locked country and largely rural with high demand for low cost transport. In addition, the exemption breaks the VAT chain and compromises its integrity and productivity.
- ***Exemption threshold***. The threshold is set at Ushs50 million (or more than USD27,000).
- ***Refund***: Excess input VAT is refundable within 1 months of the due date for the return in a particular tax period. Instead of refund, offsetting for tax liability in the following filing period is applied if the refund amount is less than Ushs5 million. However, our field interviews with the private sector indicate that honest taxpayers face the typical delay in VAT refund. The legal treatments of the VAT refund and tax arrears are not systematic: the amount of delayed refunds is not entitled to any interest, whereas tax areas are subject to an above market interest rate. The MFPED's Poverty Eradication Action Plan 2004/5-2007/8 (2004) also highlights business reports that refund has indeed become slower.
- ***VAT withholding***. Related to the issue of VAT refund is the on-going practice by which government organizations withhold the VAT payments of their purchases to the suppliers. The VAT code does not have a specific provision about VAT withholding, and thus the practice is not legally supported. In addition, government affiliates typically delay in remitting the VAT withheld to the URA and thus deny their suppliers the right to claim for credit for the VAT paid on their inputs. It is unclear about the net revenue impact of the VAT withholding practice. It is apparent, however, that the withholding creates serious cash flow problems to firms and generates a quasi-cascading issue.

Excise tax is applicable on selected commodities (such as alcohol, sugar, tobacco, matches, petroleum products, and cellular phone air time) manufactured domestically or imported. Uganda also applies a surcharge on some high income elasticity goods, such as motor vehicles. Most excisable items, except for petroleum fuels (petrol, diesel and paraffin), are also subject to VAT.

- While ***petroleum*** is exempt from the VAT, it is subject to excessively high excise: in particular, the excise rate is significantly higher than in Kenya and more than twice of that in Tanzania. A combination of VAT exemption and high excise is not an optimal policy choice. It causes a dual problem: potential cascading and administration complication due to VAT exempt status; and high incentives for smuggling and avoidance due to high excise statutory rate. On average, the excise on petroleum products has increased by 7 percent recently.

- ***Telecommunications***. The rate on mobile phones rose from 10 to 12 percent. A proposal to levy 5 percent excise on landline phones to raise revenues and to address the equity issues (use of mobile v. landline phone) is being considered.

Trade taxes. There are only three tariff bands: 0, 10, and 15 percent. The ad valorem duty is applied to the CIF values. Most goods are subject to either a rate of 7 or 15 percent. Rates from COMESA countries are reduced to 0, 4 and 6 percent, respectively. Certain beverages (e.g., mineral water, beer, whisky, and wine) are also subject to an import surtax. The import duty on petroleum accounts for the biggest share of trade tax revenues—in 2003/04 it amounts to approximately 16 percent of the total taxes, while the customs duty on other commodities accounts for just half of it (annex 1).

Assessment of tax administration

Overall progress in tax administration is impressive. As part of the comprehensive reform program, the URA was established in 1992 as a semiautonomous agency and is responsible for administering both domestic taxes and Customs. The URA has made substantial progress in organizational restructuring, human resource (HR) management and development, integrity, and simplifying processes. The private sectors' perception of the URA's effectiveness and integrity has improved, even though corruption remains a great challenge. A new tax procedural code is being updated and new reform strategic measures are being integrated in reform implementation plan.

The structure of a modern revenue administration has been framed. The organization is currently structured along functional line with five major departments: (1) domestic taxes; (2) customs; (3) corporate services; (4) legal; (5) internal audit. Some elements of taxpayer segmentation in the new organizational structure are also introduced with the establishment of a dedicated large taxpayer office (LTO) and a number of pilot medium taxpayer offices. At present, the LTO administers more than 500 companies with the collection of 68 percent of the total revenues. The new organizational structure reflects the trend for a modern revenue administration agency. It potentially allows for reducing duplication of efforts in administering different specific types of taxes, improving accountability, and encouraging specialization and concurrently cross-functional coordination.

However, administration remains weak with extremely low compliance and rising administrative costs. Compliance is extremely low. The general level of compliance in domestic taxes stands at just 38 percent. The filing rate is 28 percent for CIT, 52 percent for VAT, and 62 percent for PAYE. In customs, smuggling, diversion of goods, poor evaluation and classification remain key issues. Compliance in customs administration becomes a greater challenge due to further simplification of cross-border trade processes as required upon the establishment of the EAC customs union at the time when the URA is not yet ready. On the other hand, administrative cost has been rising in the last three years: it has increased from approximately 3.1 percent of the total revenue collection in 2003 to 3.36 and 3.42 percent in 2004 and 2005 respectively.

Multiple weaknesses and challenges in revenue administration exist. The draft URA Corporate Plan 2006-10 candidly presents key weaknesses and challenges, including inadequate staff development program; poor information exchange and coordination across departments; low levels of integrity among some individual staff; lack of infrastructure and undeveloped record management systems; lack of a strong taxpayer service function; inadequate IT function and infrastructure. In addition, poor tax culture; rigid legal framework; political interference with tax policy and administration; and corrupt external environment; and high illiteracy are cited as the key institutional challenges for the operation of the URA. In the new URA structure, a specialized taxpayer service department is still missing. The URA still lacks basic analytical capacity. Performance standard is still largely based on revenue target. In registration, no integrated strategy or action plan to identify new potential taxpayers has been prepared. Auditing and collection remain ineffective. Particularly, national audit plan has been prepared but not yet functional or tested, and third party information exchange (e.g., with banks) is not automatic. On the other hand, refund process is still delayed. MFPED (2004) highlights that duty drawback which is supposed to be resolved within 7 days may take months due to manual administration. Businesses complain that even though URA has made serious efforts to approve for VAT refund within the stipulated period of one month, actual release of funds has not improved but become slower.[132]

The URA is embarking on the next stage of revenue administration reforms. The draft URA Corporate Plan establishes a target of reaching a revenue-GDP ratio of 16 percent by the year 2011. We share the view of major donors that the target is very ambitious but not unattainable, if proper administration reforms are undertaken in tandem with an overhaul of main tax policies. The URA has shown its strong commitment to a reform strategy that has been prepared with inputs from the IMF and DFID. The URA reform plan is comprehensive, covering both domestic taxes and customs. The broad reform measures include:

- *Institutional strengthening.* The reforms are targeted to enhance governance of revenue administration, rationalize the organizational structure, improve HR management and development, and encourage stakeholder participation. An integrated good governance and anti-corruption strategy is being formulated on the basis of simplified procedures; new performance standards; disseminating of a code of conduct; and establishment of an internal audit function. The capacity building for the URA headquarters will improve its operational policy making and performance monitoring. The functioning of the LTO will be reviewed and strengthened. The reform measures are based on a new, effective, and transparent policy framework.
- *Tax and customs business process reengineering.* Process integration, and simplification will be developed to establish an effective mechanism to register new taxpayers, collect taxes, duties, and arrears with special focus on enhancing audit capacity and taxpayer service.
- *IT development.* The reform will develop and implement an integrated IT system with the key elements: (1) Full deployment of ASYCUDA++ and rollout to key sites; (2) development of Integrated Tax Administration System (ITAS) to support key functions of tax administration; and (3) building a robust data network linking the ASYCUDA++ and ITAS sites.

[132] Our field interview indicates that the issue of VAT refund creates a major tension between the administration and private sector. However it can only be resolved if both refund procedures are improved and there is a better coordination between the URA and MFPED. The refund process involves multiple stages, including auditing and approval (by URA), and payment release (partly a responsibility of the MFPED). Some efforts have been made to alleviate the delay in refund release when URA is allowed to set aside an annual budget for refunds of all taxes. However, the budget may fall short of actual refunds, as currently refund budget is prepared without any reliable estimation of refund needs.

Annex 3 of Chapter 6. Revenue collection structure in selected SSA countries (2000-04)

		Direct Income Taxes					Indirect Taxes				
Countries (2000-2004)	Total Tax Revenue	Total Income Taxes	CIT	PIT	Wealth and Property Taxes	Others	Total Indirect Taxes	General Sales Taxes, VAT	Excises	Trade Taxes	Others
Ghana	19.3	6.0	2.9	2.1	n/a	1.0	12.9	1.5	3.3	8.0	4.1
Kenya	17.8	6.1	n/a	3.1	n/a	3.0	11.7	5.2	4.0	2.4	0.1
Madagascar	9.9	1.6	0.9	0.7	0.1	0.2	7.9	2.0	0.5	5.0	0.7
Namibia	28.4	11.7	5.0	7.0	0.2	n/a	16.3	6.1	0.1	9.6	0.4
Senegal	16.1	3.8	1.4	1.8	0.4	0.5	11.7	n/a	n/a	5.8	n/a
Uganda	10.8	2.2	2.1	0.1	n/a	n/a	8.6	3.5	1.2	3.6	0.3
Averages	17.1	5.2	2.5	2.5	0.2	1.2	11.0	3.9	1.6	5.2	1.0
Sub-Saharan Africa	18.22	3.3	2.3	1.1	0.2	1.2	9.8	3.7	1.4	4.4	0.7
Low-income AFR	14.57	3.5	1.4	1.7	0.1	0.6	10.6	2.5	1.3	6.0	0.7
Low-income World	12.85	3.4	1.8	1.3	0.1	0.3	5.7	3.3	1.6	3.7	0.5

Source: IMF database; and author's estimates.

Annex 3 of Chapter . Revenue collection and fiscal architecture in selected SSA countries (2003)

Countries	Revenue/ GDP (%)	Tax/GDP (%)	GDP per capita (Constant USD 2002)	Imports and exports share in GDP (%)	Agriculture (% of GDP)	Population growth (%)	Working Age Population (%)	Literacy rate 2004 (%)	Corruption index	Shadow economy (% GDP)
Ghana	20.8	20.2	269	92.5	35.8	2.2	56.6	73.6	2.5	38.4
Kenya	18.2	17.2	418	52.2	27.8	2.1	54.0	73.6	3.5	34.3
Namibia	28.1	25.9	1,943	106.4	11.5	1.3	54.0	85.0	1.5	NA
Senegal	18.0	17.2	445	69.9	17.6	2.4	53.5	39.3	2.5	43,2
Tanzania	n/a	11.4	300	45.6	45.0	2.0	53.6	69.4	2.0	58.3
Uganda	12.1	11.9	257	39.0	32.4	3.4	47.1	66.8	2.0	43.1
Zambia	19.1	18.4	327	48.5	22.8	1.7	50.9	68.0	2.0	48.9
Average	**19.4**	**17.5**	**566**	**64.9**	**27.5**	**2.2**	**52.8**	**68.0**	**2.3**	**44.6**
Sub-Saharan Africa	27.1	21.6	914	76.4	26.7	2.2	52.5	59.6	2.0	
Low-income, AFR	28.1	22.4	279	68.3	32.2	2.5	51.8	50.4	2.0	
Low-income, World	18.8	13.2	336	72.8	31.1	2.2	59.3	57.8	1.9	

Source: WDI 2006; IMF database; Schneider, Friedrich and Klinglmair, Robert (2004): Shadow Economies around the World: What Do We Know? 1043, IZA, Bonn; and author's estimates.

Annex 4 of Chapter 6. Key Elements of EAC Corporate Income Taxes

	Uganda	Kenya	Tanzania
Corporate tax rate	30 percent (lower rates for small businesses)	30 percent (37.5 percent for foreign branches)	30 percent
Withholding tax on dividends paid to nonresidents	15 percent; 15 percent branch profit tax on repatriated income	10 percent	10 percent—applies to deemed dividends (i.e., remittances of profits) by foreign branches
Withholding tax on interest paid to nonresidents	15 percent	15 percent	15 percent
Withholding tax on Management fees paid to nonresidents	15 percent	20 percent	10 percent
Loss carryover	Indefinite	Indefinite	Indefinite
Limits on interest expense	2:1 related party debt equity ratio (excess interest not deductible)	3:1 debt equity ratio (excess interest not deductible)	None
Investment or initial allowance/deduction	50 percent initial allowance for most plant and equipment; 20 percent initial allowance for industrial building. Allowances reduce cost to be recovered by depreciation.	100 percent investment deduction (70 percent in 2003) for buildings and new machinery used for manufacturing or of hotels. Deduction reduces the cost to be recovered by depreciation.	20 percent investment allowance for industrial buildings and machinery, hotels, and telecommunication machinery and towers. Allowance does not reduce cost to be recovered by depreciation.
Depreciation	30 percent declining balance depreciation for machinery, 45 percent for computers, 20 percent for office furniture; 5 percent straight line depreciation for most buildings.	12.5 percent declining balance depreciation for most machinery and equipment (30 percent for computers, 25 percent for vehicles); 2.5 percent straight line depreciation for most buildings (4 percent for hotels)	50 percent the first year for machinery; 12.5 percent declining balance for second and subsequent years for most machinery and equipment (37.5 percent for heavy vehicles and 25 percent for other vehicles); 5 percent straight line depreciation for buildings; full expensing for mining industry with 15 percent uplift each year for unrecovered costs.
Tax holidays	None since 1997; companies enjoying holidays prior to 1997 were grandfathered	Exemption from income tax and withholding tax on payments to nonresidents for first 10 years in EPZs; 25 percent corporate rate for the next 10 years	Exemption from income tax and withholding tax on payments to nonresidents for 10 years in EPZs; 25 percent corporate rate thereafter; various individual exemptions (e.g., expatriates, corporate income, withholding taxes) are provided for in Government Notices

Source: IMF.

Annex 5 of Chapter 6. Uganda: structure of personal income tax

(i) Resident Individual

Total Income (U-Shs)	Tax (Shs)
0-1,560,000.00	0
1,560,000.00 - 2,820,000.00	10% of excess
2,820,001.00 - 4,920,000.00	126,000.00 + 20% of excess
4,920,001.00 and over	546,000 + 30% of excess

(ii) Pay as You Earn (PAYE)

Total monthly employment	Monthly PAYE Tax (Shs) Income (U-Shs)
0 - 130,000.00	0
130,001 - 235,000.00	10% excess
235,001.00 - 410,000.00	10,500 + 20% of excess
410,001 and over	45,500 + 30% of excess

* Source: Income Tax Act (1997).

Annex 6 of Chapter 6. Revenue projection (2005/06-2015/16)—low elasticity case scenario

Unit: Ush millions

Year	2004/05	2005/06	2006/07	2007/08	2008/09	2009/10	2010/11	2011/12	2012/13	2013/14	2014/15	2015/16
Direct income taxes	513,046	578,854	660,064	738,728	827,074	922,828	1,032,008	1,154,694	1,292,182	1,459,628	1,647,482	1,857,663
VAT (domestic + imports)	622,546	714,435	822,866	923,448	1,036,545	1,160,554	1,299,600	1,455,526	1,630,405	1,826,564	2,046,621	2,293,519
Excise duty (domestic + imports)	502,472	576,637	664,154	745,337	836,620	936,711	1,048,938	1,174,790	1,315,938	1,474,262	1,651,876	1,851,153
Trade taxes	233,585	268,062	308,747	346,486	388,921	435,450	487,622	546,126	611,742	685,343	767,911	860,549
Total taxes	1,871,649	2,137,989	2,455,831	2,753,999	3,089,160	3,455,544	3,868,168	4,331,136	4,850,267	5,445,798	6,113,890	6,862,884
Fees	63,228	72,225	82,963	93,035	104,358	116,735	130,674	146,314	163,851	183,970	206,539	231,842
Total revenue (taxes and fees)	1,934,877	2,210,214	2,538,794	2,847,034	3,193,518	3,572,279	3,998,842	4,477,450	5,014,119	5,629,767	6,320,429	7,094,726
Tax refunds	-47,395	-54,140	-62,189	-69,739	-78,226	-87,504	-97,953	-109,677	-122,822	-137,903	-154,821	-173,788
Total revenues (net of refunds)	1,887,482	2,156,075	2,476,605	2,777,295	3,115,292	3,484,774	3,900,889	4,367,774	4,891,296	5,491,864	6,165,608	6,920,938
GDP at market prices	15,134,299	16,999,104	19,278,519	21,335,380	23,673,685	26,218,655	29,119,695	32,351,065	35,967,623	40,363,255	45,290,214	50,717,970
Total revenues/GDP	12.47%	12.68%	12.85%	13.02%	13.16%	13.29%	13.40%	13.50%	13.60%	13.61%	13.61%	13.65%
Government Expenditures/GDP	22.73%	22.80%	20.75%	19.86%	20.31%	20.79%	21.30%	21.83%	22.41%	23.08%	23.80%	24.57%
Expenditure-Revenue gap/GDP	-10.26%	-10.11%	-7.90%	-6.84%	-7.15%	-7.49%	-7.90%	-8.33%	-8.81%	-9.47%	-10.19%	-10.92%

* URA (2006) database, and author's projection

Annex 7 of Chapter 6. Revenue projection (2005/06-2015/16)—Base case scenario

Unit: Ush millions

Year	2004/05 (actual)	2005/06	2006/07	2007/08	2008/09	2009/10	2010/11	2011/12	2012/13	2013/14	2014/15	2015/16
Direct income taxes	513,046	598,969	709,659	828,751	968,365	1,127,171	1,316,698	1,539,321	1,800,061	2,135,938	2,531,397	2,995,421
VAT (domestic + imports)	622,546	714,893	823,923	925,230	1,039,215	1,164,296	1,304,635	1,462,113	1,638,848	1,837,220	2,059,905	2,309,916
Excise duty (domestic + imports)	502,472	577,007	665,007	746,775	838,775	939,731	1,053,001	1,180,106	1,322,753	1,482,863	1,662,598	1,864,388
Trade taxes	233,585	268,234	309,143	347,154	389,923	436,854	489,511	548,598	614,911	689,341	772,895	866,701
Total taxes	1,871,649	2,159,104	2,507,732	2,847,910	3,236,277	3,668,051	4,163,845	4,730,139	5,376,573	6,145,363	7,026,795	8,036,426
Fees	63,228	72,939	84,716	96,208	109,328	123,914	140,663	159,793	181,631	207,602	237,379	271,486
Total revenue (taxes and fees)	1,934,877	2,232,042	2,592,448	2,944,117	3,345,605	3,791,965	4,304,507	4,889,932	5,558,205	6,352,965	7,264,174	8,307,912
Tax refunds	-47,395	-54,675	-63,503	-72,117	-81,952	-92,885	-105,440	-119,780	-136,150	-155,618	-177,938	-203,505
Total revenues (net of refunds)	1,887,482	2,177,368	2,528,945	2,872,000	3,263,653	3,699,080	4,199,067	4,770,152	5,422,055	6,197,348	7,086,236	8,104,407
GDP at market prices	15,134,299	16,999,104	19,278,519	21,335,380	23,673,685	26,218,655	29,119,695	32,351,065	35,967,623	40,363,255	45,290,214	50,717,970
Total revenues/GDP	12.47%	12.81%	13.12%	13.46%	13.79%	14.11%	14.42%	14.74%	15.07%	15.35%	15.65%	15.98%
Government Expenditures/GDP	22.73%	22.80%	20.75%	19.86%	20.31%	20.79%	21.30%	21.83%	22.41%	23.08%	23.80%	24.57%
Expenditure-Revenue gap/GDP	-10.26%	-9.99%	-7.63%	-6.39%	-6.52%	-6.68%	-6.88%	-7.09%	-7.33%	-7.73%	-8.16%	-8.59%

* URA (2006) database, and author's projection

Annex 8 of Chapter 6. Revenue projection (2005/06-2015/16)—high elasticity case scenario

Unit: Ush millions

Year	2004/05	2005/06	2006/07	2007/08	2008/09	2009/10	2010/11	2011/12	2012/13	2013/14	2014/15	2015/16
Direct income taxes	513,046	590,348	688,185	789,327	905,768	1,035,590	1,187,601	1,362,848	1,564,307	1,817,975	2,110,586	2,447,062
VAT (domestic + imports)	622,546	732,760	865,665	996,502	1,147,417	1,317,919	1,514,068	1,739,760	1,999,493	2,298,453	2,642,624	3,038,910
Excise duty (domestic + imports)	502,472	591,427	698,698	804,300	926,108	1,063,724	1,222,039	1,404,201	1,613,838	1,855,135	2,132,924	2,452,776
Trade taxes	233,585	274,938	324,805	373,896	430,521	494,495	568,092	652,773	750,228	862,400	991,536	1,140,226
Total taxes	1,871,649	2,189,473	2,577,353	2,964,026	3,409,814	3,911,728	4,491,800	5,159,583	5,927,865	6,833,964	7,877,670	9,078,975
Fees	63,228	73,965	87,068	100,131	115,190	132,146	151,742	174,301	200,255	230,865	266,123	306,705
Total revenue (taxes and fees)	1,934,877	2,263,438	2,664,421	3,064,157	3,525,005	4,043,874	4,643,541	5,333,884	6,128,120	7,064,828	8,143,793	9,385,680
Tax refunds	-47,395	-55,444	-65,266	-75,057	-86,346	-99,056	-113,745	-130,655	-150,110	-173,055	-199,485	-229,905
Total revenues (net of refunds)	1,887,482	2,207,994	2,599,155	2,989,099	3,438,658	3,944,818	4,529,796	5,203,229	5,978,010	6,891,773	7,944,308	9,155,775
GDP at market prices	15,134,299	16,999,104	19,278,519	21,335,380	23,673,685	26,218,655	29,119,695	32,351,065	35,967,623	40,363,255	45,290,214	50,717,970
Total revenues/GDP	12.47%	12.99%	13.48%	14.01%	14.53%	15.05%	15.56%	16.08%	16.62%	17.07%	17.54%	18.05%
Government Expenditures/GDP	22.73%	22.80%	20.75%	19.86%	20.31%	20.79%	21.30%	21.83%	22.41%	23.08%	23.80%	24.57%
Expenditure-Revenue gap/GDP	-10.26%	-9.81%	-7.27%	-5.85%	-5.78%	-5.74%	-5.74%	-5.75%	-5.79%	-6.01%	-6.26%	-6.51%

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